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Inventories**

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Life Cycle Inventories of Highly Pure Chemicals

Data v2.0 (2007)

ETH

Jürgen Sutter
ETH S + U, Zurich

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Participating consultants:	Basler & Hofmann, Zürich Bau- und Umweltchemie, Zürich Carbotech AG, Basel Chudacoff Oekoscience, Zürich Doka Life Cycle Assessments, Zürich Dr. Werner Environment & Development, Zürich Ecointsys - Life Cycle Systems Sarl. ENERS Energy Concept, Lausanne ESU-services Ltd., Uster Infras AG, Bern
Software Support:	ifu Hamburg GmbH
Project leader:	Rolf Frischknecht, ecoinvent Centre, Empa, Dübendorf

Marketing and Sales:

Annette Köhler, ecoinvent Centre, Empa,
Dübendorf

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Life Cycle Inventories of Highly Pure Chemicals

Author:	Jürgen Sutter
Reviewer:	Roland Hischier, Mischa Classen, EMPA St. Gallen
Contact address:	ecoinvent Centre Empa P.O. Box CH-8600 Dübendorf http://www.ecoinvent.org/ frischknecht@ecoinvent.org
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Zusammenfassung

Im Rahmen des Projekts „ecoinvent data v2“ wurden umfassende Lebenszyklusanalysen (Ökobilanz, LCA) für Dienstleistungen der IT- und Elektronik-Industrie erarbeitet.

Innerhalb des Elektronik-Projekts wurden folgende Ebenen unterschieden:

- IT-Dienstleistungen
- Geräte
- Komponenten
- Bauteile
- Grundmaterialien
- Produktionsmittel und -abfälle

In der untersten Ebene „Produktionsmittel und -abfälle“ finden sich eine Reihe von Chemikalien, die in hochreiner Form (highly pure, ultrapure) benötigt werden.

Das Ziel des Projekts „Life Cycle Inventories of Highly Pure Chemicals“ war, für die in der IT- und Elektronikindustrie eingesetzten hochreinen Chemikalien qualitativ hochwertige, homogenisierte Inventardaten zu erarbeiten. Eine Liste von 66 Chemikalien wurde erstellt, von denen 21 im Rahmen des Projekts bilanziert wurden. Acht weitere Datensätze wurden für wichtige Vorprodukte erarbeitet. Elf Chemikalien wurden im Rahmen anderer Projekte bilanziert, die übrigen wurden mit anderen ecoinvent-Datensätzen abgeschätzt. Fehlende Daten wurden anhand von Daten ähnlicher Prozesse oder ecoinvent-Standardswerten abgeschätzt.

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Part I

Methodological Remarks

Author: Jürgen Sutter, ETH S+U, Zurich



Introduction

Background

The chemicals and materials used to manufacture and package semiconductors and printed circuit boards are considered electronic chemicals. The more than 100 different products used in these applications are grouped into the following 13 major product categories (Ullmanns 2005):

1. acids and etchants
2. atmospheric gases
3. auxiliary materials
4. cleaners
5. dopants
6. masks
7. packaging materials
8. photoresists
9. photoresist adjuncts
10. plating chemicals
11. solvents
12. specialty gases
13. substrates

The level of chemical purity needed is extremely high because the presence of impurities even in very small proportions can have large effects on the properties of the material.

Goal and Scope

The aim of this project is to present inventories of the most important highly pure chemicals using the LCA method.

Chemicals under study

In Tab. 0.1 the datasets are given, which are described in this report.

Tab. 0.1 Datasets of highly pure chemicals.

No.	Name
1	ammonium thiocyanate, at plant, GLO
2	arsine, at plant, GLO
3	boric oxide, at plant, GLO
4	boron carbide, at plant, GLO
5	boron trifluoride, at plant, GLO
6	Brown-Schlesinger-process, GLO
7	carbon disulfide, at plant, GLO
8	chlorosilane amination, GLO
9	dimethylacetamide, at plant, GLO
10	dimethylamine borane, at plant, GLO
11	dipropylene glycol monomethyl ether, at plant, GLO
12	ethylene chlorofluorination, GLO
13	Imperial smelting furnace, GLO
14	lithium carbonate hydration, GLO
15	lithium chloride, at plant, GLO
16	lithium chloride electrolysis, GLO
17	lithium manganese oxide, at plant, GLO
18	methyl acrylate, at plant, GLO
19	methyl-3-methoxypropionate, at plant, GLO
20	modified Solvay Process, GLO
21	natural gas, to helium extraction, GLO
22	phosphane, at plant, GLO
23	phosphoryl chloride, at plant, GLO
24	potassium carbonate , at plant, GLO
25	potassium perchlorate, at plant, GLO
26	silicon hydrochloration, at plant, GLO
27	sodium perchlorate, at plant, GLO
28	sodium persulfate, at plant, GLO
29	sodium tetrahydroborate fluorination, GLO
30	Sulfuric acid digestion of spodumene, GLO
31	trichloroborane, at plant, GLO
32	trifluoromethane, at plant, GLO
33	trimethyl borate, at plant, GLO
34	water, ultrapure, at plant, GLO

General methodology

The methodology for the investigation of life cycle inventory data of different unit processes has been described in detail in the methodology report of the ecoinvent 2000 project (Frischknecht et al. 2007) and was used in this study. Thus, all unit processes are compatible with the ecoinvent datasets. For readers, who are not familiar with this methodology, it is recommended to read the methodology report before working with the unit process datasets shown in this report.

In order to comply with the quality standards of the ecoinvent database, all inventories of highly pure chemicals are established according to the guidelines presented in Frischknecht et al. (2007) and Hischier et al. (2004). Thus, all data are presented on a unit process level. Whenever possible, actual production data is used to determine inventory flows. In case such data is not available, basic information from technical reference books together with additional assumptions and simplifications was used, as described in Hischier et al. (2004). With this procedure, missing inventory data of energy consumption, process efficiency, water consumption, emissions to air and water, transports, and infrastructure can be estimated.

References

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|--------------------------|--|
| Althaus et al. 2007 | Althaus H.-J., Chudacoff M., Hischier R., Jungbluth N., Osses M. and Primas A. (2007) Life Cycle Inventories of Chemicals. Final report ecoinvent v2.0 No. 8. EMPA Dübendorf, Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org . |
| Frischknecht et al. 2007 | Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hischier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org . |
| Hischier et al. 2004 | Hischier, R, Hellweg, S, Capello, C, and Primas, A: 2004. Establishing Life Cycle Inventories of Chemicals Based on Differing Data Availability. <i>International Journal of LCA</i> . 10. (1). 59-67. |
| Ullmanns 2005 | Daigle, S., Vogelsberg, E., Lim, B., Butcher, I.: Electronic Chemicals. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2005 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: http://www.mrw.interscience.wiley.com/emrw/9783527306732/home . |

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1 Ammonium thiocyanate

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hirschier, EMPA, St. Gallen
 Last changes: 2007

1.1 Introduction

Ammonium thiocyanate (NH_4SCN ; CAS 1762-95-4) is a white, hygroscopic salt which forms stable monoclinic crystals below 90 – 92 °C. At higher temperatures it is converted to the rhombic form. Ammonium thiocyanate is readily soluble in many polar solvents. The solubilities in 100 g solvent are for liquid ammonia (25 °C) 312 g, for liquid sulfur dioxide (0 °C) 46.8 g, and for acetonitrile (18 °C) 7.5 g. Ammonium thiocyanate is readily soluble in ethanol and acetone.

For this inventory the functional unit is 1 kg of pure highly pure ammonium thiocyanate. The most important chemical and physical properties of ammonium thiocyanate used in this inventory are given here.

Tab. 1.1 Chemical and physical properties of ammonium thiocyanate (Ullmann 2004)

Property	Value	Unit	Remarks
Molecular weight	76.12	g mol^{-1}	
Specific gravity	1305	kg m^{-3}	
Boiling point	-		decomposes at 170 °C
Melting point	149	°C	

1.2 Use

Most of the ammonium thiocyanate is produced as an intermediate in the production of other inorganic thiocyanates. It is also used as the starting material for compounds such as aminothiazoles.

Large quantities of ammonium thiocyanate are used in agriculture. It is used to combat worms in cattle and, as a mixture with aminotriazole, reinforces the activity of the latter as a broad-spectrum herbicide. Ammonium thiocyanate is also used as a raw material for the production of other herbicides, such as diallates, triallates and tricyclazoles. It is also used as a raw material for fungicides, such as methyl isothiocyanate.

Ammonium thiocyanate is extensively used in photography as an accelerator with sodium thiosulfate in fixing baths, where it removes undeveloped silver (Ag^+). Addition of ammonium thiocyanate to electrolytic cells in the production of ammonium peroxodisulfate, $(\text{NH}_4)_2\text{S}_2\text{O}_8$ considerably increases the yield.

Ammonium thiocyanate is used as a solvent for cellulose, which can also be spun. In the textile industry ammonium thiocyanate is widely used to improve desired properties.

In the steel industry ammonium thiocyanate is used for coating and as a corrosion inhibitor against acidic gases. It is used as an additive for silver and tin plating baths and for etching aluminum.

Because of the neutron-absorbing properties of hafnium, only hafnium-free zirconium can be used in nuclear technology. Following their extraction, Zr and Hf are dissolved in aqueous ammonium thiocyanate so that the $\text{Hf}(\text{SCN})_4$ complex can subsequently be extracted with an organic solvent.

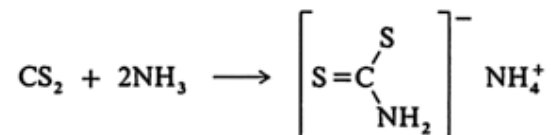
Ammonium thiocyanate can also be used as a component for energy storage in combination with heat pumps, in vulcanization, for the production of fire-extinguishing agents, and in absorbents for ammonia or steam

1.3 Production Technologies (Ullmann 2004)

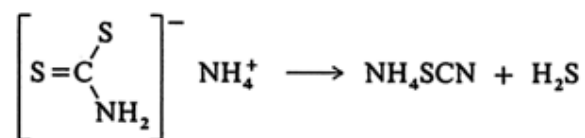
Ammonium thiocyanate is obtained from CS₂ and NH₃ or from the reaction of hydrogen cyanide or cyanides with sulfur and/or polysulfides.

Production from Ammonia and Carbon Disulfide

This continuous, pressure-free process is carried out in two steps. In the first step CS₂ and ammonia gas are reacted in the presence of water over activated charcoal at ca. 50 °C in an interface reactor to give ammonium dithiocarbamate.

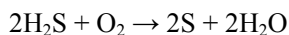


The latter is decomposed in the second step at 95 – 100 °C on activated charcoal.



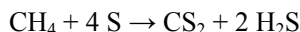
If the NH₃ and CS₂ used are of high purity, the 35 – 45 % ammonium thiocyanate solution formed can be marketed after concentration without further purification. The crystals, which are produced by vacuum evaporation and centrifugation, are of higher purity. In this process, apart from H₂S and excess CS₂, no other waste substances are formed. After CS₂ and H₂S have been separated, the former can be recycled and the latter can be used, for example, for sulfur production in the Claus process.

Claus process



From sulfur carbon disulfide is produced:

CS₂ production



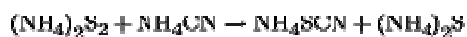
There is also a one-step variant of this process in which the reaction of NH₃ with CS₂ and decomposition take place at ca. 115 °C and 5 bar.

The whole chemical equation with included desulfurizing and CS₂-production from the recovered sulfur:



Production from Hydrogen Cyanide contained in Coke Oven Gas

Cooled coke oven gas, which has been freed from tar, contains HCN, NH₃, and H₂S. NH₃ and H₂S are reacted with an aqueous slurry of sulfur in a circulating gas scrubber forming (NH₄)₂S, which dissolves the elemental sulfur to form ammonium polysulfide. At the same time the HCN combines with the NH₃ dissolving in the washing liquid to give NH₄CN, which then reacts with the polysulfide to give ammonium thiocyanate and ammonium sulfide.



The ammonium sulfide is distilled off. A 25 – 50 % ammonium thiocyanate solution remains, which, after purification, is further processed as in the other processes.

Production from Hydrogen Cyanide, Aqueous Ammonia, and Sulfur/Ammonium Polysulfide

Hydrogen cyanide is neutralized with aqueous ammonia, and the ammonium cyanide formed is treated with elemental sulfur and/or ammonium polysulfide with warming to give ammonium thiocyanate.

1.4 System Characterisation

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg ammonium thiocyanate, at plant, global**. For this study the production of ammonium thiocyanate by means of the reaction of CS_2 and NH_3 is considered. All data in the present report are referred to 1 kg ammonium thiocyanate 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

1.5 Raw materials and auxiliaries

As there was no data available for the production of ammonium thiocyanate, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from a large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the arsine production.

Tab. 1.2 Consumption of raw materials and auxiliaries for the ammonium thiocyanate production

Input	Values	Remarks
Carbon disulfide (kg)	0.526	Calculated with a yield of 95 %
Methane (kg)	0.108	Calculated with a yield of 95 %
Ammonium (kg)	0.452	Calculated with a yield of 95 %
Cooling water (kg)	24	Estimation

1.6 Energy

There was no information available on the amount of energy used for the production process. In order not to neglect the process energy demand those values were approximated with data from a large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg⁻¹) were used as approximation for the energy consumption of the ammonium thiocyanate production. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used. A summary of the values used is given in Tab. 1.3.

Tab. 1.3 Energy consumption for the ammonium thiocyanate production

Input	per kg ammonium thiocyanate
Heat, natural gas burned in industrial furnace > 100 kW	2
Electricity (kWh)	0.333

1.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train. Tab. 1.4 summarises the total transport amounts for the ammonium thiocyanate production.

Tab. 1.4 Total transport amounts for the ammonium thiocyanate production

tkm kg ⁻¹ ammonium thiocyanate	lorry	train
Total transports	6.08E-01	3.65

1.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of ammonium thiocyanate production plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

1.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the ammonium thiocyanate production. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the input of CS₂, NH₃, CH₄ input. The remaining CH₃ is oxidated to CO₂.

Tab. 1.5 Process emissions to air from the ammonium thiocyanate production

Output	Value	Remarks
Waste heat (MJ)	1.2	calculated from electricity input
CS ₂ , to air (kg)	1.05E-03	estimated as 0.2 % of the input
CH ₄ , to air (kg)	2.22E-04	estimated as 1 % of the input
NH ₃ , to air (kg)	9.04E-04	estimated as 0.2 % of the input
CO ₂ , fossil, to air (kg)	2.85E-02	from burning of co-products

1.10 Emissions to water

The remaining amount of unreacted ammonium (4.8 % of the input) and of unreacted carbon disulfide (4.8 % of the input) was assumed to leave the production process with the waste water. Further it was assumed that this wastewater is treated in a internal wastewater plant. The removal efficiency for carbon disulfide was assumed with 90%. The carbon contained in the removed carbon disulfide was accounted as CO₂ emissions to air.

For the ammonia within the waste water a removal efficiency of 50% was assumed. The nitrogen emitted to air (N₂) and remaining in the sewage sludge was neglected.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of CS₂ in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used.

The carbon disulfide emission to water is calculated as sulfide emission.

Tab. 1.6 Process emissions to water from the ammonium thiocyanate production

Output	Value	Remarks
Sulfide (kg)	2.13E-03	calculated from mass balance
NH ₃ (kg)	3.62E-03	calculated from mass balance
COD, BOD (kg)	3.47E-03	calculated from water emissions
TOC, DOC (kg)	3.99E-04	calculated from water emissions

1.11 Life cycle inventory of ammonium thiocyanate production and data quality considerations

Tab. 1.7 shows the life cycle inventory and the data quality indicators for the ammonium thiocyanate production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 1.7 Unit process raw data for the ammonium thiocyanate production

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	ammonium thiocyanate, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
401										
662			Location				GLO			
493			InfrastructureProcess				0			
403			Unit				kg			
resource, in water	-	0	ammonium thiocyanate, at plant	GLO	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	carbon disulfide, at plant	GLO	0	kg	5.26E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	natural gas, high pressure, at consumer	RER	0	MJ	5.10E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	ammonia, liquid, at regional storehouse	RER	0	kg	4.52E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	2.00E+0	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.33E-1	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	6.08E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	3.65E+0	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	1.20E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input
	-	4	Carbon disulfide	-	-	kg	1.05E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Methane, fossil	-	-	kg	2.22E-4	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Ammonia	-	-	kg	9.04E-4	1	1.37	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	2.85E-2	1	1.30	(4,5,na,na,na,na); Estimation
emission water, river	-	4	Sulfide	-	-	kg	2.13E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Ammonium, ion	-	-	kg	3.62E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	3.47E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	3.47E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	3.99E-4	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	3.99E-4	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

1.12 Cumulative Results and Interpretati

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 1.8 shows selected LCI results and the cumulative energy demand for the ammonium thiocyanate production.

Tab. 1.8 Selected LCI results and the cumulative energy demand for the ammonium thiocyanate production

Name		ammonium thiocyanate, at plant	
Location	Unit	Unit	GLO kg 0
Infrastructure			
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	44.6
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	3.9
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.5
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.5
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1
selected LCI results			
land occupation	resource	m2a	1.2E-1
CO2, fossil	air	kg	1.9E+0
NM VOC	air	kg	1.3E-3
nitrogen oxides	air	kg	3.9E-3
sulphur dioxide	air	kg	1.9E-2
particulates, <2.5 um	air	kg	4.8E-4
BOD	water	kg	7.1E-3
cadmium	soil	kg	7.3E-10

1.13 Appendix: EcoSpold Meta Information

Type	ID	Field name, IndexNumber	33128
ReferenceFunction	401	Name	ammonium thiocyanate, at plant GLO
Geography	662	Location	0
ReferenceFunction	493	InfrastructureProcess	kg
ReferenceFunction	403	Unit	1
DataSetInformation	201	Type	1.0
	202	Version	0
	203	energyValues	en
	205	LanguageCode	de
	206	LocalLanguageCode	24
DataEntryBy	302	Person	1
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	Production of ammonium thiocyanate including materials, energy uses, infrastructure and emissions.
	402	IncludedProcesses	1
	404	Amount	Ammoniumthiocyanat, ab Werk
	490	LocalName	
	491	Synonyms	
	492	GeneralComment	The process "ammonium thiocyanate, at plant, GLO" is modelled for the production of ammonium thiocyanate from carbon disulfide and ammonia in the world. Raw materials are modelled with a stoichiometric calculation. Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	NH4SCN
	501	StatisticalClassification	
	502	CASNumber	1762-95-4
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	Amination of carbon disulfide.
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

1.14 References

Frischknecht et al. 2007

Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.

- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>
- Ullmanns 2004 Hoek, Theo van: Thiocyanates, Inorganic. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

2 Arsine

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

2.1 Introduction

Arsine (AsH_3 , CAS-No. 7784-42-1, synonym: arsenic hydride, arsane) is a colorless gas with an unpleasant, garlic-like odor. It is extremely toxic: in many cases the inhalation of small amounts has caused death. Arsenic hydride is so poisonous that a few inhalations of air containing arsenic hydride can result in death. The MAK is 0.2 mg/m³. Arsine does not ignite spontaneously but burns with a grayish-blue flame to form arsenic trioxide. If the flame is extinguished, for example, by holding a cold porcelain dish in it, arsenic is deposited as a black patch. The pure gas is stable at normal temperature. If arsenic hydride is passed through a red-hot glass tube, it decomposes into hydrogen and arsenic (arsenic mirror), 185.1 kJ being released per mole of arsenic. The arsenic is deposited just behind the heated area as a bright, metallic black mirror (Marsh test). When treated with hypochlorite solution or with alkaline hydrogen peroxide solution, the patch changes into soluble arsenate. Arsenic hydride has a strong reducing action. It precipitates metallic silver from silver nitrate solution. It reacts with solid silver nitrate to form the yellow compound $\text{AsAg}_3 \cdot 3 \text{AgNO}_3$ (Gutzeit arsenic test). For this inventory the functional unit is 1 kg of highly pure arsine. The most important chemical and physical properties of arsine used in this inventory are given here.

Tab. 2.1 Chemical and physical properties of arsine (Ullmanns 2005)

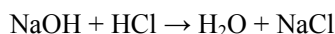
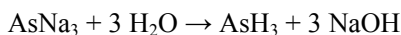
Property	Value	Unit	Remarks
Molecular weight	77.95	g mol ⁻¹	
Boiling point	-55	°C	at normale pressure
Melting point	-116.3	°C	at normale pressure

2.2 Use

Arsenic hydride is used as a dopant in the semiconductor industry.

2.3 Production Technologies (Ullmanns 2004)

Arsine cannot be prepared by direct synthesis. The most convenient method of production is the reaction between sodium arsenide and water. The sodium hydroxide is then neutralized with hydrochloric acid.



Another method to produce arsine is the reaction between zinc arsenide and dilute hydrochloric acid.

These reactions are fast and complete and have the advantage that the product is not contaminated with hydrogen. Arsenic hydride can also be produced by cathodic reduction or by reduction of arsenic compounds in acidic solution with zinc or magnesium. The last reaction is not quantitative because elemental arsenic, which does not react, precipitates. The product also has a high hydrogen content. Because arsenic is present as a common impurity in numerous metals suitable for the production of hydrogen, hydrogen produced by acid – metal reduction must be passed through a potassium permanganate solution to destroy any arsenic hydride present. Particular practical significance is attributed to

the formation of arsenic hydride in the pickling of metals when arsenic-containing sulfuric acid is used.

2.4 System Characterisation

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg arsine, at plant, global**. For this study the production of arsine by means of the reaction between sodium arsenide and water is considered. All data in the present report are referred to 1 kg arsine 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

2.5 Raw materials and auxiliaries

As there was no data available for the production of arsine, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007). The water for the reaction is calculated as ultrapure water.

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the arsine production.

Tab. 2.2 Consumption of raw materials and auxiliaries for the arsine production

Input	Values	Remarks
Sodium arsenide (kg)	1.94	Calculated with a yield of 95 %
HCl (kg)	1.46	Calculated with a yield of 95 %
Water, ultrapure (kg)	0.69	Stoichiometric calculation
Cooling water (kg)	24	Estimation

2.6 Energy

As the reaction is exothermic, no process heat is considered in this study. Electricity is needed to run the process and the waste water treatment. There was no information available on the amount of energy used for the production process. In order not to neglect the electricity consumption this value is approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf (2000)). The value for the electricity consumption per kg of product of this plant (0.333 kWh kg⁻¹) is used as approximation for the arsine production.

Tab. 2.3 Energy consumption for the arsine production

Input	per kg arsine
Electricity (kWh)	0.333

2.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train. Tab. 2.4 summarises the total transport amounts for the arsine production.

Tab. 2.4 Total transport amounts for the arsine production

tkm kg ⁻¹ arsine	lorry	train
Total transports	6.8E-01	1.65

2.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of arsine production plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

2.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the arsine production. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the HCl.

Tab. 2.5 Process emissions to air from the arsine production

Output	Value	Remarks
Waste heat (MJ)	1.2	calculated from electricity input
HCl, to air (kg)	2.91E-03	estimated as 0.2 % of the input

2.10 Emissions to water

The sodium chloride is assumed to leave the production process with the waste water.

Tab. 2.6 Process emissions to water from the arsine production

Output	Value	Remarks
sodium (kg)	8.85E-01	calculated from mass balance
chloride (kg)	1.41	calculated from mass balance
AOX (kg)	1.41	calculated from water emissions

2.11 Waste

The remaining sodium arsenide (5 % of the input) is assumed to be deposited as hazardous waste in an underground deposit.

Tab. 2.7 Wastes from the arsine production

Output	Value
disposal, hazardous waste, 0% water, to underground deposit (kg)	9.72E-02

2.12 Life cycle inventory of arsine production and data quality considerations

Tab. 2.8 shows the life cycle inventory and the data quality indicators for the arsine production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 2.8 Unit process raw data for the arsine production

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	arsine, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
401										
662			Location				GLO			
493			InfrastructureProcess				0			
403			Unit				kg			
resource, in water	-	0	arsine, at plant	GLO	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	sodium arsenide, at plant	GLO	0	kg	1.94E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	water, ultrapure, at plant	GLO	0	kg	6.92E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	hydrochloric acid, 30% in H2O, at plant	RER	0	kg	1.46E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	disposal, hazardous waste, 0% water, to underground deposit	DE	0	kg	9.72E-2	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.33E-1	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	6.80E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	1.65E+0	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	1.20E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input
emission water, river	-	4	Hydrogen chloride	-	-	kg	2.91E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Sodium, ion	-	-	kg	8.85E-1	1	5.10	(4,5,na,na,na,na); Estimation
	-	4	Chloride	-	-	kg	1.41E+0	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	AOX, Adsorbable Organic Halogen as Cl	-	-	kg	1.40E+0	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

2.13 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 2.9 shows selected LCI results and the cumulative energy demand for the arsine production.

Tab. 2.9 Selected LCI results and the cumulative energy demand for the arsine production

Name		arsine, at plant	
Location	Unit	Unit	GLO
Infrastructure			kg
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	93.2
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	21.8
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	2.6
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	2.9
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.4
selected LCI results			
land occupation	resource	m2a	8.0E-1
CO ₂ , fossil	air	kg	6.6E+0
NM VOC	air	kg	8.3E-3
nitrogen oxides	air	kg	4.6E-2
sulphur dioxide	air	kg	2.3E-2
particulates, <2.5 um	air	kg	4.4E-3
BOD	water	kg	9.2E-3
cadmium	soil	kg	3.5E-9

2.14 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	arsine, at plant
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of arsine including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Arsin, ab Werk
	491	Synonyms	arsane
	492	GeneralComment	The process "arsine, at plant, GLO" is modelled for the production of arsine from sodium arsenide in the world. Raw materials are modelled with a stoichiometric calculation. Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	inorganics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Anorganika
	499	Formula	AsH3
	501	StatisticalClassification	
	502	CASNumber	7784-42-1
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	Hydration of sodium arsenide
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	Highly Pure chemicals

2.15 References

- Friskhnecht et al. 2007 Friskhnecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschler R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.

- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>
- Ullmanns 2004 Grund, S., Hanusch K., Wolf H. Arsenic and Arsenic Compounds. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

3 Boric oxide

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

3.1 Introduction

Boric oxide (B_2O_3 ; CAS 1303-86-2; synonyms: boron trioxide, diboron trioxide) exists as vitreous glass and as two crystalline forms. The vitreous form is a colorless, hard, tough, glasslike solid, commonly prepared by dehydration of boric acid. The most common crystalline B_2O_3 has a hexagonal or α -form and is crystallized at 200 – 250 °C at ambient pressure. A less common monoclinic β -form can be crystallized at 4000 MPa and 600 °C.

For this inventory the functional unit is 1 kg of highly pure boric oxide. The most important chemical and physical properties of boric oxide used in this inventory are given here.

Tab. 3.1 Chemical and physical properties of boric oxide (Ullmanns 2004)

Property	Value	Unit
Molecular weight	69.62	g mol ⁻¹
Specific gravity	1840	kg m ⁻³
Boiling point	1860	°C
Melting point	450	°C

3.2 Reserves and Resources

The estimated world reserves of borate minerals and world production of boron products for 1982 are listed in . Of the total world borates the United States produces 62 %, Turkey 29 %, and all other sources 9 %, based on B_2O_3 . World borate consumption in the United States is 30 %, Western Europe 37 %, Japan 5 %, and all other countries 28 %. The European and Japanese borate requirements are served through imports largely from the United States and Turkey. Almost all B_2O_3 production in Turkey and about one-half the production in the United States is exported.

Tab. 3.2 Estimated world borate reserves and mine production (as B_2O_3) in 1982 (Ullmanns 2004)

Country	Reserves, 1000 t	Production
United States	70000	550
Turkey	70000	320
Soviet Union	65000	40
China	30000	5
World total	270000	949

3.3 Use

The glass and ceramic industry in the United States consumes about one-half of the total domestic B_2O_3 . This includes borosilicate glasses, insulation fiberglass, textile fiberglass, porcelain enamels, and ceramic glazes. The largest market is for insulation fiberglass, with typically 5 – 7 % of contained B_2O_3 from borax pentahydrate, ulexite, or probertite; whereas textile fiberglass typically has 8 – 9 % of contained B_2O_3 from boric acid or colemanite. Borosilicate glass has 12 – 15 % of contained B_2O_3 largely from borax pentahydrate. Borates are used as a flux in the manufacture of porcelain enamels and ceramic glazes. Of the total United States domestic B_2O_3 consumption, about 15 % is used in the

form of boric acid and metal borates as fire retardants in cellulosic insulation and plastics; 4 % as soluble borates in agriculture as a fertilizer to amend boron-deficient soils and as a herbicide at high dosages; 10 % from borax decahydrate and sodium perborate for cleaning and bleaching of laundry products; and 2 % as corrosion inhibitors in aqueous systems, especially automotive antifreeze.

In Western Europe about one-half of the B_2O_3 is used in the manufacture of glass-related products. Sodium perborate accounts for about one-third of the total boron consumption as a bleach in high-temperature laundry products.

In Japan 70 % of B_2O_3 consumption is used in glass-related products.

3.4 Production Technologies (Ullmanns 2004)

High-purity B_2O_3 (99 %) is made by fusing refined boric acid.



A 96 % grade of B_2O_3 is made commercially by fusing a mixture of sodium tetraborate plus sulfuric acid at 800 °C. Two molten layers are formed, an upper layer of boric oxide and a lower layer of sodium sulfate. The molten B_2O_3 layer is separated and cooled on chill rollers. The resulting glass is ground, sized, and packaged in moisture-proof containers. The product B_2O_3 contains 3 – 4 % sodium sulfate.

Boric oxide can also be prepared by thermal decomposition of ammonium pentaborate, $NH_4B_5O_8 \cdot 4 H_2O$, at 500 – 900 °C and through direct reaction of boron with oxygen.

3.5 System Characterisation

In this study the fusing of boric acid is considered. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

3.6 Raw materials and auxiliaries

As there was no data available for the production of boric oxide, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the boric oxide production.

Tab. 3.3 Consumption of raw materials and auxiliaries for the boric oxide production

Input	Values	Remarks
Boric acid (kg)	1.86	Calculated with a yield of 95 %
Cooling water (kg)	24	Estimation

3.7 Energy

There was no information available on the amount of energy used for the production process. In order not to neglect the process energy demand those values were approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg-1) were used as approximation for the energy consumption of boric oxide production. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used. A summary of the values used is given in Tab. 3.4.

Tab. 3.4 Energy consumption for the boric oxide production

Input	per kg boric oxide
Heat, natural gas burned in industrial furnace > 100 kW	2
Electricity (kWh)	0.333

3.8 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train. Tab. 3.5 summarises the total transport amounts for the boric oxide production.

Tab. 3.5 Total transport amounts for the boric oxide production

tkm kg ⁻¹ boric oxide	lorry	train
Total transports	1.86E-01	1.12

3.9 Infrastructure and land use

No information was readily available about infrastructure and land-use of boric oxide production plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

3.10 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the boric oxide production. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the input of boric acid. The boric acid is calculated as boron emission.

Tab. 3.6 Process emissions to air from the boric oxide production

Output	Value	Remarks
Waste heat (MJ)	1.2	calculated from electricity input
Boron, to air (MJ)	6.62E-04	calculated as 0.2 % of the input

3.11 Emissions to water

The remaining amount of unreacted boric acid (4.8 % of the input) is assumed to leave the production process with the waste water. The boric acid is calculated as boron emission.

Tab. 3.7 Process emissions to water from the boric oxide production

Output	Value	Remarks
Boron (kg)	1.59E-02	calculated from mass balance

3.12 Life cycle inventory of boric oxide production and data quality considerations

Tab. 3.8 shows the life cycle inventory and the data quality indicators for the boric oxide production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 3.8 Unit process raw data for the boric oxide production

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	boric oxide, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
401										
662			Location				GLO			
493			InfrastructureProcess				0			
403			Unit				kg			
resource, in water	-	0	boric oxide, at plant	GLO	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	boric acid, anhydrous, powder, at plant	RER	0	kg	1.86E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	2.00E+0	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.33E-1	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.86E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	1.12E+0	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	1.20E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input
emission water, river	-	4	Boron	-	-	kg	6.62E-4	1	5.10	(4,5,na,na,na,na); Estimation
	-	4	Boron	-	-	kg	1.59E-2	1	5.10	(4,5,na,na,na,na); Stoichiometric calculation

3.13 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 3.9 shows selected LCI results and the cumulative energy demand for the boric oxide production.

Tab. 3.9 Selected LCI results and the cumulative energy demand for the boric oxide production

Name		boric oxide, at plant	
Location	Unit	Unit	GLO kg 0
Infrastructure			
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	27.4
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	6.8
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	1.4
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.9
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1
selected LCI results			
land occupation	resource	m2a	4.0E-1
CO ₂ , fossil	air	kg	1.8E+0
NM VOC	air	kg	1.5E-3
nitrogen oxides	air	kg	7.4E-3
sulphur dioxide	air	kg	3.0E-2
particulates, <2.5 um	air	kg	1.2E-3
BOD	water	kg	3.1E-3
cadmium	soil	kg	1.7E-9

3.14 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	boric oxide, at plant
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of boric oxide including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Boroxid, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "boric oxide, at plant, GLO" is modelled for the production of boric oxide from boric acid in the world. Raw materials are modelled with a stoichiometric calculation, Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	inorganics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Anorganika
	499	Formula	B2O3
	501	StatisticalClassification	
	502	CASNumber	1303-86-2
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	Fusion of boric oxide
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

3.15 References

Frischknecht et al. 2007

Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.

- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>
- Ullmanns 2004 Baudis, U., Fichte, R.: Boron and Boron Alloys. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

4 Boron carbide

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

4.1 Introduction

Boron carbide (B_4C ; CAS 12069-32-8; synonyms: black diamond, tetrabor) is an exceedingly hard material, inferior in hardness only to diamond and cubic *boron* nitride. Boron carbide crystals are deep black and shiny. The crystals melt congruently. The density of B_4C having the natural isotopic composition is 2.52 g/cm³; $10B_4C$ is less dense, 2.37 g/cm³. Boron carbide belongs to the important group of nonmetallic hard materials, which includes alumina, silicon carbide, and diamond. The composition of commercial boron carbide is usually close to a boron : carbon stoichiometry of 4 : 1, the stoichiometric limit on the high-carbon side.

For this inventory the functional unit is 1 kg of highly pure boron carbide. The most important chemical and physical properties of boron carbide used in this inventory are given here.

Tab. 4.1 Chemical and physical properties of boron carbide (Ullmann 2005)

Property	Value	Unit
Molecular weight	55.255	g mol ⁻¹
Specific gravity	2465	kg m ⁻³
Boiling point	above 3500	°C
Melting point	2450	°C

4.1.1 Reserves and Resources

ESK is the world's largest producer of *boron carbide*. The granular and sintered B_4C products of this company are sold commercially under the trademark Tetrabor®. The price of B_4C is 10 – 150 \$/kg, depending on grit size. Annual world production is about 3500 t.

4.1.2 Use

Uses Based on Hardness

The major industrial use of boron carbide is as abrasive grit or powder. Commercial grits are available in particle sizes from 1 µm to 10000 µm and predominantly used as polishing, lapping, and grinding media for hard materials, especially cemented tungsten carbide and fine ceramics. Water and commercial oils serve as lubricant and coolant. Available as a substitute for loose powders is B_4C paste, which can withstand temperatures up to 350 °C.

The second category is wear-resistant engineering components made of dense, sintered shapes. Boron carbide sand-blasting nozzles are now the most common form of such a component. These nozzles are characterized by minimum wear, even with corundum or silicon carbide grit. Sintered- B_4C wheel dressing sticks are used to produce new cutting edges on surface grains of grinding wheels with minimum wear. So-called hand laps are used to render the cutting edges of cemented carbide tools keen and smooth. Mortars and pestles of boron carbide are suitable for grinding hard materials without contamination from the grinding equipment.

Uses in Nuclear Technology

Boron carbide is both cheaper and easier to fabricate than the element boron itself. As a result it has found almost exclusive usage as a control rod material, neutron poison, and a neutron shielding material.

Chemical Uses

Boron carbide powder is used to introduce boron into the surface layer of steels and other ferrous materials.

Its high boron content makes B₄C useful as a raw material for the production of other boron compounds, including the boron halides and metal borides. Much of the BCl₃ made commercially is used in the production of boron filament for reinforcement of metals and plastics.

Uses in Refractories

In carbon-bonded refractories, B₄C powder is used as an antioxidant. Small additions of B₄C to the refractory material reduce the burn out of the carbon and improve the corrosion resistance against slags and steel melts. A simultaneous addition of B₄C and a metallic antioxidant results in better oxidation resistance of the refractories than the addition of B₄C or the metallic antioxidant alone. The effect depends on the oxidation of the antioxidants and reaction with the oxides of the refractory material, which creates oxidic phases and gaseous components. Both prevent the infiltration of molten slag or steel, and the oxide phases protect the carbon against oxidation by forming a protective oxidic layer at the hot face of the refractory.

Uses in Metal Matrix Composites

Metal matrix composites (MMCs) made of B₄C and Al or Al-alloy have a higher stiffness, strength and hardness and lower weight than the metal itself. The content of B₄C in the MMCs is up to 50 wt %. This material could be used to produce magnetic disks for hard disk drives, automotive components, sports goods, etc.

Other Uses

The thermoelectric properties of B₁₃C₂ are such that B₁₃C₂ could be an interesting material for high-temperature thermoelectric conversion. Thermoelements made of the couple B₄C – C can be used for temperature measurements up to 2300 °C.

4.2 Production Technologies (Ullmanns 2004)

Powder

The most common method for large-scale production of boron carbide is carbothermic reduction of boric oxide:



The process is so strongly endothermic, 1812 kJ/mol or 9.1 kWh/kg, that it is usually carried out at 1500 – 2500 °C in an electric furnace. Either arc furnaces or resistance furnaces, analogous to Acheson furnaces for SiC, are used. The starting material is an intimate mixture of boric oxide and carbon, e.g., petroleum coke or graphite. Large quantities of carbon monoxide, 2.3 m³/kg, are generated, and boron can be lost by evaporation of B₂O₃ at the high temperatures. In the electrothermic process used by ESK, a subsidiary of Wacker Chemie, the product is cooled, and the unconverted outer-zone material is removed to leave a fused coarse-grained boron carbide of high purity: B plus C ≥ 99 wt %, total metal impurity ≤ 0.2 wt %, N plus O ≤ 0.3 wt %, B : C mole ratio of 4.0 – 4.3. This

boron carbide is obtained in the form of regular blocks, which are broken up and milled to produce the B₄C grain size appropriate for final use.

If the carbothermic reduction is carried out below the melting point of B₄C, e.g., at 1600 – 1800 °C, in a vented graphite tube furnace starting from a dehydrated mixture of H₃BO₃, acetylene black or sugar, and ethylene glycol, very fine-grained (0.5 – 5 µm), stoichiometric B₄C powders are produced, although productivity is low.

Uniformly sized submicrometer powders of boron carbide (specific surface area 20–35 m²/g) can be synthesized continuously by rapid carbothermic reduction (RCR) of boron oxide in a graphite transport reactor at ca. 2000°C.

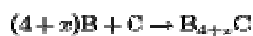
Boron carbide powders are produced directly (no need for the expensive comminution) by magnesiothermic reduction of boric oxide in the presence of carbon at 1000 – 1800 °C:



This reaction is strongly exothermic. It is carried out either directly by point ignition (thermite process) or in a carbon tube furnace under hydrogen. The product is washed free of magnesium oxide and unconsumed magnesium metal with hydrochloric or sulfuric acid and can be further purified by heating under vacuum at 1800 °C. Since the MgO acts as a grain growth inhibitor, ultrafine boron carbide particles on the order of 0.1 – 1.5 µm are obtained.

There are other methods of powder preparation:

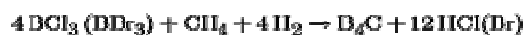
synthesis from the elements



reduction of boron trichloride by hydrogen in the presence of carbon



chemical vapor deposition [41]



However, these methods are generally used only on the laboratory scale for preparing high-purity boron carbide crystals or coatings.

Dense Shapes

Self-bonded boron carbide shapes having 100 % theoretical density, small final grain size, and high mechanical strength can be produced by hot pressing. This is accomplished by placing a loose powder or a compact preform in a graphite die and applying uniaxial pressure of 10 – 30 MPa at a temperature exceeding 2000 °C for 5 – 10 min. Inert or reducing atmospheres are necessary to protect B₄C and the graphite from oxidation. Increasing the boron content to the composition B₁₃C₂ has been found to facilitate sintering by reducing the necessary temperatures and to almost double the strength. However, for boron-rich B_{4+x}C compositions, expensive boron nitride-lined graphite molds must be used, to avoid carburization resulting in formation of carbon-rich B₄C.

In 1977, ESK developed a process for pressureless sintering of boron carbide, a process now used in everyday production. The key observation, that carbon additives are extremely effective in inducing densification of B₄C, was also made by Henney, Jones and Suzuki et al. almost simultaneously. There are two necessary conditions for obtaining a high sintered density:

The use of submicrometer B₄C powders, i.e., powders with a mean particle size below 1 μm

The use of small amounts of elemental carbon-containing additive (novolac resin), which pyrolyses to amorphous carbon during the heating cycle.

Up to the early 1990s, the carbon was added almost exclusively as novolac resin. However, after an improved method of dispersing carbon black (particle size 20 nm) was developed, the resin route was replaced by the carbon black method. Using nanosized carbon black as sintering aid gives a better sintering behavior with very fine grained materials (mean grain size of B₄C = 1 – 4 μm) having an approximate composition of 96 wt % B₄C and 4 wt % C. The properties of sintered B₄C can be improved further by near net shape injection molding and post-HIPing (HIP, hot isostatic pressing) after pressureless sintering. In this post-HIP stage, the sintered shapes are densified to > 99.5 % TD at 1950 – 2050 °C in a high-pressure autoclave (≥ 10 MPa).

The pressureless sintering/HIP route has also been used to prepare B₄C – SiC composites and B₄C – TiB₂ composites. The B₄C – SiC composites combine the good thermal shock and oxidation resistance of silicon carbide with the hardness and low density of boron carbide. In this way, a maximum strength of 550 MPa (four-point bend) was achieved for a composite of 59 wt % SiC, 37 wt % B₄C and 4 wt % C. This composite can be used in oxidizing atmospheres up to 1200 °C. Excellent tribological and wear properties were obtained by Kevorkijan et al. for B₄C – SiC composite materials, which suggest their use for heavily loaded mechanical seals and bearings.

The newly developed B₄C – TiB₂ composite materials are electroconductive and combine excellent fracture toughness, with the high hardness of boron carbide. They are used as nozzle materials for hard alumina grits in abrasive air-jet cleaning in dental laboratories. While the above-mentioned composite materials are formed by mixing the desired metal boride phases and subsequent sintering, processes have also been described in which the desired metal boride composition is only formed after a suitable reaction of the starting materials during the sintering step. Skorokhod and Krstic fabricated a 85 vol % B₄C – 15 vol % TiB₂ composite with a flexural strength of 621 MPa and a fracture toughness of 6.1 MPa · m^{1/2} by reaction hot pressing of a boron carbide powder with submicrometer particle size using additions of submicrometer-sized TiO₂ and carbon at 2000°C. The formation of uniformly distributed TiB₂ particles was in accordance with the reaction.



Another approach to produce tough and high-strength B₄C ceramics is liquid-phase sintering. Lee and Kim (quoted in Ullmanns 2005) showed that addition of alumina (Al₂O₃) promotes densification of boron carbide, and a maximum density of 96% of theoretical can be obtained with 3 wt % alumina-doped B₄C sintered at 2150°C, i.e., above the melting point of Al₂O₃. However, when the addition of Al₂O₃ exceeded 3 wt %, exaggerated grain growth occurred, which was attributed to the liquid phase. It has been reported that by hot pressing of B₄C with alumina additions of up to 5 vol % at 2000 °C the mechanical properties can be remarkably increased as compared to undoped, hot-pressed B₄C of 88 % relative density. Fracture toughness increased steadily with the addition of Al₂O₃ from ca. 3 MPa · m^{1/2} (2.5 vol % Al₂O₃) up to 3.8 MPa · m^{1/2} (10 vol % Al₂O₃). However, the achieved flexural strength was below 560 MPa.

The fracture toughness of boron carbide can be improved by addition of yttria or of yttria in combination with other oxides. The fracture toughness of B₄C can be increased to 3.9 MPa·m^{1/2} by vacuum sintering at 1900 – 1975°C of powder compacts of composition 97.5 wt % B₄C and 2.5 wt % carbon packed in a yttria grit of 0.15 – 1.4 mm grain size. The vacuum allows yttria vapor to penetrate the powder compact and promote reaction sintering of carbon-doped B₄C to full density (2.62 g/cm³). X-ray diffraction showed that yttrium boride and yttrium borocarbide were coexistent with B₄C. The method disclosed in uses liquid-phase sintering under vacuum or streaming argon of powder compacts

comprising B₄C powder (average particle size 0.6 – 3.5 µm) and 1 – 28 wt % additions of Y₂O₃ in combination with Al₂O₃ or aluminum nitride (AlN) and La₂O₃ or CeO₂.

4.3 System Characterisation

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg boron carbide, at plant, global**. For this study the production of boron carbide by means of the reduction of boric oxide is considered. All data in the present report are referred to 1 kg boron carbide 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

4.4 Raw materials and auxiliaries

As there was no data available for the production of boron carbide, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the arsine production.

Tab. 4.2 Consumption of raw materials and auxiliaries for the boron carbide production

Input	Values	Remarks
Boron oxide (kg)	2.5	Calculated with a yield of 95 %
Carbon (kg)	1.499	Calculated with a yield of 100 %
Cooling water (kg)	24	Estimation

4.5 Energy

Tab. 4.3 Energy consumption for the boron carbide production

Input	Ullmanns 2005 (per kg boron carbide)
Electricity (kWh)	9.1

4.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train. Tab. 4.4 summarises the total transport amounts for the boron carbide production.

Tab. 4.4 Total transport amounts for the boron carbide production

tkm kg ⁻¹ boron carbide	lorry	train
Total transports	4E-01	2.4

4.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of boron carbide production plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

4.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the boron carbide production. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 1% of the produced CO. The remaining CO is oxidated to CO₂.

Tab. 4.5 Process emissions to air from the boron carbide production

Output	Value	Remarks
Waste heat (MJ)	32.76	calculated from electricity input
CO, fossil, to air (kg)	2.85E-02	estimated
CO ₂ , fossil, to air (kg)	4.43	from burning of co-product

4.9 Emissions to water

As no process water is used no emissions to water are assumed.

4.10 Waste

The remaining boric oxide and carbon (5 % of the input) are assumed to be disposed as inert waste.

Tab. 4.6 Wastes from the boron carbide production

Output	Value
disposal, inert waste, 5% water, to inert material landfill (kg)	2E-01

4.11 Life cycle inventory of boron carbide production and data quality considerations

Tab. 4.7 shows the life cycle inventory and the data quality indicators for the boron carbide production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 4.7 Unit process raw data for the boron carbide production

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	boron carbide, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							GLO			
							0			
							kg			
resource, in water	401	0	boron carbide, at plant	GLO	0	kg	1.00E+0			
	662	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	493	-	boric oxide, at plant	GLO	0	kg	2.50E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	403	-	hard coal coke, at plant	GLO	0	MJ	1.50E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
		-	disposal, inert waste, 5% water, to inert material landfill	CH	0	kg	2.00E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
		-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	9.10E+0	1	1.08	(1,1,2,1,1,3); Ullmann 2005
		-	transport, lorry >16t, fleet average	RER	0	tkm	4.00E-1	1	2.09	(4,5,na,na,na,na); Standard distances
		-	transport, freight, rail	RER	0	tkm	2.40E+0	1	2.09	(4,5,na,na,na,na); Standard distances
		-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
		-								
emission air, high population density		4	Heat, waste	-	-	MJ	3.28E+1	1	1.08	(1,1,2,1,1,3); Calculated from the electricity input
		4	Carbon monoxide, fossil	-	-	kg	2.85E-2	1	5.10	(4,5,na,na,na,na); Estimation
		4	Carbon dioxide, fossil	-	-	kg	4.43E+0	1	1.30	(4,5,na,na,na,na); Estimation

4.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 4.8 shows selected LCI results and the cumulative energy demand for the boron carbide production.

Tab. 4.8 Selected LCI results and the cumulative energy demand for the boron carbide production

Name		boron carbide, at plant	
Location	Unit	Unit	GLO kg 0
Infrastructure			
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	132.4
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	56.6
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	5.2
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	6.7
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	1.0
selected LCI results			
land occupation	resource	m2a	1.2E+0
CO ₂ , fossil	air	kg	1.4E+1
NM VOC	air	kg	5.5E-3
nitrogen oxides	air	kg	2.8E-2
sulphur dioxide	air	kg	9.1E-2
particulates, <2.5 um	air	kg	4.4E-3
BOD	water	kg	1.0E-2
cadmium	soil	kg	6.6E-9

4.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	boron carbide, at plant
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of boron carbide including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Borcarbid, ab Werk
	491	Synonyms	black diamond/tetrabor
	492	GeneralComment	The process "boron carbide, at plant, GLO" is modelled for the production of boron carbide from boric oxide in the world. Raw materials are modelled with a stoichiometric calculation. Emissions are calculated estimated. Infrastructure and transports are calculated with standard values. Energy consumptions are taken from literature.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	inorganics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Anorganika
	499	Formula	BC4
	501	StatisticalClassification	
	502	CASNumber	12069-32-8
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	Reduction of boric oxide
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Emissions are calculated estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

4.14 References

Frischknecht et al. 2007

Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.

- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>
- Ullmanns 2004 Baudis, U., Fichte, R.: Boron and Boron Alloys. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

5 Boron trifluoride

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

5.1 Introduction

Boron trifluoride (BF₃; CAS 7637-07-2; synonym: trifluoroborane) is a colourless gas. For this inventory the functional unit is 1 kg of highly pure trifluoroborane. The most important chemical and physical properties of trifluoroborane used in this inventory are given here.

Tab. 5.1 Chemical and physical properties of trifluoroborane (Ullmann 2005)

Property	Value	Unit	Remarks
Molecular weight	67.8	g mol ⁻¹	
Specific gravity	1570	kg m ⁻³	
Heat of fusion	-1136	kJ mol ⁻¹	at 298 K
Boiling point	-99.9	°C	at normale pressure
Melting point	-127.1	°C	

5.1.1 Reserves and Resources

Boron trifluoride is the largest volume boron halide. The total annual production of BF₃ and its adducts is about 2300 – 4500 t/a.

5.1.2 Use

Boron trifluoride is the most widely used boron halide. Most applications take advantage of its strong Lewis acidity. The most important uses are in organic syntheses. Boron trifluoride is commonly used as a catalyst for Friedel-Crafts alkylation reactions. It also is used to catalyze the cleavage of ethers to alcohols, to catalyze esterification reactions, and in the nitration and sulfonation of aromatic compounds. Many olefin polymerization reactions use BF₃ as an initiator, in conjunction with a proton donor, such as water. Also BF₃ is used to catalyze the isomerization of alkenes and alkanes and in petroleum cracking and desulfurization. Amine complexes of BF₃ are used as epoxy curing agents.

5.2 Production Technologies (Ullmann 2005)

Boron trifluoride was first prepared by the reaction of boric acid and fluorspar (CaF₂) at red heat. Commercially, BF₃ has been produced by the stepwise addition of hydrofluoric acid and sulfuric acid to anhydrous borax (sodium tetraborate):



Another method involves addition of sulfuric acid and fluorspar and a source of boron, such as anhydrous borax or boric oxide (B₂O₃).

A process was developed in which BF₃ is obtained by the reaction of HF and boric acid, using oleum as a dehydrating agent.



Boron trifluoride is made commercially in the United States by a proprietary process from boric acid and hydrofluoric acid.

A variety of synthetic routes are available for the production of small quantities of BF_3 , including thermal decomposition of diazonium fluoroborate.

The diethyl etherate adduct, $\text{BF}_3 \cdot \text{O}(\text{C}_2\text{H}_5)_2$, produced by bubbling BF_3 into diethyl ether, provides a convenient source of BF_3 for many synthetic applications.

5.3 Quality Specifications, Chemical Analysis

Boron trifluoride is available in quantity as a 99 % pure gas.

Common impurities in BF_3 include air, silicon tetrafluoride, and sulfur dioxide. The major impurities in boron trichloride are phosgene and chlorine. Boron tribromide typically contains traces of bromine.

5.4 System Characterisation

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg trifluoroborane, at plant, global**. For this study the production of trifluoroborane by means of the reaction of boric acid and hydrofluoric acid is considered. All data in the present report are referred to 1 kg trifluoroborane 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

5.5 Raw materials and auxiliaries

As there was no data available for the production of trifluoroborane, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the boron trifluoride production.

Tab. 5.2 Consumption of raw materials and auxiliaries for the boron trifluoride production

Input	Values	Remarks
Boric acid (kg)	0.96	Calculated with a yield of 95 %
HF (kg)	0.929	Calculated with a yield of 95 %
Cooling water (kg)	24	Estimation

5.6 Energy

There was no information available on the amount of energy used for the production process. In order not to neglect the process energy demand those values were approximated with data from an large

chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg⁻¹) were used as approximation for the energy consumption of the boron trifluoride production. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used. A summary of the values used is given in Tab. 5.3.

Tab. 5.3 Energy consumption for the boron trifluoride production

Input	per kg trifluoroborane
Heat, natural gas burned in industrial furnace > 100 kW	2
Electricity (kWh)	0.333

5.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train. Tab. 5.4 summarises the total transport amounts for the production of 1 kg boron trifluoride.

Tab. 5.4 Total transport amounts for the boron trifluoride production

tkm kg ⁻¹ trifluoroborane	lorry	train
Total transports	1.89E-01	1.13

5.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of trifluoroborane production plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

5.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the trifluoroborane production. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the input of boric acid and hydrogen fluoride. The boric acid is calculated as boron emission.

Tab. 5.5 Process emissions to air from the boron trifluoride production

Output	Value	Remarks
Waste heat (MJ)	1.2	calculated from electricity input
Boron, to air (kg)	3.4E-04	estimated as 0.2 % of the input
HF, to air (kg)	1.86E-03	estimated as 0.2 % of the input

5.10 Emissions to water

The remaining amount of unreacted boric acid and hydrogen fluoride (4.8 % of the input) is assumed to leave the production process with the waste water. The boric acid is calculated as boron emission.

Tab. 5.6 Process emissions to water from the boron trifluoride production

Output	Value	Remarks
Boron (kg)	8.17E-03	calculated from mass balance
Fluoride (kg)	4.24E-02	calculated from mass balance

5.11 Life cycle inventory of boron trifluoride production and data quality considerations

Tab. 5.7 shows the life cycle inventory and the data quality indicators for the trifluoroborane production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 5.7 Unit process raw data for the boron trifluoride production

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	boron trifluoride, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							GLO 0 kg			
resource, in water	401	0	boron trifluoride, at plant	GLO	0	kg	1.00E+0			
	662	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	493	-	boric acid, anhydrous, powder, at plant	RER	0	kg	9.60E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	403	-	hydrogen fluoride, at plant	GLO	0	kg	9.29E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
		-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	2.00E+0	1	1.30	(4,5,na,na,na,na); Estimation
		-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.33E-1	1	1.30	(4,5,na,na,na,na); Estimation
		-	transport, lorry >16t, fleet average	RER	0	tkm	1.89E-1	1	2.09	(4,5,na,na,na,na); Standard distances
		-	transport, freight, rail	RER	0	tkm	1.13E+0	1	2.09	(4,5,na,na,na,na); Standard distances
		-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
		-								
emission air, high population density		4	Heat, waste	-	-	MJ	1.20E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input
emission water, river		4	Boron	-	-	kg	3.41E-4	1	5.10	(4,5,na,na,na,na); Estimation
		4	Hydrogen fluoride	-	-	kg	1.86E-3	1	1.62	(4,5,na,na,na,na); Estimation
		4	Boron	-	-	kg	8.17E-3	1	5.10	(4,5,na,na,na,na); Estimation
		4	Fluoride	-	-	kg	4.24E-2	1	1.62	(4,5,na,na,na,na); Estimation

5.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 5.8 shows selected LCI results and the cumulative energy demand for the boron trifluoride production.

Tab. 5.8 Selected LCI results and the cumulative energy demand for the boron trifluoride production

	Name	boron trifluoride, at plant	
	Location		
	Unit	Unit	GLO
	Infrastructure		kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	53.4
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	18.5
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	3.2
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	2.3
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.3
selected LCI results			
land occupation	resource	m2a	8.1E-1
CO2, fossil	air	kg	3.7E+0
NMVOC	air	kg	2.4E-3
nitrogen oxides	air	kg	1.3E-2
sulphur dioxide	air	kg	8.8E-2
particulates, <2.5 um	air	kg	1.6E-3
BOD	water	kg	6.2E-3
cadmium	soil	kg	3.7E-9

5.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	boron trifluoride, at plant
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of boron trifluoride including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Bortrifluorid, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "boron trifluoride , at plant, GLO" is modelled for the production of boron trifluoride from boric acid and hydrofluoric acid in the world. Raw materials are modelled with a stoichiometric calculation. Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	inorganics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Anorganika
	499	Formula	BF3
	501	StatisticalClassification	
	502	CASNumber	7637-07-2
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	Fluorination of boric acid
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

5.14 References

Friskhnecht et al. 2007

Friskhnecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.

- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>
- Ullmanns 2004 Brotherton, R...: Boron Compounds. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

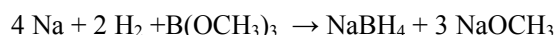
6 Brown-Schlesinger-Process

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

The Brown-Schlesinger process is one of the most important production processes for sodium tetrahydroborate.

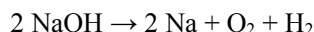
6.1 Production Technologies (Ullmanns 2004)

In the Brown-Schlesinger-Process methyl borate is added to hydrogenated sodium in mineral oil:

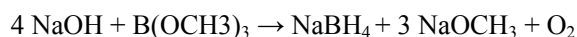


Separation of the sodium methoxide from NaBH₄ was accomplished by such methods as extraction with amines, and extraction with water followed by a counterextraction with a solvent, such as 2-aminopropane.

The sodium and hydrogen for the Brown-Schlesinger process are produced by an electrolysis of NaOH.



The overall equation for the process:



6.2 System Characterisation

In this study the Brown-Schlesinger process is described. The functional unit is 1 kg trimethyl borate. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers. Within the Brown-Schlesinger process, sodium tetrahydroborate and sodium methoxide are derived as co-product. Therefore an allocation between the two products was made. The following processes were modelled:

- Brown-Schlesinger process, GLO. Multi output process with sodium tetrahydroborate and sodium methoxide as allocated products

6.3 Sodium tetrahydroborate

Sodium tetrahydroborate (NaBH₄; CAS 16940-66-2; synonyms: sodium borohydride) is a white crystalline solid. It is stable in dry air, reacting slowly with moisture.

The most important chemical and physical properties of sodium borohydride used in this inventory are given here.

Tab. 6.1 Chemical and physical properties of sodium borohydride (Ullmanns 2004)

Property	Value	Unit
Molecular weight	37.83	g mol ⁻¹
Specific gravity	0.17	kg m ⁻³
Boiling point	-268.9	°C
Melting point	-272.2	°C

6.3.1 Use

A concentrated aqueous solution of NaBH₄ stabilized by NaOH is used to reduce hydrogen sulfite to dithionite in the bleaching of ground wood pulp. Also, NaBH₄ has extensive and important use as a reducing agent in organic synthesis. Other hydroborates can be obtained by reaction with NaBH₄ to give different reductive capabilities (e.g., NaBH₃CN).

Additional applications are extensive, e.g., electroless plating, blowing agent for cellular plastics, removal of heavy metals from waste streams, polymerization catalyst, and hydrogen source for fuel cells.

6.4 Raw materials and auxiliaries

Tab. 6.2 shows values for the use of raw materials for the Brown-Schlesinger process. As there was no data available for the process, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from a large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the chlorosilane amination.

Tab. 6.2 Consumption of raw materials and auxiliaries for the Brown-Schlesinger process

Input	Values	Remarks
Trimethyl borate (kg)	1	Functional unit
Sodium hydroxide (kg)	1.538	Calculated with a yield of 95 %
Cooling water (m3)	0.0438	Estimation

6.5 Energy

There was no information available on the amount of energy used for the production process. In order not to neglect the process energy demand those values were approximated with data from a large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg⁻¹) were used as approximation for the energy consumption of the Brown-Schlesinger process. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used.

For the sodium hydroxide electrolysis Ullmanns 2005 reports a electricity consumption of 10 kWh per kg sodium. A summary of the values used is given in Tab. 6.3.

Tab. 6.3 Energy consumption for the chlorosilane amination

Input	per kg methyl borate
Heat, natural gas burned in industrial furnace > 100 kW	3.65
Electricity (kWh)	9.45

6.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train. Tab. 6.4 summarises the total transport amounts for the Brown-Schlesinger process.

Tab. 6.4 Total transport amounts for the Brown-Schlesinger process

tkm kg ⁻¹ chlorosilane	lorry	train
Total transports	4.077E-01	1.846

6.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of Brown-Schlesinger process plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

6.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the Brown-Schlesinger process. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the trimethyl borate input.

Tab. 6.5 Process emissions to air from the Brown-Schlesinger process

Output	Value	Remarks
Waste heat (MJ)	34.04	calculated from electricity input
Trimethyl borate, to air (kg)	2E-03	estimated as 0.2 % of the input
CO ₂ , fossil, to air (kg)	4.57E-02	from wastewater treatment

6.9 Emissions to water

The remaining amount of unreacted sodium hydroxide and trimethyl borate is assumed to leave the production process with the waste water. Further it was assumed that this wastewater is treated in a internal wastewater plant. The removal efficiency for trimethyl borate is assumed with 90%. The carbon contained in the removed trimethyl borate is accounted as CO₂ emissions to air. The trimethyl borate is calculated as boron emission.

The values for COD, BOD, TOC, and DOC used in this inventory are calculated from the amount of trimethyl borate in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC is

used.

Tab. 6.6 Process emissions to water from the Brown-Schlesinger process

Output	Value	Remarks
Boron (kg)	5.1E-03	calculated from mass balances
sodium (kg)	3.54E-02	calculated from mass balance
hydroxide (kg)	2.62E-02	calculated from mass balance
BOD, COD (kg)	5.94E-02	calculated from water emissions
TOC, DOC (kg)	1.86E-02	calculated from water emissions

6.10 Allocation of sodium tetrahydroborate and sodium methoxide

Within the Brown-Schlesinger process, sodium tetrahydroborate and sodium methoxide are obtained as co-products. A stoichiometric allocation of the two products is made. The used allocation factors are given in Tab. 6.7.

Tab. 6.7 Allocation of sodium tetrahydroborate and sodium methoxide from the Brown-Schlesinger process

Process	Brown-Schlesinger process, GLO		
	Unit	per kg trimethyl borate	Allocation factor (stoichiometric)
sodium tetrahydroborate, at plant, GLO	Kg	0.362	19
sodium methoxide, at plant, GLO	kg	1.542	81

6.11 Life cycle inventory of Brown-Schlesinger process and data quality considerations

Tab. 6.8 shows the life cycle inventory and the data quality indicators for the Brown-Schlesinger process. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 6.8 Unit process raw data for the Brown-Schlesinger process

401	InputGroup	OutputGroup	Name	Location	Infrastructure	Processes	Unit	Brown-Schlesinger process	UncertaintyType	StandardDeviation95%	GeneralComment	sodium tetrahydroborate, at plant	sodium methoxide, at plant
												GLO 0 kg	GLO 0 kg
662			Location									GLO 0 kg	GLO 0 kg
493			Infrastructure										
403			Unit										
allocated products resource, in water technosphere	-	2	sodium tetrahydroborate, at plant	GLO	0	kg	3.62E-1					100	0
	-	2	sodium methoxide, at plant	GLO	0	kg	1.54E+0					0	100
	4	-	Water, cooling, unspecified natural origin	-	-	m3	4.38E-2	1	1.30	(4,5,na,na,na,na); Estimation	19.000	81.000	
	5	-	sodium hydroxide, 50% in H2O, production mix, at plant	RER	0	kg	1.54E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation	19.000	81.000	
	5	-	trimethyl borate, at plant	GLO	0	kg	1.00E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation	19.000	81.000	
	5	-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	3.65E+0	1	1.30	(4,5,na,na,na,na); Estimation	19.000	81.000	
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	9.45E+0	1	1.30	(4,5,na,na,na,na); Estimation	19.000	81.000	
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	4.08E-1	1	2.09	(4,5,na,na,na,na); Standard distance 100km	19.000	81.000	
	5	-	transport, freight, rail	RER	0	tkm	1.85E+0	1	2.09	(4,5,na,na,na,na); Standard distance 600km	19.000	81.000	
	5	-	chemical plant, organics	RER	1	unit	7.31E-10	1	3.09	(4,5,na,na,na,na); Estimation	19.000	81.000	
emission air, high population density	-	4	Heat, waste	-	-	MJ	3.40E+1	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input	19.000	81.000	
	-	4	Methyl borate	-	-	kg	2.00E-3	1	1.62	(4,5,na,na,na,na); Estimation	19.000	81.000	
	-	4	Carbon dioxide, fossil	-	-	kg	4.57E-2	1	1.30	(4,5,na,na,na,na); Estimation	19.000	81.000	
emission water, river	-	4	Boron	-	-	kg	5.08E-3	1	1.30	(4,5,na,na,na,na); Estimation	19.000	81.000	
	-	4	Sodium, ion	-	-	kg	3.54E-2	1	5.10	(4,5,na,na,na,na); Estimation	19.000	81.000	
	-	4	Hydroxide	-	-	kg	2.62E-2	1	1.62	(4,5,na,na,na,na); Estimation	19.000	81.000	
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	5.94E-2	1	1.62	(4,5,na,na,na,na); Estimation	19.000	81.000	
	-	4	COD, Chemical Oxygen Demand	-	-	kg	5.94E-2	1	1.62	(4,5,na,na,na,na); Estimation	19.000	81.000	
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	1.86E-2	1	1.62	(4,5,na,na,na,na); Estimation	19.000	81.000	
	-	4	TOC, Total Organic Carbon	-	-	kg	1.86E-2	1	1.62	(4,5,na,na,na,na); Estimation	19.000	81.000	
	-	4	TOC, Total Organic Carbon	-	-	kg	1.86E-2	1	1.62	(4,5,na,na,na,na); Estimation	19.000	81.000	

6.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 6.9 shows selected LCI results and the cumulative energy demand for the Brown-Schlesinger process.

Tab. 6.9 Selected LCI results and the cumulative energy demand for the Brown-Schlesinger process

Name			sodium tetrahydroborate, at plant	sodium methoxide, at plant
Location			GLO	GLO
Unit		Unit	kg	kg
Infrastructure			0	0
LCIA results				
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	77.2	77.2
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	29.7	29.7
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	1.9	1.9
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	3.5	3.5
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.5	0.5
selected LCI results				
land occupation	resource	m2a	3.4E-1	3.4E-1
CO2, fossil	air	kg	4.6E+0	4.6E+0
NM VOC	air	kg	4.1E-3	4.1E-3
nitrogen oxides	air	kg	8.9E-3	8.9E-3
sulphur dioxide	air	kg	1.8E-2	1.8E-2
particulates, <2.5 um	air	kg	1.2E-3	1.2E-3
BOD	water	kg	3.8E-2	3.8E-2
cadmium	soil	kg	2.8E-9	2.8E-9

6.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	Brown-Schlesinger process
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	5
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Brown-Schlesinger process including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Brown-Schlesinger-Prozess
	491	Synonyms	
	492	GeneralComment	The multioutput-process "Brown-Schlesinger process " delivers the co-products sodium tetrahydroborate and sodium methoxide. The allocation is based on stoichiometric calculation.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	inorganics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications.
Geography	663	Text	The inventory is modelled for the world..
Technology	692	Text	sodium hydrogenation of trimethyl borate
Representativeness	722	Percent	100
	724	ProductionVolume	na
	725	SamplingProcedure	Literature data
	726	Extrapolations	Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
DataGeneratorAnd	727	UncertaintyAdjustments	none
	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

6.14 References

Friskhnecht et al. 2007 Friskhnecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.

- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>
- Ullmanns 2004 Baudis, U., Fichte, R.: Boron and Boron Alloys. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

7 Carbon disulfide

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

7.1 Introduction

Carbon disulfide (CS₂; CAS 75-15-0) is a volatile, dense liquid with a wide flammability range in air (1 – 50 vol %). It has an unusually low autoignition point (100 °C) and a low flash point (– 30 °C). Its evaporation rate into air is 1.6 times that of diethyl ether. It is an excellent solvent for many organic compounds and readily dissolves sulfur, phosphorus, iodine, waxes, rubber, and resins. The commercial product has a pungent odor characteristic of sulfur compounds.

For this inventory the functional unit is 1 kg of highly pure carbon disulfide. The most important chemical and physical properties of carbon disulfide used in this inventory are given here.

Tab. 7.1 Chemical and physical properties of carbon disulfide (Ullmanns 2004)

Property	Value	Unit	Remarks
Molecular weight	76.13	g mol ⁻¹	
Specific gravity	1260	kg m ⁻³	liquide
Boiling point	46.23	°C	
Melting point	-111.6	°C	

7.2 Reserves and Resources

World production capacity for carbon disulfide in 1984 is shown in Tab. 7.2.

Tab. 7.2 World production of CS₂ in 1984 in 10³ tons (Ullmanns 2004)

Area	Charcoal process	Methane process
North America	–	335
Western Europe	–	360
Eastern Europe	40	260
USSR	40	120
Japan	–	100
India	50	–
China	20	–
Latin America	60	15
Others	15	–
Total	225	1190

The world production capacity for carbon disulfide in 1991 is shown in Tab. 7.3.

Tab. 7.3 World production of CS₂ in 1991 in 10³ tons (Ullmanns 2004)

Area	10 ³ tons
North and Central America	190
South America	54
Europe and Africa	822
Asia	221
Total	1287

7.3 Use

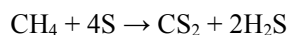
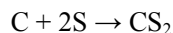
Carbon disulfide is used as a reaction and extraction solvent, as a catalyst presulfidation agent, an oil well solvent, and a cold vulcanization additive for rubber, synthetic chemicals.

7.4 Production Technologies (Ullmanns 2004)

Carbon disulfide is commercially manufactured by the reaction of sulfur with charcoal or methane. Ethane, propane, and propene have been used to a limited extent. Since the methane process was first introduced in the early 1950s, it has steadily supplanted the older charcoal process, which is no longer a factor in carbon disulfide manufacture in the United States, Europe, and Japan. In areas where natural gas or methane is not readily available or when plant size is relatively small, the charcoal process still supplies local viscose rayon requirements.

Thermodynamics and Rates of Reaction

The reactions of sulfur with either carbon or methane have been evaluated for the following equations:



The maximum yield from carbon was found to be 91 % at 720 °C. For the methane reaction, complete conversion was attained at 300 °C. In practice, higher temperature is required to provide economic reaction rates. The reactions are endothermic over most of the temperature range practiced commercially; the energy supplied is taken up mainly by vaporization and superheating of sulfur.

The thermodynamics of sulfur dissociation determine the concentration of the reactive diatomic sulfur molecule, S₂; high temperature and low pressure favor high S₂ concentrations. Equilibration rates of the vapor species, S_n (n = 1 – 8), are relatively fast so that equilibrium values of S₂ may be used in rate equation formulation.

Homogeneous and heterogeneous rate data for the reaction of methane and sulfur have been obtained for temperatures between 500 and 700 °C. The rate equation that best fits these data is one that is first order with respect to methane and S₂:

$$\text{rate} = k_{\text{CH}_4\text{S}_2}$$

where k is the reaction rate constant.

The following are the Arrhenius equations for the rate constants, based on reference. For the homogeneous reaction:

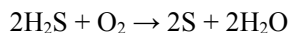
$$k = 1.9 \times 10^6 \exp(-52000/RT) \text{ mol h}^{-1} \text{ cm}^{-3} \text{ kPa}^{-2}$$

For the silica gel catalyzed reaction:

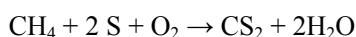
$$k = 7.7 \times 10^2 \exp(-28000/RT) \text{ mol h}^{-1} \text{ g}^{-1} \text{ catal-kPa}^{-2}$$

Other values for these rate constants as well as space velocities for the catalyzed reaction have been reported. At 600 °C, the homogeneous rate is about 1 % that for the silica gel catalyzed reaction.

After CS₂ and H₂S have been separated, the H₂S is recycled and the latter is used for sulfur production in the Claus process.



The whole chemical equation with included desulfurizing and CS₂-production from the recovered sulfur:



Production from Charcoal and Sulfur

The basic process for the reaction of charcoal with sulfur has changed little since its inception in the mid 1840s. Externally heated retorts are still being used, although equipment design has undergone much improvement as better materials of construction have become available. Details on retort design are available in references.

In retort plants molten sulfur is vaporized and superheated either before or after addition to the retort. Reaction between the superheated sulfur vapor and a stationary bed of charcoal at 850 – 900 °C takes place inside the retort at slightly above atmospheric pressure. The product gas, consisting of carbon disulfide, some sulfur, hydrogen sulfide, carbonyl sulfide, and inerts, passes through a train of condenser, scrubber, and oil absorber where the crude carbon disulfide is recovered; the pure product is obtained by further treatment in distillation columns. The tail gas, made up primarily of hydrogen sulfide, carbonyl sulfide, and inerts, is either incinerated and scrubbed with a caustic solution or routed to a sulfur recovery plant.

Typically, each retort can produce 1 – 3 t/d of carbon disulfide. High-temperature corrosion of the cast iron retorts limits their useful life to less than 1 year. The raw materials used in the retorts should be low in ash and residue to minimize the frequency of reactor cleanouts. Fouling of the retorts also restricts heat transfer and shortens retort life. Charcoal quality is extremely important because less reactive material requires higher temperature for satisfactory operation; charcoal derived from hardwood is preferred for this reason. The charcoal is normally precalcined at 500 °C to remove volatiles that could result in formation of objectionable byproducts. Thermal efficiency of retorts is ca. 25 %; carbon and sulfur yields are generally below 90 %.

Electrical resistance heating in brick-lined electrothermal reactors provides an alternative heat source for the reaction of charcoal with sulfur. Where cheap power is available, the electrical method may be more economic. Power consumption as a function of carbon disulfide produced is ca. 1200 kW h/t. Other advantages offered by electrothermal reactors are a larger capacity per reactor (up to 10 t/d) and a longer reactor life; furthermore, a wider range of solid carbon feed can be used because electrothermal reactors can generate higher temperatures.

Production from Methane and Sulfur

As demand for carbon disulfide rapidly increased in the 1940s, limitations of the charcoal processes provided the incentive for development of a more efficient large-scale method for its manufacture. Readily available cheap natural gas and favorable thermodynamics led to the successful development

of the process based on the catalyzed reaction of methane and sulfur. During the period 1943 – 1965, many patents were issued that covered various aspects of the process.

The commercial methane process is based primarily on work described by the patents issued to the Pure Oil Co. The rights to these patents were acquired by the Food Machinery Corp. and Stauffer Chemical Co., and in 1953, the first methane process plant was built by FMC in South Charleston, West Virginia. This was followed by a Stauffer plant at LeMoyné, Alabama, three years later. By the mid 1960s all carbon disulfide in the United States was manufactured by this method; many similar plants soon followed in Europe and elsewhere. Over 85 % of the world's production capacity is based on methane.

Figure 1 shows a typical methane process flow sheet. Purified natural gas containing over 95 % methane and low concentrations of propane and heavier hydrocarbons is preheated to about 250 °C in the convection section of the reaction furnace. The furnace is a typical box-type pyrolysis furnace commonly used in the petrochemical industry, with radiant and convection sections. The hot methane gas is mixed with liquid sulfur and introduced into the horizontal heating coil located in the radiant section. There the liquid sulfur is vaporized, and the temperature of the mixture is raised to 550 – 650 °C. Operating pressure is typically 400 – 700 kPa. Alternatively, the methane and sulfur may be introduced separately and at different locations of the coil. A slight excess of sulfur (5 %) is used to ensure good methane conversion and to minimize carbon and tar formation.

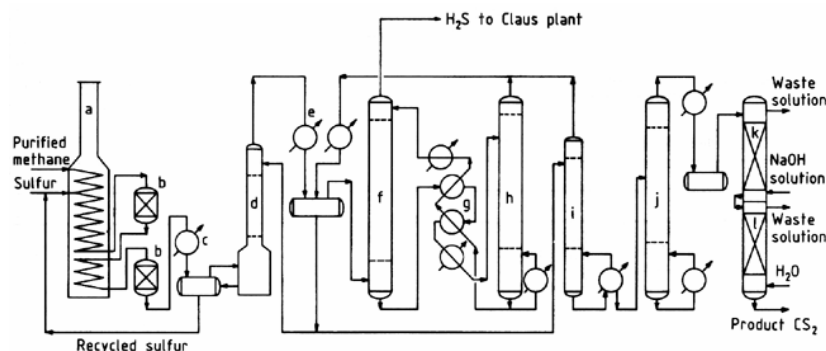


Fig. 7.1 Methane process for CS₂ production

a) Reaction furnace; b) Catalytic reactors; c) Sulfur condenser; d) CS₂ scrubber; e) CS₂ condenser; f) CS₂ absorber; g) Oil interchangers; h) CS₂ stripper; i) Stabilizer column; j) Purification column; k) Caustic wash column; l) Water wash column

To withstand the high-temperature corrosive environment, cast high-alloy stainless steel is used as tube material. Tube life is normally 2 – 3 years. Packed, brick-lined reactors are connected to the heating coil to provide additional residence time and housing for catalyst beds. Reaction takes place both in the tubes and in the reactors; almost all the hydrocarbons are converted by the time the gas leaves the last reactor.

Gas exiting the reaction furnace system passes through the sulfur condenser where most of the excess sulfur is separated. Next the cooled gas goes through a sulfur scrubber and then to the CS₂ scrubber in which the remaining sulfur is removed. The relatively sulfur-free gas then enters water-cooled condensers for bulk separation of carbon disulfide. The remaining carbon disulfide is finally removed from the hydrogen sulfide gas in the oil absorber. The hydrogen sulfide gas exiting the oil absorber contains 1 – 2 % methane and less than 1000 ppm carbon disulfide. This gas is routed to a Claus sulfur recovery plant where it is converted back to sulfur.

Crude carbon disulfide from the condensers, including that recovered in the stripper, is sent to the stabilizer column for removal of low-boiling impurities. The bottoms from the stabilizer column are next

processed in the purification column where high-boiling contaminants are separated from the product. A caustic and water wash complete the product purification operations, and the finished product is pumped to check tanks and storage.

Other Processes

The use of different feedstocks for the manufacture of carbon disulfide is a subject of much economic interest. Raw materials evaluated include the following: methane and hydrogen sulfide; coke and coal with sulfur dioxide and hydrogen sulfide; liquid hydrocarbons with sulfur; hydrocarbons with sulfur and sulfur dioxide; chlorinated hydrocarbons and sulfur; carbon monoxide and sulfur; and organosulfur compounds with sulfur. Most of these alternatives are unattractive because of poor yield, undesirable byproducts, or difficulties in processing due to tar and polymer formation.

Important refinements to the basic methane process include the use of high-pressure (2000 kPa) condensation and fractionation for product recovery and purification, and modifications to the reaction furnace system to allow the use of olefinic feedstocks.

7.5 System Characterisation

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg carbon disulfide, at plant, global**. For this study the production of carbon disulfide by means of the reaction between sulfur and methane is considered. All data in the present report are referred to 1 kg carbon disulfide 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

7.6 Raw materials and auxiliaries

As there was no data available for the production of carbon disulfide, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the arsine production.

Tab. 7.4 Consumption of raw materials and auxiliaries for the carbon disulfide production

Input	Values	Remarks
Methane (kg)	0.222	Calculated with a yield of 95 %
Sulfur (kg)	0.886	Calculated with a yield of 95 %
Cooling water (kg)	24	Estimation

7.7 Energy

There was no information available on the amount of energy used for the production process. In order not to neglect the process energy demand those values were approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg⁻¹) were used as approximation for the energy consumption of the carbon disulfide production. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used.

Tab. 7.5 Energy consumption for the carbon disulfide production

Input	Per kg carbon disulfide
Heat, natural gas burned in industrial furnace > 100 kW	2
Electricity (kWh)	0.333

7.8 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km. Tab. 7.6 summarises the total transport amounts for the carbon disulfide production.

Tab. 7.6 Total transport amounts for the carbon disulfide production

tkm kg ⁻¹ carbon disulfide	lorry	train
Total transports	1.11E-01	6.65E-01

7.9 Infrastructure and land use

No information was readily available about infrastructure and land-use of carbon disulfide production plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

7.10 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the carbon disulfide production. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 1% of the unreacted CH₄ input. The remaining CH₄ is oxidated to CO₂.

Tab. 7.7 Process emissions to air from the carbon disulfide production

Output	Value	Remarks
Waste heat (MJ)	1.2	calculated from electricity input
CH ₄ , to air (kg)	1.11E-04	estimated as 1 % of the unreacted input
CO ₂ , to air (kg)	3.02E-02	calculated from mass balance

7.11 Emissions to water

No emissions to water are considered.

7.12 Waste

The remaining sulphur (5 % of the input) is assumed to be disposed as inert waste.

Tab. 7.8 Wastes from the carbon disulfide production

Output	Value
disposal, inert waste, 5% water, to inert material landfill (kg)	4.43E-02

7.13 Life cycle inventory of carbon disulfide production and data quality considerations

Tab. 7.9 shows the life cycle inventory and the data quality indicators for the carbon disulfide production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 7.9 Unit process raw data for the carbon disulfide production

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	carbon disulfide, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							GLO 0 kg			
resource, in water	401	0	carbon disulfide, at plant	GLO	0	kg	1.00E+0			
	662	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	493	-	natural gas, high pressure, at consumer	RER	0	MJ	1.02E+1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	403	-	secondary sulphur, at refinery	RER	0	kg	8.86E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
		-	disposal, inert waste, 5% water, to inert material landfill	CH	0	kg	4.43E-2	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
		-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	2.00E+0	1	1.30	(4,5,na,na,na,na); Estimation
		-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.33E-1	1	1.30	(4,5,na,na,na,na); Estimation
		-	transport, lorry >16t, fleet average	RER	0	tkm	1.11E-1	1	2.09	(4,5,na,na,na,na); Standard distances
		-	transport, freight, rail	RER	0	tkm	6.65E-1	1	2.09	(4,5,na,na,na,na); Standard distances
		-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density		4	Heat, waste	-	-	MJ	1.20E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input
		4	Methane, fossil	-	-	kg	1.11E-4	1	1.62	(4,5,na,na,na,na); Estimation
		4	Carbon dioxide, fossil	-	-	kg	3.02E-2	1	1.30	(4,5,na,na,na,na); Estimation

7.14 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 7.10 shows selected LCI results and the cumulative energy demand for the carbon disulfide production.

Tab. 7.10 Selected LCI results and the cumulative energy demand for the carbon disulfide production

Name		carbon disulfide, at plant	
Location	Unit	Unit	GLO kg 0
Infrastructure			
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	22.6
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	2.1
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.4
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.3
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.0
selected LCI results			
land occupation	resource	m2a	9.1E-2
CO2, fossil	air	kg	8.0E-1
NM VOC	air	kg	6.3E-4
nitrogen oxides	air	kg	1.6E-3
sulphur dioxide	air	kg	2.9E-2
particulates, <2.5 um	air	kg	1.8E-4
BOD	water	kg	1.6E-3
cadmium	soil	kg	4.2E-10

7.15 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	carbon disulfide, at plant
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of carbon disulfide including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Kohlenstoffdisulfid, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "acarbon disulfide, at plant, GLO" is modelled for the production of carbon disulfide from methane and sulphur in the world. Raw materials are modelled with a stoichiometric calculation. Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	CS2
	501	StatisticalClassification	
	502	CASNumber	75-10-0
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	Sulphurization of methane
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

7.16 References

- Frischknecht et al. 2007 Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirsch R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.
- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>
- Ullmanns 2004 Lay, M., Sauerhoff, M., Saunders, D.:Carbon disulfide. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebaum D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

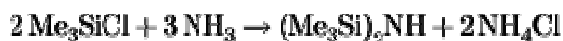
8 Chlorosilane amination

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

Chlorosilane amination is the most important process to produce hexamethyldisilazane.

8.1 Production Technologies (Ullmanns 2004)

Many chlorosilanes react with ammonia to give not stable, simple silylamines $\equiv\text{SiNH}_2$, but disubstitution products with the SiNHSi building block. The exothermic reaction can be carried out in solvents such as hexane, toluene, or isododecane, but is best performed in warm HMDS itself.



The latter variant leads to the formation of coarse-grained ammonium chloride, which can be readily stirred and filtered and gives better turnover and yield. The suggestion has been made that the salt be separated by treatment with water.

A continuous process for the production of hexamethyldisilazane consisting of the concurrent actions of continuously introducing predetermined quantities per time unit of trimethylchlorosilane and gaseous ammonia to form a mixture of hexamethyldisilazane and ammonium chloride in a first reactor while simultaneously subjecting said mixture to continuous high shear agitation, for a time period related to the volume of the reactor and the quantities of trimethylchlorosilane and ammonia introduced, to form a slurry; and simultaneously separating the ammonium chloride from said slurry to provide a solution principally comprising hexamethyldisilazane, while continuing to introduce a continuous flow of further quantities of trimethylchlorosilane and ammonia as the hexamethyldisilazane is removed.

8.2 System Characterisation

In this study the chlorosilane amination is described. The functional unit is 1 kg chlorosilane. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers. Within the chlorosilane amination, hexamethyldisilazane and ammonium chloride are derived as co-products. Therefore an allocation between the two products was made. The following processes were modelled:

- Chlorosilane amination, GLO. Multi output process with hexamethyldisilazane, at plant, GLO and ammonium chloride from chlorosilane amination, at plant, GLO as allocated products

8.3 Hexamethyldisilazane

Hexamethyldisilazane ($\text{C}_6\text{H}_{19}\text{NSi}_2$; CAS 999-97-3; synonyms: HMDS) is a clear, colorless liquid that hydrolyzes slowly upon exposure to water.

HMDS forms strong hydrogen bonds and is only weakly basic. Alkali and alkaline-earth metals form salts with it, e.g., $\text{LiN}(\text{SiMe}_3)_2$.

For this inventory the functional unit is 1 kg of pure hexamethyldisilazane. The most important chemical and physical properties of hexamethyldisilazane used in this inventory are given here.

Tab. 8.1 Chemical and physical properties of hexamethyldisilazane (Ullmanns 2004)

Property	Value	Unit	Remarks
Molecular weight	161.4	g mol ⁻¹	
Specific gravity	760	kg m ⁻³	gaseous
Boiling point	125	°C	at normale pressure
Melting point	-78	°C	under pressure

8.3.1 Use

Hexamethyldisilazane is used as a catalyst in condensation reactions of heterocyclic compounds such as in the microwave synthesis of a derivative of xanthine.

It is also used as raw material to produce amidosilanes.

8.4 Raw materials and auxiliaries

Tab. 8.2 shows values for the use of raw materials for the chlorosilane amination. The chlorosilane input is modelled with the dataset "silicon product". As there was no data available for the chlorosilane amination, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the chlorosilane amination.

Tab. 8.2 Consumption of raw materials and auxilliaries for the chlorosilane amination

Input	Values	Remarks
Chlorosilane (kg)	1	Functional unit
Ammonium (kg)	0.236	Calculated with a yield of 95 %
Cooling water (m3)	0.028	Estimation

8.5 Energy

There was no information available on the amount of energy used for the production process. In order not to neglect the process energy demand those values were approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg⁻¹) were used as approximation for the energy consumption of the chlorosilane amination. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used. A summary of the values used is given in Tab. 8.3.

Tab. 8.3 Energy consumption for the chlorosilane amination

Input	per kg ammonium
Heat, natural gas burned in industrial furnace > 100 kW	2.35
Electricity (kWh)	0.391

8.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train. Tab. 8.4 summarises the total transport amounts for the chlorosilane amination.

Tab. 8.4 Total transport amounts for the chlorosilane amination

tkm kg ⁻¹ chlorosilane	lorry	train
Total transports	1.236E-01	7.417E-01

8.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of chlorosilane amination plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

8.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the chlorosilane amination. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input.

Tab. 8.5 Process emissions to air from the chlorosilane amination

Output	Value	Remarks
Waste heat (MJ)	1.41	calculated from electricity input
Chlorosilane, to air (kg)	2E-03	estimated as 0.2 % of the input
Ammonium, to air (kg)	4.72E-04	estimated as 0.2 % of the input
CO ₂ , fossil, to air (kg)	5.28E-02	from water treatment

8.9 Emissions to water

The remaining amount of unreacted chlorosilane and ammonium (4.8 % of the input) is assumed to leave the production process with the waste water. Further it was assumed that this wastewater is treated in a internal wastewater plant. For the ammonia within the waste water a removal efficiency of 50% was assumed. The nitrogen emitted to air (N₂). The removal efficiency for chlorosilane is assumed with 90%. The carbon contained in the removed chlorosilane is accounted as CO₂ emissions to air. The remaining in the sewage sludge is neglected. The remaining chlorosilane is calculated as silicon and chloride emission.

The value AOX used in this inventory are calculated from the amount of chloride in the treated waste water.

Tab. 8.6 Process emissions to water from the chlorosilane amination

Output	Value	Remarks
silicon (kg)	1.24E-02	calculated from water emissions
chloride (kg)	1.56E-02	calculated from water emissions
ammonium (kg)	5.67E-03	calculated from mass balance
AOX (kg)	1.56E-02	calculated from water emissions

8.10 Allocation of ammonium chloride and hexamethyldisilazane

Within the chlorosilane amination, ammonium chloride and hexamethyldisilazane are obtained as co-products. A stoichiometric allocation of the two products is made. The used allocation factors are given in Tab. 8.7.

Tab. 8.7 Allocation of ammonium chloride and hexamethyldisilazane from the chlorosilane amination

Process	Chlorosilane amination, GLO		
	Unit	per kg chlorosilane	Allocation factor (stoichiometric)
Ammonium chloride from chlorosilane amination, at plant, GLO	Kg	0.486	39.7
hexamethyldisilazane, at plant, GLO	kg	0.738	60.3

8.11 Life cycle inventory of chlorosilane amination and data quality considerations

Tab. 8.8 shows the life cycle inventory and the data quality indicators for the chlorosilane amination. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 8.8 Unit process raw data for the chlorosilane amination

401	InputGroup	OutputGroup	Name	Location	InfrastructureProcesses	Unit	chlorosilane amination	UncertaintyType	StandardDeviation95 %	GeneralComment	ammonium chloride from chlorosilane, at plant	hexamethyldisilazane, at plant
662			Location				GLO				GLO	GLO
493			InfrastructureProcess				0				0	0
403			Unit				kg				kg	kg
allocated products resource, in water technosphere	-	2	ammonium chloride from chlorosilane, at plant	GLO	0	kg	4.86E-1				100	0
	-	2	hexamethyldisilazane, at plant	GLO	0	kg	7.38E-1				0	100
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.82E-2	1	1.30	(4,5,na,na,na,na); Estimation	39.700	60.300
	5	-	silicone product, at plant	RER	0	kg	1.00E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation	39.700	60.300
	5	-	ammonia, liquid, at regional storehouse	RER	0	kg	2.36E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation	39.700	60.300
	5	-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	2.35E+0	1	1.30	(4,5,na,na,na,na); Estimation	39.700	60.300
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.91E-1	1	1.30	(4,5,na,na,na,na); Estimation	39.700	60.300
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.24E-1	1	2.09	(4,5,na,na,na,na); Standard distance 100km	39.700	60.300
	5	-	transport, freight, rail	RER	0	tkm	7.42E-1	1	2.09	(4,5,na,na,na,na); Standard distance 600km	39.700	60.300
	5	-	chemical plant, organics	RER	1	unit	4.70E-10	1	3.09	(4,5,na,na,na,na); Estimation	39.700	60.300
emission air, high population density	-	4	Heat, waste	-	-	MJ	1.41E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input	39.700	60.300
	-	4	Chlorosilane, trimethyl-	-	-	kg	2.00E-3	1	1.62	(4,5,na,na,na,na); Estimation	39.700	60.300
	-	4	Ammonia	-	-	kg	4.72E-4	1	1.37	(4,5,na,na,na,na); Estimation	39.700	60.300
	-	4	Carbon dioxide, fossil	-	-	kg	5.28E-2	1	1.30	(4,5,na,na,na,na); Estimation	39.700	60.300
emission water, river	-	4	Silicon	-	-	kg	1.24E-2	1	1.30	(4,5,na,na,na,na); Estimation	39.700	60.300
	-	4	Chloride	-	-	kg	1.56E-2	1	3.09	(4,5,na,na,na,na); Estimation	39.700	60.300
	-	4	Ammonium, ion	-	-	kg	5.67E-3	1	1.62	(4,5,na,na,na,na); Estimation	39.700	60.300
	-	4	AOX, Adsorbable Organic Halogen as Cl	-	-	kg	1.56E-2	1	1.62	(4,5,na,na,na,na); from water emissions	39.700	60.300

8.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 8.9 shows selected LCI results and the cumulative energy demand for the chlorosilane amination.

Tab. 8.9 Selected LCI results and the cumulative energy demand for the chlorosilane amination

Name		hexamethyldisilazane, at plant		ammonium chloride from chlorosilane, at plant
Location	Unit	GLO	GLO	
Infrastructure		kg	kg	
		0	0	
LCIA results				
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	55.2	55.2
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	6.1	6.1
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	2.4	2.4
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	4.3	4.3
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1	0.1
selected LCI results				
land occupation	resource	m2a	2.5E-1	2.5E-1
CO ₂ , fossil	air	kg	2.8E+0	2.8E+0
NM VOC	air	kg	4.0E-3	4.0E-3
nitrogen oxides	air	kg	6.0E-3	6.0E-3
sulphur dioxide	air	kg	6.8E-3	6.8E-3
particulates, <2.5 um	air	kg	6.0E-4	6.0E-4
BOD	water	kg	3.4E-3	3.4E-3
cadmium	soil	kg	1.0E-9	1.0E-9

8.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	chlorosilane amination
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	5
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Chlorosilane amination including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Chlorsilanaminierung
	491	Synonyms	
	492	GeneralComment	The multioutput-process "Chlorosilane amination" delivers the co-products ammonium chloride and hexamethyldisilazane. The allocation is based on stoichiometric calculation.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	inorganics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Anorganika
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications.
Geography	663	Text	The inventory is modelled for the world..
Technology	692	Text	Amination of chlorosilane
Representativeness	722	Percent	100
	724	ProductionVolume	na
	725	SamplingProcedure	Literature data
	726	Extrapolations	Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

8.14 References

- Frischknecht et al. 2007 Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.

- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>
- Ullmanns 2004 Rösch, L., John, P., Reitmeier, R.: Silicon Compounds, Organic. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

9 Dimethyl acetamide

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

9.1 Introduction

Dimethylacetamide (C_4H_9NO ; CAS 127-19-5; synonyms: N,N-dimethylacetamide, DMA, DMAc, acetic acid-dimethylamide, acetyldimethylamine) is a colorless liquid which is soluble in a wide variety of solvents

For this inventory the functional unit is 1 kg of highly pure dimethylacetamide. The most important chemical and physical properties of dimethylacetamide used in this inventory are given here.

Tab. 9.1 Chemical and physical properties of Dimethylacetamide (Ullmanns 2004)

Property	Value	Unit
Molecular weight	87.1	$g\ mol^{-1}$
Specific gravity	943	$kg\ m^{-3}$
Boiling point	-165.5	$^{\circ}C$
Melting point	-20	$^{\circ}C$

9.1.1 Reserves and Resources

9.1.2 Use

Dimethylacetamide is an excellent solvent and often acts as a catalyst in halogenation, cyclization, and alkylation reactions.

Dimethylacetamide is used in the spinning of polyacrylonitrile and as a solvent in the processing of other polymers.

Dimethylacetamide lithium chelate complexes intercalate cationic sites in layered silicates.

9.2 Production Technologies (Ullmanns 2004)

N,N-Dimethylacetamide is prepared by reaction of dimethylamine with acetic acid.



Other processes are based on acetic anhydride or acetate esters. Heating dimethylamine acetate with or without a catalyst affords Dimethylacetamide. Reaction of dimethylamine with acetate esters requires a catalyst: sodium methoxide is typically used.

9.3 System Characterisation

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg dimethylacetamide, at plant, global**. For this study the production of dimethylacetamide by means of the reaction between dimethylamine and acetic acid is considered. All data in the present report are referred to 1 kg dimethylacetamide 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage

and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

9.4 Raw materials and auxiliaries

As there was no data available for the production of dimethylacetamide, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from a large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the dimethylacetamide production.

Tab. 9.2 Consumption of raw materials and auxiliaries for the dimethylacetamide production

Input	Values	Remarks
Dimethylamine (kg)	0.544	Calculated with a yield of 95 %
Acetic acid (kg)	0.726	Calculated with a yield of 95 %
Cooling water (kg)	24	Estimation

9.5 Energy

There was no information available on the amount of energy used for the production process. In order not to neglect the process energy demand those values were approximated with data from a large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg-1) were used as approximation for the energy consumption of the dimethylacetamide production. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used. A summary of the values used is given in Tab. 9.3.

Tab. 9.3 Energy consumption for the dimethylacetamide production

Input	per kg dimethylacetamide
Heat, natural gas burned in industrial furnace > 100 kW	2
Electricity (kWh)	0.333

9.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train. Tab. 9.4 summarises the total transport amounts for the dimethylacetamide production.

Tab. 9.4 Total transport amounts for the dimethylacetamide production

tkm kg ⁻¹ dimethylacetamide	lorry	train
Total transports	1.27E-01	7.62E-01

9.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of dimethylacetamide production plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

9.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of dimethylacetamide. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The assumed values for the emissions to air are given in Tab. 9.5.

Tab. 9.5 Process emissions to air from the production of 1 kg dimethylacetamide

Output	Value	Remarks
Dimethyl amine, to air (kg)	1.09E-3	Estimated as 0.2 % of input
Acetic acid, to air (kg)	1.45E-3	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	9.2E-2	from waster water treatment
Waste heat (MJ)	1.1988	calculated from electricity input

9.9 Emissions to water

The remaining amount of unreacted dimethyl amine and acetic acid (4.8 % of the input) is assumed to leave the production process with the wastewater. Further it is assumed that this wastewater is treated in a internal wastewater plant. The removal efficiency for dimethyl amine and acetic acid is assumed with 90%. The carbon contained in the removed dimethyl amine and acetic acid is accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory are calculated from the amount of dimethyl amine and acetic acid in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used.

Tab. 9.6 Process emissions to water from the production 1 kg dimethylacetamide

Output	Value	Remarks
Dimethyl amine, to water (kg)	2.61E-03	calculated from mass balance
Acetic acid, to water (kg)	3.48E-03	calculated from mass balance
COD, BOD (kg)	8.46E-03	calculated from water emissions
TOC, DOC (kg)	2.79E-03	calculated from water emissions

9.10 Life cycle inventory of dimethylacetamide production and data quality considerations

Tab. 9.7 shows the life cycle inventory and the data quality indicators for the dimethylacetamide production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 9.7 Unit process raw data for the dimethylacetamide production

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	dimethylacetamide, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
401										
662			Location				GLO			
493			InfrastructureProcess				0			
403			Unit				kg			
resource, in water	-	0	dimethylacetamide, at plant	GLO	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	dimethylamine, at plant	RER	0	kg	5.44E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	acetic acid, 98% in H2O, at plant	RER	0	kg	7.26E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	2.00E+0	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.33E-1	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.27E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	7.62E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	1.20E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input
	-	4	Dimethylamine	-	-	kg	1.09E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Acetic acid	-	-	kg	1.45E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	9.20E-2	1	1.30	(4,5,na,na,na,na); Estimation
emission water, river	-	4	Dimethylamine	-	-	kg	2.61E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Acetic acid	-	-	kg	3.48E-3	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	8.46E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	8.46E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	2.79E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	2.79E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

9.11 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 9.8 shows selected LCI results and the cumulative energy demand for the dimethylacetamide production.

Tab. 9.8 Selected LCI results and the cumulative energy demand for dimethylacetamide production

Name		dimethylacetamide, at plant	
Location	Unit	GLO	
Infrastructure		kg	0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	61.8
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	7.7
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	1.0
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	1.0
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1
selected LCI results			
land occupation	resource	m2a	2.5E-1
CO2, fossil	air	kg	2.2E+0
NM VOC	air	kg	1.1E-2
nitrogen oxides	air	kg	4.0E-3
sulphur dioxide	air	kg	5.2E-3
particulates, <2.5 um	air	kg	4.5E-4
BOD	water	kg	1.6E-2
cadmium	soil	kg	1.2E-9

9.12 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	dimethylacetamide, at plant
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of N,N-dimethylacetamide including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Dimethylazetamid, ab Werk
	491	Synonyms	DMA//DMAc//acetic acid-dimethylamide//acetyldimethylamine
	492	GeneralComment	The process "N,N-dimethylacetamide, at plant, GLO" is modelled for the production of N,N-dimethylacetamide from dimethylamine and acetic acid in the world. Raw materials are modelled with a stoichiometric calculation. Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C4H9NO
	501	StatisticalClassification	
	502	CASNumber	127-19-5
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	Amination of acetic acid.
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

9.13 References

- Frischknecht et al. 2007 Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirsch R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.
- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>
- Ullmanns 2004 Cheung, Hosea: Acetic acid. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebaum D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

10 Dimethylamine borane

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

10.1 Introduction

Dimethylamine borane ((CH₃)₂NHBH₃; CAS-No. 236-313-6) is a white solid. Most amine boranes are, stable to air and moisture. Primary and secondary amine boranes are, however, thermally unstable above 70 °C. Aliphatic amine boranes are soluble in a variety of protic and nonprotic solvents, in contrast to NaBH₄. Hydrolysis is slow under neutral conditions and increases in rate as the pH is lowered. For this inventory the functional unit is 1 kg of highly pure dimethylamine borane. The most important chemical and physical properties of dimethylamine borane used in this inventory are given here.

Tab. 10.1 Chemical and physical properties of dimethylamine borane (Ullmanns 2004)

Property	Value	Unit	Remarks
Molecular weight	58.92	g mol ⁻¹	
Specific gravity	730	kg m ⁻³	
Boiling point	61	°C	at normale pressure
Melting point	33-36	°C	

10.1.1 Reserves and Resources

Both (CH₃)₂NH·BH₃ and tert-C₄H₉NH₂·BH₃ are available in up to tonnage quantities, whereas smaller quantities of (CH₃)₃N·BH₃, (C₂H₅)₃N·BH₃, (C₂H₅)₂NH·BH₃, pyridine borane, and morpholine borane are commercially available (Callery Chemical Co., Bayer AG).

10.1.2 Use

The largest quantities of amine boranes are consumed in the electroless plating of various metals, primarily nickel. Dimethylamine borane is the preferred reductant in North America, whereas diethylamine borane is used in Europe. The nickel thus plated by chemical reduction is amorphous and contains varying amounts of boron depending on the bath formulation. This plating mainly provides corrosion protection and high solderability to electrical parts. Amine boranes have also been used to plate copper, gold, palladium, cobalt, and other metals.

Amine boranes are finding expanding use as mild reducing agents, comparable to cyanoborohydride in reducing power, but considerably less toxic. Amines of greater base strength give amine boranes of diminished reducing power; for example, (CH₃)₃N·BH₃ is a relatively weak reducer, whereas NH₃·BH₃ is a moderately strong reducer.

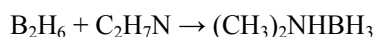
It has been shown that NH₃·BH₃ and primary and secondary amine boranes are efficient stereoselective reducing agents for aldehydes and ketones in a variety of solvents. Additionally, these reagents selectively reduce aldehydes in the presence of ketones or esters.

Chiral amine boranes have been useful in the asymmetric reduction of prochiral ketones. In the presence of BF₃ etherate, borane complexes of sodium salts of amino acids gave enantiomeric excesses up to 62 %.

10.2 Production Technologies (Ullmanns 2004)

Preparation may be by reaction of diborane or dimethyl sulfide borane with an amine, or by reacting an ammonium salt of the desired amine with NaBH_4 in an ether. The dimethyl sulfide borane route is the most convenient method of preparing small quantities of amine boranes.

Diborane route:



10.3 System Characterisation

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg dimethylamine borane, at plant, global**. For this study the production of dimethylamine borane by means of the reaction between diborane and dimethyl amine considered. All data in the present report are referred to 1 kg dimethylamine borane 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

10.4 Raw materials and auxiliaries

As there was no data available for the production of dimethylamine borane, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the dimethylamine borane production.

Tab. 10.2 Consumption of raw materials and auxiliaries for the dimethylamine borane production

Input	Values	Remarks
Diborane (kg)	0.25	Calculated with a yield of 95 %
Dimethyl amine (kg)	0.803	Calculated with a yield of 95 %
Cooling water (kg)	24	Estimation

10.5 Energy

There was no information available on the amount of energy used for the production process. In order not to neglect the process energy demand those values were approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg⁻¹) were used as approximation for the energy consumption of the dimethylamine borane production. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used. A summary of the values used is given in Tab. 10.3.

Tab. 10.3 Energy consumption for the dimethylamine borane production

Input	per kg dimethylamine borane
Heat, natural gas burned in industrial furnace > 100 kW	2
Electricity (kWh)	0.333

10.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train.

Tab. 10.4 Total transport amounts for the dimethylamine borane production

tkm kg ⁻¹ dimethylamine borane	lorry	train
Total transports	1.05E-01	6.32E-01

10.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of dimethylamine borane production plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

10.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of dimethylamine borane. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The diborane is calculated as boron emission. The assumed values for the emissions to air are given in Tab. 10.5.

Tab. 10.5 Process emissions to air from the production of 1 kg dimethylamine borane

Output	Value	Remarks
Boron, to air (kg)	8.86E-5	Estimated as 0.2 % of input
Dimethyl amine, to air (kg)	1.61E-3	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	6.78E-2	from waster water treatment
Waste heat (MJ)	1.2	calculated from electricity input

10.9 Emissions to water

The remaining amount of unreacted diborane and dimethyl amine (4.8 % of the input) is assumed to leave the production process with the wastewater. Further it is assumed that this wastewater is treated in a internal wastewater plant. The removal efficiency for dimethyl amine is assumed with 90%. The carbon contained in the removed dimethyl amine is accounted as CO₂ emissions to air. The diborane is calculated as boron emission.

The values for COD, BOD, TOC and DOC used in this inventory are calculated from the amount of

dimethyl amine in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used.

Tab. 10.6 Process emissions to water from the production 1 kg dimethylamine borane

Output	Value	Remarks
Boron, to water (kg)	2.13E-03	calculated from mass balance
Dimethyl amine, to water (kg)	3.85E-03	calculated from mass balance
COD, BOD (kg)	7.22E-03	calculated from water emissions
TOC, DOC (kg)	2.05E-03	calculated from water emissions

10.10 Life cycle inventory of dimethylamine borane production and data quality considerations

Tab. 10.7 shows the life cycle inventory and the data quality indicators for the dimethylamine borane production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 10.7 Unit process raw data for the dimethylamine borane production

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	dimethylamine borane, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							GLO 0 kg			
resource, in water	-	0	dimethylamine borane, at plant	GLO	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	diborane, at plant	GLO	0	kg	2.50E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	dimethylamine, at plant	RER	0	kg	8.03E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	2.00E+0	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.33E-1	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.05E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	6.32E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Heat, waste	-	-	MJ	1.20E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input
emission air, high population density	-	4	Boron	-	-	kg	8.86E-5	1	5.10	(4,5,na,na,na,na); Estimation
	-	4	Dimethylamine	-	-	kg	1.61E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	6.78E-2	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Boron	-	-	kg	2.13E-3	1	5.10	(4,5,na,na,na,na); Estimation
emission water, river	-	4	Dimethylamine	-	-	kg	3.85E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	7.22E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	7.22E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	2.05E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	2.05E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

10.11 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 10.8 shows selected LCI results and the cumulative energy demand for the dimethylamine borane production.

Tab. 10.8 Selected LCI results and the cumulative energy demand for dimethylamine borane production

Name		dimethylamine borane, at plant	
Location	Unit	Unit	GLO kg 0
Infrastructure			
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	47.1
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	7.3
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	1.1
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	1.0
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1
selected LCI results			
land occupation	resource	m2a	2.8E-1
CO ₂ , fossil	air	kg	2.1E+0
NM VOC	air	kg	4.5E-3
nitrogen oxides	air	kg	4.3E-3
sulphur dioxide	air	kg	1.2E-2
particulates, <2.5 um	air	kg	5.0E-4
BOD	water	kg	1.5E-2
cadmium	soil	kg	1.3E-9

10.12 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	dimethylamine borane, at plant
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of dimethylamine borane including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Dimethylaminboran, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "dimethylamine borane, at plant, GLO" is modelled for the production of dimethylamine borane from diborane and dimethyl amine in the world. Raw materials are modelled with a stoichiometric calculation. Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	(CH ₃) ₂ NHBH ₃
	501	StatisticalClassification	
	502	CASNumber	74-94-2
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	Amination of diborane.
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

10.13 References

Frischknecht et al. 2007 Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Over-

view and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.

Ullmanns 2004

Brotherton, R.: Boron Compounds. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

11 Dipropylene glycol monomethyl ether

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Niels Jungbluth, ESU Services, Uster
 Last changes: 2007

11.1 Introduction

Dipropylene glycol monomethyl ether ($\text{CH}_3(\text{OC}_3\text{H}_6)_2\text{OH}$; CAS-No. 34590-94-8; synonyms: DPM, Methoxy propoxy propanol) is a colorless liquid with a mild pleasant odor.

For this inventory the functional unit is 1 kg of pure dipropylene glycol monomethyl ether. The most important chemical and physical properties of dipropylene glycol monomethyl ether used in this inventory are given here.

Tab. 11.1 Chemical and physical properties of dipropylene glycol monomethyl ether (Ullmanns 2004)

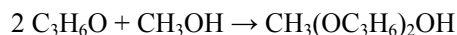
Property	Value	Unit
Molecular weight	148.2	g mol^{-1}
Specific gravity	950	kg m^{-3}
Boiling point	190	$^{\circ}\text{C}$
Melting point	-83	$^{\circ}\text{C}$

11.1.1 Use

Dipropylene glycol monomethyl ether is used as a solvent for coating systems, as additive in paint removers, as component in hydraulic fluids, and as starting material for the production of esters used as plasticizers. It reacts with acids to form esters, oxidizing agents to form aldehydes or carboxylic acids, alkali metals to form alcoholates or aldehydes to form acetals. These properties support the use in agricultural, cosmetic, electronic, ink, textile and adhesive products.

11.2 Production Technologies (Ullmanns 2004)

Dipropylene glycol monomethyl ether is produced from the reaction of propylene oxide with methanol.



11.3 System Characterisation

In this study the production dipropylene glycol monomethyl ether by the reaction of propylene oxide with methanol is considered. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

11.4 Raw materials and auxiliaries

As there was no data available for the production of dipropylene glycol monomethyl ether, the values

for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the dipropylene glycol monomethyl ether production. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the dipropylene glycol monomethyl ether production.

Tab. 11.2 Consumption of raw materials and auxiliaries for the dipropylene glycol monomethyl ether production

Input	Values	Remarks
Propylene oxide (kg)	0.83	Calculated with a yield of 95 %
Methanol (kg)	0.228	Calculated with a yield of 95 %
Cooling water (kg)	24	Estimation

11.5 Energy

There was no information available on the amount of energy used for the production process. In order not to neglect the process energy demand those values were approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg-1) were used as approximation for the energy consumption of the dipropylene glycol monomethyl ether production. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used. A summary of the values used is given in Tab. 11.3.

Tab. 11.3 Energy consumption for the dipropylene glycol monomethyl ether production

Input	per kg dipropylene glycol monomethyl ether
Heat, natural gas burned in industrial furnace > 100 kW	2
Electricity (kWh)	0.333

11.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train. Tab. 11.4 summarises the total transport amounts for the dipropylene glycol monomethyl ether production.

Tab. 11.4 Total transport amounts for the dipropylene glycol monomethyl ether production

tkm kg ⁻¹ dipropylene glycol monomethyl ether	lorry	train
Total transports	1.05E-01	6.32E-01

11.7 Infrastructure and land use

No information was available about infrastructure and land-use of dipropylene glycol monomethyl ether production plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of

50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

11.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of dipropylene glycol monomethyl ether. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The assumed values for the emissions to air are given in Tab. 11.5.

Tab. 11.5 Process emissions to air from the production of 1 kg dipropylene glycol monomethyl ether

Output	Value	Remarks
Propylene oxide, to air (kg)	1.65E-3	Estimated as 0.2 % of input
Methanol, to air (kg)	4.55E-4	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	9.46E-2	from waster water treatment
Waste heat (MJ)	1.2	calculated from electricity input

11.9 Emissions to water

The remaining amount of unreacted propylene oxide and methanol (4.8 % of the input) is assumed to leave the production process with the wastewater. Further it is assumed that this wastewater is treated in a internal wastewater plant. The removal efficiency for propylene oxide and methanol is assumed with 90%. The carbon contained in the removed propylene oxide and methanol is accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory are calculated from the amount of propylene oxide and methanol in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used.

Tab. 11.6 Process emissions to water from the production 1 kg dipropylene glycol monomethyl ether

Output	Value	Remarks
Propylene oxide, to water (kg)	3.96E-03	calculated from mass balance
Methanol, to water (kg)	1.09E-03	calculated from mass balance
COD, BOD (kg)	1.05E-02	calculated from water emissions
TOC, DOC (kg)	3.16E-03	calculated from water emissions

11.10 Life cycle inventory of dipropylene glycol monomethyl ether production and data quality considerations

Tab. 11.7 shows the life cycle inventory and the data quality indicators for the dipropylene glycol monomethyl ether production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 11.7 Unit process raw data for the production of dipropylene glycol monomethyl ether

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	dipropylene glycol monomethyl ether, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
401										
662			Location				RER			
493			InfrastructureProcess				0			
403			Unit				kg			
resource, in water	-	0	dipropylene glycol monomethyl ether, at plant	RER	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	propylene oxide, liquid, at plant	RER	0	kg	8.25E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	methanol, at plant	GLO	0	kg	2.28E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	2.00E+0	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.33E-1	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.05E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	6.32E-1	1	2.09	(4,5,na,na,na,na); Standard distances
emission air, high population density	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Heat, waste	-	-	MJ	1.20E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input
	-	4	Propylene oxide	-	-	kg	1.65E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Methanol	-	-	kg	4.55E-4	1	1.62	(4,5,na,na,na,na); Estimation
emission water, river	-	4	Carbon dioxide, fossil	-	-	kg	9.46E-2	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Propylene oxide	-	-	kg	3.96E-3	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Methanol	-	-	kg	1.09E-3	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	1.05E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	1.05E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	3.16E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	3.16E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

11.11 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 11.8 shows selected LCI results and the cumulative energy demand for the production of dipropylene glycol monomethyl ether.

Tab. 11.8 Selected LCI results and the cumulative energy demand for the dipropylene glycol monomethyl ether production

Name		dipropylene glycol monomethyl ether, at plant	
Location	Unit	Unit	RER
Infrastructure			kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	92.1
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	22.2
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	2.2
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	2.6
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.4
selected LCI results			
land occupation	resource	m2a	4.5E-1
CO ₂ , fossil	air	kg	4.1E+0
NM VOC	air	kg	6.3E-3
nitrogen oxides	air	kg	7.7E-3
sulphur dioxide	air	kg	1.1E-2
particulates, <2.5 um	air	kg	9.3E-4
BOD	water	kg	1.1E-1
cadmium	soil	kg	3.2E-9

11.12 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	dipropylene glycol monomethyl ether, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of dipropylene glycol monomethyl ether including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Dipropylenglykolmonomethylether, ab Werk
	491	Synonyms	DPM/Methoxy propoxy propanol
	492	GeneralComment	The process " dipropylene glycol monomethyl ether, at plant, RER" is modelled for the production of dipropylene glycol monomethyl ether from propylene oxide and methanol in Europe. Raw materials are modelled with a stoichiometric calculation. Energy consumptions, emissions, infrastructure are estimated. Transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	CH3(OC3H6)2OH
	501	StatisticalClassification	
	502	CASNumber	34590-94-8
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Etherification of propylene oxide
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Energy consumptions, emissions, infrastructure are estimated. Transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	24
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

11.13 References

- Frischknecht et al. 2007 Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirsch R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.
- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>
- Ullmanns 2004 Kahlich, D., Wiechern, U., Lindner, J.: Propylene Oxide. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

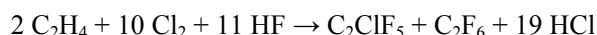
12 Ethylene chlorofluorination

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

Ethylene chlorofluorination is one of the most important processes to produce monochloropentafluoroethane.

12.1 Production Technologies (Ullmanns 2004)

In the Montedison chlorofluorination process, reaction of ethylene with chlorine and hydrogen fluoride takes place in a single step in a fluidized-bed reactor. A suitable catalyst is a combination of aluminum chloride and other metals.



The hydrochloric acid is neutralized with sodium hydroxide.

Commercial production of chlorofluoroalkanes is also possible by the electrochemical fluorination process developed by Phillips Petroleum.

12.2 Purification and Analysis

Impurities are usually removed from organic fluorine compounds by fractional distillation, fractional crystallization, or chromatographic methods. This does not apply to fluoropolymers and high-boiling perfluorinated oils, which require special measures, i.e., the use of extremely pure starting materials.

Quantitative determination of fluorine is possible in most cases by combustion and subsequent analysis of the hydrogen fluoride generated. Wet chemical methods are used to determine fluoride ions.

Because of the high volatility of organic fluorine compounds, purity can be readily determined by gas chromatography. ^{19}F -Nuclear magnetic resonance spectroscopy is a valuable tool for determining the structure of organic fluorine compounds. Structure determinations, even of mixtures, are often easier using this method than with ^1H -NMR spectra due to the larger chemical shifts of ^{19}F -NMR spectra. The ^{19}F signals can be integrated and used for quantitative analysis.

Chlorofluoroalkanes (and also the alternative HCFCs and HFCs) produced on an industrial scale are subject to stringent standards. Impurities must not exceed the following limits:

Tab. 12.1 Limits for impurities (Ullmanns 2004)

Impurity	vol. %
acids	0
moisture	< 0.001
higher-boiling fractions	< 0.05
other gases	2

12.3 System Characterisation

In this study the chlorofluorination of ethylene is considered. The functional unit is 1 kg ethylene. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product

are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers. Within the chlorofluorination of ethylene, monochloropentafluoroethane and hexafluoroethane are derived as co-products. Therefore an allocation between the two products was made. The following processes were modelled:

- Ethylene chlorofluorination, GLO. Multi output process with monochloropentafluoroethane, at plant, GLO and hexafluoroethane, at plant, GLO as allocated products

12.4 Monochloropentafluoroethane

Monochloropentafluoroethane (C₂ClF₅; CAS 76-15-3; synonyms: freon 115, CFC 115) is a colorless gas with slight ether like odor.

The most important chemical and physical properties of monochloropentafluoroethane used in this inventory are given here.

Tab. 12.2 Chemical and physical properties of monochloropentafluoroethane (Ullmanns 2004)

Property	Value	Unit	Remarks
Molecular weight	154.48	g mol ⁻¹	
Specific gravity	1291	kg m ⁻³	liquide
Boiling point	-38	°C	at normale pressure
Melting point	-106	°C	under pressure

Hexafluoroethane (C₂F₆; CAS 76-16-4; synonyms: Freon R16, R116) is a colorless gas with slight ether like odor.

The most important chemical and physical properties of hexafluoroethane used in this inventory are given here.

Tab. 12.3 Chemical and physical properties of hexafluoroethane (Ullmanns 2004)

Property	Value	Unit	Remarks
Molecular weight	138.01	g mol ⁻¹	
Specific gravity	1607	kg m ⁻³	at -78 °C
Boiling point	-78.2	°C	at normale pressure
Melting point	-100.6	°C	

12.4.1 Use

Up to the ban in the USA in 1978 chlorofluoroalkanes had been used mainly as aerosol propellants and as spraying and foam blowing agents (R 11, R 12, R 114). Further important applications up to 1996 had been in the area of refrigerants, where R 11, R 12, R 13, R 22, R 113, R 114, R 115, and the chlorine-free compounds R 23 and RC 318 were preferred. Of the drop-in alternatives to CFCs HCFC 22 is the most important compound, used as refrigerant (annual production in 1995 243 × 10³ t), followed by HCFC 141 b (113 × 10³ t in 1995) and HCFC 142 b (38 × 10³ t in 1995) used as blowing agents for closed cell foam.

Chlorofluoroalkanes, especially R 11 and R 113, were also employed as solvents and degreasing and cleaning agents for textiles; HCFC 22, CFC 113, and HCFC 142 b will be important intermediates for the production of fluoroolefins also in the future.

Higher molecular mass perchlorofluoroalkanes are used as oils, greases, and waxes, as lubricants, hydraulic fluids, damping oils, heat-transfer media, impregnating agents, and plasticizers. Oligomers of chlorotrifluoroethylene have achieved special importance in this area.

12.5 Raw materials and auxiliaries

Tab. 12.4 shows values for the use of raw materials of the chlorofluorination of ethylene. As there was no data available for the process, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007). The produced hydrochloric acid is then neutralized with sodium hydroxide.

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the chlorofluorination of ethylene.

Tab. 12.4 Consumption of raw materials and auxiliaries for the chlorofluorination of ethylene

Input	Values	Remarks
Ethylene (kg)	1	Functional unit
Chlorine (kg)	12.5	Calculated with a yield of 95 %
Hydrogen fluoride (kg)	6.48	Calculated with a yield of 95 %
Sodium hydroxide (kg)	12.89	Stoichiometric calculation
Cooling water (kg)	64.68	Estimation

12.6 Energy

There was no information available on the amount of energy used for the chlorofluorination of ethylene. In order not to neglect the process energy demand those values were approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg-1) were used as approximation for the energy consumption of the chlorofluorination of ethylene. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used. A summary of the values used is given in Tab. 12.5.

Tab. 12.5 Energy consumption for the chlorofluorination of ethylene

Input	per kg ethylene
Heat, natural gas burned in industrial furnace > 100 kW	9.9
Electricity (kWh)	1.65

12.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train.

Tab. 12.6 Total transport amounts for the chlorofluorination of ethylene

tkm kg ⁻¹ ethylene	lorry	train
Total transports	4.56	17.97

12.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of ethylene chlorofluorination plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

12.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the ethylene chlorofluorination. The remaining amount of unreacted ethylene (5 % of the input) is assumed to be emitted to air. The ethylene emissions are incinerated to CO₂ with a assumed destruction efficiency of 99 %. The chlorine emissions are estimated to 0.2% of the input.

Tab. 12.7 Process emissions to air from the chlorofluorination of ethylene

Output	Value	Remarks
Waste heat (MJ)	5.94	calculated from electricity input
Ethylene, to air (kg)	5E-04	estimated as 0.05 % of the input
Chlorine, to air (kg)	2.5E-02	estimated as 0.2 % of the input
CO₂, fossil, to air (kg)	1.56E-01	from burning of ethylene

12.10 Emissions to water

The remaining amount of unreacted chlorine (4.8 % of the input) and sodium chloride from the HCl neutralization is assumed to leave the production process with the waste water.

Tab. 12.8 Process emissions to water from the chlorofluorination of ethylene

Output	Value	Remarks
chlorine (kg)	6E-01	calculated from mass balance
sodium (kg)	7.41E-01	calculated from mass balance
chloride (kg)	1.128	calculated from mass balance
AOX (kg)	1.73	calculated from water emissions

12.11 Allocation of monochloropentafluoroethane and hexafluoroethane

Within the chlorofluorination of ethylene monochloropentafluoroethane and hexafluoroethane are obtained as co-products. A stoichiometric allocation of the two products is made. The used allocation factors are given in Tab. 12.9.

Tab. 12.9 Allocation of monochloropentafluoroethane and hexafluoroethane from the chlorofluorination of ethylene

Process	Chlorofluorination of ethylene, GLO		
	Unit	per kg ethylene	Allocation factor (stoichiometric)
monochloropentafluoroethane, at plant, GLO	Kg	2.695	52.7
hexafluoroethane, at plant, GLO	kg	2.415	47.3

12.12 Life cycle inventory of chlorofluorination of ethylene and data quality considerations

Tab. 12.10 shows the life cycle inventory and the data quality indicators for the chlorofluorination of ethylene. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 12.10 Unit process raw data for the chlorofluorination of ethylene

	401	InputGroup	OutputGroup	Name	Location	Infrastructure	Processes	Unit	ethylene chlorofluorination	UncertaintyType	StandardDeviation95%	GeneralComment	monochloropentafluoroethane, at plant	hexafluoroethane, at plant
	662			Location					GLO				GLO	GLO
	493			InfrastructureProcess					0				0	0
	403			Unit					kg				kg	kg
allocated products resource, in water technosphere	-	2	monochloropentafluoroethane, at plant	GLO	0	kg		2.61E+0					100	0
	-	2	hexafluoroethane, at plant	GLO	0	kg		2.34E+0					0	100
	4	-	Water, cooling, unspecified natural origin	-	-	m3		1.19E-1	1	1.30	(4,5,na,na,na,na); Estimation	52.740	47.260	
	5	-	ethylene, average, at plant	RER	0	kg		1.00E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation	52.740	47.260	
	5	-	chlorine, liquid, production mix, at plant	RER	0	kg		1.25E+1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation	52.740	47.260	
	5	-	hydrogen fluoride, at plant	GLO	0	kg		6.48E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation	52.740	47.260	
	5	-	sodium hydroxide, 50% in H2O, production mix, at plant	RER	0	kg		1.29E+1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation	52.740	47.260	
	5	-	disposal, inert waste, 5% water, to inert material landfill	CH	0	kg		2.07E+1	1	1.30	(4,5,na,na,na,na); Estimation	52.740	47.260	
	5	-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ		9.91E+0	1	1.30	(4,5,na,na,na,na); Estimation	52.740	47.260	
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh		1.65E+0	1	1.30	(4,5,na,na,na,na); Estimation	52.740	47.260	
	5	-	transport, lorry >16t, fleet average	RER	0	tkm		4.58E+0	1	2.09	(4,5,na,na,na,na); Standard distance 100km	52.740	47.260	
	5	-	transport, freight, rail	RER	0	tkm		1.80E+1	1	2.09	(4,5,na,na,na,na); Standard distance 600km	52.740	47.260	
	5	-	chemical plant, organics	RER	1	unit		1.98E-9	1	3.09	(4,5,na,na,na,na); Estimation	52.740	47.260	
	emission air, high population density	-	4	Heat, waste	-	-	MJ		5.94E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input	52.740	47.260
		-	4	Ethene	-	-	kg		5.00E-4	1	1.62	(4,5,na,na,na,na); Estimation	52.740	47.260
-		4	Chlorine	-	-	kg		2.50E-2	1	1.62	(4,5,na,na,na,na); Estimation	52.740	47.260	
-		4	Carbon dioxide, fossil	-	-	kg		1.56E-1	1	1.30	(4,5,na,na,na,na); Estimation	52.740	47.260	
emission water, river	-	4	Chlorine	-	-	kg		6.00E-1	1	3.09	(4,5,na,na,na,na); Estimation	52.740	47.260	
	-	4	Sodium, ion	-	-	kg		7.41E-1	1	5.10	(4,5,na,na,na,na); Estimation	52.740	47.260	
	-	4	Chloride	-	-	kg		1.13E+0	1	3.09	(4,5,na,na,na,na); Estimation	52.740	47.260	
	-	4	AOX, Adsorbable Organic Haloqen as Cl	-	-	kg		1.73E+0	1	1.62	(4,5,na,na,na,na); from water emissions	52.740	47.260	

12.13 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 12.11 shows selected LCI results and the cumulative energy demand for the chlorofluorination of ethylene.

Tab. 12.11 Selected LCI results and the cumulative energy demand for the chlorofluorination of ethylene

Name		hexafluorethane, at plant		monochloropentafluoroethane, at plant	
Location	Unit	GLO	kg	GLO	kg
Infrastructure		0		0	
LCIA results					
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	145.1		145.1
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	63.8		63.8
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0		0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	7.0		7.0
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-		-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0		0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	7.7		7.7
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	1.1		1.1
selected LCI results					
land occupation	resource	m2a	1.6E+0		1.6E+0
CO2, fossil	air	kg	1.0E+1		1.0E+1
NM VOC	air	kg	5.2E-3		5.2E-3
nitrogen oxides	air	kg	2.6E-2		2.6E-2
sulphur dioxide	air	kg	1.2E-1		1.2E-1
particulates, <2.5 um	air	kg	3.1E-3		3.1E-3
BOD	water	kg	1.1E-2		1.1E-2
cadmium	soil	kg	1.0E-8		1.0E-8

12.14 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	ethylene chlorofluorination
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformatic	201	Type	5
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Chlorofluorination of ethylene including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Ethylenchlorfluorinierung
	491	Synonyms	
	492	GeneralComment	The multioutput-process "ethylene chlorofluorination" delivers the co-products monochloropentafluoroethane and hexafluoroethane. The allocation is based on stoichiometric calculation.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications.
Geography	663	Text	The inventory is modelled for the world..
Technology	692	Text	Chlorofluorination of ethylene
Representative	722	Percent	100
	724	ProductionVolume	na
	725	SamplingProcedure	Literature data
	726	Extrapolations	Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAn	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

12.15 References

- Frischknecht et al. 2007 Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.
- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>

Ullmanns 2004

Siegemund, G., Schwertfeger, W., Feiring, A., Smart, B., Behr, F., Vogel, H., McKusick, B.: Fluorine Compounds, Organic. Published online: 2002. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

13 Imperial smelting furnace

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, Mischa Classen, EMPA, St. Gallen
 Last changes: 2007

The Imperial smelting furnace is a process from the zinc production and one of the most important processes to produce sodium arsenide.

For zinc production see also Althaus et al. (2004).

13.1 Production Technologies (Ullmanns 2004)

Sodium arsenide is produced as co-product in the zinc production. The pyrometallurgical route is used for mixed zinc/lead concentrates and secondary material and uses the Imperial Smelting Furnace (ISF), a special form of a blast furnace. The electrothermic furnace is used in other parts of the world. Zinc produced in the Imperial Smelting Furnace may contain varying amounts of cadmium, lead, copper, arsenic, antimony and iron and the process uses a refining stage. Zinc from the ISF is refined by reflux distillation in columns containing a large number of refractory trays (New Jersey Distillation). The lower ends of the columns are heated externally by natural gas. The upper ends are not heated and run cool enough to reflux the higher boiling point metals before vapours pass to a condenser. The New Jersey distillation column is also used for secondary zinc materials. Distillation proceeds in two stages; first the separation of zinc and cadmium from lead and then separation of cadmium from zinc. In the first stage, molten zinc is fed into a column where all the cadmium and a high proportion of the zinc is distilled. The mixture is condensed and fed directly to a second column. This column is operated at a slightly lower temperature to distil mainly cadmium, which is condensed as a zinc-cadmium alloy. The alloy is transferred to a cadmium refinery. The metal run-off from the bottom of the second column is high-grade zinc (SHG) of 99.995% purity. The run-off metal from the first stage is zinc with lead, tin, arsenic, iron, antimony and copper impurities. This alloy is cooled to separate lead, which is recycled to the ISF splash condenser and an intermetallic compound of iron, zinc and arsenic, which is recycled to the ISF. The zinc is then treated with sodium to remove residual arsenic and antimony as sodium arsenides and antimonides, which can be separated or also recycled to the ISF. The zinc produced in this way is of a lower grade (GOB), but free of cadmium, and is used mainly for galvanising (IPPC (2001)).

13.2 System Characterisation

In this study the production of sodium arsenide from the pyrometallurgical production of zinc (Imperial Smelting Furnace) is considered. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers. Within the Imperial Smelting Furnace, zinc and sodium arsenide are derived as co-products. Therefore an allocation between the two products was made. The following processes were modelled:

- Imperial smelting furnace, GLO. Multi output process with zinc from Imperial smelting furnace, at plant, GLO and sodium arsenide, at plant, GLO as allocated products

13.3 Sodium arsenide

Sodium arsenide (AsNa_3 ; CAS 12044-25-6, Molecular weight: 143.9 g mol⁻¹) is a mineral, which occurs in nature like other arsenides.

Some arsenides occur have a defined composition, others are probably mixtures. Naturally occurring arsenides include Cu_3As (domeykite), FeAs_2 (löllingite), NiAs_2 (chloanthite), NiAs (niccolite), and CoAs_2 (smaltite). There are many minerals in which sulfur is partially replaced by arsenic.

13.3.1 Use

The *arsenides* themselves have little commercial significance. Those produced are obtained by direct synthesis. The alkali metal *arsenides* are also obtained by reaction in liquid ammonia. However, in this case polyarsenides can form in addition to the simple *arsenides*. The *arsenides* of the alkali metals, alkaline earth metals, zinc, etc., are rapidly decomposed by water and acids with the formation of arsine. They may be regarded as derivatives of arsenic hydride. Most of the heavy metal *arsenides*, whose composition often does not reflect any simple relationship to arsenic hydride, are intermetallic phases and are highly resistant to acids. Gallium and indium *arsenides* are important in the semiconductor industry

13.4 Raw materials and auxiliaries

Tab. 13.1 shows values for the use of raw materials of the Imperial smelting furnace.

Tab. 13.1 Consumption of raw materials and auxiliaries for the Imperial smelting furnace

Input	Ayers et al. 2002 (per kg zinc)	Xiao et al. 2003 (per kg zinc)	This study (per kg concentrate)
Concentrate (kg)	2.526		1
Cooling water (m ³)		0.13	0.0515
Diesel oil (kg) ¹		8E-04	1.35E-02
Gasoline (kg) ¹		1.7E-03	2.89E-02
steam (kg) ²		5.3E-01	0.21
oxygen (kg)		5.4E-01	0.214

1: inventoried as diesel, building machine

13.5 Energy

Tab. 13.2 Energy consumption for the Imperial smelting furnace

Input	IPPC 2001 (per kg zinc)	This study (per kg concentrate)
total coal (kg)	7.85E-04	3.11E-04
natural gas (m ³)	0.16	6.33E-02
electricity (kWh)	0.75	2.97E-01

13.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train. For 75% of the zinc concentrate a transport distance of 5000 km per ship is assumed. Tab. 13.3 summarises the total transport amounts for the Imperial smelting furnace.

Tab. 13.3 Total transport amounts for the Imperial smelting furnace

tkm kg ⁻¹ concentrate	ship	lorry	train
Total transports	3.75	2.561E-02	1.537E-01

13.7 Infrastructure and land use

The infrastructure and land use is included in the dataset "zinc concentrate".

13.8 Emissions to air

The emissions to air can escape the process either as stack emissions or as fugitive emissions depending on the age of the plant and the technology used. Stack emissions are normally monitored continuously or periodically and reported. The main emissions to air from zinc production are sulphur dioxide (SO₂), other sulphur compounds and acid mists; oxides of nitrogen (NO_x) and other nitrogen compounds; metals and their compounds; dust; VOC and dioxins (IPPC (2001)).

Tab. 13.4 Process emissions to air from the Imperial smelting furnace

Emissions to air	CEPMEIP (per kg zinc)	Xiao et al. 2003 (per kg zinc)	EEA 1999 (per kg zinc)	This study (per kg concentrate)
Waste heat (MJ)				1.067
PM < 2.5 (kg)	3E-04			1.19E-04
PM 2.5-10 (kg)	1E-04			3.96E-05
PM > 10 (kg)	1E-04			3.96E-05
Zn (kg)			1E-04	3.96E-05
SO ₂ (kg)		1.1E-03		4.35E-04
Pb (kg)			5E-06	1.98E-06
Hg (kg)		3.47E-05		1.37E-05

1: calculated from the electricity input

13.9 Emissions to water

Main emissions to water are metals and their compounds and materials in suspension are the main pollutants emitted to water. The metals concerned are Zn, Cd, Pb, Hg, Se, Cu, Ni, As, Co and Cr. Other significant substances are fluorides, chlorides and sulphates (IPPC (2001)).

Tab. 13.5 Process emissions to water from the Imperial smelting furnace

Output	Xiao et al. 2003 (per kg zinc)	This study (per kg concentrate)
Hg (kg)	6.9E-07	2.73E-07
Cd (kg)	8E-06	3.17E-06
As (kg)	1.9E-06	7.52E-07
Pb (kg)	1.38E-04	5.46E-05
Zn (kg)	2.29E-04	9.07E-05
Fluoride (kg)	1.49E-04	5.9E-05

13.10 Wastes

The ISF or direct smelting furnaces are significant sources of solid slag. This slag has been subjected to high temperatures and generally contains low levels of leachable metals, consequently may then be used in construction. Solid residues also arise as the result of the treatment of liquid effluents. The main waste stream is gypsum waste (CaSO₄) and metal hydroxides that are produced at the wastewater neutralisation plant. These wastes are considered to be a crossmedia effect of these treatment techniques but many are recycled to pyrometallurgical process to recover the metals. Dust or sludge from the treatment of gases are used as raw materials for the production of other metals such as Ge, Ga, In and As etc or can be returned to a smelter or into the leach circuit for the recovery of lead and zinc. Hg/Se residues arise at the pre-treatment of mercury or selenium streams from the gas cleaning stage. This solid waste stream amounts to approximately 40 - 120 t/y in a typical plant. Hg and Se can be recovered from these residues depending on the market for these metals (IPPC (2001)).

Tab. 13.6 Wastes from the Imperial smelting furnace

Output	Xiao et al. 2003 (per kg zinc)	This study (per kg concentrate)
disposal, inert waste, 5% water, to inert material landfill	0.582	0.23

13.11 Allocation of zinc and sodium arsenide

Within the Imperial smelting furnace, zinc and sodium arsenide are obtained as co-products. As allocation factor the weight of the two products was used. The used allocation factors are given in Tab. 13.7.

Ayers et al. (2002) reports a value of 0.396 kg zinc per kg concentrate. Ullmanns (2004) reports a content of 0.8% As in the concentrate.

Tab. 13.7 Allocation of zinc and sodium arsenide from the Imperial smelting furnace

Process	Imperial smelting furnace, GLO		
	Unit	per kg zinc concentrate	Allocation factor
Zinc, from Imperial smelting furnace, at plant, GLO	kg	0.396	96.265
Sodium arsenide, from Imperial smelting furnace, at plant, GLO	kg	0.0154	3.735

13.12 Life cycle inventory of ethylene hydration and data quality considerations

Tab. 13.8 shows the life cycle inventory and the data quality indicators for the Imperial smelting furnace. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 13.8 Unit process raw data for the production of ethanol from ethylene

401	InputGroup OutputGroup	Name	Location	InfrastructureProcess	s	Unit	Imperial smelting furnace	UncertaintyType	StandardDeviation95 %	GeneralComment	zinc , from Imperial smelting furnace	sodium arsenide, at plant
662		Location					GLO				GLO	GLO
493		InfrastructureProcess					0				0	0
403		Unit					kg				kg	kg
allocated products resource, in water technosphere	- 2	zinc , from Imperial smelting furnace	GLO	0	kg	3.96E-1					100	0
	- 2	sodium arsenide, at plant	GLO	0	kg	1.54E-2					0	100
	4	- Water, cooling, unspecified natural origin	-	-	m3	5.15E-2	1	1.33	(3,3,2,1,3,5); Xiao 2003		96.265	3.735
	5	- zinc concentrate, at beneficiation	GLO	0	kg	1.00E+0	1	1.33	(3,3,2,1,3,5); Ayres 2002		96.265	3.735
	5	- diesel, burned in building machine	GLO	0	MJ	1.81E+0	1	1.33	(3,3,2,1,3,5); Xiao 2003		96.265	3.735
	5	- oxygen, liquid, at plant	RER	0	kg	2.14E-1	1	1.33	(3,3,2,1,3,5); Xiao 2003		96.265	3.735
	5	- disposal, inert waste, 5% water, to inert material landfill	CH	0	kg	2.30E-1	1	1.33	(3,3,2,1,3,5); Xiao 2003		96.265	3.735
	5	- steam, for chemical processes, at plant	RER	0	kg	2.10E-1	1	1.33	(3,3,2,1,3,5); Xiao 2003		96.265	3.735
	5	- hard coal, burned in industrial furnace 1-10MW	RER	0	MJ	9.80E-3	1	1.33	(3,3,2,1,3,5); IPCC 2001		96.265	3.735
	5	- natural gas, burned in industrial furnace low-NOx >100kW	RER	0	MJ	2.30E+0	1	1.33	(3,3,2,1,3,5); IPCC 2001		96.265	3.735
	5	- electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	2.97E-1	1	1.33	(3,3,2,1,3,5); IPCC 2001		96.265	3.735
	5	- transport, transoceanic freight ship	OCE	0	tkm	3.75E+0	1	2.10	(4,3,2,1,3,3); Standard distance 5000km		96.265	3.735
	5	- transport, lorry >16t, fleet average	RER	0	tkm	2.56E-2	1	2.10	(4,3,2,1,3,3); Standard distance 100km		96.265	3.735
	5	- transport, freight, rail	RER	0	tkm	1.54E-1	1	2.10	(4,3,2,1,3,3); Standard distance 600km		96.265	3.735
emission air, high population density	- 4	Heat, waste	-	-	MJ	1.07E+0	1	1.33	(3,3,2,1,3,5); Calculated from the electricity input		96.265	3.735
	- 4	Particulates, < 2.5 um	-	-	kg	1.19E-4	1	3.11	(3,3,2,1,3,5); Xiao 2003		96.265	3.735
	- 4	Particulates, > 2.5 um, and < 10um	-	-	kg	3.96E-5	1	2.11	(3,3,2,1,3,5); Xiao 2003		96.265	3.735
	- 4	Particulates, > 10 um	-	-	kg	3.96E-5	1	1.64	(3,3,2,1,3,5); Xiao 2003		96.265	3.735
	- 4	Zinc	-	-	kg	3.96E-5	1	5.12	(3,3,2,1,3,5); EEA 1999		96.265	3.735
	- 4	Sulfur dioxide	-	-	kg	4.35E-4	1	1.33	(3,3,2,1,3,5); Xiao 2003		96.265	3.735
	- 4	Lead	-	-	kg	1.98E-6	1	5.12	(3,3,2,1,3,5); EEA 1999		96.265	3.735
	- 4	Mercury	-	-	kg	1.37E-5	1	5.12	(3,3,2,1,3,5); Xiao 2003		96.265	3.735
	- 4	Mercury	-	-	kg	2.73E-7	1	1.33	(3,3,2,1,3,5); Xiao 2003		96.265	3.735
	- 4	Cadmium, ion	-	-	kg	3.17E-6	1	3.11	(3,3,2,1,3,5); Xiao 2003		96.265	3.735
emission water, river	- 4	Arsenic, ion	-	-	kg	7.52E-7	1	5.12	(3,3,2,1,3,5); Xiao 2003		96.265	3.735
	- 4	Lead	-	-	kg	5.46E-5	1	5.12	(3,3,2,1,3,5); Xiao 2003		96.265	3.735
	- 4	Zinc, ion	-	-	kg	9.07E-5	1	5.12	(3,3,2,1,3,5); Xiao 2003		96.265	3.735
	- 4	Fluoride	-	-	kg	5.90E-5	1	1.64	(3,3,2,1,3,5); Xiao 2003		96.265	3.735

13.13 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 13.9 shows selected LCI results and the cumulative energy demand for the Imperial smelting furnace.

Tab. 13.9 Selected LCI results and the cumulative energy demand for the Imperial smelting furnace

Name			sodium arsenide, at plant	zinc, from Imperial smelting furnace
Location Unit Infrastructure		Unit	GLO kg 0	GLO kg 0
LCIA results				
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	35.7	35.7
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	6.1	6.1
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.6	0.6
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.8	0.8
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1	0.1
selected LCI results				
land occupation	resource	m2a	2.3E-1	2.3E-1
CO ₂ , fossil	air	kg	2.6E+0	2.6E+0
NM/OC	air	kg	3.8E-3	3.8E-3
nitrogen oxides	air	kg	2.2E-2	2.2E-2
sulphur dioxide	air	kg	8.8E-3	8.8E-3
particulates, <2.5 µm	air	kg	2.0E-3	2.0E-3
BOD	water	kg	3.7E-3	3.7E-3
cadmium	soil	kg	6.2E-10	6.2E-10

13.14 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	Imperial smelting furnace
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	5
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Imperial smelting furnace including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Zinkverhüttung
	491	Synonyms	
	492	GeneralComment	The multioutput-process "Imperial smelting furnace" delivers the co-products zinc and sodium arsenide. The allocation is based on mass calculation.
	494	InfrastructureIncluded	1
	495	Category	metals
	496	SubCategory	extraction
	497	LocalCategory	Metalle
	498	LocalSubCategory	Gewinnung
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	2000
	602	EndDate	2007
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications.
Geography	663	Text	The inventory is modelled for the world..
Technology	692	Text	Imperial smelting furnace
Representativeness	722	Percent	100
	724	ProductionVolume	na
	725	SamplingProcedure	Literature data
	726	Extrapolations	Some transports are calculated with standard distances.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

13.15 References

- Althaus et al. 2004 Althaus H.-J., Blaser S., Classen M. and Jungbluth N. (2004) Life Cycle Inventories of Metals. Final report ecoinvent 2000 No. 10. EMPA Dübendorf, Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: www.ecoinvent.ch.
- Ayres et al. 2002 Ayres R. U., Ayres L. W. and Råde I. (2002) The Life Cycle of Copper, its Co-Products and Byproducts. MMSD - Mining, Minerals and Sustainable Development
- CEPMEIP CEPMEIP Database. http://www.air.sk/tno/cepmeip/em_factors_results.php?, accessed 08.01.03

- EEA 1999 EEA (ed.) (1999b) Primary Zinc Production (B335-1). In: EMEP / CORINAIR Emission Inventory Guidebook. 3rd edition October 2002 Update Edition. European Environmental Agency, Online-Version under: http://reports.eea.eu.int/technical_report_2001_3/en/en/tab_content_RLR
- Frischknecht et al. 2007 Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirsch R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.
- IPPC 2001 IPPC (2001) Integrated Pollution Prevention and Control (IPPC); Reference Document on Best Available Techniques in the Non Ferrous Metals Industries. Retrieved 15.01 from <http://www.jrc.es/pub/english.cgi/0/733169>.
- Ullmanns 2004 Grund, S., Hanusch K., Wolf H. Arsenic and Arsenic Compounds. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebaum D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>
- Xiao et al. 2003 Xiao X., Songwen X., Xueyi G., Kelong H. and Ryoichi Y. (2003) LCA Case Study of Zinc Hydro and Pyro-Metallurgical Process in China. In: International Journal for Life Cycle Assessment, Online first, pp. 1 - 5.

14 Lithium carbonate hydration

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

Lithium carbonate hydration is one of the most important processes to produce lithium hydroxide.

14.1 Production Technologies (Ullmanns 2004)

Lithium carbonate is hydrated by calcium hydroxide:



This yields solutions containing up to 3.5 % lithium hydroxide. Attempts to work at higher concentration lead to loss of lithium carbonate in the calcium carbonate residue, because the solubility of lithium carbonate is reduced by the presence of lithium hydroxide. The reaction between these sparingly soluble starting materials requires intensive agitation and gives yields just over 95 % after many hours. The insoluble residue (mainly calcium carbonate) is removed, and lithium hydroxide monohydrate is crystallized from the mother liquor by evaporation, under vacuum if required. It is then separated and dried at 130–140 °C. Further heating to 150–180 °C produces calcined lithium hydroxide (LiOH). The removal of water can be accelerated by use of vacuum.

Other production processes that have been proposed include the reaction of lithium sulfate solution with barium hydroxide and the reaction of lithium carbonate with silicic acid in the molten state followed by hydrolysis of the lithium silicate formed. These processes are, however, of little more than academic interest.

14.2 System Characterisation

In this study the reaction between lithium carbonate and calcium hydroxide is considered. The functional unit is 1 kg lithium carbonate. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers. Within the lithium hydroxide production, calcium carbonate is derived as co-product. Therefore an allocation between the two products was made. The following processes were modelled:

- Lithium carbonate hydration, GLO. Multi output process with lithium hydroxide, at plant, GLO and calcium carbonate from lithium carbonate hydration, at plant, GLO as allocated products

14.3 Introduction

Lithium hydroxide (LiOH; CAS 1310-65-2) is a white hygroscopic crystalline material. It is soluble in water, and slightly soluble in ethanol. There are commercial forms of the hydrated and anhydrous variations of lithium hydroxide. The most important chemical and physical properties of lithium hydroxide used in this inventory are given here.

Tab. 14.1 Chemical and physical properties of Lithium hydroxide (Ullmanns 2004)

Property	Value	Unit
Molecular weight	23.95	g mol ⁻¹
Specific gravity	1460	kg m ⁻³
Boiling point	924	°C
Melting point	462	°C

14.3.1 Use

Lithium hydroxide is used mainly for the production of lubricating greases. Over 50 % of the greases used today in motor vehicles, aircraft, and heavy machinery contain lithium stearates, mainly lithium 12-hydroxystearate. These greases have good viscosity properties up to 200 °C and are practically insoluble in water.

Lithium hydroxide is also used in the production of dyes [73]. It has attained some importance as an additive to nickel – iron accumulators and, in its calcined form (98 % LiOH), as an absorbent for carbon dioxide in breathing apparatus, especially for crews of submarines and space vehicles. One gram of anhydrous lithium hydroxide absorbs 450 mL of carbon dioxide, so that ca. 750 g of lithium hydroxide is sufficient to absorb the carbon dioxide generated by one person in one day.

Lithium hydroxide is also used in the building industry. As an additive to the mix water, it helps prevent the alkali – silica reaction, which is a cause of deterioration (cracking, disintegration) in portland cement concrete structures.

14.4 Raw materials and auxiliaries

Tab. 14.2 shows values for the use of raw materials of the lithium carbonate hydration. This study is based on an yield of 95 % kg per kg lithium carbonate and calcium hydroxide.

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the lithium carbonate hydration.

Tab. 14.2 Consumption of raw materials and auxiliaries for the lithium carbonate hydration

Input	Values	Remarks
Lithium carbonate (kg)	1	Functional unit
Calcium hydroxide (kg)	1	Calculated with a yield of 95 %
Cooling water (kg)	45.6	Estimation

14.5 Energy

There was no information available on the amount of energy used for the lithium carbonate hydration. In order not to neglect the process energy demand those values were approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg⁻¹) were used as approximation for the energy consumption of the lithium carbonate hydration. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be

natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used. A summary of the values used is given in Tab. 14.3.

Tab. 14.3 Energy consumption for the lithium carbonate hydration

Input	per kg lithium carbonate
Heat, natural gas burned in industrial furnace > 100 kW	3.8
Electricity (kWh)	0.633

14.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train. Tab. 14.4 summarises the total transport amounts for the lithium carbonate hydration.

Tab. 14.4 Total transport amounts for the lithium carbonate hydration

tkm kg ⁻¹ lithium carbonate	lorry	train
Total transports	2E-01	1.2

14.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of lithium carbonate hydration plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

14.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

Tab. 14.5 Process emissions to air from the lithium carbonate hydration

Output	Value	Remarks
Waste heat (MJ)	2.287E-01	calculated from electricity input

14.9 Emissions to water

The remaining amount of unreacted lithium carbonate and calcium hydroxide (5 % of the input) is assumed to leave the production process with the waste water.

The values for TOC and DOC used in this inventory are calculated from the amount of carbonate in the waste water. For the calculation of the values for DOC the worst case scenario TOC = DOC was used.

Tab. 14.6 Process emissions to water from the lithium carbonate hydration

Output	Value	Remarks
lithium (kg)	9.46E-03	calculated from mass balance
carbonate (kg)	4.05E-02	calculated from mass balance
calcium (kg)	2.7E-02	calculated from mass balance
hydroxide (kg)	2.3E-02	calculated from mass balance
TOC, DOC (kg)	9.86E-03	calculated from water emissions

14.10 Allocation of lithium hydroxide and calcium carbonate

Within the lithium carbonate hydration, lithium hydroxide and calcium carbonate are obtained as co-products. A stoichiometric allocation of the two products is made. The used allocation factors are given in Tab. 14.7.

Tab. 14.7 Allocation of lithium hydroxide and calcium carbonate from the lithium carbonate hydration

Process	modified Solvay process, GLO		
	Unit	per kg lithium carbonate	Allocation factor (stoichiometric)
lithium hydroxide, at plant, GLO	Kg	0.616	32.4
calcium carbonate from lithium carbonate hydration, at plant, GLO	kg	1.284	67.6

14.11 Life cycle inventory of lithium carbonate hydration and data quality considerations

Tab. 14.8 shows the life cycle inventory and the data quality indicators for the lithium carbonate hydration. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 14.8 Unit process raw data for the lithium carbonate hydration

	InputGroup	OutputGroup	Name	Location	InfrastructureProcesses	Unit	lithium carbonate hydration	UncertaintyType	StandardDeviation95 %	GeneralComment	lime from lithium carbonate hydration	lithium hydroxide, at plant
401												
662			Location				GLO				GLO	GLO
493			InfrastructureProcess				0				0	0
403			Unit				kg				kg	kg
allocated products resource, in water technosphere	-	2	lime from lithium carbonate hydration	GLO	0	kg	1.28E+0				100	0
	-	2	lithium hydroxide, at plant	GLO	0	kg	6.16E-1				0	100
	4	-	Water, cooling, unspecified natural origin	-	-	m3	4.56E-2	1	1.30	(4,5,na,na,na,na); Estimation	67.568	32.432
	5	-	lithium carbonate, at plant	GLO	0	kg	1.00E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation	67.568	32.432
	5	-	lime, hydrated, loose, at plant	CH	0	kg	1.00E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation	67.568	32.432
	5	-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	3.80E+0	1	1.30	(4,5,na,na,na,na); Estimation	67.568	32.432
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	6.33E-1	1	1.30	(4,5,na,na,na,na); Estimation	67.568	32.432
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	2.00E-1	1	2.09	(4,5,na,na,na,na); Standard distance 100km	67.568	32.432
	5	-	transport, freight, rail	RER	0	tkm	1.20E+0	1	2.09	(4,5,na,na,na,na); Standard distance 600km	67.568	32.432
	5	-	chemical plant, organics	RER	1	unit	7.60E-10	1	3.09	(4,5,na,na,na,na); Estimation	67.568	32.432
emission air, high population density emission water, river	-	4	Heat, waste	-	-	MJ	2.28E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input	67.568	32.432
	-	4	Lithium, ion	-	-	kg	9.46E-3	1	1.62	(4,5,na,na,na,na); Estimation	67.568	32.432
	-	4	Carbonate	-	-	kg	4.05E-2	1	3.09	(4,5,na,na,na,na); Estimation	67.568	32.432
	-	4	Calcium, ion	-	-	kg	2.70E-2	1	3.09	(4,5,na,na,na,na); Estimation	67.568	32.432
	-	4	Hydroxide	-	-	kg	2.30E-2	1	1.62	(4,5,na,na,na,na); Estimation	67.568	32.432
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	9.86E-3	1	1.62	(4,5,na,na,na,na); from water emissions	67.568	32.432
	-	4	TOC, Total Organic Carbon	-	-	kg	9.86E-3	1	1.62	(4,5,na,na,na,na); from water emissions	67.568	32.432

14.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 14.9 shows selected LCI results and the cumulative energy demand for the lithium carbonate hydration.

Tab. 14.9 Selected LCI results and the cumulative energy demand for the lithium carbonate hydration

Name			lithium hydroxide, at plant	lime from lithium carbonate hydration
Location	Unit	Unit	GLO	GLO
Infrastructure			kg	kg
			0	0
LCIA results				
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	14.0	14.0
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	3.4	3.4
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.7	0.7
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.6	0.6
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1	0.1
selected LCI results				
land occupation	resource	m2a	1.8E-1	1.8E-1
CO ₂ , fossil	air	kg	1.2E+0	1.2E+0
NM/OC	air	kg	6.6E-4	6.6E-4
nitrogen oxides	air	kg	2.4E-3	2.4E-3
sulphur dioxide	air	kg	5.5E-3	5.5E-3
particulates, <2.5 µm	air	kg	3.5E-4	3.5E-4
BOD	water	kg	9.5E-4	9.5E-4
cadmium	soil	kg	8.2E-10	8.2E-10

14.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	lithium carbonate hydration
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	5
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Hydration of lithium carbonate including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Lithiumcarbonathydrirung
	491	Synonyms	
	492	GeneralComment	The multioutput-process "lithium carbonate hydration" delivers the co-products lime and lithium hydroxide. The allocation is based on stoichiometric calculation.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	inorganics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications.
Geography	663	Text	The inventory is modelled for the world..
Technology	692	Text	Hydration of lithium carbonate
Representativeness	722	Percent	100
	724	ProductionVolume	na
	725	SamplingProcedure	Literature data
	726	Extrapolations	Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

14.14 References

- Frischknecht et al. 2007 Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschler R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.
- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>

Ullmanns 2004

Wietelmann, U., Bauer, R.: Lithium and Lithium Compounds. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under:
<http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

15 Lithium chloride

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hirschier, EMPA, St. Gallen
 Last changes: 2007

15.1 Introduction

Lithium chloride (LiCl; CAS 7447-41-8) is white crystalline salt. It is hygroscopic and highly soluble in water, and is highly polar. It is more soluble in polar organic solvents such as methanol and acetone than is sodium chloride or potassium chloride. Its solubility in ethanol at 25 °C is 2.5 g/100 g, and in methanol, 17 g/100 g.

For this inventory the functional unit is 1 kg of highly pure lithium chloride. The most important chemical and physical properties of lithium chloride used in this inventory are given here.

Tab. 15.1 Chemical and physical properties of Lithium chloride (Ullmanns 2004)

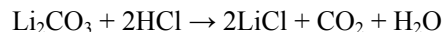
Property	Value	Unit
Molecular weight	42.4	g mol ⁻¹
Specific gravity	2070	kg m ⁻³
Boiling point	1382	°C
Melting point	614	°C

15.1.1 Use

Lithium chloride is extremely hygroscopic; thus, like lithium bromide, it is used for drying gases and in air conditioners of the absorber type. Because molten lithium chloride readily dissolves metallic oxides, it is used as a component of fluxes and dipping baths for welding and brazing aluminum and light metal alloys. Most of the lithium chloride produced is used as starting material for the production of lithium metal by molten salt electrolysis.

15.2 Production Technologies (Ullmanns 2004)

The known methods of producing lithium chloride from lithium ores have not been adopted commercially. At present, it is produced exclusively by reaction of lithium carbonate and hydrochloric acid, with special steel or nickel equipment because of the extreme corrosivity of lithium chloride.



Concentration of the solution (e.g., in a vacuum evaporator) causes lithium chloride to crystallize. It is then separated from the mother liquor, dried, and packed in moisture-proof containers.

15.3 System Characterisation

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg lithium chloride, at plant, global**. For this study the production of lithium chloride by means of the reaction of lithium carbonate and hydrochloric acid considered. All data in the present report are referred to 1 kg lithium chloride 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing

plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

15.4 Raw materials and auxiliaries

Tab. 15.2 shows values for the use of raw materials of the production of lithium chloride. As there was no data available for the production of lithium chloride, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the lithium chloride production.

Tab. 15.2 Consumption of raw materials and auxiliaries for the lithium chloride production

Input	Values	Remarks
Lithium carbonate (kg)	0.927	Calculated with a yield of 95 %
HCl, 30% in H ₂ O (kg)	0.902	Calculated with a yield of 95 %
Cooling water (kg)	24	Estimation

15.5 Energy

Tab. 15.3 Energy consumption for the lithium chloride production (Kim 2003)

Input	per kg lithium chloride
Electricity (kWh)	5.694E-01

15.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train. Tab. 15.4 summarises the total transport amounts for the lithium chloride production.

Tab. 15.4 Total transport amounts for the lithium chloride production

tkm kg ⁻¹ lithium chloride	lorry	train
Total transports	3.93E-01	8.571E-01

15.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of lithium chloride production plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

15.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the lithium chloride production. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the hydrogen chloride input. The carbon dioxide emissions are calculated from the co-produced CO₂ and the water treatment.

Tab. 15.5 Process emissions to air from the lithium chloride production

Output	Value	Remarks
Waste heat (MJ)	2.05	calculated from electricity input
HCl, to air (kg)	1.805E-03	estimated as 0.2 % of the input
CO ₂ , fossil, to air (kg)	5.24E-01	co-product + from water treatment

15.9 Emissions to water

The remaining amount of unreacted lithium carbonate (5 % of the input) and hydrogen chloride (4.8 % of the input) is assumed to leave the production process with the waste water.

The values for TOC and DOC used in this inventory were calculated from the amount of carbonate in the treated waste water. For the calculation of the values for DOC the worst case scenario TOC = DOC was used. AOX is calculated from the chloride emissions.

Tab. 15.6 Process emissions to water from the lithium chloride production

Output	Value	Remarks
lithium (kg)	8.77E-03	calculated from mass balance
carbonate (kg)	3.76E-02	calculated from mass balance
chloride (kg)	4.21E-02	calculated from mass balance
TOC, DOC (kg)	1.74E-02	calculated from water emissions
AOX (kg)	4.21E-02	calculated from water emissions

15.10 Life cycle inventory of lithium chloride production and data quality considerations

Tab. 15.7 shows the life cycle inventory and the data quality indicators for the lithium chloride production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 15.7 Unit process raw data for the lithium chloride production

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	lithium chloride, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
401										
662			Location				GLO			
493			InfrastructureProcess				0			
403			Unit				kg			
resource, in water	-	0	lithium chloride, at plant	GLO	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	lithium carbonate, at plant	GLO	0	kg	9.27E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	hydrochloric acid, 30% in H2O, at plant	RER	0	kg	9.02E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	5.69E-1	1	1.13	(1,2,3,1,1,3); Kim 2003
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	3.93E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	8.57E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	2.05E+0	1	1.13	(1,2,3,1,1,3); Calculated from the electricity input
	-	4	Hydrogen chloride	-	-	kg	1.80E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	5.24E-1	1	1.30	(4,5,na,na,na,na); Estimation
emission water, river	-	4	Lithium, ion	-	-	kg	8.77E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Carbonate	-	-	kg	3.76E-2	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Chloride	-	-	kg	4.21E-2	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	1.74E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	1.74E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	AOX, Adsorbable Organic Halogen as Cl	-	-	kg	4.21E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

15.11 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 15.8 shows selected LCI results and the cumulative energy demand for the lithium chloride production.

Tab. 15.8 Selected LCI results and the cumulative energy demand for the lithium chloride production

Name	lithium chloride, at plant		
	Location		
	Unit	Unit	GLO
	Infrastructure		kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	27.1
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	10.3
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	1.7
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	1.3
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.2
selected LCI results			
land occupation	resource	m2a	4.1E-1
CO2, fossil	air	kg	2.5E+0
NMVOC	air	kg	1.1E-3
nitrogen oxides	air	kg	5.2E-3
sulphur dioxide	air	kg	1.2E-2
particulates, <2.5 um	air	kg	7.8E-4
BOD	water	kg	2.3E-3
cadmium	soil	kg	2.3E-9

15.12 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	lithium chloride, at plant
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of lithium chloride including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Lithiumchlorid, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "lithium chloride, at plant, GLO" is modelled for the production of lithium chloride from lithium carbonate and chlorine in the world. Raw materials are modelled with a stoichiometric calculation, Energy consumptions are calculated with literature data. Emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	inorganics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Anorganika
	499	Formula	LiCl
	501	StatisticalClassification	
	502	CASNumber	7447-41-8
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	Chlorination of lithium carbonate
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Emissions are estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

15.13 References

Frischknecht et al. 2007

Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.

Gendorf 2000

Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>

- Kim 2003 Kim S., Overcash M.: Energy in chemical manufacturing processes: gate-to-gate information for life cycle assessment. In: Journal of Chemical Technology & Biotechnology vol. 78, no. 9: 995-1005(11). 2003
- Ullmanns 2004 Wietelmann, U., Bauer, R.: Lithium and Lithium Compounds. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under:
<http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

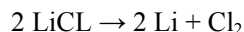
16 Lithium chloride electrolysis

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

Lithium chloride electrolysis is the only process to produce lithium metal.

16.1 Production Technologies (Ullmanns 2004)

Lithium metal is produced by the electrolysis of molten lithium chloride, whose melting point of 614 °C is lowered by addition of potassium chloride. The lithium chloride – potassium chloride eutectic, with 44.3 % lithium chloride, melts at 352 °C. The salt mixtures used in industry contain 45 – 55 % lithium chloride, which allows electrolysis to be carried out at 400 – 460 °C.



The electrolytic cells most commonly employed resemble the Downs cell used for the production of sodium.

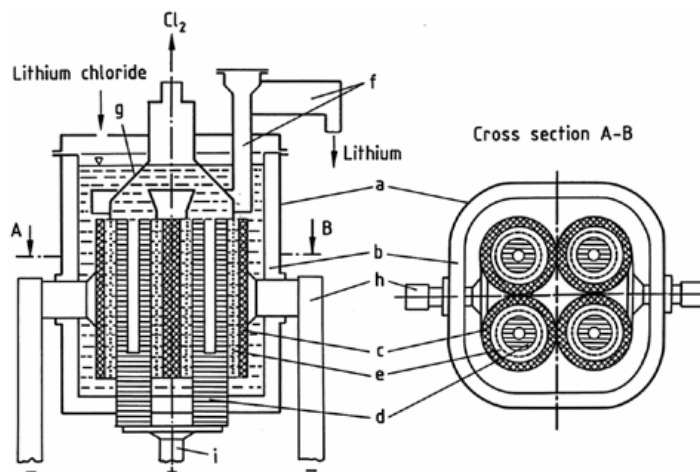


Fig. 16.1 Schematic of an electrolytic cell for the production of lithium

a) Steel casing; b) Ceramic lining; c) Cathode with supporting arm; d) Anode; e) Diaphragm; f) Annular collection zone with riser pipe and overflow for lithium; g) Dome for chlorine collection and removal; h) Cathode conductor rail; i) Anode conductor rail

The cell and the cathode are constructed from steel, whereas the anode is made of graphite. Four cathode tubes are combined to form a unit, with each tube enclosing an anode cylinder. The anodes enter the electrolysis space from below, and the cathodes are supported by arms attached to the cell sides. The anode and cathode are separated by a wire-mesh steel diaphragm. The cell is equipped with an insulating inner lining, because both electrodes are introduced below the surface of the melt. Lithium metal is formed at the cathode, rises to the top, and is collected under an argon atmosphere.

In the United States, lithium cells are sometimes operated without a diaphragm or ceramic insulating lining. This greatly prolongs the lifetime of the cells but results in greater power consumption.

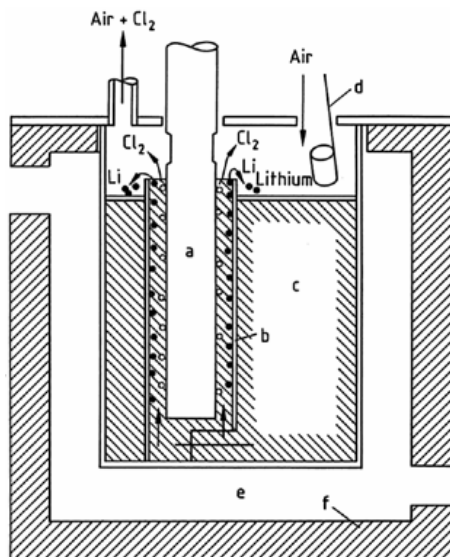


Fig. 16.2 Schematic of an American lithium electrolysis cell

a) Graphite anode; b) Steel cathode; c) Melt; d) Ladle; e) Heating chamber; f) Insulation

Another lithium electrolysis cell has neither ceramic insulation nor a diaphragm. In this cell, a molten salt mixture containing the lithium metal formed rises continuously in the electrode space and is collected in an annular trough at the top of the cathode near the surface of the melt. It is then transferred through a siphon-like tube to a separation vessel connected to the cell, but isolated from the chlorine gas. Here, the lithium metal is separated from the electrolyte under an inert-gas atmosphere; the electrolyte is recycled, and the lithium metal is recovered.

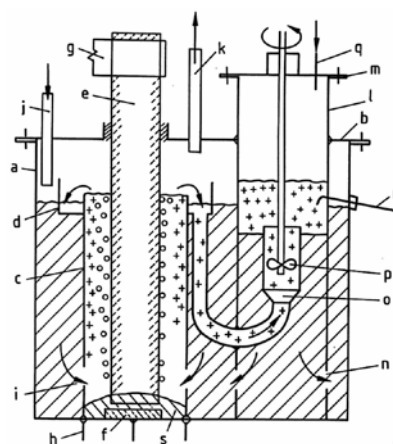


Fig. 16.3 Schematic of a special electrolytic cell for production of lithium

a) Electrolysis cell; b) Cover plate; c) Cathode; d) Collecting trough; e) Graphite anode; f) Insulating material; g) Anode connection; h) Cathode connection; i) Openings in cathode wall; j) Charging pipe for lithium chloride; k) Chlorine outlet; l) Separating cylinder; m) Cover plate; n) Openings in wall of separating cylinder; o) Pipe connecting collecting trough to separating cylinder; p) Agitator; q) Inlet for inert gas; r) Outlet pipe for lithium; s) Solidified melt

The output of an individual cell is usually 200 – 300 kg of lithium metal per day. The decomposition voltage of lithium chloride is ca. 3.68 V, but in practice 6.0 – 6.8 V is usually required. The theoretical power consumption is 14.2 kWh/kg of lithium, but the actual figure is 28 – 32 kWh/kg of lithium. The current efficiency is ca. 90 %.

Lithium metal is shipped in steel drums or cans under an argon atmosphere or under a coating of oil. Lithium metal can be stored indefinitely in the absence of air.

Molten lithium can be cast in low-carbon steel molds without special precautions. Its low hardness enables it to be easily extruded or rolled. However, to maintain the silvery luster of lithium, the operation must be carried out in air of very low moisture content or under an inert-gas atmosphere.

16.2 Quality Specifications and Analysis

Lithium metal of standard grade has a purity in excess of 99 % and contains ca. 0.5 % sodium. Battery grade lithium contains less than 200 ppm of sodium. Standard quality lithium is usually produced in the form of bricks or cylinders 0.1 – 1 kg in weight, rods with a diameter of 10 – 20 mm, or granules with a diameter of ca. 2.5 mm. Battery grade lithium is supplied as foil with a thickness down to 0.03 mm.

Lithium is detected qualitatively by its characteristic red flame coloration; this is also used for quantitative determination by flame emission spectroscopy, in which the emission line at 670.8 nm is measured. Analysis is carried out by using diffraction or refraction spectrometers. In the presence of interfering ions, especially other alkali metals, a calibration curve should be used to obtain adequate accuracy (± 1 %).

Lithium can also be determined by gravimetric, titrimetric, and colorimetric methods, after precipitation as the periodate, phosphate, zinc uranyl acetate, aluminate, or fluoride. Lithium sulfate can be used as a primary standard. Lithium hydroxide and organolithium compounds can be determined easily by acidimetric methods. Indirect methods for the determination of lithium salts are also used, e.g., separation of the lithium by ion exchange followed by titration of the acid produced and, for lithium salts of organic acids, by titration with perchloric acid in nonaqueous media.

Analysis of lithium hydrides can be carried out by measuring the volume of gas evolved on hydrolysis. For low concentrations of lithium, ion-exchange chromatography is most suitable.

16.3 System Characterisation

In this study the direct electrolysis of lithium chloride is considered. The functional unit is 1 kg lithium chloride. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers. Within the electrolysis of lithium chloride, lithium and chlorine are derived as co-products. Therefore an allocation between the two products was made. The following processes were modelled:

- Lithium chloride electrolysis, GLO. Multi output process with lithium, at plant, GLO and chlorine, gaseous, lithium chloride, at plant, GLO as allocated products

16.4 Lithium

Lithium (Li; CAS 7439-93-2), a silvery white metal, is the lightest element that is solid at normal temperature. Its natural isotopes, ^6Li , and ^7Li , ^7Li are not radioactive. The artificial isotopes ^8Li , ^9Li , and ^{11}Li have half-lives of 842, 178, and 8.7 ms, respectively. The electronic configuration of the free atom is $1s^2 2s^1$. The atomic radius is 0.155 nm (in the metal), and the ionic radius is 0.086 nm. Lithium is a member of group 1 of the periodic table and thus has a valence of + 1. In many re-

spects (e.g., the strong alkalinity of lithium hydroxide solutions and the reactivity of the metal) it has the character of an alkali metal, but it also exemplifies the diagonal relationship in the periodic table in its resemblance to magnesium: the metal forms only the oxide on combustion (not the peroxide), and with nitrogen it forms the nitride directly.

For this inventory the functional unit is 1 kg of pure lithium. The most important chemical and physical properties of lithium used in this inventory are given here.

Tab. 16.1 Chemical and physical properties of Lithium (Ullmanns 2004)

Property	Value	Unit	Remarks
⁶ Li: Abundance of the isotope	7.42	%	
⁷ Li: Abundance of the isotope	92.58	%	
Molecular weight		g mol ⁻¹	
Heat of fusion	431.4	J g ⁻¹	
Specific gravity	535	kg m ⁻³	at 20 °C
Boiling point	1342	°C	
Melting point	180-54	°C	

16.4.1 Reserves and Resources

The world market for lithium and lithium compounds is now supplied mainly from production facilities in the United States and from the Salar de Atacama in Chile, and from production by Chemetall GmbH.

The consumption of lithium compounds (calculated as lithium carbonate) in the Western world was ca. 3.2×104 t in 1989, with an annual rate of increase of ca. 3 – 4 %. Approximately 40 % of the lithium production was consumed by the United States, where, together with Chile, the main ore and brine deposits are to be found. The capacity available at present together with the projected expansion will ensure that the market remains satisfied. About 25 % of the lithium produced is used in the lubrication industry, 25 % in the aluminum industry, 25 % in the enamel, glass, and ceramic industries, and the rest for miscellaneous uses such as chemicals, pharmaceuticals, catalysts for the rubber industry, additives for welding materials, and absorbers in air conditioners.

16.4.2 Use

The largest application of lithium metal is the production of organolithium compounds, principally butyllithium, and lithium hydride.

Lithium is used as a reducing agent in organic chemistry, as a solution in liquid ammonia for Birch reductions, and in the synthesis of vitamins.

A rapidly growing application of lithium metal is in sacrificial anodes in batteries. Lithium has a high electrochemical equivalent (3.86 A·h/g) and a low electrochemical potential (– 3.045 V). The main advantages of lithium batteries are high power output, extremely good storage properties (low rate of self-discharge), and relatively small power loss at low temperature. Commercially produced primary cells have counterelectrodes made of manganese dioxide, chromium oxide (CrOx), or graphite loaded with sulfur dioxide or thionyl chloride, and electrolytes consisting of lithium salts dissolved in organic solvents such as propylene carbonate or dimethoxyethane. They have cell voltages up to 3 V and energy densities up to 300 W h/kg. Secondary lithium cells, some based on aluminum – lithium alloys, are being investigated extensively.

Lithium metal is becoming increasingly important as an alloying component; it is also used as a de-oxidizing and desulfurizing agent, especially for copper, nickel, and alloy steels.

Lithium and its compounds play a number of important roles in the nuclear industry, e.g., in the production of tritium, as a heat-exchange cooling medium, as a shielding material, and in the form of a molten salt mixture, as a solvent for other nuclear fuels.

These applications depend on the extremely large difference in neutron capture cross section and absorption distance between the two stable lithium isotopes.

16.5 Raw materials and auxiliaries

Tab. 16.2 shows values for the use of raw materials for the lithium chloride electrolysis. The values for the consumption of the raw materials are calculated with an yield of 90 % (Ullmann 2004).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the lithium chloride electrolysis.

Tab. 16.2 Consumption of raw materials and auxiliaries for the lithium chloride electrolysis

Input	Values	Remarks
Lithium chloride (kg)	1	Functional unit
Cooling water (m3)	0.0216	Estimation

16.6 Energy

The data for the energy consumption are taken from Ullmanns 2005. For this study an average value of 30 kWh/kg lithium is is used.

Tab. 16.3 Energy consumption for the lithium chloride electrolysis

Input	Ullmanns 2005 (per kg lithium)	per kg lithium chloride
Electricity (kWh)	28-32	27

16.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train. Tab. 16.4 summarises the total transport amounts for the lithium chloride electrolysis.

Tab. 16.4 Total transport amounts for the lithium chloride electrolysis

tkm kg ⁻¹ ammonium	lorry	train
Total transports	1E-01	6E-01

16.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of lithium chloride electrolysis plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

16.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

Tab. 16.5 Process emissions to air from the lithium chloride electrolysis

Output	Value	Remarks
Waste heat (MJ)	97.2	calculated from electricity input

16.10 Emissions to water

The remaining amount of unreacted lithium chloride (10 % of the input) is assumed to leave the production process with the waste water.

Tab. 16.6 Process emissions to water from the lithium chloride electrolysis

Output	Value	Remarks
Lithium (kg)	1.67E-02	calculated from mass balance
Chloride (kg)	8.33E-02	calculated from mass balance
AOX (kg)	0.1	calculated from water emissions

16.11 Allocation of ammonium lithium and chlorine

Within the lithium chloride electrolysis lithium and chlorine are obtained as co-products. A stoichiometric allocation of the two products is made. The used allocation factors are given in Tab. 16.7.

Tab. 16.7 Allocation of ammonium chloride and sodium carbonate from the lithium chloride electrolysis

Process	Lithium chloride electrolysis, GLO		
	Unit	per kg lithium chloride	Allocation factor (stoichiometric)
Lithium, at plant, GLO	Kg	0.15	16.7
Chlorine, gaseous, lithium chloride, at plant, GLO	kg	0.75	83.3

16.12 Life cycle inventory of lithium chloride electrolysis and data quality considerations

Tab. 16.8 shows the life cycle inventory and the data quality indicators for the lithium chloride electrolysis. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 16.8 Unit process raw data for the production of lithium chloride electrolysis

401	InputGroup	OutputGroup	Name	Location	Infrastructure	Processes	Units	lithium chloride electrolysis	Uncertainty Type	Standard Deviation %	General Comment	lithium, at plant	chlorine, gaseous, lithium chloride electrolysis, at plant
662			Location					GLO				GLO	GLO
493			Infrastructure					0				0	0
403			Unit					kg				kg	kg
allocated products	-	2	lithium, at plant	GLO	0		kg	1.50E-1				100	0
	-	2	chlorine, gaseous, lithium chloride electrolysis, at plant	GLO	0		kg	7.50E-1				0	100
resource, in water technosphere	4	-	Water, cooling, unspecified natural origin	-	-		m3	2.16E-2	1	1.30	(4,5,na,na,na,na); Estimation	16.667	83.333
	5	-	lithium chloride, at plant	GLO	0		kg	1.00E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation	16.667	83.333
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0		kWh	2.70E+1	1	1.08	(1,1,2,1,1,3); Ullmanns 2005	16.667	83.333
	5	-	transport, lorry >16t, fleet average	RER	0		tkm	1.00E-1	1	2.09	(4,5,na,na,na,na); Standard distance 100km	16.667	83.333
	5	-	transport, freight, rail	RER	0		tkm	6.00E-1	1	2.09	(4,5,na,na,na,na); Standard distance 600km	16.667	83.333
	5	-	chemical plant, organics	RER	1		unit	3.60E-10	1	3.09	(4,5,na,na,na,na); Estimation	16.667	83.333
	-	4	Heat, waste	-	-		MJ	9.72E+1	1	1.08	(1,1,2,1,1,3); Calculated from the electricity input	16.667	83.333
emission air, high population density emission water, river	-	4	Lithium, ion	-	-		kg	1.67E-2	1	1.30	(4,5,na,na,na,na); Estimation	16.667	83.333
	-	4	Chloride	-	-		kg	8.33E-2	1	3.09	(4,5,na,na,na,na); Estimation	16.667	83.333
	-	4	AOX, Adsorbable Organic Halogen as Cl	-	-		kg	1.00E-1	1	1.62	(4,5,na,na,na,na); from water emissions	16.667	83.333

16.13 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 16.9 shows selected LCI results and the cumulative energy demand for the lithium chloride electrolysis.

Tab. 16.9 Selected LCI results and the cumulative energy demand for the lithium chloride electrolysis

Name	Location Unit Infrastructure	Unit	lithium, at plant		chlorine, gaseous, lithium chloride electrolysis, at plant
			GLO	GLO	
			kg	kg	
			0	0	
LCIA results					
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	225.6	225.6	
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	139.5	139.5	
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0	0.0	
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	6.4	6.4	
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-	-	
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0	0.0	
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	15.9	15.9	
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	2.6	2.6	
selected LCI results					
land occupation	resource	m2a	8.4E-1	8.4E-1	
CO2, fossil	air	kg	1.8E+1	1.8E+1	
NMVOG	air	kg	4.5E-3	4.5E-3	
nitrogen oxides	air	kg	3.3E-2	3.3E-2	
sulphur dioxide	air	kg	6.4E-2	6.4E-2	
particulates, <2.5 um	air	kg	4.7E-3	4.7E-3	
BOD	water	kg	9.0E-3	9.0E-3	
cadmium	soil	kg	9.7E-9	9.7E-9	

16.14 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	lithium chloride electrolysis
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	5
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Lithium chloride electrolysis including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Lithiumchlorideelektrolyse
	491	Synonyms	
	492	GeneralComment	The multioutput-process "Lithium chloride electrolysis" delivers the co-products lithium and chlorine. The allocation is based on stoichiometric calculation.
	494	InfrastructureIncluded	1
	495	Category	metals
	496	SubCategory	extraction
	497	LocalCategory	Metalle
	498	LocalSubCategory	Gewinnung
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications.
Geography	663	Text	The inventory is modelled for the world..
Technology	692	Text	Lithium chloride electrolysis
Representativeness	722	Percent	100
	724	ProductionVolume	na
	725	SamplingProcedure	Literature data
	726	Extrapolations	Emissions are estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

16.15 References

- Frisknecht et al. 2007 Frisknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.
- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>
- Ullmanns 2004 Wietelmann, U., Bauer, R.: Lithium and Lithium Compounds. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition,

2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under:
<http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

17 Lithium manganese oxide

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hirschier, EMPA, St. Gallen
 Last changes: 2007

17.1 Introduction

Lithium manganese oxide (LiMn_2O_4 ; CAS 39457-42-6) is a ternary spinel-type double oxide. It is a cheap candidate for cathodes in high-power secondary lithium ion batteries. Practical specific capacities are in the range 90 – 120 mA h/g.

For this inventory the functional unit is 1 kg of pure lithium manganese oxide. The most important chemical and physical properties of lithium manganese oxide used in this inventory are given here.

Tab. 17.1 Chemical and physical properties of lithium manganese oxide (Ullmanns 2004)

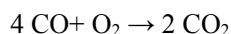
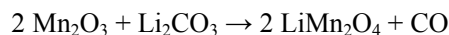
Property	Value	Unit
Molecular weight	180.82	g mol^{-1}
Specific gravity	4280	kg m^{-3}

17.1.1 Use

Manganese oxide is a material which has been used from long ago as a cell active material, and a lithium manganese composite oxide which is a composite material of manganese and lithium, is a material which has attracted an attention in recent years as a positive electrode active material for a lithium secondary cell.

17.2 Production Technologies (Iwata 2002)

Lithium manganese oxide is commonly prepared by sintering manganese oxide (MnO_2 or Mn_2O_3) with Li_2CO_3 at high temperatures. The produced carbon monoxide is oxidated to carbon dioxide.



17.3 System Characterisation

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg lithium manganese oxide, at plant, global**. For this study the production of lithium manganese oxide by sintering manganese oxide is considered.. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

17.4 Raw materials and auxiliaries

As there was no data available for the production of lithium manganese oxide, the values for the con-

sumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007). As approximation for manganese oxide, the dataset "Manganese concentrate" is used.

Tab. 17.2 Consumption of raw materials and auxiliaries for the lithium manganese oxide production

Input	Values	Remarks
Manganese concentrate, 42.4 % Mn (kg)	1.509	Calculated from Mn content with a yield of 95%
Lithium carbonate (kg)	2.15E-01	Calculated with a yield of 95%
Cooling water (m3)	4E-04	Approximation from the dataset "sinter, iron, at plant"

17.5 Energy

As no information about production of lithium manganese oxide by sintering manganese oxide is available, data from the dataset "sinter, iron, at plant" have been used as approximation (Althaus et al. 2004). A summary of the values used is given in Tab. 17.3.

Tab. 17.3 Energy consumption for the lithium manganese oxide production

Input	per kg lithium manganese oxide
Heat, natural gas burned in industrial furnace > 100 kW (MJ)	1.43
Electricity (kWh)	0.01

17.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials. As approximation a distance of 50 km by lorry 32t taken from the dataset "ferromanganese, high-coal, 74.5% Mn, at regional storage" is used (Althaus et al. 2004). Tab. 17.4 summarises the total transport amounts for the production of 1 kg lithium manganese oxide.

Tab. 17.4 Total transport amounts for the lithium manganese oxide production

tkm kg ⁻¹ lithium manganese oxide	lorry
Total transports	8.62E-02

17.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of lithium manganese oxide production plants. Therefore, in this study, the infrastructure is estimated based on the dataset "aluminum oxide plant" (Althaus et al. 2004). For this study, the estimated value is 2.5 E-11 units per kg of produced chemical.

17.8 Emissions to air

As no information about emissions from production of lithium manganese oxide by sintering manganese oxide is available, data from the dataset "sinter, iron, at plant" have been used as approximation. As approximation for the carbon monoxide emission, it is assumed that 5% of the carbon monoxide produced in the sintering process is emitted to air. The remaining 95 % are assumed to be oxidated to carbon dioxide.

Tab. 17.5 Process emissions to air from the lithium manganese oxide production

Output	Value	Remarks
Waste heat (MJ)	3.6E-02	calculated from electricity input
Dust fine, to air (kg)	2.06E-04	approximated from iron sinter production
Cd, to air (kg)	1.93E-08	approximated from iron sinter production
Cr, to air (kg)	2.52E-08	approximated from iron sinter production
Cu, to air (kg)	7.66E-08	approximated from iron sinter production
Hg, to air (kg)	7.57E-08	approximated from iron sinter production
Mn, to air (kg)	1.93E-07	approximated from iron sinter production
Ni, to air (kg)	1.93E-08	approximated from iron sinter production
Pb, to air (kg)	3.23E-06	approximated from iron sinter production
Tl, to air (kg)	1.60E-08	approximated from iron sinter production
V, to air (kg)	1.15E-08	approximated from iron sinter production
Zn, to air (kg)	8.26E-07	approximated from iron sinter production
HCl, to air (kg)	3.76E-05	approximated from iron sinter production
HF, to air (kg)	2.25E-06	approximated from iron sinter production
NOx, to air (kg)	5.27E-04	approximated from iron sinter production
SO ₂ , to air (kg)	1.26E-03	approximated from iron sinter production
VOC, to air (kg)	1.37E-04	approximated from iron sinter production
PAH, to air (kg)	4.72E-07	approximated from iron sinter production
PCB, to air (kg)	6.42E-09	approximated from iron sinter production
Dioxin, to air (kg)	7.00E-12	approximated from iron sinter production
CO, to air (kg)	0.003867403	estimated as 5 % of the output
CO ₂ , fossil, to air (kg)	0.115469613	from oxidation of CO

17.9 Waste

The remaining manganese concentrate and lithium carbonate (5 % of the input) are assumed to be disposed as inert waste.

Tab. 17.6 Wastes from the boron carbide production

Output	Value
disposal, inert waste, 5% water, to inert material landfill (kg)	5.67E-02

17.10 Life cycle inventory of lithium manganese oxide production and data quality considerations

Tab. 17.7 shows the life cycle inventory and the data quality indicators for the lithium manganese oxide production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 17.7 Unit process raw data for the lithium manganese oxide production

	InputGroup	OutputGroup	Name	Location	Infrastructure	Unit	lithium manganese oxide, at plant	Uncertainty Type	Standard Deviation95%	GeneralComment
	401									
	662		Location				GLO			
	493		InfrastructureProcess				0			
	403		Unit				kg			
resource, in water	-	0	lithium manganese oxide, at plant	GLO	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	4.00E-4	1	1.30	(4,5,na,na,na,na); Approximated from iron sintering
	5	-	manganese concentrate, at beneficiation	GLO	0	kg	1.51E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	lithium carbonate, at plant	GLO	0	kg	2.15E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	disposal, inert waste, 5% water, to inert material landfill	CH	0	kg	5.67E-2	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	1.43E+0	1	1.30	(4,5,na,na,na,na); Approximated from iron sintering
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	1.00E-2	1	1.13	(1,2,3,1,1,3); Approximated from iron sintering
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	8.62E-2	1	2.09	(4,5,na,na,na,na); Approximated from iron sintering
	5	-	chemical plant, organics	RER	1	unit	2.50E-11	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Heat, waste	-	-	MJ	3.60E-2	1	1.13	(1,2,3,1,1,3); Calculated from the electricity input
emission air, high population density	-	4	Particulates, < 2.5 um	-	-	kg	2.06E-4	1	3.09	(4,5,na,na,na,na); Approximated from iron sintering
	-	4	Cadmium	-	-	kg	1.93E-8	1	5.10	(4,5,na,na,na,na); Approximated from iron sintering
	-	4	Chromium	-	-	kg	2.52E-8	1	5.10	(4,5,na,na,na,na); Approximated from iron sintering
	-	4	Copper	-	-	kg	7.66E-8	1	5.10	(4,5,na,na,na,na); Approximated from iron sintering
	-	4	Mercury	-	-	kg	7.57E-8	1	5.10	(4,5,na,na,na,na); Approximated from iron sintering
	-	4	Manganese	-	-	kg	1.93E-7	1	5.10	(4,5,na,na,na,na); Approximated from iron sintering
	-	4	Nickel	-	-	kg	1.93E-8	1	5.10	(4,5,na,na,na,na); Approximated from iron sintering
	-	4	Lead	-	-	kg	3.23E-6	1	5.10	(4,5,na,na,na,na); Approximated from iron sintering
	-	4	Thallium	-	-	kg	1.60E-8	1	5.10	(4,5,na,na,na,na); Approximated from iron sintering
	-	4	Vanadium	-	-	kg	1.15E-8	1	5.10	(4,5,na,na,na,na); Approximated from iron sintering
	-	4	Zinc	-	-	kg	8.26E-7	1	5.10	(4,5,na,na,na,na); Approximated from iron sintering
	-	4	Hydrogen chloride	-	-	kg	3.76E-5	1	1.62	(4,5,na,na,na,na); Approximated from iron sintering
	-	4	Hydrogen fluoride	-	-	kg	2.25E-6	1	1.62	(4,5,na,na,na,na); Approximated from iron sintering
	-	4	Nitrogen oxides	-	-	kg	5.27E-4	1	1.62	(4,5,na,na,na,na); Approximated from iron sintering
	-	4	Sulfur dioxide	-	-	kg	1.26E-3	1	1.30	(4,5,na,na,na,na); Approximated from iron sintering
	-	4	Hydrocarbons, aliphatic, alkanes, cyclic	-	-	kg	1.37E-4	1	1.62	(4,5,na,na,na,na); Approximated from iron sintering
	-	4	PAH, polycyclic aromatic hydrocarbons	-	-	kg	4.72E-7	1	3.09	(4,5,na,na,na,na); Approximated from iron sintering
	-	4	Polychlorinated biphenyls	-	-	kg	6.42E-9	1	3.09	(4,5,na,na,na,na); Approximated from iron sintering
	-	4	Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin	-	-	kg	7.00E-12	1	3.09	(4,5,na,na,na,na); Approximated from iron sintering
	-	4	Carbon monoxide, fossil	-	-	kg	3.87E-3	1	5.10	(4,5,na,na,na,na); Approximated from iron sintering
	-	4	Carbon dioxide, fossil	-	-	kg	1.15E-1	1	1.30	(4,5,na,na,na,na); Approximated from iron sintering

17.11 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 17.8 shows selected LCI results and the cumulative energy demand for the lithium manganese oxide production.

Tab. 17.8 Selected LCI results and the cumulative energy demand for the lithium manganese oxide production

Name		lithium manganese oxide, at plant	
Location	Unit	Unit	GLO kg 0
Infrastructure			
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	4.7
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	0.7
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.2
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.1
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.0
selected LCI results			
land occupation	resource	m2a	8.3E-2
CO2, fossil	air	kg	4.2E-1
NM VOC	air	kg	3.8E-4
nitrogen oxides	air	kg	1.6E-3
sulphur dioxide	air	kg	3.2E-3
particulates, <2.5 um	air	kg	3.3E-4
BOD	water	kg	3.3E-4
cadmium	soil	kg	2.2E-10

17.12 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	lithium manganese oxide, at plant
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of lithium manganese oxide including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Lithiummanganoxid, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "lithium manganese oxide, at plant, GLO" is modelled for the production of lithium manganese oxide from manganese oxide and lithium carbonate in the world. Raw materials are modelled with a stoichiometric calculation, Energy consumptions, emissions, infrastructure and transports are approximated with a similar process.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	inorganics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Anorganika
	499	Formula	LiMn2O4
	501	StatisticalClassification	
	502	CASNumber	39457-42-6
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	Sintering of manganese oxide with lithium carbonate
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Energy consumptions, emissions, infrastructure and transports are approximated with a similar process.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

17.13 References

Althaus et al. 2004

Althaus H.-J., Blaser S., Classen M. and Jungbluth N. (2004) Life Cycle Inventories of Metals. Final report ecoinvent 2000 No. 10. EMPA Dübendorf, Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: www.ecoinvent.ch.

Frischknecht et al. 2007

Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Over-

- view and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.
- Iwata et al. 2002 Iwata, E., Maeda, K., Suzuki, N., Ohsaki, Y. Kwaguchi, T., Mori, T.: Lithium manganese oxide, and process for its production and secondary cell employing it. United States Patent 6475672. Published 2002-11-05. Online under: <http://www.freepatentsonline.com/6475672.html>
- Ullmanns 2004 Wietelmann, U., Bauer, R.: Lithium and Lithium Compounds. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

18 Methyl-3-methoxypropionate

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hirschier, EMPA, St. Gallen
 Last changes: 2007

18.1 Introduction

Methyl-3-methoxypropionate ($C_5H_{10}O_3$; CAS 3852-09-3; synonym: 3-Methoxypropionic acid methyl ester) is a colourless liquid.

For this inventory the functional unit is 1 kg of highly pure methyl-3-methoxypropionate. The most important chemical and physical properties of methyl-3-methoxypropionate used in this inventory are given here.

Tab. 18.1 Chemical and physical properties of methyl-3-methoxypropionate (Green 1988)

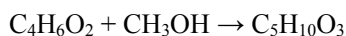
Property	Value	Unit	Remarks
Molecular weight	118.13	$g\ mol^{-1}$	
Specific gravity	1010	$kg\ m^{-3}$	
Boiling point	-142	$^{\circ}C$	at normale pressure
Melting point	-47	$^{\circ}C$	

18.2 Use

Methyl-3-methoxypropionate is sometimes used as solvent in some special chemical reactions.

18.3 Production Technologies (Green 1988)

Methyl-3-methoxypropionate is produced by an reaction of methyl acrylate with methanol using sodium hydroxide as catalyst.



18.4 System Characterisation

In this study the reaction of methyl acrylate with methanol is considered. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

18.5 Raw materials and auxiliaries

Tab. 18.2 shows values for the use of raw materials of the production of methyl-3-methoxypropionate. As there was no data available for the production of methyl-3-methoxypropionate, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not

to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the methyl-3-methoxypropionate production.

Tab. 18.2 Consumption of raw materials and auxiliaries for the methyl-3-methoxypropionate production

Input	Values	Remarks
Methyl acrylate (kg)	0.767	Calculated with a yield of 95 %
Methanol (kg)	0.285	Calculated with a yield of 95 %
Cooling water (kg)	24	Estimation

18.6 Energy

There was no information available on the amount of energy used for the production process. In order not to neglect the process energy demand those values were approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg-1) were used as approximation for the energy consumption of the methyl-3-methoxypropionate production. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used. A summary of the values used is given in Tab. 18.3.

Tab. 18.3 Energy consumption for the methyl-3-methoxypropionate production

Input	per kg methyl-3-methoxypropionate
Heat, natural gas burned in industrial furnace > 100 kW	2
Electricity (kWh)	0.333

18.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train. Tab. 18.4 summarises the total transport amounts for the methyl-3-methoxypropionate production.

Tab. 18.4 Total transport amounts for the methyl-3-methoxypropionate production

tkm kg ⁻¹ methyl-3-methoxypropionate	lorry	train
Total transports	1.05E-01	6.32E-01

18.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of methyl-3-methoxypropionate production plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

18.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the methyl-3-methoxypropionate production. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The assumed values for the emissions to air are given in Tab. 18.5.

Tab. 18.5 Process emissions to air from the methyl-3-methoxypropionate production

Output	Value	Remarks
Methyl acrylate, to air (kg)	1.53E-3	Estimated as 0.2 % of input
Methanol, to air (kg)	5.71E-4	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	8.48E-2	from waster water treatment
Waste heat (MJ)	1.2	calculated from electricity input

18.10 Emissions to water

The remaining amount of unreacted methyl acrylate and methanol (4.8 % of the input) is assumed to leave the production process with the wastewater. Further it is assumed that this wastewater is treated in a internal wastewater plant. The removal efficiency for methyl acrylate and methanol is assumed with 90%. The carbon contained in the removed methyl acrylate and methanol is accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory are calculated from the amount of methyl acrylate and methanol in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used.

Tab. 18.6 Process emissions to water from the methyl-3-methoxypropionate production

Output	Value	Remarks
Methyl acrylate, to water (kg)	3E-02	calculated from mass balance
Methanol, to water (kg)	1.31E-02	calculated from mass balance
COD, BOD (kg)	6.27E-02	calculated from water emissions
TOC, DOC (kg)	2.21E-02	calculated from water emissions

18.11 Life cycle inventory of methyl-3-methoxypropionate production and data quality considerations

Tab. 18.7 shows the life cycle inventory and the data quality indicators for the methyl-3-methoxypropionate production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 18.7 Unit process raw data for the methyl-3-methoxypropionate production

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	methyl-3-methoxypropionate, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
401										
662			Location				GLO			
493			InfrastructureProcess				0			
403			Unit				kg			
resource, in water	-	0	methyl-3-methoxypropionate, at plant	GLO	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	methyl acrylate, at plant	GLO	0	kg	7.67E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	methanol, at plant	GLO	0	kg	2.85E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	2.00E+0	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.33E-1	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.05E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	6.32E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	1.20E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input
	-	4	Methyl acrylate	-	-	kg	1.53E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Methanol	-	-	kg	5.71E-4	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	8.48E-2	1	1.30	(4,5,na,na,na,na); Estimation
emission water, river	-	4	Methyl acrylate	-	-	kg	3.00E-2	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Methanol	-	-	kg	1.31E-2	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	6.27E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	6.27E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	2.21E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	2.21E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

18.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 18.8 shows selected LCI results and the cumulative energy demand for the methyl-3-methoxypropionate production.

Tab. 18.8 Selected LCI results and the cumulative energy demand for the methyl-3-methoxypropionate production

Name		methyl-3-methoxypropionate, at plant	
Location	Unit	GLO	
Infrastructure		kg	0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	79.3
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	4.8
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.9
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.6
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1
selected LCI results			
land occupation	resource	m2a	2.2E-1
CO ₂ , fossil	air	kg	2.9E+0
NM VOC	air	kg	6.6E-3
nitrogen oxides	air	kg	3.9E-3
sulphur dioxide	air	kg	3.8E-3
particulates, <2.5 um	air	kg	3.2E-4
BOD	water	kg	6.9E-2
cadmium	soil	kg	9.5E-10

18.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	methyl-3-methoxypropionate, at plant
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of methyl-3-methoxypropionate including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Methyl-3-methoxypropionat, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "methyl-3-methoxypropionate, at plant, GLO" is modelled for the production of methyl-3-methoxypropionate from methyl acrylate and methanol in the world. Raw materials are modelled with a stoichiometric calculation. Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C5H10O3
	501	StatisticalClassification	
	502	CASNumber	3852-09-3
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	Reaction of methyl acrylate and methanol
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.D19
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

18.14 References

Frischknecht et al. 2007 Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Over-

view and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.

Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>

Green 1988 Green, Michael: Chemical process involving ethers and amines. United States Patent 4730075. Online-Version under: <http://www.freepatentsonline.com/4730075.html>

19 Methyl acrylate

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

19.1 Introduction

Methyl acrylate (C₄H₆O₂; CAS 96-33-3) is a flammable, volatile, mildly toxic, colorless liquid. It is soluble in water and forms azeotropes with water.

For this inventory the functional unit is 1 kg of highly pure methyl acrylate. The most important chemical and physical properties of methyl acrylate used in this inventory are given here.

Tab. 19.1 Chemical and physical properties of methyl acrylate (Ullmanns 2004)

Property	Value	Unit	Remarks
Molecular weight	86.09	g mol ⁻¹	
Specific heat	0.48	kJ mol ⁻¹ K ⁻¹	
Specific gravity	951	kg m ⁻³	
Boiling point	80.3	°C	at 101.3 kPa
Melting point	-76	°C	under pressure

19.1.1 Reserves and Resources

Western world production capacities of acrylic acid esters in 1993 were around 1.200.500 t/a. Plants based on other processes, such as the ketene and cyanohydrin methods, were shut down during the past decade because of inefficiency.

Tab. 19.2 World production of acrylic acid esters 1993 (Ullmanns 2004)

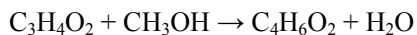
Production	Acrylic acid esters (t a ⁻¹)
USA	587.000
Western Europe	312.000
Japan	206.000
World	1.200.00

19.1.2 Use

Acrylic esters are used exclusively for the production of polymers. The polymers are used mainly for coatings, paints, adhesives, and binders for leather, paper, and textiles. About 80 % of the methyl ester produced is used as a copolymer component of acrylic fibers.

19.2 Production Technologies (Ullmanns 2004)

Although acrylic acid can be esterified in the vapor phase, the liquid phase esterification is industrially more important. Two types of acid catalyst are used: a strong acid, such as sulfuric acid or p-toluenesulfonic acid, or a solid acid, such as a cation-exchange resin.



Although sulfuric acid is superior to ion-exchange resins, its use causes problems in waste disposal. In general, cation-exchange resins are favored for esterification using such alcohols as methanol and

ethanol, whereas sulfuric acid is favored for higher alcohols having slower rates of esterification (e.g., pentanols and octanols).

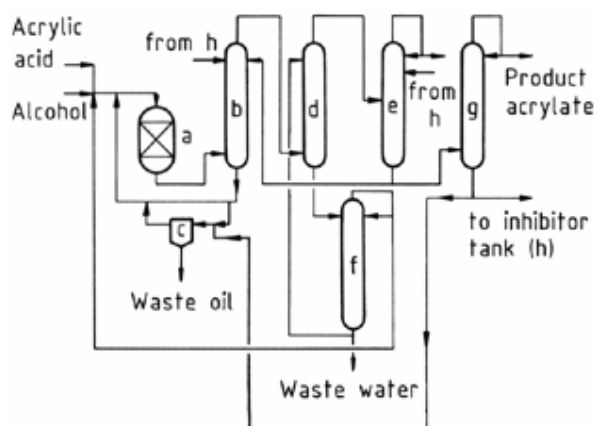
Acrylic acid and a small excess (10 – 30 %) of an alcohol are fed into the fixed-bed reactor (a) which is packed with a cation-exchange resin and operated at a temperature of 60 to 80 °C. The reaction liquid then goes to the ester stripper (b) where the desired ester, water, and unreacted alcohol are removed overhead using part of the bottoms from the light-ends column (e) as reflux. The bottom liquid from b contains unreacted acid and is recycled to the reactor. Part of the recycled liquid is fed into the bottom stripper (c), where high-boiling materials, such as inhibitors, impurities, and polymers, are removed to prevent their accumulation in the reaction system.

The acid-free mixture of ester and alcohol distilled from the ester stripper (b) is fed into the extraction column (d), where the alcohol is extracted with water fed from the top of the column. The raffinate from the top of the column goes into the light-ends cut column (e), where light-ends such as water, acetate, and alcohol are separated overhead.

The extract from the bottom of the extraction column is fed into the alcohol recovery column (f), where the alcohol is recovered for reuse in the reaction. Part of the bottom liquid is reused as extracting water; the rest is taken out as waste, concentrated, and either treated biologically or incinerated.

Crude ester from the bottom of the light-ends column is distilled in the product column (g) to obtain acrylate of high purity. The bottom liquid from the product column is recycled (via the inhibitor tank) to the ester stripper (b) and light-ends cut column (e) to be reused as an inhibitor. However, a part of it is sent to the bottom stripper (c) to recover ester and separate high-boiling materials such as polymers.

Polymerization inhibitors, such as hydroquinone or phenothiazine, are added to each column. The light-ends cut column and the product column are operated at reduced pressure to permit lower distillation temperatures.



Esterification – lower alkyl acrylate process

a) Esterification reactor; b) Ester stripper; c) Bottom stripper; d) Extraction column; e) Light-ends cut column; f) Alcohol-recovery column; g) Product column

This process for making alkyl acrylates is quite economical because only a small excess of alcohol is applied and the inhibitor is reused; this leads to low energy and inhibitor consumptions. The yield reaches 95 % based on acrylic acid and 97 % based on alcohol. The purity of the product exceeds 99.5 wt %.

19.3 System Characterisation

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg methyl**

acrylate, at plant, global. For this study the production of methyl acrylate by means of the esterification of acrylic acid is considered. All data in the present report are referred to 1 kg methyl acrylate 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

19.4 Raw materials and auxiliaries

Tab. 19.3 shows values for the use of raw materials of the production of methyl acrylate. This study is based on an yield of 95 % kg per kg acrylic acid and 97 % per kg methanol (Ullmann 2004).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the methyl acrylate production.

Tab. 19.3 Consumption of raw materials and auxiliaries for the methyl acrylate production

Input	Values	Remarks
Acrylic acid (kg)	0.881	Calculated with a yield of 95 %
Methanol (kg)	0.384	Calculated with a yield of 97 %
Cooling water (kg)	24	Estimation

19.5 Energy

As no data for the production of methyl acrylate are available, the values for the energy use have been estimated with data from Kim 2003, who gives data for the production of ethyl acetate from ethanol esterification. This process is considered to be similar to the esterification of acrylic acid with methanol.

The value for the steam input used in this inventory is the difference between the steam and the potential recovered energy, because this recovered energy is usually fed into the heating systems.

Tab. 19.4 Energy consumption for the production of 1 kg methyl acrylate

Input	Esterification 1	Inventory 2
Steam (MJ)	10.5	8.84
Electricity (kWh)	0.00725	7.25E-03
Potential recovered energy (MJ)	- 1.66	
Sum of consumed energy (MJ)	8.87	

1 Source: Kim 2003

2 Values used in this inventory

19.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used:

100 km by lorry 32t and 600 km by train for ethylene, H₃PO₄, Benzene, and NaOH, and 100 km by lorry 32t and 200 km by train for HCl. Tab. 19.5 summarises the total transport amounts for the methyl acrylate production.

Tab. 19.5 Total transport amounts for the methyl acrylate production

tkm kg ⁻¹ ethylene	lorry	train
Total transports	1.265E-01	7.589E-01

19.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of methyl acrylate production plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

19.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the methyl acrylate production. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the input of acrylic acid and methanol. The assumed values for the emissions to air are given in Tab. 19.6.

Tab. 19.6 Process emissions to air from the methyl acrylate production

Output	Value	Remarks
Acrylic acid, to air (kg)	1.76E-3	Estimated as 0.2 % of input
Methanol, to air (kg)	7.67E-4	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	8.31E-2	from waster water treatment
Waste heat (MJ)	2.61E-02	calculated from electricity input

19.9 Emissions to water

The remaining amount of unreacted acrylic acid and methanol is assumed to leave the production process with the wastewater. Further it is assumed that this wastewater is treated in a internal wastewater plant. The removal efficiency for acrylic acid and methanol is assumed with 90%. The carbon contained in the removed acrylic acid and methanol is accounted as CO₂ emissions to air. The acrylic cid is calculated as acrylate emission.

The values for COD, BOD, TOC and DOC used in this inventory are calculated from the amount of acrylic acid and methanol in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used.

Tab. 19.7 Process emissions to water from the methyl acrylate production

Output	Value	Remarks
Acrylic acid, to water (kg)	4.17E-03	calculated from mass balance
Methanol, to water (kg)	1.07E-03	calculated from mass balance
COD, BOD (kg)	6.5E-03	calculated from water emissions
TOC, DOC (kg)	2.52E-03	calculated from water emissions

19.10 Life cycle inventory of methyl acrylate production and data quality considerations

Tab. 19.8 shows the life cycle inventory and the data quality indicators for the methyl acrylate production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 19.8 Unit process raw data for the methyl acrylate production

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	methyl acrylate, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
401										
662			Location				GLO			
493			InfrastructureProcess				0			
403			Unit				kg			
resource, in water	-	0	methyl acrylate, at plant	GLO	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	acrylic acid, at plant	RER	0	kg	8.81E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	methanol, at plant	GLO	0	kg	3.84E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	8.84E+0	1	1.35	(3,3,3,1,3,5); Estimation
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	7.25E-3	1	1.35	(3,3,3,1,3,5); Estimation
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.26E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	7.59E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	2.61E-2	1	1.35	(3,3,3,1,3,5); Calculated from the electricity input
	-	4	Acrylic acid	-	-	kg	1.76E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Methanol	-	-	kg	7.67E-4	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	8.31E-2	1	1.30	(4,5,na,na,na,na); Estimation
emission water, river	-	4	Acrylate, ion	-	-	kg	4.17E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Methanol	-	-	kg	1.07E-3	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	6.50E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	6.50E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	2.52E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	2.52E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

19.11 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 19.9 shows selected LCI results and the cumulative energy demand for the production methyl acrylate production.

Tab. 19.9 Selected LCI results and the cumulative energy demand for the methyl acrylate production

Name	methyl acrylate, at plant		
	Location		GLO
	Unit	Unit	kg
Infrastructure			0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	81.3
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	3.7
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.7
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.4
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.0
selected LCI results			
land occupation	resource	m2a	1.7E-1
CO2, fossil	air	kg	2.8E+0
NM VOC	air	kg	5.2E-3
nitrogen oxides	air	kg	3.6E-3
sulphur dioxide	air	kg	3.4E-3
particulates, <2.5 um	air	kg	2.7E-4
BOD	water	kg	7.5E-3
cadmium	soil	kg	7.0E-10

19.12 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	methyl acrylate, at plant
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of methyl acrylate including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Methylacrylat, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "methyl acrylate, at plant, GLO" is modelled for the production of methyl acrylate from acrylic acid in the world. Raw materials are modelled with a stoichiometric calculation. Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C4H6O2
	501	StatisticalClassification	
	502	CASNumber	96-33-3
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	Esterification of acrylic acid
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

19.13 References

Frischknecht et al. 2007

Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Over-

- view and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.
- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>
- Kim 2003 Kim S., Overcash M.: Energy in chemical manufacturing processes: gate-to-gate information for life cycle assessment. In: Journal of Chemical Technology & Biotechnology vol. 78, no. 9: 995-1005(11). 2003
- Ullmanns 2004 Ohara, T., Sato, T., Shimizu, N., Prescher, G., Schwind, H., Weiberg, O., Marten, K., Greim, H.: Acrylic acid and derivatives. Published online: 2003. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

20 Modified Solvay process

Author: Jürgen Sutter, ETH S+U, Zurich
Review: Roland Hirschler, EMPA, St. Gallen
Last changes: 2007

The Modified Solvay process is one of the most important processes to produce ammonium chloride.

20.1 Production Technologies (Ullmanns 2004)

In the Solvay process, ammonia and carbon dioxide are dissolved in aqueous sodium chloride to produce sparingly soluble sodium bicarbonate, which is calcined to sodium carbonate. The ammonia is recovered from the mother liquor by reaction with lime, this reaction also producing the calcium chloride. The source of the lime and the carbon dioxide is limestone. The net reaction is the conversion of the feedstocks rock salt and limestone into sodium carbonate and the byproduct calcium chloride.

Water, carbon dioxide, and ammonia are added only to the extent necessary to compensate for plant losses. The feedstock and product dictate the location of a Solvay plant. For this reason, the plants normally function as largely independent units.

The modified Solvay process (ammonium chloride – soda ash process) is one of the oldest examples of integrated industrial production of two substances. It differs from the Solvay process in that ammonium chloride is also precipitated from the mother liquor. The net reaction is



The amounts of ammonium chloride and soda ash produced are almost equal: two moles (107 g) of NH_4Cl are produced for each mole (106 g) of Na_2CO_3 . The modified process, requiring external NH_3 and CO_2 , must be incorporated into an integrated system of plants.

Furthermore, the treatment of ammonia-containing waste gases is carried out for environmental reasons, rather than to reduce NH_3 and CO_2 losses. Finally, the energy balance in the modified process is entirely different from that in the traditional Solvay process because of the different feedstocks and products.

Process Description

As a rule, the manufacture of NH_4Cl and Na_2CO_3 is carried out in a continuously operated recycle process in which the concentrations within the cycle may be adjusted according to whether ammonium chloride or sodium carbonate is the primary product. BASF originally developed the process in which NH_4Cl is the primary product.

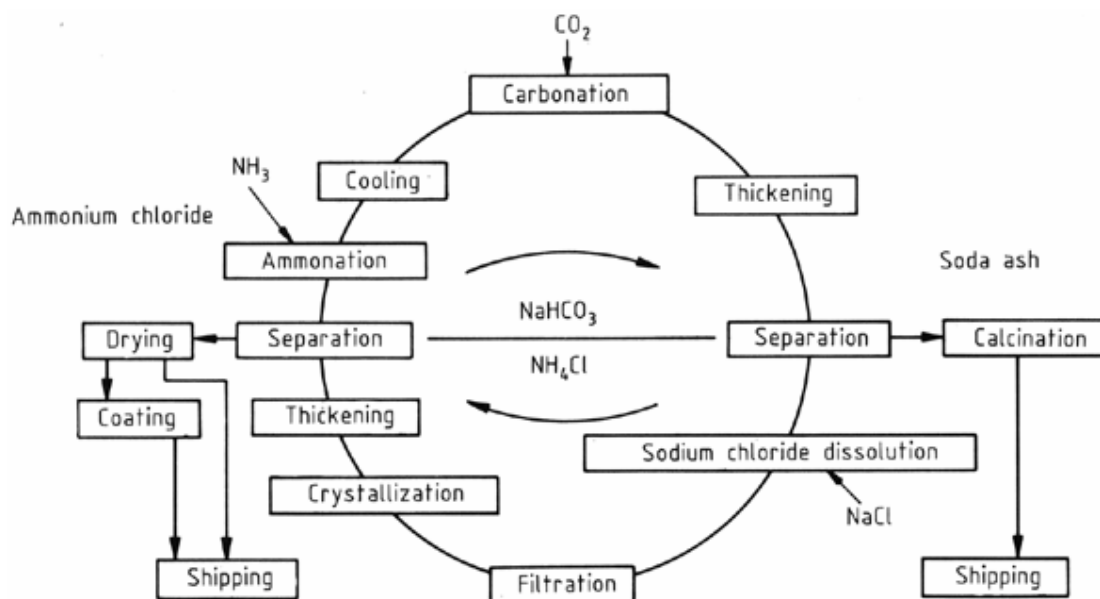


Fig. 20.1 Ammonium chloride – soda ash process

The cycle begins with the addition of ammonia and carbon dioxide to the aqueous recycle solution. The ammonium bicarbonate intermediate reacts further with the sodium chloride present in the solution to yield ammonium chloride and sparingly soluble sodium bicarbonate. After thickening, the latter is separated out with a centrifuge; it is washed during centrifugation. It is calcined to produce sodium carbonate.

The sodium bicarbonate remaining in the mother liquor is decomposed by heating to 337.2 K. The very soluble sodium carbonate does not interfere with the crystallization of ammonium chloride. The high temperature of the solution at this point is utilized to dissolve the rock salt feedstock rapidly. This rock salt is introduced via a series of mixing tanks, the quantity dictated by the material balance over the entire cycle. Any insoluble impurities in the rock salt are filtered off. The major impurities are sulfates and other salts of iron, calcium, and magnesium. Disposal of the residues represent an increasing problem. Therefore, pure salt is replacing rock salt as raw material.

Ammonium chloride is recovered from the clear solution in a two-stage crystallization by indirect cooling. The suspension from the crystallizer is thickened in hydrocyclones, and the solid is separated and washed in a centrifuge. The solid from the centrifuge containing 5 – 7 wt % water is dried in a current drier to a moisture content less than 0.1 %. The final product can be coated with an anticaking agent. After the removal of the ammonium chloride, the cycle is complete, and the mother liquor can be reused.

A process that produces soda ash as the primary product has been developed by Asahi Glass.

Ammonia is added to the mother liquor prior to the crystallization of ammonium chloride. Crushed raw salt is added with or without washing and dissolved in the solution. The solution is then cooled below 20 °C to crystallize ammonium chloride. The crystals are separated with a centrifugal separator, and the mother liquor is recycled to the carbonation section. The ammonium chloride crystals are dried to a moisture content below 0.3 % in a rotary or fluidized bed drier. Crystal sizes can be kept between 1 and 2 mm for granule or about 200 µm in diameter for high-grade compound fertilizer.

In the carbonation section, the solution is sent into a carbonating tower for the precipitation of sodium bicarbonate through the reaction with blown-in carbon dioxide at 30 – 40 °C. The resulting slurry is

separated into sodium bicarbonate crystals and the mother liquor. The mother liquor is sent to the ammonia absorption section. The separated crystals are calcined and converted to dense soda ash.

This process is capable of producing bicarbonate as pure as that from the conventional Solvay process and fertilizer-grade ammonium chloride. The purity of ammonium chloride exceeds 97 wt %.

In the Asahi Glass the ammonia is added after separation of the sodium bicarbonate. The heat of solution of ammonia serves to provide the heat needed to decompose the residual sodium bicarbonate. However, the energy saving is offset to some extent by the higher energy need in the crystallization because ammonium chloride is more soluble in the ammonia-rich solution. A washing step for the rock salt feed, prior to solution, avoids the filtration before crystallization. With or without washing this process utilizes the raw salt to the fullest extent, an important consideration in Japan where the salt must be imported.

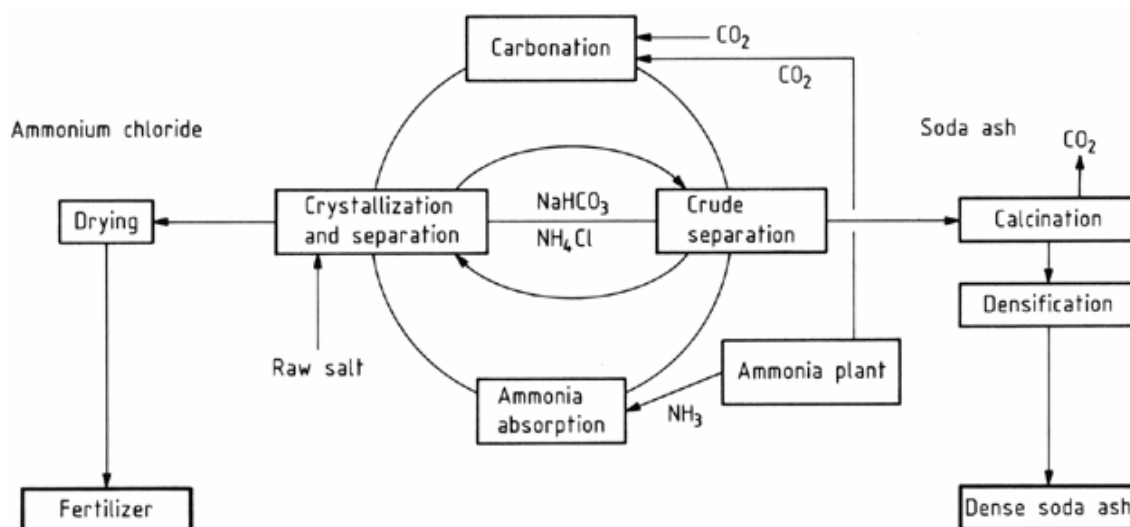


Fig. 20.2 Ammonium chloride co-production process

Equipment Variations

Little is known about the process details of the individual process variants. Nevertheless, general characteristics of the process technology, adequate for plant selection, can be provided.

The individual steps of several other important process variants are described below.

If the feed gases are available at sufficiently high pressure, they can be mixed with the recycle solution by gas – liquid contacting devices, such as jet injectors. In the case of ammonia the amount of excess gas is reduced, and the solution is homogenized rapidly. In the case of CO_2 the nucleation rate and crystal growth of NaHCO_3 are improved.

Further parameters influencing the size of the NaHCO_3 crystals are temperature and residence time in the carbonation step. The various carbonation columns used in the production of NaHCO_3 differ in the number and type of internal parts such as mixing devices and additional heat exchangers, the positioning and type of the CO_2 nozzles, and the volume needed to produce a given NaHCO_3 equivalent. The large NaHCO_3 crystals may be separated and washed using a band filter, or after thickening they may be separated by centrifugation.

The important differences in NH_4Cl crystallization lie in the number of stages and type of cooling. Direct cooling during the crystallization, in comparison with indirect cooling, uses more energy and requires the use of additional equipment for refrigerant condensation and recovery. However, it avoids

problems associated with indirect cooling, i.e., the operation of heat exchangers, from which a NaCl crust must be periodically removed. Special plate heat exchangers have been developed to solve the fouling problem, and these are to be preferred to shell and tube heat exchangers and coil coolers.

NaHCO₃ and NH₄Cl crystallization and removal are the keys to high conversion per pass and product purity.

Ammonium chloride is frequently dried in a drum drier instead of a pneumatic drier.

Pollution Problems

The pollution problems associated with plant operation are not critical. By careful operation the production of wastewater streams can be almost entirely eliminated. However, because of its high ammonia content, any wastewater produced will require treatment.

The scrubbing of waste gases with hydrochloric acid to produce additional ammonium chloride offers a neat solution to the problem of gaseous emissions. Generally ammonia conversions of 98 % and rock salt conversions of 95 % are achieved, the exact values depending on the type of process and the environmental measures.

Grades and Capacity

After closure of several production plants due to reduced market demand, the capacity for NH₄Cl by the modified Solvay process for the western hemisphere and Japan was estimated at ca. 100000 t/a in 1997. Only limited information is available on total capacities and applied technology in China. In Asia, a granular product containing more than 97 % NH₄Cl is usually produced. The ammonia nitrogen content of this fertilizer grade NH₄Cl always exceeds 25 wt %. These plants are primarily for the production of soda ash.

However, if the production of NH₄Cl is of primary interest, the process can be carried out in such a way as to give purities exceeding 99.7 wt % NH₄Cl with < 0.25 wt % NaCl and < 3 ppm Fe, a quality adequate for most industrial purposes. Reagent-grade NH₄Cl can be produced on a large scale with special operating techniques but without additional processing steps. This ammonium chloride contains < 0.01 wt % NaCl and corresponds to the ACS specification.

20.2 System Characterisation

In this study the modified Solvay process is considered. The functional unit is 1 kg ammonium. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers. Within the ammonium chloride production, sodium carbonate is derived as co-product. Therefore an allocation between the two products was made. The following processes were modelled:

- Modified Solvay process, GLO. Multi output process with ammonium chloride, at plant, GLO and sodium carbonate, from modified solvay process, plant, GLO as allocated products

20.3 Ammonium chloride

Ammonium chloride (NH₄Cl; CAS 12125-02-9; synonyms: sal ammoniac) is a clear white water-soluble crystalline salt with a biting, slightly sour taste. The aqueous ammonium chloride solution is

mildly acidic. In nature it occurs in volcanic regions, forming on volcanic rocks near fume-releasing vents. Ammonium chloride is very soluble in liquid NH₃ but virtually insoluble in acetone and pyridine. At 292.7 K its solubility in methanol is 3.24 wt %, and its solubility in ethanol is 0.64 wt %.

Ammonium chloride has two modifications. The transformation between the two is reversible at 457.6 K (184.5 °C):



The most important chemical and physical properties of ammonium chloride used in this inventory are given here.

Tab. 20.1 Chemical and physical properties of ammonium chloride (Ullmanns 2004)

Property	Value	Unit
Molecular weight	53.49	g mol ⁻¹
Specific gravity	1527	kg m ⁻³
Specific heat	1.63	kJ kg ⁻³
Boiling point	520	°C
Melting point	338	°C

20.3.1 Reserves and Resources

Although ammonium chloride occurs naturally in volcanic material, production from natural sources is of no significance. It is produced commercially by two processes:

- Modified Solvay process (ammonia – soda ash process or ASAP)
- Direct reaction between HCl and NH₃

A third process, the reaction of reciprocal pairs of salts, has not yet been commercially applied, but is still of interest.

20.3.2 Use

Ammonium chloride is widely used as an effective nitrogen fertilizer for paddy and upland rice, wheat, and other crops in Japan, China, and South East Asia. Most of the annual production of NH₄Cl in Japan is used as high-grade compound fertilizers: chloro-ammonium phosphate, chloro-potash-ammonium phosphate, magnesia-chloro-potash-ammonium phosphate, and nitrogen-potash. Further fertilizer uses are limited by the acidity and high chlorine content.

The industrial uses of technical-grade ammonium chloride are, in order of importance: solid electrolytes in dry cell batteries, one component of quarrying explosives (an especially fine form is used), hardeners for formaldehyde-based adhesives, one component of etching solutions in the manufacture of printed circuit boards, and a component, along with zinc chloride, of fluxes in tin and zinc plating. Minor uses include a rapid fixer additive in photography, cleaner additives, and a nutrient in yeast cultures. Ammonium chloride is also used in tanning, refining of precious metals, textile printing and dyeing, and in the rubber industry. The addition of ammonium chloride to tiles and bricks prior to firing was proposed. The ammonium chloride serves to control product porosity and accelerates the firing process. The use of NH₄Cl as hardener for formaldehyde-based adhesives has been reduced significantly due to substitution by chloride-free products.

High-purity ammonium chloride is employed in the food and pharmaceutical industries and in a few chemical syntheses.

20.4 Raw materials and auxiliaries

Tab. 20.2 shows values for the use of raw materials for the modified Solvay process. As there was no data available for the production process, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the ammonium chloride production.

Tab. 20.2 Consumption of raw materials and auxiliaries for the modified Solvay process

Input	Values	Remarks
Ammonium (kg)	1	Functional unit
Sodium chloride (kg)	1.71	Calculated with a yield of 95 %
Carbon dioxide (kg)	2.588	Calculated with a yield of 95 %
Water, ultrapure (kg)	0.503	Stoichiometric calculation
Cooling water (m3)	0.142	Estimation

20.5 Energy

There was no information available on the amount of energy used for the production process. In order not to neglect the process energy demand those values were approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg-1) were used as approximation for the energy consumption of the modified Solvay process. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used. A summary of the values used is given in Tab. 20.3.

Tab. 20.3 Energy consumption for the modified Solvay process

Input	per kg ammonium
Heat, natural gas burned in industrial furnace > 100 kW	11.8
Electricity (kWh)	1.97

20.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train. Tab. 20.4 summarises the total transport amounts for the modified Solvay process.

Tab. 20.4 Total transport amounts for the modified Solvay process

tkm kg ⁻¹ ammonium	lorry	train
Total transports	5.294E-01	3.176

20.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of modified Solvay process plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

20.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the modified Solvay process. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input.

Tab. 20.5 Process emissions to air from the modified Solvay process

Output	Value	Remarks
Waste heat (MJ)	7.1	calculated from electricity input
Ammonium, to air (kg)	2E-03	estimated as 0.2 % of the input
CO ₂ , fossil, to air (kg)	0.129E-01	calculated from mass balance

20.9 Emissions to water

The remaining amount of unreacted ammonium (4.8 % of the input) and of unreacted sodium chloride (5 % of the input) was assumed to leave the production process with the waste water. Further it was assumed that this wastewater is treated in a internal wastewater plant. For the ammonia within the waste water a removal efficiency of 50% was assumed. The nitrogen emitted to air (N₂) and remaining in the sewage sludge was neglected. AOX is calculated from chloride emission.

Tab. 20.6 Process emissions to water from the modified Solvay process

Output	Value	Remarks
ammonium (kg)	2.4E-02	calculated from mass balance
sodium (kg)	3.38E-02	calculated from mass balance
chloride (kg)	5.15E-02	calculated from mass balance
AOX (kg)	5.15E-02	calculated from water emissions

20.10 Allocation of ammonium chloride and sodium carbonate

Within the modified Solvay process ammonium chloride and sodium carbonate are obtained as co-products. A stoichiometric allocation of the two products is made. The used allocation factors are given in Tab. 20.7.

Tab. 20.7 Allocation of ammonium chloride and sodium carbonate from the modified Solvay process

Process	modified Solvay process, GLO		
	Unit	per kg ammonium	Allocation factor (stoichiometric)
Ammonium chloride, at plant, GLO	Kg	3.055	50
Sodium carbonate, from modified Solvay process, at plant, GLO	kg	3.055	50

20.11 Life cycle inventory of the modified Solvay process and data quality considerations

Tab. 20.8 shows the life cycle inventory and the data quality indicators for the modified Solvay process. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 20.8 Unit process raw data for the modified Solvay process

	InputGroup	OutputGroup	Name	Location	Infrastructure	Processes	Unit	modified Solvay process	Uncertainty Type	Standard Deviation %	General Comment	ammonium chloride, at plant	sodium carbonate from ammonium chloride production, at plant
	401												
	662		Location					GLO				GLO	GLO
	493		Infrastructure					0				0	0
	403		Unit					kg				kg	kg
allocated products resource, in water technosphere	-	2	ammonium chloride, at plant	GLO	0	kg	3.06E+0					100	0
	-	2	sodium carbonate from ammonium chloride production, at plant	GLO	0	kg	3.06E+0					0	100
	4	-	Water, cooling, unspecified natural origin	-	-	m3	1.42E-1	1	1.30	(4,5,na,na,na,na); Estimation		50.000	50.000
	5	-	ammonia, liquid, at regional storehouse	RER	0	kg	1.00E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation		50.000	50.000
	5	-	sodium chloride, powder, at plant	RER	0	kg	1.71E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation		50.000	50.000
	5	-	carbon dioxide liquid, at plant	RER	0	kg	2.59E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation		50.000	50.000
	5	-	water, ultrapure, at plant	GLO	0	kg	5.03E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation		50.000	50.000
	5	-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	1.18E+1	1	1.30	(4,5,na,na,na,na); Estimation		50.000	50.000
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	1.97E+0	1	1.30	(4,5,na,na,na,na); Estimation		50.000	50.000
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	5.29E-1	1	2.09	(4,5,na,na,na,na); Standard distance 100km		50.000	50.000
emission air, high population density	5	-	transport, freight, rail	RER	0	tkm	3.18E+0	1	2.09	(4,5,na,na,na,na); Standard distance 600km		50.000	50.000
	5	-	chemical plant, organics	RER	1	unit	2.37E-9	1	3.09	(4,5,na,na,na,na); Estimation		50.000	50.000
	-	4	Heat, waste	-	-	MJ	7.10E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input		50.000	50.000
	-	4	Ammonia	-	-	kg	2.00E-3	1	1.37	(4,5,na,na,na,na); Estimation		50.000	50.000
emission water, river	-	4	Carbon dioxide, fossil	-	-	kg	1.29E-1	1	1.30	(4,5,na,na,na,na); Estimation		50.000	50.000
	-	4	Ammonium, ion	-	-	kg	2.40E-2	1	1.62	(4,5,na,na,na,na); Estimation		50.000	50.000
	-	4	Sodium, ion	-	-	kg	3.38E-2	1	5.10	(4,5,na,na,na,na); calculated from mass balance		50.000	50.000
	-	4	Chloride	-	-	kg	5.15E-2	1	3.09	(4,5,na,na,na,na); calculated from mass balance		50.000	50.000
	-	4	AOX, Adsorbable Organic Halogen as Cl	-	-	kg	5.15E-2	1	1.62	(4,5,na,na,na,na); from water emissions		50.000	50.000

20.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 20.9 shows selected LCI results and the cumulative energy demand for the modified Solvay process.

Tab. 20.9 Selected LCI results and the cumulative energy demand for the modified Solvay process

Name	Location	Unit	ammonium	sodium
			chloride, at	carbonate from
			plant	ammonium
Unit	Unit	GLO	GLO	
Infrastructure	Infrastructure	kg	kg	
		0	0	
LCIA results				
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	17.5	17.5
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	3.1	3.1
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.6	0.6
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.4	0.4
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1	0.1
selected LCI results				
land occupation	resource	m2a	1.6E-1	1.6E-1
CO2, fossil	air	kg	1.1E+0	1.1E+0
NM VOC	air	kg	6.1E-3	6.1E-3
nitrogen oxides	air	kg	1.7E-3	1.7E-3
sulphur dioxide	air	kg	2.4E-3	2.4E-3
particulates, <2.5 um	air	kg	2.7E-4	2.7E-4
BOD	water	kg	1.6E-3	1.6E-3
cadmium	soil	kg	8.1E-10	8.1E-10

20.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	modified Solvay process
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	5
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Modified Solvay process including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Modifizierter Solvayprozess
	491	Synonyms	
	492	GeneralComment	The multioutput-process "modified Solvay process" delivers the co-products ammonium chloride and sodium carbonate. The allocation is based on stoichiometric calculation.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	inorganics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications.
Geography	663	Text	The inventory is modelled for the world..
Technology	692	Text	Chlorination of ammonium
Representativeness	722	Percent	100
	724	ProductionVolume	na
	725	SamplingProcedure	Literature data
	726	Extrapolations	Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

20.14 References

Frischknecht et al. 2007

Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.

- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>
- Ullmanns 2004 Zapp, K., Wostbrock, K., Schäfer, M., Sato, K., Seiter, H., Zwick, W., Creutziger, R., Leiter, H.: Ammonium Compounds. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

21 Natural gas, to helium extraction

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

The most important process to produce helium is the extraction from natural gas.

21.1 Production Technologies (Ullmann 2005)

Of the noble gases contained in natural gas, only helium is suitable for extraction. The other noble gases are present in such small concentrations that they are not worth separating, but with a content of 0.005 – 8 % the helium offers a good source for helium production. Extraction of helium by itself is only viable at contents above 1 %, and it is only worth extracting helium at lower concentrations if other products are also obtained from the natural gas.

In industrial plants a crude helium stream is first separated from the natural gas. This is then followed by one or more purification stages if highly pure helium is required. The crude helium extraction and the purification systems are often interconnected by recycle streams.

Crude Helium Extraction

Crude helium is usually produced by partial condensation at low temperatures. Increasing consideration has also recently been given to the membrane process, or to a combination of cryogenic and membrane processes.

Low-Temperature Fractionation

Partial condensation exploits the extremely low liquefaction temperature of helium. Natural gas is cooled in stages until practically all the components with the exception of helium have condensed. The residual gas phase is the crude helium product, with a helium concentration which is determined by the pressure and the lowest condensation temperature. Maximum helium concentrations of ca. 99 % can be reached at temperatures close to the melting point of nitrogen.

Industrially, helium extraction is usually combined with other processing steps for natural gas, particularly with partial condensation. Helium separation then becomes a subprocess of a main process which determines the process structure and this naturally makes a significant improvement in the cost-effectiveness of the extraction process. Very often this is carried out during the separation of nitrogen from nitrogen-rich natural gases to improve the calorific value, because a high helium content is usually associated with a high nitrogen content.

In natural gas liquefaction, the helium collects during condensation in a gas residue, which can be readily concentrated and purified. The first stages of the gas treatment and partial condensation are also already performed during the extraction of ethane and heavier hydrocarbons from natural gas, thus relieving any subsequent helium separation of this task.

An important factor in all processes is the initial concentration of helium in the natural gas. With low helium concentrations great care has to be exercised in the recovery of the helium dissolved in the condensates in order to achieve good yields.

An important part is also played by the natural gas impurities, especially CO₂, H₂S, and heavy hydrocarbons. These components can solidify at low temperatures and lead to deposits. If their concentrations exceed certain limits they must therefore be removed before they enter a low-temperature section. Great efforts are made to use skilful process technology to pass the highest possible percentage of impurities safely through the system and to avoid preceding purification steps. It is, however, impossible to avoid removing the water vapor.

The required output pressure of the helium-free natural gas is the decisive factor determining the energy requirement of the system. If the treated natural gas can be delivered under reduced pressure, then the refrigerating requirements can sometimes be covered by the Joule – Thomson effect alone. In other cases additional refrigerating cycles or recompression of the natural gas is necessary.

A simple, but typical modern process for helium recovery combined with nitrogen removal from natural gas is shown in Esterification – lower alkyl acrylate process

. The clean natural gas is cooled against cold product streams in heat exchanger (a) and is fed to the bottom of the high-pressure column (b). The condenser (c) provides reflux for this column and for the low-pressure column (d). That part which is not condensable contains most of the helium. It is drawn from the crude helium separator (e) and leaves the plant as crude helium product after warming.

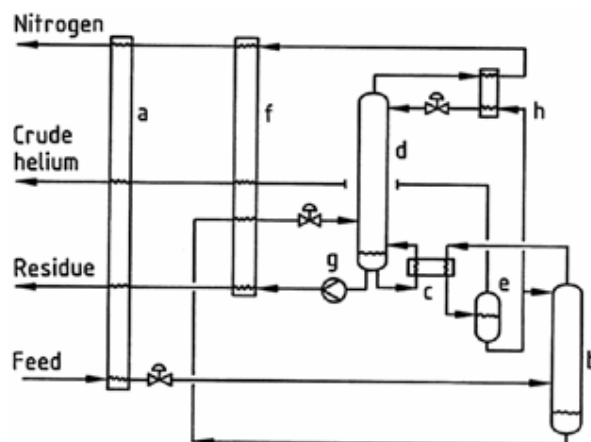


Fig. 21.1 Nitrogen rejection and crude helium recovery from natural gas (taken from Ullmanns 2005, Figure 21)

a) Feed/product exchanger; b) High-pressure column; c) Reboiler/condenser; d) Low-pressure column; e) Crude helium separator; f) Feed subcooler; g) Residue pump; h) Reflux subcooler

The bottoms from the high-pressure column, having some of the nitrogen removed, form the feed stream for the low-pressure column (d), where the final nitrogen – methane separation is performed. The overhead is the nitrogen stream, which is rewarmed against the incoming feed stream. A reboiler in the bottom of the column provides a liquid, methane-rich residual product, which is pumped to an elevated pressure, vaporized and rewarmed against the incoming feed.

The reboiler duty in the low-pressure column necessary to provide a residue with low nitrogen content is provided by condensing nitrogen at the top of the high-pressure column. This is possible because of the different pressures of the two columns. The condensation temperature of nitrogen in the overhead of the high-pressure column (b) is higher than the boiling temperature of the methane-rich liquid in the low pressure column (d).

A significantly more complicated process is used in a plant in the United States. The following products, besides a crude helium fraction, are recovered from the natural gas: a C_{2+} fraction, a methane sales gas, a fuel gas fraction, and a nitrogen fraction. The composition of the feed gas and the products is given here.

Tab. 21.1 Composition of the feed gas and products (Ullmann 2005)

Composition, mol %	Feed gas	NGL	Sales gas	Fuel gas	He	N ₂
Helium	0.5		0.01		56.94	0.17
Nitrogen	10.2		2.99	23.48	42.99	99.55
Methane	73.02	0.7	96.52	76.51	0.07	0.28
Ethane	6.94	41.21	0.47	0.01		
Propane	5.16	32.07	0.01			
C ₄ +	4.18	26.02				
Flow, kmol/h	2988	480.1	2019.4	299.1	25.4	164
Pressure, bar	44	9	8	8	27	1
Temperature, K	311	262	306	306	306	306
Gross wet calorific value, MJ/kmol	989.7	2154.5	852.7	670.2		

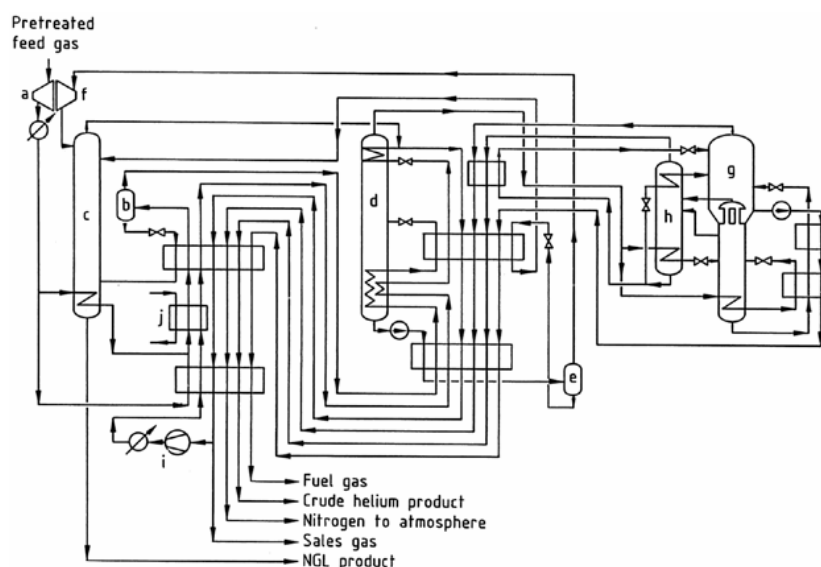


Fig. 21.2 Schematic of a plant for recovery of helium from natural gas, with integrated separation of nitrogen, methane, and a C₂+ fraction (taken from Ullmanns 2005, Figure 22)

a) Expander booster compressor; b) Cold separator; c) Demethanizer; d) Nitrogen enrichment column; e) Expander suction scrubber; f) Turbine; g) Low pressure column; h) Helium column; i) Recycle compressor; j) Propane refrigeration

Pretreated natural gas is compressed to 50 bar in an expander booster compressor (a) and is cooled to 230 K by heat exchange with low-temperature propane and with cold products. The condensate is separated in the separator (b) and conveyed to the demethanizer (c).

After partial condensation, the gas from the separator (b) enters the nitrogen enrichment column (d), which operates at 31 bar. The bottom product from this column is pumped to 40 bar and partially vaporized, and the resulting phases are separated in the expander suction scrubber (e). Both phases are fed to the demethanizer (c)—the liquid phase after partial vaporization, the gas phase after expansion in the turbine (f). The liquid produced in the turbine serves as a reflux for the column.

The bottom product from the demethanizer (the C₂+ fraction) and the top product (the sales gas, mainly methane), leaves the system, the top product having been warmed up again against the feed gas and the recycle stream.

The top product from column (d), enriched with nitrogen, is fed to a double column (g). After it has been subcooled, the bottom product from the lower, high-pressure section of the column forms the feed for the upper, low-pressure section of the column. The gaseous top product of the high-pressure column, in which almost all the helium has collected, and part of the top condensate are fed into the helium column (h) where the crude helium is concentrated to 57 % and, after being warmed up, leaves the system.

After subcooling, the bottom product from the helium column (h) serves as a reflux for the low-pressure section of the double column (g), from the top of which the nitrogen fraction is drawn off and discharged to the atmosphere after warming up. The bottom product is pumped to about 9 bar and, after warming up, used as fuel gas.

The necessary heating and cooling of the nitrogen enrichment column (d) is provided by a recycle with part of the sales gas via the recycle compressor (i).

Crude Helium Extraction by Permeation Processes

Also suitable for separating helium from natural gas are processes based on the principle of permeation. Gas permeation through homogeneous membranes occurs by solution – diffusion. The permeability coefficient is equal to the product of Henry's law constant S_i , and the diffusivity coefficient D_i , of component i in the membrane material. A membrane's ability to separate a gas mixture can be obtained from the ratio of the permeation coefficients: $\alpha_{ij} = P_i/P_j$, where α is the selectivity or separation factor of gaseous components i and j . Homogeneous membranes have a high permeability for helium compared to the remaining components of natural gas, i.e., nitrogen, methane, carbon dioxide and others.

The first applications of membranes for the extraction of helium date from 1958 when helium was extracted from natural gas by means of tubular silicate and quartz glass membranes. Commercial use of membrane processes for extracting hydrogen began in the early 1970s and since 1985 they have also been used for helium separation from natural gas for the production of crude helium.

The heart of a membrane process is the separation module. Commercially available separation modules contain either hollow fiber bundles (Permea, Ube) or spiral-wound gas permeation elements (Grace, Separex).

Fig. 21.3 shows a schematic of a helium concentration process. In this case the two stages of the membrane system consists of a row of separators each containing a hollow fiber bundle physically similar to a shell and tube heat exchanger. The membrane is an aromatic polyimide polymer membrane with a membrane area of 9000 m² per m³ of volume. The membrane works satisfactorily in the temperature range – 80 to + 150 °C and tolerates natural gas accompanying compounds such as water (up to saturation), hydrogen sulfide (up to 5 vol %) as well as C₅+ (up to saturation). Fig. 21.3 Helium concentration process (taken from Ullmanns 2004, Figure 23)

Tab. 21.2 gives some operating figures of the first stage of a membrane helium recovery system for helium-rich natural gas.

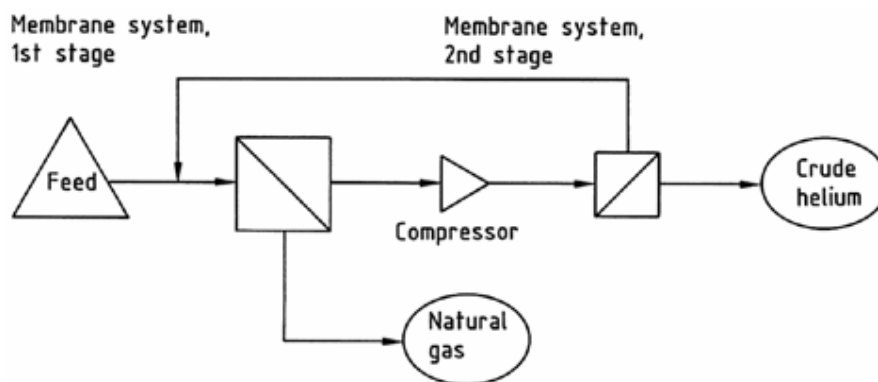


Fig. 21.3 Helium concentration process (taken from Ullmanns 2004, Figure 23)

Tab. 21.2 Operating figures of the first stage of a membrane helium-recovery unit (taken from Ullmanns 2005, Tab. 19)

	Feed	1st Stage permeate	Crude He
Pressure, bar	35	1.05	1.05
He recovery, %			95
Composition, mol %			
He	2.40	11.2	59.53
CO ₂	0.07	0.3	0.76
Ar			
N ₂	24.54	22.3	10.02
CH ₄	72.71	66.1	29.68
C ₂ +	0.28	0.1	0.01

By their nature membrane processes are more suitable for coarse separation of helium from natural gas; i.e., it is not possible to achieve a high level of helium purity combined with a high helium yield in a simple membrane process. If this is required it is necessary to change to a system of several membrane process steps connected in series. One or more compressor stations would then be needed between the process steps and for returning residual gases which are produced during the membrane process from retentates and permeates to be recycled. This would overextend the entire process and make it uneconomical. In such cases there is much to recommend a combination of cryo-process and membrane process, or of membrane process and pressure swing adsorption (PSA) process.

Several hybrid membrane – cryogenic or membrane – cryogenic – adsorption processes have been suggested. The most advantageous seems to be the hybrid process with the following characteristics (Fig. 24). Helium-containing natural gas (ca. 2 % He, 26 % N₂, 64 % CH₄) at ca. 32 bar is cooled to ca. – 50 °C to condense all the heavier hydrocarbons. Then the gas is further cooled to ca. – 150 °C to condense all the methane and some of the nitrogen. The gas from these separation stages contains 30 – 35 vol % helium. This helium-rich stream is fed to the membrane system, in which a number of membrane stages can be configured to maximize the separation efficiency and to produce the desired 95 % stream. Any unrecovered helium is recycled to the cryogenic unit. To obtain grade A helium (purity 4.5), a PSA unit may be integrated. The process can be optimized by varying the helium content of the helium-enriched stream leaving the cryogenic stage and hence the membrane area to He content ratio. It is claimed that the power required by the hybrid process is only 54 % of that of a purely cryogenic process.

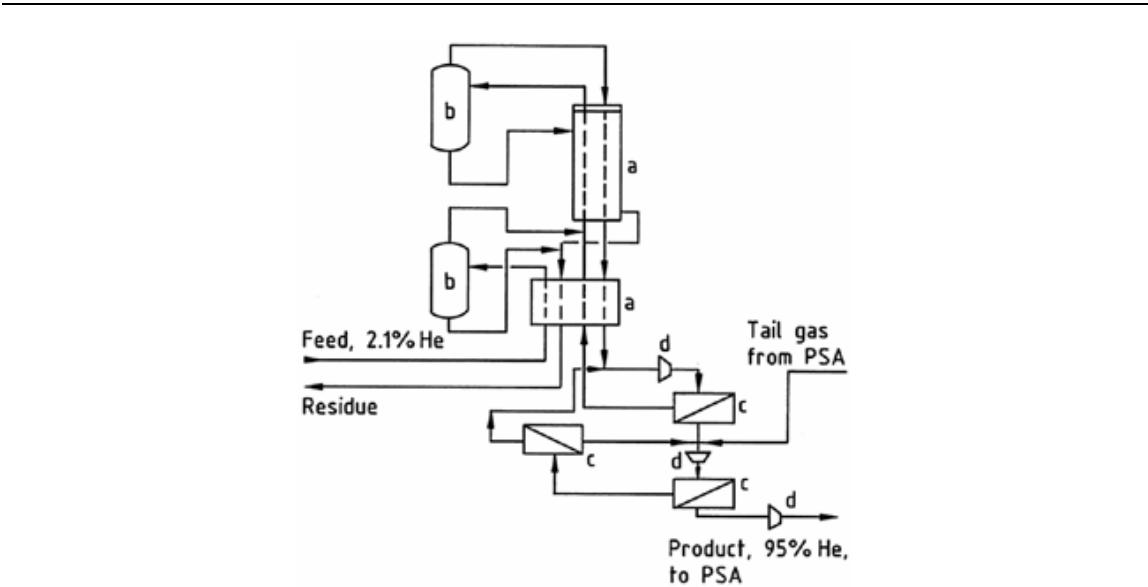


Fig. 21.4 Cryogenic – membrane hybrid process (taken from Ullmanns 2005, Figure 24)

a) Heat exchangers; b) Separators; c) Membrane stages; d) Compressors

Purification of Helium

Besides adsorptive processes also cryogenic and chemical processes are used for the purification of crude helium. Different process steps are frequently linked with one another in case the required purity cannot be achieved in a single step.

Purification of Helium by Pressure Swing Adsorption

Helium is retained only weakly by adsorbents so it is easy to purify by adsorption processes. In principle this involves processes with thermal regeneration or pressure swing adsorption (PSA) processes. In processes with thermal regeneration heat is supplied to the process externally for desorption, but in PSA processes the regeneration is carried out without the supply of external heat, merely by pressure release and purging at low temperature. Generally, processes with thermal regeneration are confined to the removal of trace impurities or readily adsorbed components. At ambient temperature these processes are used for removing water and carbon dioxide, and at low temperatures also for removing, for example, traces of nitrogen, oxygen, neon, and hydrogen. With PSA systems it is possible to remove less readily adsorbed components at ambient temperature and even at quite high concentrations.

During the extraction of helium from natural gas the hydrocarbons and the majority of the nitrogen are separated in the low-temperature system. Nitrogen is virtually the only helium impurity requiring secondary purification in the PSA system. Zeolitic molecular sieve and narrow-pored activated carbon are suitable adsorbents for nitrogen. These adsorbents give final purities of 99.999 – 99.9999 vol % He without additional purification steps.

Pressure swing adsorption systems for helium purification are comparable in principle with systems for hydrogen purification; in this process helium behaves in a similar way to hydrogen. Adsorption systems with four or less adsorber vessels are normally used because the feed gas flows to be processed are smaller than in the hydrogen systems (see Fig. 21.5). In such a system with 4 adsorbers the individual adsorbers pass cyclically through the following process steps: adsorption, pressure equalization with another adsorber, further pressure release in parallel flow to the adsorption step to output purge gas for another adsorber, counter-current pressure release to output tail gas, purging with decompression gas from another adsorber, pressure build-up by pressure equalization with yet another

adsorber, and further pressure build-up to the adsorption pressure with helium product. A helium yield of about 80 % can be achieved with an adsorption pressure of 15 – 20 bar.

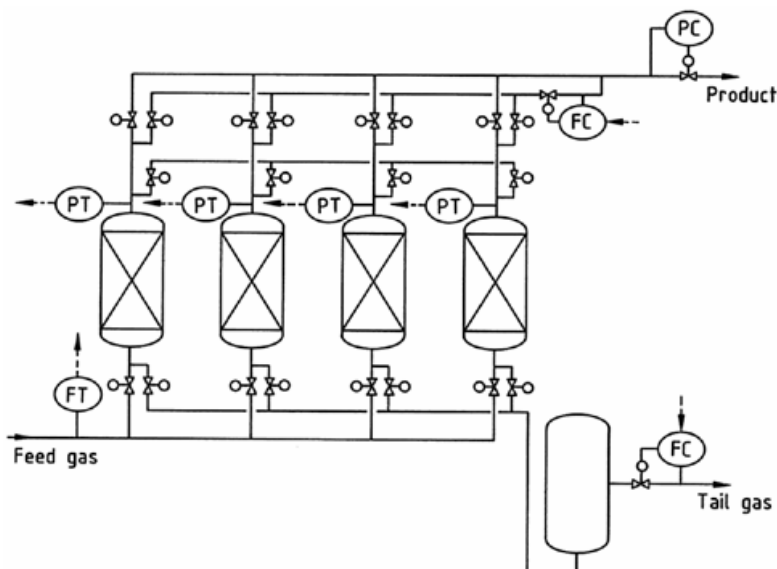


Fig. 21.5 Schematic of a typical four-adsorber PSA unit (taken from Ullmanns 2005, Figure 25)

The tail gas produced in PSA processes still contains considerable quantities of helium so, unlike the procedure in the majority of hydrogen PSA systems, it is compressed again because of the substantially higher value of helium and returned to the cryogenic unit, resulting in a helium yield of practically 100 % for the entire system.

In principle a PSA system can remove nitrogen contents of up to 50 vol % and produce ultrapure helium. The N₂ loading at the adsorption pressures normally used is virtually linear over a fairly large partial pressure range so the adsorber size only increases slightly with increasing N₂ concentration in the crude helium, while the helium yield falls with increasing adsorber size.

Due to the recycling of tail gas, the quantity of crude helium fed to the PSA system increases in inverse proportion to the helium yield of the PSA. The capital costs and the energy consumption for re-compression of the tail gas to the cryogenic unit shows the same behavior. On economic grounds it has therefore been found that an N₂ content of 5 – 15 vol % in the crude helium to the PSA is the most cost-effective solution for combination with low-temperature separation.

Other Processes for Purification of Helium

Except for hydrogen, virtually all components can be separated from helium by pressure swing adsorption. Normally there is no hydrogen in helium, but if hydrogen is present, e.g., through the addition of industrial gases, then other processes have to be used.

Hydrogen is removed by catalytic combustion. The crude helium is mixed with air and the mixture is passed over a catalyst. If the resulting heat of reaction is too high then a recycle system can be used to reduce the hydrogen content in the feed gas to the reactor. The water formed is condensed out by cooling (provided that this is possible from the point of view of the partial pressures) and the remaining water can be separated together with the other impurities in a pressure swing adsorption system.

Purification processes based on hydride-forming metal alloys are also suitable for producing high-purity helium and for purification of the other noble gases. Noninert gas constituents (H₂O, CO₂, O₂, CO, C_mH_n) are combined on the surface of highly active getter alloys, or, as in the case of hydrogen,

they diffuse into the metal lattice. When the partial pressure reaches the so-called hydride formation plateau the hydrogen is combined as metal hydride.

Highly effective getter alloys consist of Ti, V, Fe, and Mn in varying molar ratios depending on the application; the H_2 partial pressure can be lowered to 5×10^{-6} bar by $TiV1.5Fe0.4Mn0.1$. The gas flow is passed directly through the getter beds (pressure drop < 1 bar), which can be regenerated by raising the temperature (200 – 500 °C) or returned to the manufacturer after use (e.g., HWT, Germany). A helium purity of 6.0 (99.9999 vol %) is achieved; a purity of 7.0 can be attained if the gas is passed over a high-temperature getter bed at 350 – 450 °C, which allows practically no foreign gas constituents to pass. This results in an oxygen content, for example, which can no longer be detected by current analytical methods.

Example of a Complete Process

The two plants at Alfortville in France where nitrogen is separated from Groningen natural gas and helium is extracted are examples of plants which perform crude helium extraction as well as helium purification.

The helium content in the Groningen natural gas is relatively low compared with many natural gases in the United States or in Poland for which systems have also been built for helium extraction. For this reason, particular emphasis was placed here on minimizing the helium losses in the other product streams. Another particular feature is that the CO_2 is not removed in spite of its relatively high content of 0.9 % in the crude gas. Fig. 21.6 shows the process layout.

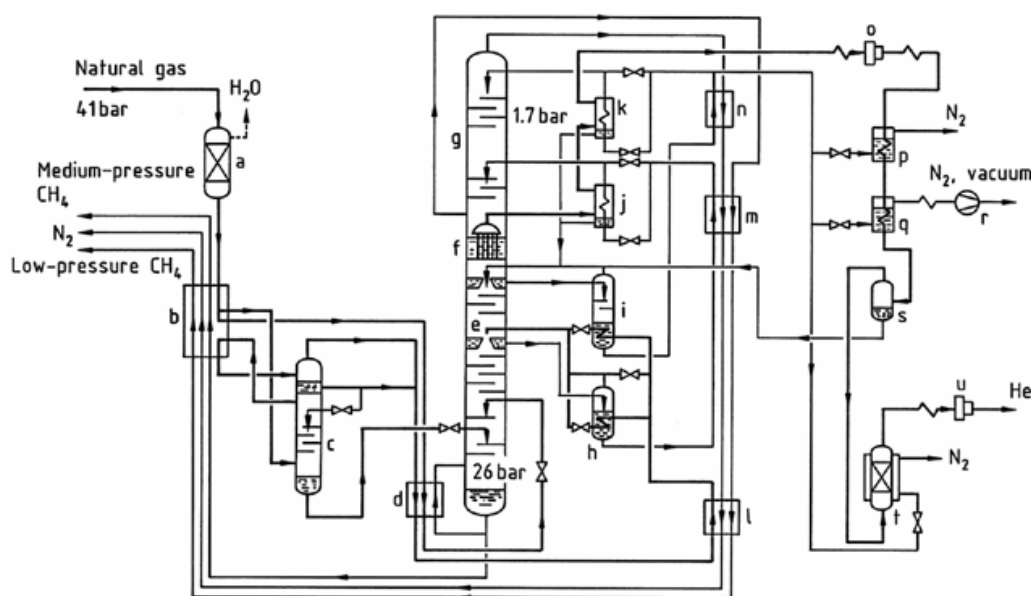


Fig. 21.6 Plant for the extraction of pure helium from Groningen natural gas (Alfortville) (taken from Ullmanns 2005, Figure 26)

a) Adsorber; b) Main heat exchanger; c) Carbon dioxide column; d) Reboiler; e) Medium-pressure column; f) Condenser; g) Low-pressure column; h), i) Helium strippers; j), k) Helium concentrators; l), m), n) Low-temperature coolers; o) Helium compressor; p) Nitrogen evaporator; q) Vacuum nitrogen evaporator; r) Vacuum pump; s) Nitrogen separator; t) Adsorber; u) Helium compressor

After removing aromatic hydrocarbons and water in the adsorber (a), CO_2 -free top product and a CO_2 -rich bottom product is obtained by rectification of a partial stream of the natural gas. The bottom product and the untreated natural gas stream condensed in the reboiler (d) are fed into the lowest sec-

tion of the medium-pressure column (e). After condensation in the reboiler (d) and in the heat exchanger (l), the CO₂-free top product from column (c) is fed into a higher section of column (e) to extract the CO₂ from the vapor rising from the lower section. In this way all the CO₂ accumulates in the bottom product of column (e) which, due to the presence of C₂+ hydrocarbons concentrated here, has sufficient dissolving capacity for CO₂ to carry it out of the system without causing any problems.

All the helium introduced by the natural gas flows with the nitrogen to the top of column (e) and is enriched in condenser (f) during the nitrogen liquefaction. Part of the helium is dissolved in the liquid trickling down and is lost with the liquid fractions which are fed from column (e) to the final nitrogen/methane separation stage in the low-pressure column (g). In order to prevent this and to reduce helium losses to a minimum the liquids drawn off are stripped in the reboiler (h) or the column (i). The helium driven off during this procedure is returned to the column (e). In this way, practically all the helium can be extracted at the top of condenser (f) at a concentration of ca. 10 %. In the two following heat exchangers (j) and (k), the helium concentrators, the helium is cooled by evaporating liquids (crude methane and nitrogen) and is enriched to 86 to 90 %.

For further concentration, the crude helium is first compressed by the diaphragm compressor (o) to 200 bar and then cooled until close to the melting point of nitrogen. The refrigeration is provided by liquid nitrogen, which vaporizes in heat exchanger (p) under atmospheric pressure and in heat exchanger (q) under vacuum. The condensate produced is separated in vessel (s) and returned together with the condensates from the concentrators (j) and (k) to the medium-pressure column (e) for recovery of dissolved helium. The helium, which now only contains 0.5 % N₂, is transferred to the final purification stage through the nitrogen-cooled molecular sieve adsorber (t). It leaves this with less than 5 vol ppm impurities. In compressor (u) it is finally compressed to the storage pressure of 350 bar.

During normal operation the system requires no external energy. The fractionation work and refrigeration requirements are covered simply by lowering the pressure of the natural gas. Nevertheless, 72 % of the combustible components can be produced as a medium-pressure fraction at 25 bar.

21.2 System Characterisation

In this study the extraction and purification of helium from natural gas is considered. The functional unit is 1 MJ of natural gas input. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers. Within the helium extraction, natural gas liquids (the C₂+ fraction) and sales gas are derived as co-products. Therefore an allocation between the three products was made. The following processes were modelled:

- Helium extraction, GLO. Multi output process with helium, liquid, at plant, GLO, natural gas liquids from helium extraction, at plant, GLO, and sales gas from helium extraction, at plant, GLO as allocated products

21.3 Helium

Helium (He; CAS 7440-59-7) is a colorless, odorless, tasteless gas. It occurs in two isotopes: ³He and ⁴He. Helium is an inert noble gas. Its boiling and melting points are the lowest among the elements. Except in extreme conditions, it exists only as a gas. At temperatures near absolute zero, it is a superfluid. The most important chemical and physical properties of helium used in this inventory are given here.

Tab. 21.3 Chemical and physical properties of Helium (Ullmann 2005)

Property	Value	Unit	Remarks
³ He: Abundance of the isotope	1.37E-4	%	
⁴ He: Abundance of the isotope	99.999863	%	
Molecular weight	4.0026	g mol ⁻¹	
Specific gravity	124.8	kg m ⁻³	liquide
Specific gravity	0.17	kg m ⁻³	gaseous
Boiling point	-268.9	°C	at normale pressure
Melting point	-272.2	°C	under pressure

21.3.1 Reserves and Resources

Helium can be produced in two different ways. Either, by liquification of air, or by separation from natural gas. Most industrial ethanol is manufactured by the separation from natural gas, because the concentration of helium is much higher in natural gas than in air (Ullmann 2004). The economically exploited helium reserves are concentrated in the USA (see Tab. 21.4)

Tab. 21.4 World helium resources in 1996 (Ullmann 2005)

Country	Helium resource (10 ⁶ m ³)	Share (%)
USA	11850	43.5
GUS	8900	32.7
Algeria	2000	7.3
Canada	2000	7.3
China	1000	3.7
Poland	800	2.9
Netherlands	672	2.5

21.3.2 Use

Helium is used on account of its special properties when use of another gas is impossible or only partially feasible.

Liquid helium is used in cryogenic technology as a refrigerant, for example, for superconducting magnets:

- For magnetic resonance imaging tomography in medical diagnosis,
- In NMR spectrometers,
- In particle accelerators,
- In magnetic separators, and
- For research in solid-state and high-energy physics.

Other uses include cryopumps for ultrahigh vacuum, superconducting sensors (SQUID's), superconducting memories and circuit elements in computers, and low noise amplifiers. Interesting potential uses are the cooling of superconducting power station generators, electromotors for ships, and nuclear fusion reactors.

Gaseous helium is used in rocketry for conveying liquid fuel and for inerting rocket motors.

In diving, helium is used instead of nitrogen to avoid problems of decompression. In medicine it is used to improve the ease of breathing of asthma sufferers, and helium-containing gas mixtures are used to investigate the functioning of the lungs.

Helium is used as a carrier gas in gas chromatography, as a filler gas for thermometers, and in counter tubes for the measurement of high-energy radiation. Helium is of importance in leak detection in plant construction and vacuum technology. It can be used to detect even the smallest leaks. This application exploits the high rate of diffusion of helium and the high sensitivity of helium mass spectrometers.

It is also used for filling balloons and airships. Its high buoyancy and nonflammability make it especially suitable for this purpose.

In high-temperature nuclear reactors it is used as a coolant. ^4He is the only element that gives no radioactive products upon high-energy irradiation. The high thermal conductivity of helium is important in this application.

Helium is essential to many high-technology processes, an example being the production of optical glass fibers, in which it serves as a coolant and as a carrier gas for the chlorine used for the removal of hydroxyl groups. Finally, helium is the major component in the gas mixture used in the CO_2 laser.

The European consumption of helium in 1997 is given in Tab. 21.5.

Tab. 21.5 European helium consumption in 1997 (Ullmann 2005)

Use	Share (%)
Low-temperature technology	36
Welding, cutting	14
Optical fibers	8
Breathing mixtures, diving	6
Analysis	14
Leak detection	9
Balloons	7
Other uses	6

21.4 Raw materials and auxiliaries

Tab. 21.6 shows values for the use of raw materials and auxiliaries of the helium extraction.

The amount of cooling water used within the helium extraction was estimated with data from "air separation, cryogenic, RER".

Tab. 21.6 Consumption of raw materials and auxiliaries for the helium extraction (Ullmanns 2005)

Input	Values	Remarks
Natural gas (MJ)	1	Functional unit
Cooling water (m^3)	5.87E-05	estimated from "air separation, cryogenic, RER"

21.5 Energy

The fuel gas is used as energy source. A value of 0.0755 MJ/MJ natural gas is calculated from the data, which are given by Ullmanns (2005). The burning of the fuel gas is calculated with data from "natural gas, burned in gas motor, for storage, GLO".

21.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train. Tab. 21.7 summarises the total transport amounts for the helium extraction.

Tab. 21.7 Total transport amounts for the helium extraction

tkm MJ ⁻¹ natural gas	lorry	train
Total transports	2.17E-03	1.3E-02

21.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of helium extraction plants. Therefore, in this study, the infrastructure is estimated based on the dataset "air separation, cryogenic, RER".

21.8 Emissions to air

The emissions from the burning of the fuel gas are calculated with emission factors from "natural gas, burned in gas motor, for storage, GLO".

Other emissions to air is methane from crude helium purification. It is assumed, that the whole methane from crude helium is emitted to air with the nitrogen. The nitrogen is also emitted to air but not listed in this study.

Tab. 21.8 Process emissions to air from the helium extraction

Output	Value
methane, fossil (kg)	1.39E-06
CO, fossil (kg)	4.53E-06
CO ₂ , fossil (kg)	4.23E-3
NM VOC (kg)	6.04E-07
NO _x (kg)	1.51E-06
P < 2.5 (kg)	1.51E-07
SO ₂ (kg)	4.15E-08
Hg (kg)	1.13E-11
dioxin (kg)	2.19E-18
waste heat (MJ)	8.31E-02

21.9 Emissions to water

No emissions to water are considered.

21.10 Wastes

No wastes are considered.

21.11 Allocation of helium, NGL, and sales gas

Within the helium extraction, natural gas liquids and sales gas are obtained as co-products. As allocation factor the mass of the products is used. The used allocation factors are given in Tab. 21.9.

Tab. 21.9 Allocation of co-products from the helium extraction

Process	Helium extraction, GLO		
	kg per MJ natural gas	MJ per MJ natural gas	Allocation factor
Helium, at plant, GLO	2.18E-05	- ¹	0.123
Natural gas liquids from helium extraction, at plant, GLO	5.261E-03	3.897E-01	29.593
Sales gas from helium extraction, at plant, GLO	1.25E-02	6.487E-01	70.284

1: No energy content because helium is a noble gas.

21.12 Life cycle inventory of helium extraction and data quality considerations

Tab. 21.10 shows the life cycle inventory and the data quality indicators for the helium extraction. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 21.10 Unit process raw data for the helium extraction

InputGroup	OutputGroup	Name	Location	InfrastructureProcesses	Unit	natural gas, to helium extraction	UncertaintyType	StandardDeviation95%	GeneralComment	helium, at plant	sales gas, from natural gas, helium extraction	natural gas liquids, from natural gas, helium extraction
401												
662		Location				GLO				GLO	GLO	GLO
493		InfrastructureProcess				0				0	0	0
403		Unit				kg				kg	kg	kg
allocated products	- 2	helium, at plant	GLO	0	kg	2.18E-5				100	0	0
	- 2	sales gas, from natural gas, helium extraction	GLO	0	kg	5.26E-3				0	100	0
	- 2	natural gas liquids, from natural gas, helium extraction	GLO	0	kg	1.25E-2				0	0	100
resource, in water technosphere	4	- Water, cooling, unspecified natural origin	-	-	m3	5.87E-5	1	1.30	(4,5,na,na,na,na); Estimation	0.123	29.593	70.284
	5	- natural gas, high pressure, at consumer	RER	0	MJ	1.00E+0	1	1.14	(2,1,3,1,1,3); Ullmanns 2004	0.123	29.593	70.284
	5	- gas motor 206kW	RER	1	unit	4.95E-10	1	3.02	(2,1,3,1,1,3); Ullmanns 2004	0.123	29.593	70.284
	5	- transport, lorry >16t, fleet average	RER	0	tkm	2.17E-3	1	2.09	(4,5,na,na,na,na); Standard distance 100km	0.123	29.593	70.284
	5	- transport, freight, rail	RER	0	tkm	1.30E-2	1	2.09	(4,5,na,na,na,na); Standard distance 600km	0.123	29.593	70.284
	5	- air separation plant	RER	1	unit	1.37E-11	1	3.09	(4,5,na,na,na,na); Estimation	0.123	29.593	70.284
emission air, high population density	- 4	Heat, waste	-	-	MJ	8.31E-2	1	1.14	(2,1,3,1,1,3); Calculated from the energy input	0.123	29.593	70.284
	- 4	Methane, fossil	-	-	kg	1.39E-6	1	1.62	(4,5,na,na,na,na); Estimation	0.123	29.593	70.284
	- 4	Carbon monoxide, fossil	-	-	kg	4.53E-6	1	5.10	(4,5,na,na,na,na); Estimation	0.123	29.593	70.284
	- 4	Carbon dioxide, fossil	-	-	kg	4.23E-3	1	1.30	(4,5,na,na,na,na); Estimation	0.123	29.593	70.284
	- 4	NM VOC, non-methane volatile organic compounds, unspecified origin	-	-	kg	6.04E-7	1	1.62	(4,5,na,na,na,na); Estimation	0.123	29.593	70.284
	- 4	Nitrogen oxides	-	-	kg	1.51E-6	1	1.62	(4,5,na,na,na,na); Estimation	0.123	29.593	70.284
	- 4	Particulates, < 2.5 um	-	-	kg	1.51E-7	1	3.09	(4,5,na,na,na,na); Estimation	0.123	29.593	70.284
	- 4	Sulfur dioxide	-	-	kg	4.15E-8	1	1.30	(4,5,na,na,na,na); Estimation	0.123	29.593	70.284
	- 4	Mercury	-	-	kg	1.13E-11	1	5.10	(4,5,na,na,na,na); Estimation	0.123	29.593	70.284
	- 4	Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin	-	-	kg	2.19E-18	1	3.09	(4,5,na,na,na,na); Estimation	0.123	29.593	70.284

21.13 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 21.11 shows selected LCI results and the cumulative energy demand for the helium extraction.

Tab. 21.11 Selected LCI results and the cumulative energy demand for the helium extraction

Name		Unit	helium, at plant	sales gas, from natural gas, helium extraction	natural gas liquids, from natural gas, helium extraction
Location	Unit		GLO kg 0	GLO kg 0	GLO kg 0
Infrastructure					
LCIA results					
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	67.1	66.9	66.9
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	0.3	0.3	0.3
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0	0.0	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.0	0.0	0.0
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-	-	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0	0.0	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.1	0.1	0.1
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.0	0.0	0.0
selected LCI results					
land occupation	resource	m2a	6.1E-3	6.1E-3	6.1E-3
CO ₂ , fossil	air	kg	7.0E-1	7.0E-1	7.0E-1
NM VOC	air	kg	1.4E-3	1.4E-3	1.4E-3
nitrogen oxides	air	kg	1.7E-3	1.7E-3	1.7E-3
sulphur dioxide	air	kg	1.4E-3	1.4E-3	1.4E-3
particulates, <2.5 µm	air	kg	6.2E-5	6.2E-5	6.2E-5
BOD	water	kg	3.4E-4	3.3E-4	3.3E-4
cadmium	soil	kg	5.7E-11	5.7E-11	5.7E-11

21.14 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	natural gas, to helium extraction
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformatic	201	Type	5
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Extraction of helium from natural gas including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Erdgas, Heliumextraktion
	491	Synonyms	
	492	GeneralComment	The multioutput-process "natural gas, helium extraction" of delivers the co-products helium, natural gas liquids and sales gas. The allocation is based on mass calculation.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	extraction
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Gewinnung
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	2000
	602	EndDate	2007
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications.
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	Helium extraction from natural gas
Representative	722	Percent	100
	724	ProductionVolume	na
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAn	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

21.15 References

Frischknecht et al. 2007

Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.

Ullmanns 2004

Häussinger, P., Glatthaar, R., Rhode, W., Kick, H., Benkmann, C., Weber, J., Wunschel, H.-J., Stenke, V., Leicht, E., Stenger, H.: Noble gases. Published online: 2001. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

22 Phosphane

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

22.1 Introduction

Phosphane (PH_3 ; CAS 7803-51-2; synonyms: methylphosphine, phosphine, phosphamine, phosphorus hydride) is a colorless, very poisonous gas with an intense and penetrating odor. The odor threshold is extremely dependent on the purity of the gas. High-purity PH_3 (5 N), as used in the semiconductor industry, is detectable by smell only when the concentration is at a dangerously toxic level (> 1000 ppm), but when PH_3 is produced by the hydrolysis of AlP or Mg_3P_2 , it can be detected even at ca. 20 ppb. Phosphane is very toxic. The LC_{50} and LD_{50} values show that the acute toxicity decreases with increasing molecular mass. Above 375°C , phosphine decomposes into the elements. Phosphine – air mixtures are explosive over a wide range. The lower explosion limit is 1.6 %, and the upper limit is close to 100 %. Pure phosphine should presumably therefore be regarded as spontaneously flammable like phosphine that contains P_2H_4 as an impurity. With strong acids, phosphine forms phosphonium salts; e.g., with hydrogen iodide, phosphonium iodide, PH_4I , is formed, from which pure phosphine may be liberated by adding water or solutions of alkalimetal hydroxides.

For this inventory the functional unit is 1 kg of highly pure phosphane. The most important chemical and physical properties of phosphane used in this inventory are given here.

Tab. 22.1 Chemical and physical properties of phosphane (Ullmanns 2004)

Property	Value	Unit
Molecular weight	34	g mol^{-1}
Specific gravity	0.14	kg m^{-3}
Boiling point	-88	$^\circ\text{C}$
Melting point	-134	$^\circ\text{C}$

22.1.1 Reserves and Resources

World production of phosphane is ca. 1500 t/a.

22.1.2 Use

The phosphane is mostly used captively as a starting material in the production of organophosphorus compounds, mainly by addition of aldehydes or alkenes to the P–H bonds. Phosphine is also used for pest control where foodstuffs and animal feeds are stored. Almost all dry products of this type are suitable for gassing with phosphane without any harmful effects on such properties as smell, taste, and appearance. The main area of use is in gassing grain silos. In the semiconductor industry, specially purified phosphane (arsine-free in particular) is used as a dopant.

22.2 Production Technologies (Ullmanns 2004)

Phosphine can be synthesized by a number of different reactions. On the laboratory scale, it may be prepared simply by the hydrolysis of metal phosphides or phosphonium salts. The two most common industrial methods are based on disproportionation reactions of elemental phosphorus catalyzed by alkalis or acids. A well-known example is the reaction of white phosphorus with sodium hydroxide so-

lution, which is used to produce both phosphane and sodium hypophosphite, NaH_2PO_2 (sodium phosphinite).



In sodium hypophosphite production plants, the phosphane byproduct is usually burnt off. In the presence of alcohols, the reaction may be controlled such that the product formed in addition to phosphane is almost entirely sodium phosphite Na_2HPO_3 . At the same time, the maximum yield based on the amount of phosphorus used increases from 25 % to 50 %.

In acid-catalyzed disproportionation, white phosphorus is converted into phosphane and phosphoric acid at ca. 280 °C, red phosphorus being formed as an intermediate.



This reaction gives the highest yield of PH_3 based on the amount of phosphorus used (theoretical yield 62.5 % for phosphane and 37.5 % for phosphoric acid). Due to the high reaction temperature and the corrosiveness of the medium, the reactor must be made of graphite.

The phosphoric acid is then neutralized with sodium hydroxide.

22.3 System Characterisation.

In this study the production of phosphane by means of the hydration of phosphorus is considered. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

22.4 Raw materials and auxiliaries

Tab. 22.2 shows values for the use of raw materials of the production of phosphane. As there was no data available for the production of phosphane, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007). The produced phosphoric acid is then neutralized with sodium hydroxide.

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of phosphane production.

Tab. 22.2 Consumption of raw materials and auxiliaries for the phosphane production

Input	Values	Remarks
Phosphorus (kg)	1.54	Calculated with a yield of 95%
Sodium hydroxide (kg)	0.706	Stoichiometric calculation
Process water (kg)	1.271	Stoichiometric calculation
Cooling water (kg)	24	Estimation

22.5 Energy

As the reaction is exothermic, no process heat is considered in this study. Electricity is needed to run the process and the waste water treatment. There was no information available on the amount of energy used for the production process. In order not to neglect the electricity consumption this value is approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf (2000)). The value for the electricity consumption per kg of product of this plant (0.333 kWh kg⁻¹) is used as approximation for the phosphane production.

Tab. 22.3 Energy consumption for the phosphane production

Input	per kg phosphane
Electricity (kWh)	0.333

22.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train. Tab. 22.4 summarises the total transport amounts for the phosphane production.

Tab. 22.4 Total transport amounts for the phosphane production

tkm kg ⁻¹ phosphane	lorry	train
Total transports	4.22E-01	2.53

22.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of phosphane production plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

22.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

Since the phosphane is a very poisonous gas, the whole system is a closed loop. Therefore no emissions to air are considered.

Tab. 22.5 Process emissions to air from the phosphane production

Output	Value	Remarks
Waste heat (MJ)	1.1988	calculated from electricity input

22.9 Emissions to water

The sodium phosphate from the phosphoric acid neutralization is assumed to leave the production process with the wastewater. Further it is assumed that this wastewater is treated in a internal wastewater plant. The removal efficiency for phosphate is assumed with 51.8%. The removed phosphate is accounted as inert waste.

Tab. 22.6 Process emissions to water from the phosphane production

Output	Value	Remarks
sodium (kg)	4.06E-01	calculated from mass balance
phosphate (kg)	2.69E-01	calculated from mass balance

22.10 Waste

The remaining phosphorus (5 % of the input) and the removed phosphate are assumed to be disposed as inert waste.

Tab. 22.7 Wastes from the phosphane production

Output	Value
disposal, inert waste, 5% water, to inert material landfill (kg)	3.66E-01

22.11 Life cycle inventory of phosphane production and data quality considerations

Tab. 22.8 shows the life cycle inventory and the data quality indicators for the phosphane production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 22.8 Unit process raw data for the phosphane production

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	phosphane, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							GLO 0 kg			
resource, in water	401	0	phosphane, at plant	GLO	0	kg	1.00E+0			
	662	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	493	-	phosphorus, white, liquid, at plant	RER	0	kg	1.54E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	403	-	water, ultrapure, at plant	GLO	0	kg	1.27E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
		-	sodium hydroxide, 50% in H2O, production mix, at plant	RER	0	kg	7.06E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
		-	disposal, inert waste, 5% water, to inert material landfill	CH	0	kg	3.66E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
		-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.33E-1	1	1.30	(4,5,na,na,na,na); Estimation
		-	transport, lorry >16t, fleet average	RER	0	tkm	4.22E-1	1	2.09	(4,5,na,na,na,na); Standard distances
		-	transport, freight, rail	RER	0	tkm	2.53E+0	1	2.09	(4,5,na,na,na,na); Standard distances
		-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density		4	Heat, waste	-	-	MJ	1.20E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input
emission water, river		4	Sodium, ion	-	-	kg	4.06E-1	1	5.10	(4,5,na,na,na,na); Estimation
		4	Phosphate	-	-	kg	2.69E-1	1	1.62	(4,5,na,na,na,na); Estimation

22.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 22.9 shows selected LCI results and the cumulative energy demand for the phosphane production.

Tab. 22.9 Selected LCI results and the cumulative energy demand for the phosphane production

Name		phosphane, at plant	
Location	Unit	Unit	GLO kg 0
Infrastructure			
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	246.1
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	109.4
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.4
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	8.7
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	12.6
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	2.0
selected LCI results			
land occupation	resource	m2a	1.3E+0
CO ₂ , fossil	air	kg	1.5E+1
NM VOC	air	kg	5.3E-3
nitrogen oxides	air	kg	3.2E-2
sulphur dioxide	air	kg	4.9E-2
particulates, <2.5 um	air	kg	9.6E-3
BOD	water	kg	1.7E-2
cadmium	soil	kg	9.6E-9

22.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	phosphane, at plant
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of phosphane including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Phosphin, ab Werk
	491	Synonyms	phosphine//phosphamine//phosphorus hydride
	492	GeneralComment	The process "phosphane, at plant, GLO" is modelled for the production of phosphane from phosphorus in the world. Raw materials are modelled with a stoichiometric calculation. Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	inorganics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Anorganika
	499	Formula	PH3
	501	StatisticalClassification	
	502	CASNumber	7803-51-2
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	Hydration of phosphorus
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

22.14 References

Frischknecht et al. 2007

Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Over-

view and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.

Gendorf 2000

Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>

Ullmanns 2004

Svara, J., Weferling, N.: Phosphorus compounds, organic. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

23 Phosphoryl chloride

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Niels Jungbluth, ESU Services, Uster
 Last changes: 2007

23.1 Introduction

Phosphoryl chloride (POCl_3 ; CAS 10025-87-3) is a colorless liquid which fumes in air and has a pungent odor. For this inventory the functional unit is 1 kg of pure phosphoryl chloride. The most important chemical and physical properties of phosphoryl chloride used in this inventory are given here.

Tab. 23.1 Chemical and physical properties of phosphoryl chloride (Ullmanns 2004)

Property	Value	Unit
Molecular weight	153.33	g mol^{-1}
Specific gravity	1645	kg m^{-3}
Boiling point	105.8	$^{\circ}\text{C}$
Melting point	1.25	$^{\circ}\text{C}$

23.1.1 Use

Phosphoryl chloride is used on a large scale to produce phosphate esters by reaction with alcohols and phenols. It is an ionizing solvent, dissociating to a small extent according to:



23.2 Production Technologies (Ullmanns 2004)

Phosphoryl chloride is obtained by oxidizing phosphorus trichloride with air or oxygen.



It can be produced industrially either batchwise or in a continuous process. A finely divided stream of oxygen is reacted with phosphorus trichloride until the freezing point of the product is 1°C . The reaction is accelerated by carrying it out under pressure. The presence of orthophosphoric acid has a catalytic effect, but traces of metallic impurities such as copper and cobalt, or of sulfur compounds, decrease the reaction rate. In a Hoechst process, phosphorus trichloride and oxygen are passed in cocurrent through a stainless steel water-cooled packed column. The oxidation process, i.e., the uptake of oxygen, can be followed by the characteristic temperature changes. The first product, which leaves the column at a maximum temperature of 35°C , has a freezing point of ca. 0.9°C , and is taken from the top of the column into a second column where it is again treated with oxygen. This procedure avoids large temperature increases. The excess oxygen leaving the plant passes through a cold trap where entrained phosphoryl chloride is condensed out and recovered. The off-gas is then scrubbed with water. Commercial phosphoryl chloride contains ca. 99.5 % POCl_3 and traces of phosphorus trichloride.

23.3 System Characterisation

In this study the oxidation of phosphorus trichloride is considered. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during

stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

23.4 Raw materials and auxiliaries

As there was no data available for the production of phosphoryl chloride, the values for the consumption of phosphorus trichloride and oxygen are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the phosphoryl chloride production.

Tab. 23.2 Consumption of raw materials and auxiliaries for the phosphoryl chloride production

Input	Values	Remarks
Phosphorus trichloride (kg)	0.942	Calculated with a yield of 95 %
Oxygen (kg)	0.111	Calculated with a yield of 95 %
Cooling water (kg)	24	Estimation

23.5 Energy

As the reaction is exothermic, no process heat is considered in this study. Electricity is needed to run the process and the waste water treatment. There was no information available on the amount of energy used for the production process. In order not to neglect the electricity consumption this value is approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf (2000)). The value for the electricity consumption per kg of product of this plant (0.333 kWh kg⁻¹) is used as approximation for the phosphoryl chloride production.

Tab. 23.3 Energy consumption for the phosphoryl chloride

Input	per kg phosphoryl chloride
Electricity (kWh)	0.333

23.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train for phosphorus trichloride and 50 km by lorry 32t and 100 km by train for oxygen. Tab. 23.4 summarises the total transport amounts for the production of phosphoryl chloride.

Tab. 23.4 Total transport amounts for the phosphoryl chloride production

tkm kg ⁻¹ phosphoryl chloride	lorry	train
Total transports	9.97E-02	5.76E-01

23.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of phosphoryl chloride production plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

23.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air. No other emissions to air are considered.

Tab. 23.5 Process emissions to air from the phosphoryl chloride production

Output	Value	Remarks
Waste heat (MJ)	1.2	calculated from electricity input

23.9 Emissions to water

The remaining amount of unreacted phosphorus trichloride (5 % of the input) is assumed to leave the production process with the wastewater. The emission of phosphorus trichloride is calculated as emission of phosphorus and chloride.

Tab. 23.6 Process emissions to water from the phosphoryl chloride production

Output	Value	Remarks
phosphorus (kg)	1.07E-02	calculated from mass balance
chloride (kg)	3.64E-02	calculated from mass balance
AOX (kg)	3.64E-02	calculated from water emissions

23.10 Life cycle inventory of phosphoryl chloride production and data quality considerations

Tab. 23.7 shows the life cycle inventory and the data quality indicators for the phosphoryl chloride production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 23.7 Unit process raw data for the production of phosphoryl chloride

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	phosphoryl chloride, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							RER 0 kg			
resource, in water	401	0	phosphoryl chloride, at plant	RER	0	kg	1.00E+0			
	662	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	493	-	phosphorous chloride, at plant	RER	0	kg	9.42E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	403	-	oxygen, liquid, at plant	RER	0	kg	1.11E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
		-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.33E-1	1	1.30	(4,5,na,na,na,na); Estimation
		-	transport, lorry >16t, fleet average	RER	0	tkm	9.97E-2	1	2.09	(4,5,na,na,na,na); Standard distances
		-	transport, freight, rail	RER	0	tkm	5.76E-1	1	2.09	(4,5,na,na,na,na); Standard distances
		-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
		-	Heat, waste	-	-	MJ	1.20E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input
		-	Phosphorus	-	-	kg	1.07E-2	1	1.62	(4,5,na,na,na,na); Estimation
emission air, high population density		-	Chloride	-	-	kg	3.64E-2	1	3.09	(4,5,na,na,na,na); Estimation
emission water, river		-	AOX, Adsorbable Organic Halogen as Cl	-	-	kg	3.64E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

23.11 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 23.8 shows selected LCI results and the cumulative energy demand for the production of phosphoryl chloride.

Tab. 23.8 Selected LCI results and the cumulative energy demand for the phosphoryl chloride production

Name		phosphoryl chloride, at plant	
Location	Unit	Unit	RER
Infrastructure			kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	50.1
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	23.8
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.1
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	2.3
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	2.8
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.4
selected LCI results			
land occupation	resource	m2a	4.3E-1
CO ₂ , fossil	air	kg	3.2E+0
NM VOC	air	kg	1.3E-3
nitrogen oxides	air	kg	7.1E-3
sulphur dioxide	air	kg	1.1E-2
particulates, <2.5 um	air	kg	1.7E-3
BOD	water	kg	3.5E-3
cadmium	soil	kg	2.8E-9

23.12 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	phosphoryl chloride, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of phosphoryl chloride including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Phosphorylchlorid, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "phosphoryl chloride, at plant, RER" is modelled for the production of phosphoryl chloride from phosphorus trichloride in Europe. Raw materials are modelled with a stoichiometric calculation. Emissions, energy consumptions, and infrastructure and transports are estimated. Transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	inorganics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Anorganika
	499	Formula	POCl ₃
	501	StatisticalClassification	
	502	CASNumber	10025-87-3
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Oxidation of phosphorus trichloride
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Emissions, infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	24
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

23.13 References

Frischknecht et al. 2007

Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.

- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>
- Ullmanns 2004 Bettermann, G., Krause, W., Riess, G., Hofmann, T.: Phosphorus Compounds, Inorganic Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

24 Potassium carbonate

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

24.1 Introduction

Anhydrous potassium carbonate (K_2CO_3 ; CAS 584-08-7; synonyms: potash, pearlash) is a white, hygroscopic, powdery material that deliquesces in moist air. It is readily soluble in water with the formation of an alkaline solution. K_2CO_3 forms several hydrates, of which $K_2CO_3 \cdot 1.5 H_2O$ is the stable phase in contact with the saturated solution from 0 °C to ca. 110 °C. This hydrate crystallizes in glassy, virtually dust-free crystals. It is also hygroscopic and deliquesces in moist air. It is completely dehydrated at 130 – 160 °C.

For this inventory the functional unit is 1 kg of highly pure anhydrous potassium carbonate. The most important chemical and physical properties of potassium carbonate used in this inventory are given here.

Tab. 24.1 Chemical and physical properties of potassium carbonate (Ullmanns 2004)

Property	Value	Unit	Remarks
Molecular weight	138.21	$g\ mol^{-1}$	
Specific gravity	2428	$kg\ m^{-3}$	
Boiling point	-		decomposes
Melting point	-272.2	°C	

24.1.1 Use

The glass industry is the most important consumer of K_2CO_3 . Large amounts are also required for potassium silicate manufacture.

Potassium carbonate is used for many organic syntheses. Numerous inorganic and organic potassium salts are manufactured from potassium carbonate (potassium phosphate, bromide, iodide, dichromate, cyanide, and ferrocyanide); in addition it is a starting material for drying, neutralization, and condensation agents. As a regenerable absorbent for carbon dioxide, hydrogen sulfide, and sulfur oxides, it is attaining importance in environmental protection. Potassium carbonate is used as a fertilizer for acidic soil.

Other users are the electrical industry, the dye industry, the printing trade, the textile industry, the leather goods industry, and the ceramic industry. Soft soap manufacture has lost its earlier importance as a customer. Potash solutions are used as fire retardants and as cooling brines (freezing point – 36 °C at 576 g/L=40.5 wt % K_2CO_3).

The food industry uses potassium carbonate as a leavening agent in baked goods, as a debitterizing agent for cocoa beans, and as an additive for drying raisins. Potash in DAB quality is frequently used in the pharmaceutical industry as a raw material and auxiliary.

The most important producer countries for potassium carbonate are the CIS, France, Germany, the United States, and Japan. Other producers are Israel, Spain, India, South Korea, Belgium, and Italy.

24.2 Production Technologies (Ullmanns 2004)

From Caustic Potash and Carbon Dioxide

The most important process for the manufacture of potassium carbonate begins with electrolytically produced potassium hydroxide solution. The almost chemically pure solution obtained by the mercury process is reacted with carbon dioxide or CO₂-containing off-gases (flue gas, lime kiln gas).



Solid potassium carbonate is then obtained by crystallization (under vacuum and with cooling) from liquors or in the fluidized-bed process.

In the continuous crystallization process, the filtered, fresh carbonate solution is mixed with mother liquor and concentrated in several preliminary evaporators connected in series until the hydrate K₂CO₃ · 1.5 H₂O finally precipitates in the crystallizer after cooling under vacuum. The mother liquor is separated from the crystal suspension in hydrocyclones and centrifuges, filtered, and fed back to the process. The crystals are dried at ca. 110 – 120 °C in rotary kilns or fluidized-bed dryers and packed for sale as potash hydrate, or they are calcined at 200 – 350 °C to give 98 – 100 % K₂CO₃. Impurities such as soda, sulfate, silicic acid, and iron that concentrate in the mother liquors can be partially removed by removing a partial stream of the mother liquor, which is either used for brine purification in the electrolysis process or sold as a low-grade potassium carbonate solution, or by crystallizing the double-salt NaKCO₃ at elevated temperature in a separate crystallization and drying process.

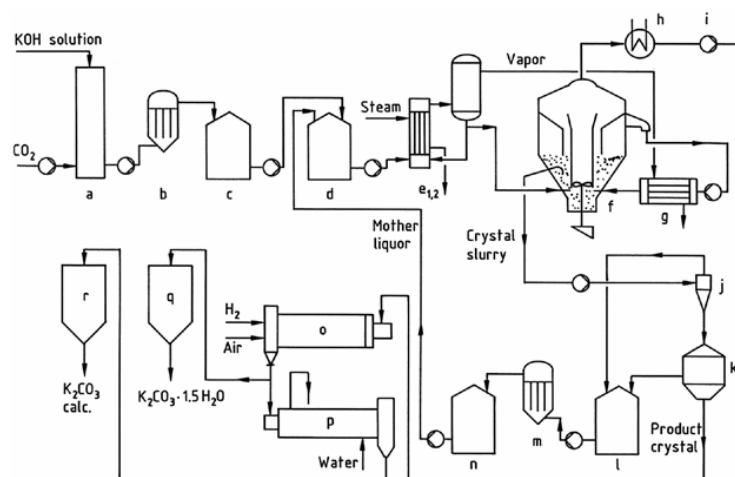


Fig. 24.1 Preparation of potassium carbonate with continuous crystallization

a) Carbonization; b) Crude liquor filter; c) Fresh liquor tank; d) Mixed liquor tank; e₁), e₂) Preliminary evaporation; f) Vacuum/cooling crystallization (Chemietechnik Messo system); g) Preheater; h) Vapor condenser; i) Vacuum pump; j) Hydrocyclone; k) Centrifuge; l) Centrifuge liquor tank; m) Filter for mother liquor; n) Mother liquor tank; o) Drying or calcining rotary kiln; p) Cooling device for calcined K₂CO₃; q) Storage for hydrated potash; r) Storage for calcined potash

The resulting potassium carbonate is very pure and meets the requirements of USP, BP, DAB, and JP if the process is operated in appropriate manner.

Starting from potassium carbonate solution, prills can be produced in a combined reactor, in which spray drying and fluidized-bed granulation take place simultaneously.

In the fluidized-bed process, aqueous potassium hydroxide solution is sprayed into a fluidized-bed reactor from above and exposed to a countercurrent of CO₂-containing hot gas. Carbonization and calcination take place in the same reactor. Hard, spherical potassium carbonate prills are formed having a high packing density. The prills are discharged and sieved. The coarse grains are ground and returned to the reactor together with the very fine grains, where they act as crystallization seeds. The salable, dust-free, medium grains are cooled and packed. Because no mother liquor is formed, the quality of the potassium carbonate depends on that of the raw materials. Compared to the crystallization process the chloride, soda, and sulfate contents are usually higher, but the investment and production costs are lower.

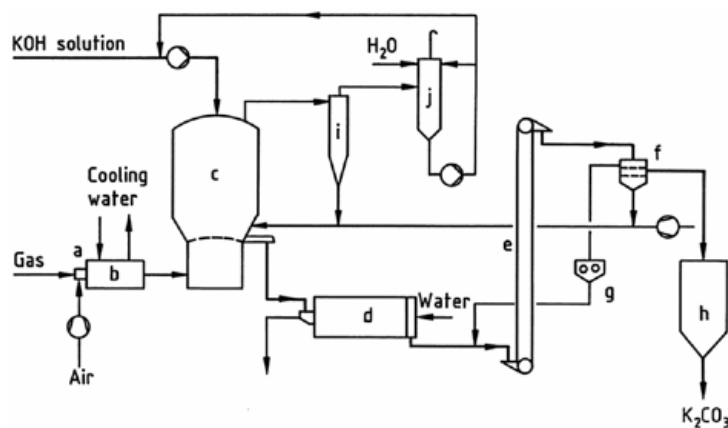


Fig. 24.2 Production of potassium carbonate by the fluidized-bed process

a) Burner; b) Gas cooler; c) Fluidized-bed reactor; d) Cooler; e) Elevator; f) Screen; g) Mill; h) Silo; i) Cyclone; j) Exhaust gas scrubber

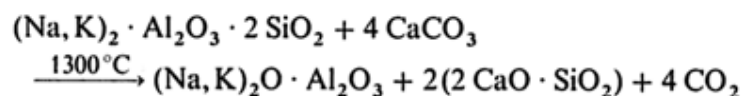
Amine Process

In the Mines de Potasse d'Alsace process, potassium chloride is reacted under pressure in autoclaves with carbon dioxide in precarbonated isopropylamine solution. Potassium hydrogencarbonate precipitates and is filtered off, carefully purified of amine by intensive washing, and dried. It can be converted to potassium carbonate by calcination. Free amine, containing carbon dioxide, is recovered from the mother liquor by distillation and recycled. The chloride, predominantly present in the mother liquor as amine chlorohydrate, is reacted with hydrated lime to give free amine and an aqueous solution of calcium chloride.

The use of triethylamine, hexamethylenimine, or piperidine is also patented. All the processes have the disadvantage that calcium chloride liquor is obtained, which can be utilized today only to a small extent and therefore represents an environmental pollutant.

Nepheline Decomposition Process

In the CIS, considerable amounts of potassium carbonate are formed as a byproduct in the nepheline decomposition process for aluminum hydroxide production. The mineral nepheline is decomposed with limestone by sintering at 1300 °C:



Alumina, portland cement, soda, and potash are obtained from the product in a complex process. The sinter product is leached with an $\text{Na}_2\text{CO}_3 - \text{NaOH}$ solution. After filtration, a filter cake is obtained that is processed to give portland cement and an aluminate solution containing silicic acid. After precipitation of the silicic acid as alkaline aluminum silicate the purified aluminate solution is reacted with carbon dioxide:



The aluminum oxide hydrate is filtered off, and the carbonate solution is concentrated by fractional crystallization in a three-stage process and separated into sodium carbonate and $\text{K}_2\text{CO}_3 \cdot 1.5 \text{H}_2\text{O}$.

Feldspar (KAlSi_3O_8) and leucite (KAlSi_2O_6) can also be decomposed analogously and used for alumina, cement, and potassium carbonate manufacture.

The magnesita process (Engel – Precht process) is of limited interest:



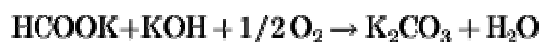
In hot water the double salt ($\text{MgCO}_3 \cdot \text{KHCO}_3 \cdot 4 \text{H}_2\text{O}$) decomposes under pressure into magnesium carbonate and dissolved potassium carbonate.

Other processes

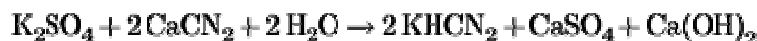
Le Blanc process:



Formate process:



“Piesteritz” process:



These processes are uneconomical today because of high energy consumption and poor product quality, and are no longer used.

In the decomposition of chromium ores with potassium hydroxide solution, a chromate-containing potassium carbonate is obtained as byproduct. The production of potassium permanganate yields considerable amounts of potassium carbonate solution.

Organic raw materials, such as sunflower stalks, molasses, and suint, are used to a small extent for potash manufacture. They are ashed, leached with water, and processed to potash by fractional crystallization and calcination.

Ion-Exchange Process

An acidic ion exchanger loaded with ammonium ions is charged with KCl solution, K^+ being absorbed and an ammonium chloride solution running off. The ion exchanger is then eluted with an excess of ammonium carbonate solution (regeneration of the exchanger). The eluate, a $K_2CO_3 - (NH_4)_2CO_3$ solution, is separated by thermal cleavage to give ammonia and carbon dioxide. The ammonium chloride is reacted with magnesium hydroxide to give magnesium chloride and ammonia, which is recycled.

24.3 System Characterisation

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg potassium carbonate, at plant, global**. For this study the production of potassium carbonate by means of the reaction of potassium hydroxide and carbon dioxide is considered. All data in the present report are referred to 1 kg potassium carbonate 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

24.4 Raw materials and auxiliaries

As there was no data available for the production of potassium carbonate, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the potassium carbonate production.

Tab. 24.2 Consumption of raw materials and auxiliaries for the potassium carbonate production

Input	Values	Remarks
Potassium hydroxide (kg)	0.854	Calculated with a yield of 95 %
Carbon dioxide (kg)	0.336	Calculated with a yield of 95 %
Cooling water (kg)	24	Estimation

24.5 Energy

There was no information available on the amount of energy used for the production process. In order not to neglect the process energy demand those values were approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg-1)

were used as approximation for the energy consumption of the potassium carbonate production. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used. A summary of the values used is given in Tab. 24.3.

Tab. 24.3 Energy consumption for the potassium carbonate production

Input	per kg potassium carbonate
Heat, natural gas burned in industrial furnace > 100 kW	2
Electricity (kWh)	0.333

24.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train. Tab. 24.4 summarises the total transport amounts for the potassium carbonate production.

Tab. 24.4 Total transport amounts for the potassium carbonate production

tkm kg ⁻¹ potassium carbonate	lorry	train
Total transports	1.19E-01	7.14E-01

24.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of potassium carbonate production plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

24.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

The unreacted carbon dioxide is assumed to be emitted to air.

Tab. 24.5 Process emissions to air from the potassium carbonate production

Output	Value	Remarks
Waste heat (MJ)	1.2	calculated from electricity input
CO ₂ , fossil, to air (kg)	1.68E-02	calculated from mass balance

24.9 Emissions to water

The remaining amount of unreacted potassium hydroxide (4.8 % of the input) is assumed to leave the production process with the wastewater.

Tab. 24.6 Process emissions to water from the potassium carbonate production

Output	Value	Remarks
potassium (kg)	2.97E-02	calculated from mass balance
hydroxide (kg)	1.3E-02	calculated from mass balance

24.10 Life cycle inventory of potassium carbonate production and data quality considerations

Tab. 24.7 shows the life cycle inventory and the data quality indicators for the potassium carbonate production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 24.7 Unit process raw data for the potassium carbonate production

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	potassium carbonate, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							GLO 0 kg			
resource, in water	401	0	potassium carbonate, at plant	GLO	0	kg	1.00E+0			
	662	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	493	-	potassium hydroxide, at regional storage	RER	0	kg	8.54E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	403	-	carbon dioxide liquid, at plant	RER	0	kg	3.36E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
		-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	2.00E+0	1	1.30	(4,5,na,na,na,na); Estimation
		-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.33E-1	1	1.30	(4,5,na,na,na,na); Estimation
		-	transport, lorry >16t, fleet average	RER	0	tkm	1.19E-1	1	2.09	(4,5,na,na,na,na); Standard distances
		-	transport, freight, rail	RER	0	tkm	7.14E-1	1	2.09	(4,5,na,na,na,na); Standard distances
		-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
		-								
emission air, high population density		4	Heat, waste	-	-	MJ	1.20E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input
emission water, river		4	Carbon dioxide, fossil	-	-	kg	1.68E-2	1	1.30	(4,5,na,na,na,na); Estimation
		4	Potassium, ion	-	-	kg	2.97E-2	1	5.10	(4,5,na,na,na,na); Estimation
		4	Hydroxide	-	-	kg	1.30E-2	1	1.62	(4,5,na,na,na,na); Estimation

24.11 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 24.8 shows selected LCI results and the cumulative energy demand for the potassium carbonate production.

Tab. 24.8 Selected LCI results and the cumulative energy demand for the potassium carbonate production

Name		potassium carbonate, at plant	
Location	Unit	Unit	GLO kg 0
Infrastructure			
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	32.7
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	10.2
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	1.2
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	1.3
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.2
selected LCI results			
land occupation	resource	m2a	2.8E-1
CO ₂ , fossil	air	kg	2.2E+0
NM VOC	air	kg	5.4E-3
nitrogen oxides	air	kg	4.3E-3
sulphur dioxide	air	kg	5.5E-3
particulates, <2.5 um	air	kg	4.9E-4
BOD	water	kg	2.1E-3
cadmium	soil	kg	1.4E-9

24.12 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	potassium carbonate, at plant
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of potassium carbonate including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Kaliumcarbonat, ab Werk
	491	Synonyms	potash/pearlash
	492	GeneralComment	The process "potassium carbonate, at plant, GLO" is modelled for the production of potassium carbonate from potassium hydroxide in the world. Raw materials are modelled with a stoichiometric calculation. Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	inorganics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Anorganika
	499	Formula	K ₂ CO ₃
	501	StatisticalClassification	
	502	CASNumber	584-08-7
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	Carbonation of potassium hydroxide
Representative	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAn	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

24.13 References

Frischknecht et al. 2007

Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.

- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>
- Ullmanns 2004 Schultz, H., Bauer, G., Schachl, E., Hagedorn, F., Schmittinger, P.: Potassium Compounds. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

25 Potassium perchlorate

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

25.1 Introduction

Potassium perchlorate (KClO_4 ; CAS 7778-74-7; synonyms: perchlorated potassium) is a strong oxidizer. It is a colorless, crystalline substance that melts at about 610 °C. It is one of the most common oxidizers used in fireworks, used variously in propellants, flash compositions, stars, and sparklers. It has been used as a solid rocket propellant, though in that application it has mostly been replaced by the higher performance ammonium perchlorate. Perchlorates are generally well soluble in water and such organic solvents as alcohols, ketones, or esters.

For this inventory the functional unit is 1 kg of highly pure potassium perchlorate. The most important chemical and physical properties of potassium perchlorate used in this inventory are given here.

Tab. 25.1 Chemical and physical properties of potassium perchlorate (Ullmanns 2004)

Property	Value	Unit
Molecular weight	138.55	g mol^{-1}
Specific gravity	2520	kg m^{-3}
Boiling point	610	°C
Melting point	400	°C

25.1.1 Reserves and Resources

Statistical data about the production and use of perchlorates are not easily accessible because of their strategic importance. The United States' production in 1974 was reported to be ca. 25 000 t, or ca. 12 % of the chlorate production. On account of the increasing developments of artificial satellites and space shuttle programs, perchlorate production will increase, due to their major use in solid propellants.

25.1.2 Use of perchlorates

The most outstanding property of ammonium perchlorate is its high oxygen concentration (54.5 % O_2) and the fact that it decomposes without leaving a solid residue. Therefore, it is used as an oxidizing component in solid rocket propellants.

Lithium perchlorate is used in lithium – nickel sulfide dry batteries, and sodium perchlorate in electrochemical machining; potassium perchlorate is a component of pyrotechnics and an ignition ingredient in flash bulbs. Magnesium perchlorate is known as a very effective drying agent.

Ammonium perchlorate mixed with an epoxy resin forms a temporary adhesive between two metallic surfaces, such as two steel plates. These can be separated whenever desired by heating at ca. 300 °C because of the self-sustained combustion of the adhesive layer. Ammonium perchlorate has been tried as a feed supplement to increase the weight of livestock, with an optimum dose of 2 – 5 mg/kg. Perchlorates have also proven to be helpful in oxygen-regenerating systems to be used in enclosed environments, such as submarines and spacecraft, and in breathing equipment. Potassium values from enriched bittern, obtained from the Dead Sea or the Great Salt Lake, can be recovered by precipitating potassium perchlorate with sodium perchlorate.

25.2 Production Technologies (Ullmanns 2004)

Various methods for the production of perchlorates are described. In practice, commercial production is based on the following steps:

1. Electrochemical production of sodium chlorate from sodium chloride
2. Electrochemical oxidation of sodium chlorate in aqueous solution to sodium perchlorate



3. Conversion of sodium perchlorate into another perchlorate, like potassium perchlorate:



In the last step, the corresponding sulfate, which causes fewer corrosion problems, can be used instead of the chloride.

This indirect method for the production of perchlorates other than NaClO_4 is advantageous over their direct electrosynthesis from the corresponding chlorates, because of their relatively small solubility in water. Attempts have been made to make sodium perchlorate directly from sodium chloride, by a single electrolysis. The advantage, however, seems debatable, because the resulting current efficiency is only slightly better than 50 % under optimum conditions.

25.3 System Characterisation

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg potassium perchlorate, at plant, global**. For this study the production of potassium perchlorate by means of the conversion of sodium perchlorate is considered. All data in the present report are referred to 1 kg potassium perchlorate 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

25.4 Raw materials and auxiliaries

As there was no data available for the production of potassium perchlorate, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the potassium perchlorate production.

Tab. 25.2 Consumption of raw materials and auxiliaries for the potassium perchlorate production

Input	Values	Remarks
Sodium perchlorate (kg)	0.931	Calculated with a yield of 95 %
Potassium chloride (kg)	0.564	Calculated with a yield of 95 %
Cooling water (kg)	24	Estimation

25.5 Energy

There was no information available on the amount of energy used for the production process. In order not to neglect the process energy demand those values were approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg⁻¹) were used as approximation for the energy consumption of the potassium perchlorate production. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used. A summary of the values used is given in Tab. 25.3.

Tab. 25.3 Energy consumption for the potassium perchlorate production

Input	per kg N,N-dimethylacetamide
Heat, natural gas burned in industrial furnace > 100 kW	2
Electricity (kWh)	0.333

25.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train. Tab. 25.4 summarises the total transport amounts for the potassium perchlorate production.

Tab. 25.4 Total transport amounts for the potassium perchlorate production

tkm kg ⁻¹ potassium perchlorate	lorry	train
Total transports	1.65E-01	9.89E-01

25.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of potassium perchlorate production plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

25.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

Tab. 25.5 Process emissions to air from the potassium perchlorate production

Output	Value	Remarks
Waste heat (MJ)	1.2	calculated from electricity input

25.9 Emissions to water

The remaining amount of unreacted sodium perchlorate and potassium chloride (4.8 % of the input) is assumed to leave the production process with the wastewater.

Tab. 25.6 Process emissions to water from the potassium perchlorate production

Output	Value	Remarks
perchlorate (kg)	3.78E-02	calculated from mass balance
potassium (kg)	1.49E-02	calculated from mass balance
chloride (kg)	2.67E-01	calculated from mass balance
sodium (kg)	1.75E-01	calculated from mass balance
AOX (kg)	2.59E-01	calculated from water emissions

25.10 Life cycle inventory of potassium perchlorate production and data quality considerations

Tab. 25.7 shows the life cycle inventory and the data quality indicators for the potassium perchlorate production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 25.7 Unit process raw data for the potassium perchlorate production

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	potassium perchlorate, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
401										
662			Location				GLO			
493			InfrastructureProcess				0			
403			Unit				kg			
resource, in water	-	0	potassium perchlorate, at plant	GLO	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	sodium perchlorate, at plant	GLO	0	kg	9.31E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	potassium chloride, as K2O, at regional storehouse	RER	0	kg	7.17E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	2.00E+0	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.33E-1	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.65E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	9.89E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	1.20E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input
emission water, river	-	4	Perchlorate, ion	-	-	kg	3.78E-2	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Potassium, ion	-	-	kg	1.49E-2	1	5.10	(4,5,na,na,na,na); Estimation
	-	4	Chloride	-	-	kg	2.67E-1	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Sodium, ion	-	-	kg	1.75E-1	1	5.10	(4,5,na,na,na,na); Estimation
	-	4	AOX, Adsorbable Organic Halogen as Cl	-	-	kg	2.59E-1	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

25.11 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 25.8 shows selected LCI results and the cumulative energy demand for the potassium perchlorate production.

Tab. 25.8 Selected LCI results and the cumulative energy demand for the potassium perchlorate production

Name		potassium perchlorate, at plant	
Location	Unit	Unit	GLO kg 0
Infrastructure			
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	65.4
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	35.1
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	2.5
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	4.1
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.6
selected LCI results			
land occupation	resource	m2a	4.9E-1
CO ₂ , fossil	air	kg	4.8E+0
NM VOC	air	kg	1.8E-3
nitrogen oxides	air	kg	9.9E-3
sulphur dioxide	air	kg	2.7E-2
particulates, <2.5 um	air	kg	1.3E-3
BOD	water	kg	4.9E-3
cadmium	soil	kg	3.4E-9

25.12 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	potassium perchlorate, at plant
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of potassium perchlorate including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Kaliumperchlorat, ab Werk
	491	Synonyms	perchlorated potassium
	492	GeneralComment	The process "potassium perchlorate, at plant, GLO" is modelled for the production of potassium perchlorate from sodium perchlorate in the world. Raw materials are modelled with a stoichiometric calculation. Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	inorganics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Anorganika
	499	Formula	KClO ₄
	501	StatisticalClassification	
	502	CASNumber	7778-74-7
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	Reaction of sodium perchlorate and potassium chloride
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

25.13 References

Frischknecht et al. 2007 Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Over-

view and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.

Gendorf 2000

Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>

Ullmanns 2004

Vogt, H., Balej, J., Bennett, J., Wintzer, P., Akbar Sheikh, S., Gallone, P.: Chlorine Oxides and Chlorine Oxygen Acids. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

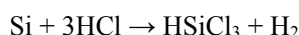
26 Silicon hydrochloration

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Niels, Jungbluth, ESU Services, Uster
 Last changes: 2007

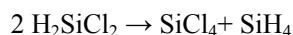
Hydrochloration of silicon and disproportionation of trichlorosilane are the most important processes to produce silicon tetrahydride.

26.1 Production Technologies (Ullmann 2005)

Industrially, silicon tetrahydride is produced from metallurgical grade silicon (MG-silicon) in a two-step process. In the first step, powdered silicon is reacted with hydrochloric acid at about 300 °C to produce trichlorosilane, HSiCl_3 , along with hydrogen gas, according to the chemical equation:



The hydrogen is re-used in other silicon purification processes and therefore not considered as co-product. In the second step, catalytic disproportionation of trichlorosilane results in SiH_4 and SiCl_4 . For a mixture of trichloro- and dichlorosilane, copper, a mixture of silica and alumina at 300 °C, and quaternary ammonium groups bonded to carbon in an ion-exchange resin have been reported as catalysts for the main reaction leading to silicon tetrahydride.



Purification

In-process purification is achieved by contacting part of the *silicon tetrahydride* produced with the starting mixture of chlorosilanes, which contains the impurities as chlorides, chiefly those of arsenic, phosphorus, and boron. This procedure converts the contaminants into their more volatile hydrides, which are removed with the off-gas.

Removal of boron compounds from silanes that are to be used for the production of high-purity silicon is very important. When SiH_4 is produced from alkali metal hydrides, it is recommended that water should be added to hydrolyze boron hydrides under the alkaline reaction conditions. Water vapor is added during distillation for the same reason. Boron compounds can be removed from SiH_4 by adsorption from the gas phase onto activated carbon or silica carriers loaded with compounds that form complexes with boron compounds (AlCl_3 , FeCl_3 , $\text{C}_2\text{H}_5\text{CN}$, NH_2CN , PCl_3).

If alkali metal hydrides MH are reacted in organic solvents, it is often unnecessary to include a special stage for removing boron compounds from the silicon compounds, as the former react to form boranate complexes under the following conditions:



26.2 System Characterisation

In this study the production of silicon tetrahydride by hydrochloration of silicon to trichlorosilane and disproportionation of trichlorosilane is considered. The functional unit is 1 kg MG-silicon. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For

the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers. Within the catalytic disproportionation of trichlorosilane, silicon tetrahydride and tetrachlorosilane are derived as co-products. Therefore an allocation between the two products was made. The following processes were modelled:

- Silicon hydrochloration, GLO. Multi output process with silicon tetrahydride, at plant, GLO and tetrachlorosilane, at plant, GLO as allocated products

26.3 Data sources

As no data are available for the silicon tetrahydride production by catalytic disproportionation of trichlorosilane, the values for energy consumptions, and emissions are estimated with data from the similar process production of electronic grade silicon (Jungbluth et al. 2007).

26.4 Raw materials and auxiliaries

Tab. 26.1 shows values for the use of raw materials of the silicon hydrochloration. This study is based on a yield of 95 % kg per kg MG-silicon.

The amount of cooling water is estimated from the production of electronic grade silicon (Jungbluth et al. 2007). In this dataset an amount of 43.5 m³ water per kg MG-silicon is used. Thus value is also used here.

Tab. 26.1 Consumption of raw materials and auxiliaries for the silicon hydrochloration process

Input	Values	Remarks
MG-silicon (kg)	1	Functional unit
HCl (kg)	23.4	Calculated with a yield of 95 %
Cooling water (m ³)	43.5	Estimation

26.5 Energy

The energy consumption for the silicon hydrochloration process is estimated from the production of electronic grade silicon (Jungbluth et al. 2007). In this dataset an amount of 150 kWh electricity and 160 MJ heat per kg EG-silicon are considered. These values are also used here. The electricity is calculated with the mix that is considered in Jungbluth et al. (2007).

Tab. 26.2 Energy consumption for the silicon hydrochloration process

Input	per kg MG-silicon
Heat, at cogen 1MWe lean burn, allocation exergy (MJ)	122
Electricity, at cogen 1MWe lean burn, allocation exergy (kWh)	86.6
Electricity, hydropower, at run-of-river power plant (kWh)	27.4

26.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train for the raw materials. Tab. 26.3 summarises the total transport amounts for the silicon hydrochloration process.

Tab. 26.3 Total transport amounts for the silicon hydrochloration process

tkm kg ⁻¹ silicon	lorry	train
Total transports	0.38	2.27

26.7 Infrastructure and land use

The infrastructure for the silicon hydrochloration process is estimated from the production of electronic grade silicon (Jungbluth et al. 2007). There the infrastructure is estimated based on the module "silicon plant". An amount of 1E-11 units per kg MG-silicon is considered. This value is also used here.

26.8 Emissions to air

In Jungbluth (2007) the release of waste heat is calculated from the electricity input minus 180 MJ/kg EG-silicon. This calculated is also done here.

Tab. 26.4 Process emissions to air from the silicon hydrochloration process

Output	Value
Waste heat (MJ)	274

26.9 Emissions to water

Emissions to water from the silicon hydrochloration process are estimated from the production of electronic grade silicon (Jungbluth et al. 2007).

Tab. 26.5 Process emissions to water from the silicon hydrochloration process

Output	Value
Chloride	2.51E-02
Copper, ion	7.15E-08
Nitrogen	1.45E-04
Phosphate	1.96E-06
Sodium, ion	2.36E-02
Zinc, ion	1.37E-06
Iron, ion	3.92E-06
BOD	1.43E-04
COD	1.41E-03
TOC	6.35E-04
TOC	6.35E-04
AOX	8.81E-06

26.10 Allocation of silicon tetrahydride and tetrachlorosilane

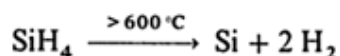
Within the catalytic silicon hydrochloration, GLO silicon tetrahydride and tetrachlorosilane are obtained as co-products. A stoichiometric allocation of the two products is made. The used allocation factors are given in Tab. 26.6. The input of hydrochloric acid and the output of chloride emissions are allocated to tetrachlorosilane.

Tab. 26.6 Allocation of silicon tetrahydride and tetrachlorosilane from the silicon hydrochloration

Process	Silicon hydrochloration, GLO		
	Unit	per kg MG-silicon	Allocation factor (stoichiometric)
Tetrachlorosilane, at plant, GLO	Kg	9.172E-01	94
Silicon tetrahydride, at plant, GLO	kg	5.824E-02	6

26.11 Silicon tetrahydride

Silicon tetrahydride (SiH₄; CAS-No. 803-62-5; synonyms: monosilane) is a gas. Above 600°C, silane decomposes into silicon and hydrogen. The thermal decomposition of silicon tetrahydride takes place as follows:



For this inventory the functional unit is 1 kg of pure silicon tetrahydride. The most important chemical and physical properties of silicon tetrahydride used in this inventory are given here.

Tab. 26.7 Chemical and physical properties of silicon tetrahydride (Ullmann 2005)

Property	Value	Unit	Remarks
Molecular weight	32.12	g mol ⁻¹	
Specific gravity	0.68	kg m ⁻³	gaseous
Boiling point	-112	°C	at normale pressure
Melting point	-185	°C	under pressure

26.11.1 Use (Ullmann 2005)

Direct use of silicon hydrides involves their decomposition on surfaces, leading to deposition of elemental silicon or silicon compounds. Polycrystalline films of elemental silicon of controllable grain size are deposited on glass or silica-coated surfaces by heating them to 450 – 800 °C in an atmosphere consisting of silicon tetrahydride and tetrabromo- or tetraiodosilane. Chemical vapor deposition is generally followed by annealing and recrystallization with an energy beam. This is used to deposit silicon on insulator surfaces. Amorphous silicon photoconducting layers on electrophotographic plates are made by using a plasma for silane decomposition and silicon deposition. Liquid crystal displays, based on thin film transistors, for screens in monitoring devices, are made from amorphous silicon which is produced on alkali-free glass surfaces in situ by an electric discharge in an atmosphere of silicon tetrahydride at reduced pressure. Residual bonds are saturated by hydrogen atoms to avoid diminishing the semiconductor effect. On the surfaces of transition metals, cleaned by sputtering in an argon atmosphere, silicon tetrahydride and the higher silicon hydrides can be thermally decomposed at ca. 300 °C to form metal silicides. Exposure to air leads to the formation of silicon oxides.

26.12 Life cycle inventory of silicon hydrochloration process and data quality considerations

Tab. 26.8 shows the life cycle inventory and the data quality indicators for the silicon hydrochloration process. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 26.8 Unit process raw data for the production of silicon hydrochloration process

	401	InputGroup OutputGroup	Name	Location	InfrastructureProcess	Units	silicon hydrochloration	UncertaintyType StandardDeviation95 %	GeneralComment	tetrachlorosila ne, at plant	silicon tetrahydride, a plant	
662			Location				GLO			GLO	RER	
493			InfrastructureProcess				0			0	0	
403			Unit				kg			kg	kg	
allocated products resource, in water technosphere	-	2	tetrachlorosilane, at plant	GLO	0	kg	9.17E-1			100	0	
	-	2	silicon tetrahydride, at plant	RER	0	kg	5.82E-2			0	100	
	4	-	Water, cooling, unspecified natural origin	-	-	m3	4.35E+1	1	1.30	(4,5,na,na,na,na); Estimated from Jungbluth (2007)	94.030	5.970
	5	-	MG-silicon, at plant	NO	0	kg	1.00E+0	1	1.38	(4,5,1,1,1,1,5); Stoichiometric calculation	94.030	5.970
	5	-	hydrochloric acid, 30% in H2O, at plant	RER	0	kg	8.36E-1	1	1.38	(4,5,1,1,1,1,5); Stoichiometric calculation	100.000	-
	5	-	heat, at cogen 1MWc lean burn, allocation	RER	0	MJ	1.22E+2	1	1.30	(4,5,na,na,na,na); Estimated from Jungbluth (2007)	94.030	5.970
	5	-	electricity, at cogen 1MWc lean burn, allocation	RER	0	kWh	8.66E+1	1	1.30	(4,5,na,na,na,na); Estimated from Jungbluth (2007)	94.030	5.970
	5	-	electricity, hydropower, at run-of-river power plant	RER	0	kWh	2.74E+1	1	1.30	(4,5,na,na,na,na); Estimated from Jungbluth (2007)	94.030	5.970
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	3.79E-1	1	2.09	(4,5,na,na,na,na); Estimated from Jungbluth (2007)	94.030	5.970
emission air, high population density emission water, river	5	-	transport, freight, rail	RER	0	tkm	2.27E+0	1	2.09	(4,5,na,na,na,na); Standard distance 600km	94.030	5.970
	5	-	silicone plant	RER	1	unit	1.00E-11	1	3.09	(4,5,na,na,na,na); Estimated from Jungbluth (2007)	94.030	5.970
	-	4	Heat, waste	-	-	MJ	2.74E+2	1	1.30	(4,5,na,na,na,na); Estimated from Jungbluth (2007)	94.030	5.970
	-	4	Chloride	-	-	kg	2.51E-2	1	3.09	(4,5,na,na,na,na); Estimated from Jungbluth (2007)	100.000	-
	-	4	Copper, ion	-	-	kg	7.15E-8	1	3.09	(4,5,na,na,na,na); Estimated from Jungbluth (2007)	94.030	5.970
	-	4	Nitrogen	-	-	kg	1.45E-4	1	1.62	(4,5,na,na,na,na); Estimated from Jungbluth (2007)	94.030	5.970
	-	4	Phosphate	-	-	kg	1.96E-6	1	1.62	(4,5,na,na,na,na); Estimated from Jungbluth (2007)	94.030	5.970
	-	4	Sodium, ion	-	-	kg	2.36E-2	1	5.10	(4,5,na,na,na,na); Estimated from Jungbluth (2007)	94.030	5.970
	-	4	Zinc, ion	-	-	kg	1.37E-6	1	5.10	(4,5,na,na,na,na); Estimated from Jungbluth (2007)	94.030	5.970
	-	4	Iron, ion	-	-	kg	3.92E-6	1	5.10	(4,5,na,na,na,na); Estimated from Jungbluth (2007)	94.030	5.970
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	1.43E-4	1	1.62	(4,5,na,na,na,na); Estimated from Jungbluth (2007)	94.030	5.970
	-	4	COD, Chemical Oxygen Demand	-	-	kg	1.41E-3	1	1.62	(4,5,na,na,na,na); Estimated from Jungbluth (2007)	94.030	5.970
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	6.35E-4	1	1.62	(4,5,na,na,na,na); Estimated from Jungbluth (2007)	94.030	5.970
-	4	TOC, Total Organic Carbon	-	-	kg	6.35E-4	1	1.62	(4,5,na,na,na,na); Estimated from Jungbluth (2007)	94.030	5.970	
-	4	AOX, Adsorbable Organic Halogen as Cl	-	-	kg	8.81E-6	1	1.62	(4,5,na,na,na,na); Estimated from Jungbluth (2007)	100.000	-	

26.13 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 26.9 shows selected LCI results and the cumulative energy demand for the silicon hydrochloration process.

Tab. 26.9 Selected LCI results and the cumulative energy demand for the silicon hydrochloration process

Name	Location Unit Infrastructure	Unit	tetrachlorosilane, at plant	silicon tetrahydride, at plant
			GLO kg 0	RER kg 0
LCIA results				
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	972.1	960.8
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	14.5	9.6
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	26.7	26.0
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	148.8	148.2
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.5	0.4
selected LCI results				
land occupation	resource	m2a	1.8E+0	1.7E+0
CO2, fossil	air	kg	5.3E+1	5.3E+1
NM VOC	air	kg	2.9E-2	2.9E-2
nitrogen oxides	air	kg	1.1E-1	1.1E-1
sulphur dioxide	air	kg	3.9E-2	3.6E-2
particulates, <2.5 um	air	kg	2.1E-3	1.8E-3
BOD	water	kg	1.6E-2	1.5E-2
cadmium	soil	kg	4.4E-9	3.3E-9

26.14 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	silicon hydrochloration
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	5
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Silicon hydrochloration and trichlorosilane hydrogenation including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Siliciumhydrochlorierung
	491	Synonyms	
	492	GeneralComment	The multioutput-process "silicon hydrochloration" delivers the co-products tetrachlorosilane and silicon tetrahydride. The allocation is based on stoichiometric calculation.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	inorganics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Gewinnung
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	2000
	602	EndDate	2007
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications.
Geography	663	Text	The inventory is modelled for the world..
Technology	692	Text	Hydrochloration of silicon and disproportionation of trichlorosilane.
Representativeness	722	Percent	100
	724	ProductionVolume	na
	725	SamplingProcedure	Literature data
	726	Extrapolations	Energy consumptions, emissions, infrastructure and transports are estimated from a similar process.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	41
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	24
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

26.15 References

- Friskhnecht et al. 2007 Friskhnecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.

- Jungbluth et al. 2007 Jungbluth N. & Tuchschnid M. (2007) Photovoltaics. In Dones, R. (Ed.) et al., Sachbilanzen von Energiesystemen: Grundlagen für den ökologischen Vergleich von Energiesystemen und den Einbezug von Energiesystemen in Ökobilanzen für die Schweiz. Final report ecoinvent v2.0 No. 6-XII, Paul Scherrer Institut Villigen, ecoinvent Centre, Dübendorf, CH.
- Ullmanns 2004 Simmer, Walter: Silicon Compounds, Inorganic. Published online: 2002. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under:
<http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

27 Sodium perchlorate

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hirschier, EMPA, St. Gallen
 Last changes: 2007

27.1 Introduction

Sodium perchlorate (NaClO_4 , CAS-No. 7601-89-0) is a white crystalline solid. It is hygroscopic and soluble in water and in alcohol. It usually comes as a monohydrate, which has a rhombic crystal structure.

For this inventory the functional unit is 1 kg of highly pure sodium perchlorate. The most important chemical and physical properties of sodium perchlorate used in this inventory are given here.

Tab. 27.1 Chemical and physical properties of sodium perchlorate (Ullmann 2005)

Property	Value	Unit
Molecular weight	122.44	g mol^{-1}
Specific gravity	2020	kg m^{-3}
Boiling point	610	$^{\circ}\text{C}$
Melting point	400	$^{\circ}\text{C}$

27.1.1 Reserves and Resources

Statistical data about the production and use of perchlorates are not easily accessible because of their strategic importance. The United States' production in 1974 was reported to be ca. 25 000 t, or ca. 12 % of the chlorate production. On account of the increasing developments of artificial satellites and space shuttle programs, perchlorate production will increase, due to their major use in solid propellants.

27.1.2 Use of perchlorates

The most outstanding property of ammonium perchlorate is its high oxygen concentration (54.5 % O_2) and the fact that it decomposes without leaving a solid residue. Therefore, it is used as an oxidizing component in solid rocket propellants.

Lithium perchlorate is used in lithium – nickel sulfide dry batteries, and sodium perchlorate in electrochemical machining; potassium perchlorate is a component of pyrotechnics and an ignition ingredient in flash bulbs. Magnesium perchlorate is known as a very effective drying agent.

Ammonium perchlorate mixed with an epoxy resin forms a temporary adhesive between two metallic surfaces, such as two steel plates. These can be separated whenever desired by heating at ca. 300 $^{\circ}\text{C}$ because of the self-sustained combustion of the adhesive layer. Ammonium perchlorate has been tried as a feed supplement to increase the weight of livestock, with an optimum dose of 2 – 5 mg/kg. Perchlorates have also proven to be helpful in oxygen-regenerating systems to be used in enclosed environments, such as submarines and spacecraft, and in breathing equipment. Potassium values from enriched bittern, obtained from the Dead Sea or the Great Salt Lake, can be recovered by precipitating potassium perchlorate with sodium perchlorate.

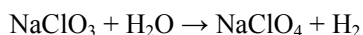
27.2 Production Technologies (Ullmann 2005)

Various methods for the production of perchlorates are described. In practice, commercial production is based on the following steps:

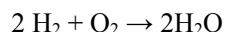
1. Electrochemical production of sodium chlorate from sodium chloride
2. Electrochemical oxidation of sodium chlorate in aqueous solution to sodium perchlorate

Anodic Oxidation

The anodic oxidation of chlorate occurs according to the following overall reaction:



The hydrogen is oxidated to water.



The standard potential of the anodic process (1.19 V) is very close to that of water oxidation (1.228 V), which gives rise to oxygen evolution. Both reactions compete strongly, independent of the acidity of the solution. To obtain a satisfactory current efficiency, the anode potential should be as large as possible, because a high polarization enhances the chlorate oxidation rate more than the oxygen evolution. This is achieved by selecting a suitable anode material and a high current density. Several reaction mechanisms have been proposed for chlorate oxidation to perchlorate.

Cell design and operating conditions must be selected to optimize electricity consumption, which depends on cell voltage and current efficiency. Both are affected by a number of variables in conflicting ways. In particular, the current efficiency is improved by increasing the current density, but this also increases the cell voltage. A temperature rise operates in reverse in that not only the voltage, but also the current efficiency is diminished. A high sodium chlorate concentration in the effluent liquor improves current efficiency as well as voltage, but requires more expensive procedures and higher chlorate consumption to obtain a sufficiently pure perchlorate solution.

Industrial Cells

Contrary to electrochemical chlorate production, the residence time of the electrolyte in the perchlorate cell is not a critical parameter. The cell is undivided and hydrogen gas evolution provides sufficient agitation to decrease the sodium chlorate concentration gradient between the cell inlet and outlet. Because of the relatively high heat dissipation, the cell is normally provided with internal cooling.

Smooth platinum anodes yield the highest oxygen overpotential and, hence, the highest current efficiency (95 – 97 %). Platinum is either used in foils or cladded onto rods of some other metal, such as tantalum or copper. Lead dioxide anodes are also used; they consist of a lead dioxide deposit plated on some conductive substrate, such as graphite rods or plates. They are less expensive than platinum, which more than offsets their lower current efficiency (85 %). More recently, platinate-coated titanium has been claimed to substitute profitably for massive smooth platinum in perchlorate manufacture

Cathodes are made from carbon steel, chromium – nickel steel, nickel, or bronze. As in chlorate electrosynthesis, sodium dichromate is added to the solution to minimize the current efficiency loss caused by cathodic reduction of perchlorate. At the same time, dichromate inhibits corrosion of the steel parts that are directly exposed to the electrolyte, in addition to their cathodic protection. However, dichromate cannot be used with lead dioxide anodes, because it catalyzes oxygen evolution. In that case, the most suitable cathode material is nickel or chromium – nickel steel.

Fig. 27.1 shows some typical cell models. In the Bitterfeld cell, each anode, made from platinum foil, works between a pair of cathodes made from perforated steel plates. The cell is kept at 35 °C by cooling water running through a steel pipe bundle.

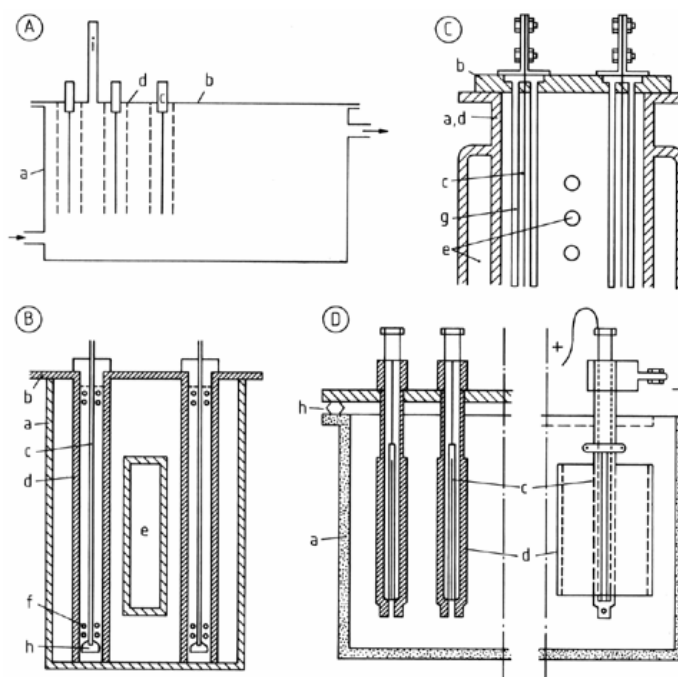


Fig. 27.1 Some typical models of perchlorate cells

A) I. G. Farbenindustrie, Bitterfeld; B) Cardox Corp.; C) American Potash and Chemical Corp.; D) Pechiney, Chedde a) Cell tank; b) Cover; c) Anode; d) Cathode; e) Cooling system; f) Circulation holes; g) Glass rods; h) Porcelain insulators; i) Gas outlet

In the Cardox cell, the anodes consist of platinum cladded onto copper rods 1.3 cm in diameter. Each anode is surrounded by a cathodic tube 7.6 cm in diameter, which is made from steel and perforated at both ends. This allows hydrogen gas to escape into the interelectrode space and ensures convective circulation.

In the model of American Potash and Chemicals (formerly Western Electrochemical Co.), a set of anodic platinum foils is arranged around the inner wall of the cylindrical cell body performing as the cathode; the interelectrode gap is stabilized by a number of glass rod spacers. The temperature is kept at 40 – 45 °C by internal cooling and by a water jacket surrounding the cylindrical cell tank.

In the Pechiney model, the anodes are platinum foils and the cathodes are plates made from bronze.

Operation

Sodium perchlorate can be separated from the cell effluent either as hydrate or as anhydrous salt. Depending on concentration, separation is carried out by cooling alone or by evaporation, followed by cooling. The monohydrate precipitates from the solution below ca. 52 °C; above this temperature, the perchlorate crystallizes as the anhydrous salt. In either case, the mother liquor still contains a large amount of perchlorate. After enrichment with sodium chlorate, this solution is recycled to the electrolysis process.

When the sodium perchlorate is used for conversion to other perchlorates, it need not be separated from the cell effluent. Sodium chlorate is destroyed by chemical treatment and dichromate precipitated as insoluble chromic hydroxide. The purified and concentrated solution is then ready for conversion to other salts.

27.3 System Characterisation

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg sodium perchlorate, at plant, global**. For this study the production of sodium perchlorate by means of the oxidation of sodium chlorate is considered. All data in the present report are referred to 1 kg sodium perchlorate 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

27.4 Raw materials and auxiliaries

As there was no data available for the production of sodium perchlorate, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007). The consumption of platinum for the anode is taken from Ullmann (2005).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the sodium perchlorate production.

Tab. 27.2 Consumption of raw materials and auxiliaries for the sodium perchlorate production

Input	Values	Remarks
Sodium chlorate (kg)	0.915	Calculated with a yield of 95 %
Process water (kg)	0.148	Stoichiometric calculation
Platinum (kg)	4.5E-06	Ullmann 2005
Cooling water (kg)	24	Estimation

27.5 Energy

The consumption of energy is taken from Ullmann (2005). Only electricity is needed.

Tab. 27.3 Energy consumption for the sodium perchlorate production

Input	per kg sodium perchlorate
Electricity (kWh)	2.74

27.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train. Tab. 27.4 summarises the total transport amounts for the sodium perchlorate production.

Tab. 27.4 Total transport amounts for the sodium perchlorate production

tkm kg ⁻¹ sodium perchlorate	lorry	train
Total transports	9.15E-02	5.49E-01

27.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of sodium perchlorate production plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

27.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

Tab. 27.5 Process emissions to air from the sodium perchlorate production

Output	Value	Remarks
Waste heat (MJ)	9,864	calculated from electricity input

27.9 Emissions to water

The remaining amount of unreacted sodium chlorate (5 % of the input) is assumed to leave the production process with the wastewater.

Tab. 27.6 Process emissions to water from the sodium perchlorate production

Output	Value	Remarks
sodium (kg)	9.92E-03	calculated from mass balance
chlorate (kg)	3.58E-02	calculated from mass balance
AOX (kg)	1.52E-02	calculated from water emissions

27.10 Life cycle inventory of sodium perchlorate production and data quality considerations

Tab. 27.7 shows the life cycle inventory and the data quality indicators for the sodium perchlorate production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 27.7 Unit process raw data for the sodium perchlorate production

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	sodium perchlorate, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							GLO 0 kg			
resource, in water	401	0	sodium perchlorate, at plant	GLO	0	kg	1.00E+0			
	662	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	493	-	sodium chlorate, powder, at plant	RER	0	kg	9.15E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	403	-	water, ultrapure, at plant	GLO	0	kg	1.48E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
		-	platinum, at regional storage	RER	0	kg	4.50E-6	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
		-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	2.74E+0	1	1.30	(4,5,na,na,na,na); Estimation
		-	transport, lorry >16t, fleet average	RER	0	tkm	9.15E-2	1	2.09	(4,5,na,na,na,na); Standard distances
		-	transport, freight, rail	RER	0	tkm	5.49E-1	1	2.09	(4,5,na,na,na,na); Standard distances
		-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
		-	Heat, waste	-	-	MJ	9.86E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input
emission air, high population density		4	Sodium, ion	-	-	kg	9.92E-3	1	5.10	(4,5,na,na,na,na); Estimation
emission water, river		4	Chlorate	-	-	kg	3.58E-2	1	3.09	(4,5,na,na,na,na); Estimation
		4	AOX, Adsorbable Organic Halogen as Cl	-	-	kg	1.52E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

27.11 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 27.8 shows selected LCI results and the cumulative energy demand for the sodium perchlorate production.

Tab. 27.8 Selected LCI results and the cumulative energy demand for the sodium perchlorate production

Name		sodium perchlorate, at plant	
Location	Unit	Unit	GLO kg 0
Infrastructure			
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	56.1
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	35.0
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	1.9
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	4.0
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.6
selected LCI results			
land occupation	resource	m2a	3.0E-1
CO ₂ , fossil	air	kg	4.3E+0
NM VOC	air	kg	1.2E-3
nitrogen oxides	air	kg	8.2E-3
sulphur dioxide	air	kg	2.7E-2
particulates, <2.5 um	air	kg	1.2E-3
BOD	water	kg	4.3E-3
cadmium	soil	kg	2.8E-9

27.12 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	sodium perchlorate, at plant
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of sodium perchlorate including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Natriumperchlorat, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "sodium perchlorate, at plant, GLO" is modelled for the production of sodium perchlorate from carbon sodium chlorate in the world. Raw materials are modelled with a stoichiometric calculation. Emissions are estimated. Energy consumptions are taken from literature. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	inorganics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Anorganika
	499	Formula	NaClO ₄
	501	StatisticalClassification	
	502	CASNumber	7601-89-0
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	Anodic oxidation of sodium chlorate
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Emissions are estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

27.13 References

Frischknecht et al. 2007

Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.

- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>
- Ullmanns 2004 Vogt, H., Balej, J., Bennett, J., Wintzer, P., Akbar Sheikh, S., Gallone, P.: Chlorine Oxides and Chlorine Oxygen Acids. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

28 Sodium persulfate

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

28.1 Introduction

Sodium persulfate ($\text{Na}_2\text{S}_2\text{O}_8$; CAS 7775-27-1; synonyms: sodium peroxydisulfate) is a white, odorless salt. It is a strong oxidizer. For this inventory the functional unit is 1 kg of highly pure sodium persulfate. The most important chemical and physical properties of sodium persulfate used in this inventory are given here.

Tab. 28.1 Chemical and physical properties of sodium persulfate (Ullmanns 2004)

Property	Value	Unit
Molecular weight	73.89	g mol^{-1}
Specific gravity	2110	kg m^{-3}
Boiling point	1310	$^{\circ}\text{C}$
Melting point	723	$^{\circ}\text{C}$

28.1.1 Use

Sodium persulfate is used as a bleach, both standalone (particularly in hair cosmetics) and as a detergent component.

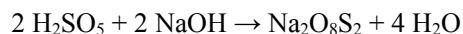
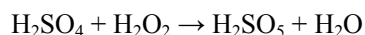
Sodium persulfate is a replacement for ammonium persulfate in etching mixtures for zinc and printed circuit boards, and is used for pickling of copper and some other metals.

Sodium persulfate is a source of free radicals, making it useful as an initiator for eg. emulsion polymerization reactions and for accelerated curing of low formaldehyde adhesives.

Sodium persulfate is also used as a soil conditioner and in manufacture of dyestuffs, modification of starch, bleach activator, desizing agent for oxidative desizing, etc.

28.2 Production Technologies (Ullmanns 2004)

The starting materials are sulfuric acid or oleum, hydrogen peroxide, and sodium hydroxide. In the Peroxid-Chemie process, peroxomonosulfuric acid is produced from H_2SO_4 (90 – 100 wt %) and H_2O_2 (30 – 90 wt %), and then continuously fed, together with a concentrated solution of sodium hydroxide.



The water introduced with the raw materials and formed in the neutralization reaction is removed by passing the reaction mixture through vacuum treatment equipment at ca. 25 $^{\circ}\text{C}$ and 1 kPa. Some of the circulating liquor is continuously removed and cooled to ca. 5 $^{\circ}\text{C}$ in a crystallizing vessel, resulting in precipitation of the triple salt. The salt is isolated by centrifugation, and the mother liquor is fed back into the circulating solution. The filter cake is washed while in the centrifuge, and the wet salt is dried in a fluidized-bed dryer.

28.3 System Characterisation

In this study the production of sodium persulfate by means of the reaction of sulfuric acid with hydrogen peroxide and sodium hydroxide is considered. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

28.4 Raw materials and auxiliaries

As there was no data available for the production of sodium persulfate, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the sodium persulfate production.

Tab. 28.2 Consumption of raw materials and auxiliaries for the sodium persulfate production

Input	Values	Remarks
Sulfuric acid (kg)	0.867	Calculated with a yield of 95 %
Hydrogen peroxide (kg)	0.301	Calculated with a yield of 95 %
Sodium hydroxide (kg)	0.354	Calculated with a yield of 95 %
Cooling water (kg)	24	Estimation

28.5 Energy

There was no information available on the amount of energy used for the production process. In order not to neglect the process energy demand those values were approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg-1) were used as approximation for the energy consumption of the sodium persulfate production. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used. A summary of the values used is given in Tab. 28.3.

Tab. 28.3 Energy consumption for the sodium persulfate production

Input	per kg sodium persulfate
Heat, natural gas burned in industrial furnace > 100 kW	2
Electricity (kWh)	0.333

28.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train. Tab. 28.4 summarises the total transport amounts for the production of 1 kg sodium persulfate.

Tab. 28.4 Total transport amounts for the sodium persulfate production

tkm kg ⁻¹ sodium persulfate	lorry	train
Total transports	1.88E-01	1.13

28.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of sodium persulfate production plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

28.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the sodium persulfate production. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the sulfuric acid input. Sulfuric acid is calculated as sulfate emission

Tab. 28.5 Process emissions to air from the sodium persulfate production

Output	Value	Remarks
Waste heat (MJ)	1.2	calculated from electricity input
Sulfate, to air (kg)	1.7E-03	estimated as 0.2 % of the input

28.9 Emissions to water

The remaining amount of unreacted sulfuric acid, hydrogen peroxide, and sodium hydroxide is assumed to leave the production process with the wastewater. Sulfuric acid is calculated as sulfate emission. Sodium hydroxide is calculated as sodium emission.

Tab. 28.6 Process emissions to water from the sodium persulfate production

Output	Value	Remarks
sulfate (kg)	4.08E-02	calculated from mass balance
hydrogen peroxide (kg)	1.5E-02	calculated from mass balance
sodium (kg)	1.02E-02	calculated from water emissions

28.10 Life cycle inventory of sodium persulfate production and data quality considerations

Tab. 28.7 shows the life cycle inventory and the data quality indicators for the sodium persulfate production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 28.7 Unit process raw data for the sodium persulfate production

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	sodium persulfate, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
401										
662			Location				GLO			
493			InfrastructureProcess				0			
403			Unit				kg			
resource, in water	-	0	sodium persulfate, at plant	GLO	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	sulphuric acid, liquid, at plant	RER	0	kg	8.67E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	hydrogen peroxide, 50% in H2O, at plant	RER	0	kg	3.01E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	sodium hydroxide, 50% in H2O, production mix, at plant	RER	0	kg	3.54E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	2.00E+0	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.33E-1	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.88E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	1.13E+0	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	1.20E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input
emission water, river	-	4	Sulfate	-	-	kg	1.70E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Sulfate	-	-	kg	4.08E-2	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Hydrogen peroxide	-	-	kg	1.50E-2	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Sodium, ion	-	-	kg	1.02E-2	1	5.10	(4,5,na,na,na,na); Estimation

28.11 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 28.8 shows selected LCI results and the cumulative energy demand for the sodium persulfate production.

Tab. 28.8 Selected LCI results and the cumulative energy demand for the sodium persulfate production

Name		sodium persulfate, at plant	
Location	Unit	Unit	GLO kg 0
Infrastructure			
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	19.5
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	6.2
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.9
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.8
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1
selected LCI results			
land occupation	resource	m2a	2.1E-1
CO ₂ , fossil	air	kg	1.3E+0
NM VOC	air	kg	9.0E-4
nitrogen oxides	air	kg	3.0E-3
sulphur dioxide	air	kg	1.4E-2
particulates, <2.5 um	air	kg	3.3E-4
BOD	water	kg	1.7E-3
cadmium	soil	kg	1.2E-9

28.12 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	sodium persulfate, at plant
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of sodium persulfate including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Natriumpersulfat, ab Werk
	491	Synonyms	sodium peroxydisulfate
	492	GeneralComment	The process "asodium persulfate, at plant, GLO" is modelled for the production of sodium persulfate from sulfuric acid and sodium hydroxide in the world. Raw materials are modelled with a stoichiometric calculation. Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	inorganics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Anorganika
	499	Formula	Na2O8S2
	501	StatisticalClassification	
	502	CASNumber	7775-27-1
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	Reaction of sulfuric acid, hydrogen peroxide, and sodium peroxide.
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

28.13 References

- Frischknecht et al. 2007 Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirsch R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.
- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>
- Ullmanns 2004 Bertsch-Frank, B., Engel, D., Kleinschmit, P., Lehmann, T., Panster, P., Steiner, N., Jacobi, S.: Peroxo Compounds, Inorganic. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

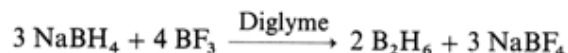
29 Sodium tetrahydroborate fluorination

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hirschier, EMPA, St. Gallen
 Last changes: 2007

Sodium tetrahydroborate fluorination is one of the most important processes to produce diborane.

29.1 Production Technologies (Ullmanns 2004)

The method preferred for the commercial preparation of diborane is the addition of boron trifluoride to a stirred solution of sodium borohydride in diethylene glycol dimethyl ether (diglyme):



Diborane is shipped either as a compressed gas (ambient) or as a liquid (cylinder over packaged with dry ice). It is packaged as a hazardous material regulated by the DOT.

29.2 System Characterisation

In this study the production of diborane by means of the reaction of boron trifluoride and sodium borohydride is considered. The functional unit is 1 kg sodium tetrahydroborate. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers. Within the diborane production, sodium tetrafluoroborate is derived as co-product. Therefore an allocation between the two products was made. The following processes were modelled:

- Sodium tetrahydroborate fluorination, GLO. Multi output process with diborane, at plant, GLO and sodium tetrafluoroborate, at plant, GLO as allocated products

29.3 Diborane

Diborane (B_2H_6 ; CAS 19287-45-7) is a colorless gas at room temperature with a repulsively sweet odor. B_2H_6 mixes well with air, easily forming explosive mixtures. Diborane will ignite spontaneously in moist air at room temperature.

The most important chemical and physical properties of diborane used in this inventory are given here.

Tab. 29.1 Chemical and physical properties of diborane (Ullmanns 2004)

Property	Value	Unit	Remarks
Molecular weight	27.69	g mol ⁻¹	
Specific gravity	1.18	kg m ⁻³	gaseous
Boiling point	-92.6	°C	at normale pressure
Melting point	-164.9	°C	

29.3.1 Use

Diborane serves as a strong but selective reducing agent in organic chemistry. It is an electrophilic reagent and reacts markedly different from the nucleophilic alkali-metal boron and aluminum hydrides. The rates of reduction with diborane were found to decrease in the order: carboxylic acids > olefins > ketones > nitriles > epoxides > esters > acid chlorides.

Hydroboration, the addition of diborane or borane to olefins, has found great significance in preparative chemistry. Hydroboration is applicable to multiple bonds between C–C, C–O, C–N, N–N, N–O. Diborane catalyzed by the presence of an ether adds, in an anti-Markovnikov mode, virtually quantitatively to the carbon – carbon double bond of olefins, forming an alkyl borane.

Other important derivatives of diborane are Lewis base adducts, carbonyl adducts, and oxygen-containing products. Further areas of application for diborane are the doping of semiconductor silicon and germanium. Diborane is readily decomposed at high temperature (600 – 700 °C) to give fine-powder, high-purity elemental boron. Copyrolysis of diborane with hydrocarbons yields boron carbide, and with NH₃ yields boron nitride. These materials, deposited on metal substrates as thin, hard layers, are used in ceramics, tools, and wear-resistant parts. Boron fibers are formed from the decomposition of diborane on substrates and used for light-weight structural components.

The large cross-sectional area of the ¹⁰B isotope for neutron capture makes these compounds especially useful in nuclear applications. Borane is also supplied as stable liquid adducts (CH₃)₂S · BH₃ (Callery Chemical Co., Aldrich Chemical Co.), and as a 1 M solution in tetrahydrofuran (THF) (Aldrich Chemical Co., Ventron Div. of Morton-Thiokol Inc.).

29.4 Raw materials and auxiliaries

As there was no data available for the sodium tetrahydroborate fluorination, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the sodium tetrahydroborate fluorination.

Tab. 29.2 Consumption of raw materials and auxiliaries for the sodium tetrahydroborate fluorination

Input	Values	Remarks
NaBH ₄ (kg)	1	Calculated with a yield of 95 %
Boron trifluoride (kg)	2.386	Calculated with a yield of 95 %
Cooling water (kg)	77.2	Estimation

29.5 Energy

There was no information available on the amount of energy used for the production process. In order not to neglect the process energy demand those values were approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg⁻¹) were used as approximation for the energy consumption of the sodium tetrahydroborate fluorination. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used. A summary of the values used is given in Tab. 29.3.

Tab. 29.3 Energy consumption for the sodium tetrahydroborate fluorination

Input	per kg sodium tetrahydroborate
Heat, natural gas burned in industrial furnace > 100 kW	6.43
Electricity (kWh)	1.07

29.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train. Tab. 29.4 summarises the total transport amounts for the sodium tetrahydroborate fluorination.

Tab. 29.4 Total transport amounts for the sodium tetrahydroborate fluorination

tkm kg ⁻¹ sodium tetrahydroborate	lorry	train
Total transports	3.386E-01	2.032

29.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of sodium tetrahydroborate fluorination plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

29.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the sodium tetrahydroborate fluorination. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. Sodium tetrahydrobortae is calculated as boron emission.

Tab. 29.5 Process emissions to air from the sodium tetrahydroborate fluorination

Output	Value	Remarks
Waste heat (MJ)	3.86	calculated from electricity input
Boron, to air (kg)	2.89E-05	estimated as 0.2 % of the input
Boron trifluoride, to air (kg)	2.39E-04	estimated as 0.2 % of the input

29.9 Emissions to water

The remaining amount of unreacted sodium tetrahydroborate and boron trifluoride (4.8 % of the input) are assumed to leave the production process with the waste water. They are counted as emission of boron, sodium, and fluoride.

Tab. 29.6 Process emissions to water from the sodium tetrahydroborate fluorination

Output	Value	Remarks
boron (kg)	3.37E-02	calculated from mass balance
sodium (kg)	3.02E-02	calculated from mass balance
fluoride (kg)	9.98E-02	calculated from mass balance

29.10 Allocation of diborane and sodium tetrafluoroborate

Within the sodium tetrahydroborate fluorination diborane and sodium tetrafluoroborate are obtained as co-products. A stoichiometric allocation of the two products is made. The used allocation factors are given in Tab. 29.7.

Tab. 29.7 Allocation of ammonium chloride and sodium carbonate from the sodium tetrahydroborate fluorination

Process	Sodium tetrahydroborate fluorination, GLO		
	Unit	per kg sodium tetrahydroborate	Allocation factor (stoichiometric)
diborane, at plant, GLO	Kg	0.486	14.5
sodium tetrafluoroborate, at plant, GLO	kg	2.866	85.5

29.11 Life cycle inventory of sodium tetrahydroborate fluorination and data quality considerations

Tab. 29.8 shows the life cycle inventory and the data quality indicators for the sodium tetrahydroborate fluorination. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 29.8 Unit process raw data for the sodium tetrahydroborate fluorination

401	InputGroup	OutputGroup	Name	Location	Infrastructure	Processes	Unit	sodium tetrahydroborate fluorination	UncertaintyType	StandardDeviation95%	GeneralComment	diborane, at plant	sodium tetrafluoroborate, at plant
662			Location					GLO				GLO	GLO
493			InfrastructureProcess					0				0	0
403			Unit					kg				kg	kg
allocated products resource, in water technosphere	- 2	diborane, at plant	GLO	0	kg	8.55E-1						100	0
	- 2	sodium tetrafluoroborate, at plant	GLO	0	kg	1.45E-1						0	100
	4	- Water, cooling, unspecified natural origin	-	-	m3	7.72E-2	1	1.30	(4,5,na,na,na,na); Estimation		14.508	85.492	
	5	- sodium tetrahydroborate, at plant	GLO	0	kg	1.00E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation		14.508	85.492	
	5	- boron trifluoride, at plant	GLO	0	kg	2.39E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation		14.508	85.492	
	5	- heat, natural gas, at industrial furnace >100kW	RER	0	MJ	6.43E+0	1	1.30	(4,5,na,na,na,na); Estimation		14.508	85.492	
	5	- electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	1.07E+0	1	1.30	(4,5,na,na,na,na); Estimation		14.508	85.492	
	5	- transport, lorry >16t, fleet average	RER	0	tkm	3.39E-1	1	2.09	(4,5,na,na,na,na); Standard distance 100km		14.508	85.492	
	5	- transport, freight, rail	RER	0	tkm	2.03E+0	1	2.09	(4,5,na,na,na,na); Standard distance 600km		14.508	85.492	
	5	- chemical plant, organics	RER	1	unit	1.29E-9	1	3.09	(4,5,na,na,na,na); Estimation		14.508	85.492	
emission air, high population density	- 4	Heat, waste	-	-	MJ	3.86E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input		14.508	85.492	
	- 4	Boron	-	-	kg	2.89E-5	1	5.10	(4,5,na,na,na,na); Estimation		14.508	85.492	
	- 4	Boron trifluoride	-	-	kg	2.39E-4	1	1.62	(4,5,na,na,na,na); Estimation		14.508	85.492	
emission water, river	- 4	Boron	-	-	kg	3.37E-2	1	5.10	(4,5,na,na,na,na); Estimation		14.508	85.492	
	- 4	Sodium, ion	-	-	kg	3.02E-2	1	5.10	(4,5,na,na,na,na); Estimation		14.508	85.492	
	- 4	Fluoride	-	-	kg	9.98E-2	1	1.62	(4,5,na,na,na,na); Estimation		14.508	85.492	

29.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 29.9 shows selected LCI results and the cumulative energy demand for the sodium tetrahydroborate fluorination.

Tab. 29.9 Selected LCI results and the cumulative energy demand for the sodium tetrahydroborate fluorination

Name			diborane, at plant	sodium tetrafluoroborate, at plant
Location			GLO	GLO
Unit		Unit	kg	kg
Infrastructure			0	0
LCIA results				
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	38.2	1'327.1
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	13.5	469.7
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	1.8	62.4
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0	0.1
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	1.7	57.9
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.2	8.5
selected LCI results				
land occupation	resource	m2a	4.3E-1	1.5E+1
CO2, fossil	air	kg	2.5E+0	8.8E+1
NM VOC	air	kg	1.8E-3	6.3E-2
nitrogen oxides	air	kg	7.3E-3	2.5E-1
sulphur dioxide	air	kg	3.9E-2	1.4E+0
particulates, <2.5 um	air	kg	9.1E-4	3.2E-2
BOD	water	kg	9.1E-3	3.2E-1
cadmium	soil	ka	2.2E-9	7.6E-8

29.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	sodium tetrahydroborate fluorination
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	5
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Fluorination of sodium tetrahydroborate including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Natriumtetrahydroboratfluorinierung
	491	Synonyms	
	492	GeneralComment	The multioutput-process "sodium tetrahydroborate fluorination" delivers the co-products diborane and sodium tetrafluoroborate. The allocation is based on stoichiometric calculation.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	inorganics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications.
Geography	663	Text	The inventory is modelled for the world..
Technology	692	Text	Fluorination of sodium tetrahydroborate
Representativeness	722	Percent	100
	724	ProductionVolume	na
	725	SamplingProcedure	Literature data
	726	Extrapolations	Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

29.14 References

Frischknecht et al. 2007 Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Over-

view and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.

Gendorf 2000

Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>

Ullmanns 2004

Brotherton, R.: Boron Compounds. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

30 Sulfuric acid digestion of spodumene

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hirschier, EMPA, St. Gallen
 Last changes: 2007

Sulfuric acid digestion of spodumene is the most important production process of lithium carbonate.

30.1 Production Technologies (Ullmanns 2004)

The sulfuric acid digestion process can be used for all lithium ores. It is usually carried out at 250 – 400 °C, although 850 – 900 °C is recommended for amblygonite because this leads to the formation of insoluble aluminum phosphate which facilitates removal of the insoluble residue.

Sulfuric acid digestion gives a product that contains an insoluble residue, together with lithium sulfate which is leached from the crushed product with hot water. To remove alkaline-earth metals, aluminum, and iron, the solution is neutralized with lime and soda and then filtered. Lithium carbonate is precipitated by adding concentrated sodium carbonate solution. This is carried out near the boiling point, to take advantage of the decreasing solubility of lithium carbonate at higher temperature.

The sulfuric acid process gives good yields of lithium (ca. 90 %) and has a relatively favorable energy balance compared with other processes for the production of lithium compounds from ores. Digestion by hydrochloric acid or hydrogen chloride gas at 935 °C has been suggested for the production (by sublimation) of lithium chloride from lepidolite. This process has, however, little significance compared to the sulfuric acid process.

Spodumene is first crushed to a particle size of 20 – 80 mm. Alternatively, flotation concentrate, which is already finely divided, is used. The crushed ore is fed into an oil-, coal-, or gas-fired rotary decrepitation kiln, which attains the required temperature (1050 – 1100 °C) over at least 10 m of its length. The spodumene takes ca. 4 h to pass through the rotary kiln, the dwell time in the decrepitation zone being 15 – 30 min, which results in practically complete transformation into β -spodumene.

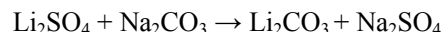
The heat efficiency is improved by using a countercurrent flow of material and hot gas. The exhaust gases are used to preheat the ore, to heat the roasting furnace, and to concentrate the liquors by evaporation. The spodumene is quite hot as it leaves the rotary kiln, and sometimes a rotary cooler, which can also be used to preheat the combustion air, is necessary.

If the starting material is in lump form, the β -spodumene produced by decrepitation is crushed to a grain size of less than 0.1 mm. Sulfuric acid (93 %, in 35 % excess) is then added either by a mixer conveyor or by spraying onto a band conveyor fitted with a deflector plate. The digestion process, in which lithium in the ore is converted into soluble lithium sulfate, is carried out at ca. 250 °C in a lined rotary furnace or on a conveyor heated by exhaust gases, which carries the spodumene through a tunnel directly to the leaching stage. Digestion is usually complete in less than 1 h.

The lithium sulfate formed is leached out of the reaction product with hot water in Pachuca tanks agitated by compressed air or in stirred vessels. A continuous leaching process, in which the material and the water move countercurrently through several stirred vessels in series, has proved advantageous. After leaching, aluminum silicate residues are removed by a rotary vacuum filter, and the filtrate is treated with lime and soda to precipitate alkaline-earth metals and iron. After repeated filtration, the solution is neutralized with sulfuric acid. During neutralization and in the subsequent evaporation, which concentrates the solution to 200 – 250 g/L of lithium sulfate, aluminum hydroxide precipitates and is removed together with any purifying additives (e.g., activated charcoal).

The precipitation of lithium carbonate by concentrated sodium carbonate solution is carried out near the boiling point. Lithium carbonate is removed by filtration or centrifugation and washed with hot

water to remove the adhering mother liquor (mainly sodium sulfate). It is then dried with a vacuum plate dryer, a band dryer, or a vacuum screw conveyor dryer.



The particle size of lithium carbonate depends strongly on the concentration during precipitation, the temperature, and the rate of addition of sodium carbonate.

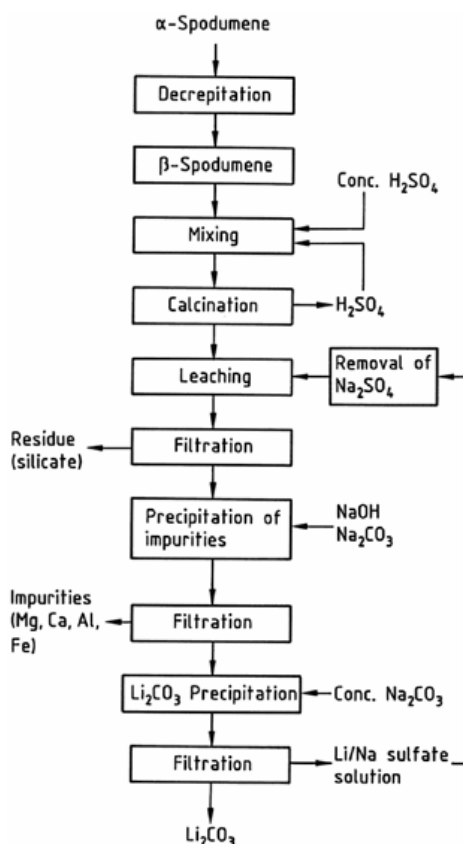


Fig. 30.1 Flow sheet for lithium carbonate production from spodumene

Mother liquor from lithium carbonate precipitation always contains ca. 1 g of lithium per liter and is therefore recycled. Before the liquor is returned to the leaching stage, sodium sulfate is removed by crystallization at low temperature. This is a valuable byproduct and improves the economic viability of the process.

The use of fluidized-bed furnaces has been proposed for the decrepitation process, enabling it to be carried out under mild conditions. Recycling of excess sulfuric acid from the dissolution process has also been proposed.

30.2 System Characterisation

In this study the sulfuric acid digestion of spodumene is considered. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is

assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers. Within the sulfuric acid digestion of spodumene, lithium carbonate and sodium sulfate are derived as co-products. Therefore an allocation between the two products was made. The following processes were modelled:

- Sulfuric acid digestion of spodumene, GLO. Multi output process with lithium carbonate, at plant, GLO and sodium sulfate from spodumene, at plant, GLO as allocated products

30.3 Lithium carbonate

Lithium carbonate (Li_2CO_3 , CAS 554-13-2) crystallizes as colorless monoclinic prisms that melt with slight decomposition. The technical product is a fine white powder with a bulk density of ca. 0.8 kg/L. Lithium carbonate is only slightly soluble in water. Its solubility decreases with increasing temperature. For this inventory the functional unit is 1 kg of pure lithium carbonate. The most important chemical and physical properties of lithium carbonate used in this inventory are given here.

Tab. 30.1 Chemical and physical properties of Lithium carbonate (Ullmanns 2004)

Property	Value	Unit
Molecular weight	73.89	g mol^{-1}
Specific gravity	2110	kg m^{-3}
Boiling point	1310	$^{\circ}\text{C}$
Melting point	723	$^{\circ}\text{C}$

30.3.1 Use

Lithium carbonate is the starting material for the industrial production of all other lithium compounds and is itself used in industry in large quantities. It is an additive in glass, enamels, and ceramics (as are many lithium ores, e.g., spodumene and petalite). The lithium oxide generated by lithium carbonate is a very active flux for silica. It is increasingly replacing fluorspar (CaF_2) for environmental reasons. Its function is generally based on reduction of melting point (particularly important for enamels), reduction of viscosity of molten glass (leading to higher throughput and smaller wall thickness), increased surface tension (giving improved reflectivity to enamels and glazes), and improved chemical resistance. Lithium aluminum silicates with compositions in the range LiAlSiO_4 – $\text{LiAlSi}_4\text{O}_{10}$ are produced that have extremely low or negative thermal expansion coefficients. These compounds and high-lithium glasses are used in heat-resistant ovenware.

The addition of lithium carbonate to cement or concrete leads to quicker setting. Amounts of 1 – 5 % are used to control setting times. In quick-setting tile adhesives, lithium carbonate leads to faster adhesion of the tiles.

Lithium carbonate is also used as an additive to molten salt baths for the electrolytic production of aluminum. It reacts with aluminum trifluoride in the melt to form lithium fluoride. The concentration used is 2 – 5 %, preferably 3.5 %, of lithium fluoride. The following effects are produced: reduction of melting point (by 10 – 25 $^{\circ}\text{C}$), density, and viscosity of the melt and an increase in electrical conductivity of ca. 10 %. This results in a reduced operating temperature, reduction in fluorine emission (up to 50 %), and savings in energy (2 – 3 %) and anode carbon (ca. 2 %). It is possible to operate existing electrolytic cells at higher loading (ca. 10 %), thus increasing capacity without additional capital investment.

Specially prepared high-purity lithium carbonate (conforming to the national pharmacopoeias, e.g., U.S.P., DAB, E.P., B.P., J.P.) is used increasingly for the treatment of manic-depressive conditions; daily doses of ca. 150 – 500 mg of lithium carbonate are used. The successful treatment of herpes, alcoholism, and cancer by lithium carbonate assisted chemotherapy has also been reported.

Mixtures of lithium and potassium carbonate are used as electrolytes in molten carbonate fuel cells (MCFCs). The MCFCs operating at high temperatures (600 – 650 °C) are especially suited for modern coal-fired baseload electric utility plants and natural gas units. The high-quality waste heat can be used for industrial cogeneration or internal reforming.

30.4 Raw materials and auxiliaries

Tab. 30.2 shows values for the use of raw materials for the sulfuric acid digestion of spodumene. The use of spodumene is calculated with a yield of 90% (Ullmann 2004).

As there was no data available for the consumption of sulfuric acid and sodium carbonate of ammonium thiocyanate, these values are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the sulfuric acid digestion of spodumene.

Tab. 30.2 Consumption of raw materials and auxiliaries for the sulfuric acid digestion of spodumene

Input	Values	Remarks
Spodumene (kg)	1	Calculated with a yield of 90 %
Sulfuric acid (kg)	3.57E-01	Calculated with a yield of 95 %
Sodium carbonate (kg)	3.86E-01	Calculated with a yield of 95 %
Cooling water (kg)	1.79E-02	Estimation

30.5 Energy

Tab. 30.3 Energy consumption for the sulfuric acid digestion of spodumene (Kim 2003)

Input	per kg spodumene
Heat, natural gas burned in industrial furnace > 100 kW	1.241
Electricity (kWh)	2.098E-01

30.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train. Tab. 30.4 summarises the total transport amounts for the sulfuric acid digestion of spodumene.

Tab. 30.4 Total transport amounts for the sulfuric acid digestion of spodumene

t km kg ⁻¹ spodumene	lorry	train
Total transports	1.743E-01	1.046

30.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of spodumene sulfuric acid digestion plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical

plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

30.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the sulfuric acid digestion of spodumene. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the sulfuric acid input. It is calculated as sulfate emission.

Tab. 30.5 Process emissions to air from the sulfuric acid digestion of spodumene

Output	Value	Remarks
Waste heat (MJ)	7.552E-01	calculated from electricity input
Sulfate, to air (kg)	3.5E-05	estimated as 0.2 % of the input

30.9 Emissions to water

The remaining amount of unreacted sulfuric acid and sodium carbonate (4.8 % of the input) is assumed to leave the production process with the waste water. The sulfuric acid is calculated as sulfate emission.

Tab. 30.6 Process emissions to water from the sulfuric acid digestion of spodumene

Output	Value	Remarks
sulfate (kg)	1.75E-02	calculated from mass balance
sodium (kg)	8.38E-03	calculated from mass balance
carbonate (kg)	1.09E-02	calculated from mass balance

30.10 Wastes

The silicate residues are calculated as "disposal, inert waste, 5% water, to inert material landfill", due to lack of further information about the composition of the wastes mentioned.

Tab. 30.7 Waste from the sulfuric acid digestion of spodumene

Output	
disposal, municipal solid waste (kg)	9.238E-01

30.11 Allocation of lithium carbonate and sodium sulfate

Within the sulfuric acid digestion of spodumene, lithium carbonate and sodium sulfate are obtained as co-products. A stoichiometric allocation of the two products is made. The used allocation factors are given in Tab. 30.8.

Tab. 30.8 Allocation of lithium carbonate and sodium sulfate from the sulfuric acid digestion of spodumene

Process	Sulfuric acid digestion of spodumene, GLO		
	Unit	per kg spodumene	Allocation factor (stoichiometric)
lithium carbonate, at plant, GLO	Kg	2.562E-01	34.3
Sodium sulfate, from spodumene, at plant, GLO	kg	4.915E-01	65.7

30.12 Life cycle inventory of sulfuric acid digestion of spodumene and data quality considerations

Tab. 30.9 shows the life cycle inventory and the data quality indicators for the sulfuric acid digestion of spodumene. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 30.9 Unit process raw data for the sulfuric acid digestion of spodumene

401	InputGroup OutputGroup	Name	Location	InfrastructureProces	s	Unit	sulphuric acid digestion of spodumene	Uncertainty Type	StandardDeviation95 %	GeneralComment	lithium carbonate, at plant	sodium sulphate from sulfuric acid digestion of spodumene
662		Location					GLO				GLO	GLO
493		InfrastructureProcess					0				0	0
403		Unit					kg				kg	kg
allocated	- 2	lithium carbonate, at plant	GLO	0	kg	2.56E-1					100	0
products	- 2	sodium sulphate from sulfuric acid digestion of spodumene	GLO	0	kg	4.92E-1					0	100
resource, in water technosphere	4	- Water, cooling, unspecified natural origin	-	-	m3	1.79E-2	1	1.30	(4,5,na,na,na,na); Estimation		34.259	65.741
	5	- spodumene, at plant	RER	0	kg	1.00E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation		34.259	65.741
	5	- sulphuric acid, liquid, at plant	RER	0	kg	3.57E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation		34.259	65.741
	5	- soda, powder, at plant	RER	0	kg	3.86E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation		34.259	65.741
	5	- heat, natural gas, at industrial furnace >100kW	RER	0	MJ	1.24E+0	1	1.13	(1,2,3,1,1,3); Kim 2003		34.259	65.741
	5	- electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	2.10E-1	1	1.13	(1,2,3,1,1,3); Kim 2003		34.259	65.741
	5	- disposal, inert waste, 5% water, to inert material landfill	CH	0	kg	9.24E-1	1	1.30	(4,5,na,na,na,na); Estimation		34.259	65.741
	5	- transport, lorry >16t, fleet average	RER	0	tkm	1.74E-1	1	2.09	(4,5,na,na,na,na); Standard distance 100km		34.259	65.741
	5	- transport, freight, rail	RER	0	tkm	1.05E+0	1	2.09	(4,5,na,na,na,na); Standard distance 600km		34.259	65.741
	5	- chemical plant, organics	RER	1	unit	2.99E-10	1	3.09	(4,5,na,na,na,na); Estimation		34.259	65.741
emission air, high population density	- 4	Heat, waste	-	-	MJ	7.55E-1	1	1.13	(1,2,3,1,1,3); Calculated from the electricity input		34.259	65.741
emission water, river	- 4	Sulfate	-	-	kg	3.50E-5	1	1.30	(4,5,na,na,na,na); Estimation		34.259	65.741
	- 4	Sulfate	-	-	kg	1.75E-2	1	1.30	(4,5,na,na,na,na); Estimation		34.259	65.741
	- 4	Sodium, ion	-	-	kg	8.38E-3	1	5.10	(4,5,na,na,na,na); Estimation		34.259	65.741
	- 4	Carbonate	-	-	kg	1.09E-2	1	3.09	(4,5,na,na,na,na); Estimation		34.259	65.741

30.13 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 30.10 shows selected LCI results and the cumulative energy demand for the sulfuric acid digestion of spodumene.

Tab. 30.10 Selected LCI results and the cumulative energy demand for the sulfuric acid digestion of spodumene

Name			lithium carbonate, at plant	sodium sulphate from sulfuric acid digestion of spodumene
Location	Unit	Unit	GLO	GLO
Infrastructure			kg 0	kg 0
LCIA results				
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	10.7	10.7
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	2.6	2.6
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.6	0.6
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.4	0.4
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.0	0.0
selected LCI results				
land occupation	resource	m2a	1.7E-1	1.7E-1
CO ₂ , fossil	air	kg	7.6E-1	7.6E-1
NM/OC	air	kg	5.0E-4	5.0E-4
nitrogen oxides	air	kg	2.4E-3	2.4E-3
sulphur dioxide	air	kg	8.3E-3	8.3E-3
particulates, <2.5 µm	air	kg	4.2E-4	4.2E-4
BOD	water	kg	9.6E-4	9.6E-4
cadmium	soil	kg	7.9E-10	7.9E-10

30.14 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	sulphuric acid digestion of spodumene
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	5
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Sulfuric acid digestion of spodumene including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Spodumennassveraschung
	491	Synonyms	
	492	GeneralComment	The multioutput-process "Sulfuric acid digestion of spodumene" delivers the co-products lithium carbonate and sodium sulfate. The allocation is based on stoichiometric calculation. Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	inorganics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Gewinnung
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	2000
	602	EndDate	2007
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications.
Geography	663	Text	The inventory is modelled for the world..
Technology	692	Text	Sulfuric acid digestion of spodumene
Representativeness	722	Percent	100
	724	ProductionVolume	na
	725	SamplingProcedure	Literature data
	726	Extrapolations	Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

30.15 References

Frischknecht et al. 2007 Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Over-

- view and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.
- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>
- Kim 2003 Kim S., Overcash M.: Energy in chemical manufacturing processes: gate-to-gate information for life cycle assessment. In: Journal of Chemical Technology & Biotechnology vol. 78, no. 9: 995-1005(11). 2003
- Ullmanns 2004 Wietelmann, U., Bauer, R.: Lithium and Lithium Compounds. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

31 Trichloroborane

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

31.1 Introduction

Trichloroborane (BCl_3 ; CAS 10294-34-5; synonyms: boron trichloride, boron chloride) is a colourless gas. For this inventory the functional unit is 1 kg of highly pure trichloroborane. The most important chemical and physical properties of trichloroborane used in this inventory are given here.

Tab. 31.1 Chemical and physical properties of trichloroborane (Ullmanns 2004)

Property	Value	Unit	Remarks
Molecular weight	117.19	g mol^{-1}	
Specific gravity	1359	kg m^{-3}	liquide
Heat of fusion	-403	kJ mol^{-1}	at 298 K
Boiling point	12.5	$^{\circ}\text{C}$	at normale pressure
Melting point	-107	$^{\circ}\text{C}$	under pressure

31.1.1 Reserves and Resources

The annual production of BCl_3 is about 225 – 450 t/a.

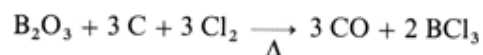
31.1.2 Use

One of the most important uses of BCl_3 is in the preparation of boron fibers. Typically an electrically heated tungsten filament is passed through a chamber containing BCl_3 and hydrogen. The BCl_3 is reduced, and boron is deposited on the filament, producing a stiff, strong boron fiber.

Boron trichloride, like the trifluoride, has been used as a Lewis acid catalyst in organic synthesis in the polymerization of olefins and phosphazines, as well as in catalysis of other organic reactions. Boron trichloride is also used in plasma etching of aluminum and silicon, in semiconductor manufacturing, and as a source of boron for chemical vapor deposition. Steel is boronized by contacting it with a reactive mixture of hydrogen, hydrocarbons, and BCl_3 at high temperatures.

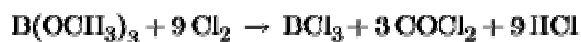
31.2 Production Technologies (Ullmanns 2004)

Trichloroborane can be prepared by the reaction of boric oxide, a reducing agent, and chlorine in a melt at high temperatures, for example:



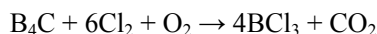
The product BCl_3 is recovered by fractional condensation. Sodium tetraborate $\text{Na}_2\text{B}_4\text{O}_7$ can be used in place of B_2O_3 . Molten sodium chloride, formed as a byproduct, settles to the bottom of the reactor. A molten combination of aluminum chloride and an alkali-metal chloride has also been used as a medium for reductive chlorination of borates.

Free-radical chlorination of borate esters also yields BCl_3 :



A modification of this process, in which a free-radical initiator is used, has been reported. A low reaction temperature, 40 – 90 °C, and the use of an initiator decrease the consumption of Cl₂ and prevent the formation of phosgene, COCl₂.

Commercially, boron trichloride is prepared by the reaction of boron carbide with chlorine in a borate melt. This reaction can also be done with dry B₄C in a fluidized-bed reactor using a transition-metal halide as a catalyst.



Contamination of BCl₃ with COCl₂ from the chlorination of carbon oxides is a serious problem in BCl₃ manufacturing. Because of similar vapor pressure – temperature curves, BCl₃ and COCl₂ cannot be separated by fractional distillation. Thermal or photochemical conversion of COCl₂ to CO and Cl₂ followed by fractional distillation has been used to purify BCl₃.

Boron trichloride can be prepared in the laboratory by the reaction of boron trifluoride and aluminum chloride.

31.3 Quality Specifications, Chemical Analysis

Boron trichloride is typically sold as a 99.9 % pure liquid. The major impurities in boron trichloride are phosgene and chlorine.

The boron halides may be analyzed by hydrolysis followed by analysis of B₂O₃ and the halide, or by base titration. The concentration of phosgene (COCl₂) in BCl₃ may be determined by infrared spectroscopy. The trifluoride, trichloride, and tribromide may be purified by fractional distillation.

31.4 System Characterisation

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg trichloroborane, at plant, global**. For this study the production of trichloroborane by means of the reaction of boron carbide with chlorine is considered. All data in the present report are referred to 1 kg trichloroborane 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

31.5 Raw materials and auxiliaries

As there was no data available for the production of trichloroborane, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the trichloroborane production.

Tab. 31.2 Consumption of raw materials and auxiliaries for the trichloroborane production

Input	Values	Remarks
Boron carbide (kg)	0.127	Calculated with a yield of 95 %
Chlorine (kg)	0.953	Calculated with a yield of 95 %
Cooling water (kg)	24	Estimation

31.6 Energy

There was no information available on the amount of energy used for the production process. In order not to neglect the process energy demand those values were approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg⁻¹) were used as approximation for the energy consumption of the trichloroborane production. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used. A summary of the values used is given in Tab. 31.3.

Tab. 31.3 Energy consumption for the trichloroborane production

Input	per kg trichloroborane
Heat, natural gas burned in industrial furnace > 100 kW	2
Electricity (kWh)	0.333

31.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train. Tab. 31.4 summarises the total transport amounts for the trichloroborane production.

Tab. 31.4 Total transport amounts for the trichloroborane production

tkm kg ⁻¹ trichloroborane	lorry	train
Total transports	1.08E-01	6.48E-01

31.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of trichloroborane production plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

31.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the trichloroborane production. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the boron carbide and chlorine input. The boron carbide is calculated as boron emission.

Tab. 31.5 Process emissions to air from the trichloroborane production

Output	Value	Remarks
Waste heat (MJ)	1.2	calculated from electricity input
Boron, to air (kg)	9.98E-06	estimated as 0.2 % of the input
Cl ₂ , to air (kg)	1.9E-03	estimated as 0.2 % of the input

31.10 Emissions to water

The remaining amount of unreacted chlorine (4.8 % of the input) is assumed to leave the production process with the waste water.

Tab. 31.6 Process emissions to water from the trichloroborane production

Output	Value	Remarks
Cl ₂ (kg)	4.57E-02	calculated from mass balance
AOX (kg)	4.57E-02	calculated from water emissions

31.11 Waste

The remaining boric carbide (4.8 % of the input) is assumed to be disposed as inert waste.

Tab. 31.7 Wastes from the trichloroborane production

Output	Value
disposal, inert waste, 5% water, to inert material landfill (kg)	6.34E-03

31.12 Life cycle inventory of trichloroborane production and data quality considerations

Tab. 31.8 shows the life cycle inventory and the data quality indicators for the trichloroborane production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 31.8 Unit process raw data for the trichloroborane production

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	trichloroborane	Uncertainty Type	StandardDeviation95%	GeneralComment
							, at plant			
	401									
	662		Location				GLO			
	493		InfrastructureProcess				0			
	403		Unit				kg			
resource, in water	-	0	trichloroborane, at plant	GLO	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	boron carbide, at plant	GLO	0	kg	1.27E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	chlorine, liquid, production mix, at plant	RER	0	kg	9.53E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	disposal, inert waste, 5% water, to inert material landfill	CH	0	kg	6.34E-3	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	2.00E+0	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.33E-1	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.08E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	6.48E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	1.20E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input
	-	4	Boron	-	-	kg	9.98E-6	1	5.10	(4,5,na,na,na,na); Estimation
	-	4	Chlorine	-	-	kg	1.91E-3	1	1.62	(4,5,na,na,na,na); Estimation
emission water, river	-	4	Chlorine	-	-	kg	4.57E-2	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	AOX, Adsorbable Organic Halogen as Cl	-	-	kg	4.57E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

31.13 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 31.9 shows selected LCI results and the cumulative energy demand for the trichloroborane production.

Tab. 31.9 Selected LCI results and the cumulative energy demand for the trichloroborane production

Name		trichloroborane, at plant	
Location	Unit	GLO	
Infrastructure	Unit	kg	0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	36.1
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	16.5
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	1.6
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	2.0
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.3
selected LCI results			
land occupation	resource	m2a	3.7E-1
CO2, fossil	air	kg	3.2E+0
NM VOC	air	kg	1.3E-3
nitrogen oxides	air	kg	6.4E-3
sulphur dioxide	air	kg	1.6E-2
particulates, <2.5 um	air	kg	9.5E-4
BOD	water	kg	2.4E-3
cadmium	soil	kg	2.3E-9

31.14 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	trichloroborane, at plant
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of trichloroborane including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Trichlorboran, ab Werk
	491	Synonyms	boron trichloride//boron chloride
	492	GeneralComment	The process "trichloroborane, at plant, GLO" is modelled for the production of trichloroborane from boron carbide and chlorine in the world. Raw materials are modelled with a stoichiometric calculation. Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	inorganics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Anorganika
	499	Formula	BCl3
	501	StatisticalClassification	
	502	CASNumber	10294-34-5
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	Chlorination of boron carbide
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

31.15 References

- Frischknecht et al. 2007 Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschler R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.
- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>

Ullmanns 2004

Brotherton, R.: Boron Compounds. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

32 Trifluoromethane

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

32.1 Introduction

Trifluoromethane (CHF_3 ; CAS 75-46-7; synonyms: fluoroform, carbon trifluoride, freon 23, R-23, FE-13) is a haloform. For this inventory the functional unit is 1 kg of highly pure trifluoromethane. The most important chemical and physical properties of trifluoromethane used in this inventory are given here.

Tab. 32.1 Chemical and physical properties of trifluoromethane (Gasencyclopedia 2007)

Property	Value	Unit	Remarks
Molecular weight	70.01	g mol^{-1}	
Specific gravity	1246	kg m^{-3}	at -34 °C
Boiling point	-82.1	°C	at normale pressure
Melting point	-160	°C	at normale pressure

32.1.1 Use

CHF_3 is used in the semiconductor industry in plasma etching of silicon oxide and silicon nitride.

As a refrigerant, CHF_3 is known as R-23 or HFC-23.

HFC-23 is also used as a replacement for Halon 1301 in fire suppression systems as a total flooding gaseous fire suppression agent. When used as a fire suppressant, the chemical carries the DuPont trade name, FE-13. CHF_3 is recommended for this application because of its low toxicity, its low reactivity, and its high density.

32.2 Production Technologies (Ullmann 2005)

Trifluoromethane is produced by the exchange of chlorine for fluorine using chloroform. The novel commercially interesting hydrofluorocarbons are produced by processes, that have been developed for the production of chlorofluorocarbons and hydrofluorocarbons and optimized during the last decades.

Higher temperatures and higher hydrogen fluoride : substrate ratios are necessary to achieve complete replacement of all chlorine atoms in the starting chlorocompounds by fluorine. Both liquid-phase halogen exchange in the presence of catalysts such as antimony(V) or tin(IV) chlorofluorides and vapor phase reactions using solid-phase catalysts based on chromium are employed. The starting material is chloroform.

32.3 System Characterisation

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg fluoroform, at plant, global**. For this study the production of fluoroform by means of the fluorination of chloroform is considered. All data in the present report are referred to 1 kg fluoroform 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also

not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

32.4 Raw materials and auxiliaries

As there was no data available for the production of fluoroform, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the fluoroform production.

Tab. 32.2 Consumption of raw materials and auxiliaries for the trifluoromethane production

Input	Values	Remarks
Chloroform(kg)	1.774	Calculated with a yield of 95 %
HF (kg)	0.902	Calculated with a yield of 95 %
Cooling water (kg)	24	Estimation

32.5 Energy

There was no information available on the amount of energy used for the production process. In order not to neglect the process energy demand those values were approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg-1) were used as approximation for the energy consumption of the fluoroform production. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used. A summary of the values used is given in Tab. 32.3.

Tab. 32.3 Energy consumption for the trifluoromethane production

Input	per kg fluoroform
Heat, natural gas burned in industrial furnace > 100 kW	2
Electricity (kWh)	0.333

32.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train. Tab. 32.4 summarises the total transport amounts for the fluoroform production.

Tab. 32.4 Total transport amounts for the trifluoromethane production

tkm kg ⁻¹ fluoroform	lorry	train
Total transports	2.68E-01	1.61

32.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of fluoroform production plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

32.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the fluoroform production. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the chloroform and HF input.

Tab. 32.5 Process emissions to air from the trifluoromethane production

Output	Value	Remarks
Waste heat (MJ)	1.2	calculated from electricity input
CHCl ₃ , to air (kg)	3.55E-03	estimated as 0.2 % of the input
HF, to air (kg)	1.8E-03	estimated as 0.2 % of the input
CO ₂ , fossil, to air (kg)	2.86E-02	from water treatment

32.9 Emissions to water

The remaining amount of unreacted chloroform and HF (4.8 % of the input) is assumed to leave the production process with the waste water. The hydrogen fluoride is calculated as fluoride emission.

Further it is assumed that this wastewater is treated in a internal wastewater plant. The removal efficiency for chloroform is assumed with 90%. The carbon contained in the removed chloroform is accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory are calculated from the amount of chloroform in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. The value for AOX is calculated from the chloroform emission.

Tab. 32.6 Process emissions to water from the trifluoromethane production

Output	Value	Remarks
CHCl ₃ (kg)	8.52E-03	calculated from mass balance
fluoride (kg)	4.11E-02	calculated from mass balance
BOD, COD (kg)	1.85E-02	calculated from water emissions
TOC, DOC (kg)	8.57E-04	calculated from water emissions
AOX (kg)	7.59E-03	calculated from water emissions

32.10 Life cycle inventory of trifluoromethane production and data quality considerations

Tab. 32.7 shows the life cycle inventory and the data quality indicators for the trifluoromethane production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 32.7 Unit process raw data for the trifluoromethane production

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	trifluoromethane, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							GLO 0 kg			
resource, in water	-	0	trifluoromethane, at plant	GLO	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	trichloromethane, at plant	RER	0	kg	1.77E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	hydrogen fluoride, at plant	GLO	0	kg	9.02E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	2.00E+0	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.33E-1	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	2.68E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	1.61E+0	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Heat, waste	-	-	MJ	1.20E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input
emission air, high population density	-	4	Chloroform	-	-	kg	3.55E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Hydrogen fluoride	-	-	kg	1.80E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	2.86E-2	1	1.30	(4,5,na,na,na,na); from water treatment
emission water, river	-	4	Chloroform	-	-	kg	8.52E-3	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Fluoride	-	-	kg	4.11E-2	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	1.85E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	1.85E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	8.57E-4	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	8.57E-4	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	AOX, Adsorbable Organic Halogen as Cl	-	-	kg	7.59E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

32.11 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 32.8 shows selected LCI results and the cumulative energy demand for the trifluoromethane production.

Tab. 32.8 Selected LCI results and the cumulative energy demand for the trifluoromethane production

Name		trifluoromethane, at plant	
Location	Unit	Unit	GLO
Infrastructure			kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	108.3
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	31.1
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	4.1
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	3.9
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.6
selected LCI results			
land occupation	resource	m2a	9.8E-1
CO2, fossil	air	kg	6.4E+0
NM VOC	air	kg	2.7E-2
nitrogen oxides	air	kg	1.5E-2
sulphur dioxide	air	kg	7.9E-2
particulates, <2.5 um	air	kg	1.7E-3
BOD	water	kg	2.5E-2
cadmium	soil	kg	5.4E-9

32.12 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	trifluoromethane, at plant
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformatic	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of trifluoromethane including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Trifluormethan, ab Werk
	491	Synonyms	fluoroform/carbon trifluoride/freon 23//R-23//FE-13
	492	GeneralComment	The process "trifluoromethane, at plant, GLO" is modelled for the production of trifluoromethane from trichloromethane and hydrogen fluoride in the world. Raw materials are modelled with a stoichiometric calculation. Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	CHF3
	501	StatisticalClassification	
	502	CASNumber	75-46-7
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	Fluorination of trichloromethane
Representativene	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAn	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

32.13 References

Frischknecht et al. 2007

Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Over-

- view and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.
- Gasencyclopedia 2007 Gasencyclopedia 2007. Air Liquide. Online-Version under
<http://encyclopedia.airliquide.com/Encyclopedia.asp?GasID=66>
- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf,
Burgkirchen as pdf-File under:
<http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>
- Ullmanns 2004 Siegemund, G., Schwertfeger, W., Feiring, A., Smart, B., Behr, F., Vogel, H.,
McKusick, B.: Fluorine Compounds, Organic. Published online: 2002. In: Ull-
mann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic
Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.).
7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version
under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

33 Trimethyl borate

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
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33.1 Introduction

Trimethyl borate ((CH₃O)₃B; CAS 121-43-7; synonyms: methyl borate, boric acid trimethyl ester, boron methoxide) is a clear liquid. It is soluble in organic solvents and dissolves in water with hydrolysis to boric acid and the methanol.

For this inventory the functional unit is 1 kg of highly pure trimethyl borate. The most important chemical and physical properties of trimethyl borate used in this inventory are given here.

Tab. 33.1 Chemical and physical properties of trimethyl borate (Ullmanns 2004)

Property	Value	Unit
Molecular weight	103.09	g mol ⁻¹
Specific gravity	920	kg m ⁻³
Boiling point	68.0	°C
Melting point	-29	°C

33.1.1 Reserves and Resources

No published data are available on manufacturing processes for boric acid esters or on their annual production rates. Commercial usage is very limited except for the internal production of methyl borate by the Ventron Division of Morton-Thiokol. This ester is a key intermediate in the Ventron process for sodium borohydride. Some additional quantities of methyl borate and its methanol azeotrope are used in welding and brazing applications. Other commercially available boric acid esters include ethyl borate, n-propyl borate, isopropyl borate, n-butyl borate, and cresyl borate. A number of boric acid esters prepared from butylene and hexylene glycols are also available commercially, as well as methyl and isopropyl boroxines. Probably, all of these compounds are prepared by the standard reactions of boric acid with the appropriate alcohols or phenols.

Prices for the commercially available boric acid esters are relatively high because of their limited usage. At higher production volumes these prices would be reduced significantly. Methyl borate was available in 1985 at ca. \$ 3/kg, and the other esters are generally in the range of \$ 3 to \$ 10/kg.

33.1.2 Use

Methyl borate is the boric acid ester of greatest commercial interest because it is used internally for the production of sodium borohydride (and diborane). Methyl borate azeotrope is used as a gaseous flux for brazing and welding, particularly in Europe.

Although numerous patents have been issued on the use of boric acid esters as polymer additives, this application apparently has remained small unless there are proprietary uses that have not been disclosed. Application of small quantities of several derivatives as epoxy resin curing agents has continued with little measurable growth. Despite wide patent coverage, boric acid esters have not been used in any quantity for polymer stabilization. In the 1950s and 1960s boric acid esters derived from glycols were used extensively as gasoline additives. A mixture of these glycol derivatives is being manufactured by U.S. Borax & Chemical Corp. under the trade name Biobor JF for the control of microorganisms in fuel systems containing water. This product can be used successfully in jet fuels, diesel fuels, and fuel oil systems that have water contamination.

Borate esters have also been patented for use as flame retardants for cotton batting, modifiers for the oxidation of hydrocarbons, starting materials for the preparation of trialkylboranes, such as triethylborane and triphenylborane, wood preservatives, hydraulic fluids (particularly alkylated glycol borates), and lubricants.

33.2 Production Technologies (Ullmanns 2004)

Boric acid esters have been prepared in the laboratory by a variety of methods, but the standard approach involves the reaction of boric acid with alcohols or phenols:



This equilibrium reaction, which favors the hydrolysis products boric acid and ROH, is displaced to the right by using excess alcohol and removing water by azeotropic distillation with hydrocarbon solvents. This procedure is complicated in the case of methyl and ethyl borates by the formation of azeotropes between the alcohols and the boric acid esters. A closely related method in which boric oxide is substituted for boric acid has also been used.

The relatively easy transesterification of boric acid esters results in ready displacement of lower boiling alcohols and the preparation of higher molecular mass alkyl and aryl derivatives. A number of less practical laboratory procedures for boric acid esters have been reported, starting with boron halides, trisaminoboranes, and other boron-containing compounds.

33.3 System Characterisation

In this study the esterification of boric acid with methanol is considered. All data in the present report are referred to 1 kg trimethyl borate 100%. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

33.4 Raw materials and auxiliaries

Tab. 33.2 shows values for the use of raw materials of the production of trimethyl borate. As there was no data available for the production of trimethyl borate, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the trimethyl borate production.

Tab. 33.2 Consumption of raw materials and auxiliaries for the trimethyl borate production

Input	Values	Remarks
Boric acid (kg)	0.628	Calculated with a yield of 95 %
Methanol (kg)	0.972	Calculated with a yield of 95 %
Cooling water (kg)	24	Estimation

33.5 Energy

As no data for the production of trimethyl borate are available, the values for the energy use have been estimated with data from Kim 2003, who gives data for the production of ethyl acetate from ethanol esterification. This process is considered to be similar to the esterification of boric acid with methanol.

The value for the steam input used in this inventory is the difference between the steam and the potential recovered energy, because this recovered energy is usually fed into the heating systems.

Tab. 33.3 Energy consumption for the production of 1 kg trimethyl borate

Input	Kim 2003	This study
Steam (MJ)	10.5	8.84
Electricity (kWh)	0.00725	7.25E-03
Potential recovered energy (MJ)	- 1.66	
Sum of consumed energy (MJ)	8.87	

33.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. 2007 are used: 100 km by lorry 32t and 600 km by train. Tab. 33.4 summarises the total transport amounts for the trimethyl borate production.

Tab. 33.4 Total transport amounts for the trimethyl borate production

tkm kg ⁻¹ trimethyl borate	lorry	train
Total transports	1.6E-01	9.6E-01

33.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of ethylene hydration plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

33.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the trimethyl borate production. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2 % of the input of boric acid (calculated as boron emissions) and methanol.

Tab. 33.5 Process emissions to air from the trimethyl borate production

Output	Value	Remarks
Waste heat (MJ)	0.0261	calculated from electricity input
Boron, to air (kg)	2.23E-04	estimated as 0.2 % of the input
Methanol, to air (kg)	1.943E-03	estimated as 0.2 % of the input
CO ₂ , fossil, to air (kg)	5.77E-02	from water treatment

33.9 Emissions to water

The remaining amount of unreacted boric acid and methanol (4.8 % of the input) is assumed to leave the production process with the wastewater. Further it is assumed that this wastewater is treated in a internal wastewater plant. The removal efficiency for methanol is assumed with 90%. The carbon contained in the removed methanol is accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory are calculated from the amount of methanol in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used.

Tab. 33.6 Process emissions to water from the trimethyl borate production

Output	Value	Remarks
Boron (kg)	5.34E-03	calculated from mass balance
Methanol (kg)	4.66E-03	calculated from mass balance
COD, BOD (kg)	6.71E-03	calculated from water emissions
TOC, DOC (kg)	1.75E-03	calculated from water emissions

33.10 Life cycle inventory of trimethyl borate production and data quality considerations

Tab. 33.7 shows the life cycle inventory and the data quality indicators for the trimethyl borate production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 33.7 Unit process raw data for the production of trimethyl borate production

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	trimethyl borate, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							GLO 0 kg			
resource, in water	401	0	trimethyl borate, at plant	GLO	0	kg	1.00E+0			
	662	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	493	-	boric acid, anhydrous, powder, at plant	RER	0	kg	6.28E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	403	-	methanol, at plant	GLO	0	kg	9.72E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
		-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	8.84E+0	1	1.30	(4,5,na,na,na,na); Estimation
		-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	7.25E-3	1	1.35	(3,3,3,1,3,5); Estimation
		-	transport, lorry >16t, fleet average	RER	0	tkm	1.60E-1	1	2.09	(4,5,na,na,na,na); Standard distances
		-	transport, freight, rail	RER	0	tkm	9.60E-1	1	2.09	(4,5,na,na,na,na); Standard distances
		-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
		-								
emission air, high population density		4	Heat, waste	-	-	MJ	2.61E-2	1	1.35	(3,3,3,1,3,5); Calculated from the electricity input
		4	Boron	-	-	kg	2.23E-4	1	5.10	(4,5,na,na,na,na); Estimation
		4	Methanol	-	-	kg	1.94E-3	1	1.62	(4,5,na,na,na,na); Estimation
		4	Carbon dioxide, fossil	-	-	kg	5.77E-2	1	1.30	(4,5,na,na,na,na); Estimation
emission water, river		4	Boron	-	-	kg	5.34E-3	1	5.10	(4,5,na,na,na,na); Estimation
		4	Methanol	-	-	kg	4.66E-3	1	3.09	(4,5,na,na,na,na); Estimation
		4	BOD5, Biological Oxygen Demand	-	-	kg	6.71E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
		4	COD, Chemical Oxygen Demand	-	-	kg	6.71E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
		4	DOC, Dissolved Organic Carbon	-	-	kg	1.75E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
		4	TOC, Total Organic Carbon	-	-	kg	1.75E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

33.11 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 33.8 shows selected LCI results and the cumulative energy demand for the trimethyl borate production.

Tab. 33.8 Selected LCI results and the cumulative energy demand for the trimethyl borate production

Name		trimethyl borate, at plant	
Location	Unit	Unit	GLO kg 0
Infrastructure			
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	55.8
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	2.7
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.7
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.4
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.0
selected LCI results			
land occupation	resource	m2a	1.9E-1
CO ₂ , fossil	air	kg	1.9E+0
NM VOC	air	kg	3.8E-3
nitrogen oxides	air	kg	4.0E-3
sulphur dioxide	air	kg	1.1E-2
particulates, <2.5 um	air	kg	4.8E-4
BOD	water	kg	8.3E-3
cadmium	soil	kg	7.9E-10

33.12 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	trimethyl borate, at plant
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of trimethylborate including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Methylborat, ab Werk
	491	Synonyms	methyl borate//boric acid trimethyl ester// boron methoxide
	492	GeneralComment	The process "trimethylborate, at plant, GLO" is modelled for the production of trimethylborate from boric acid in the world. Raw materials are modelled with a stoichiometric calculation. Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	(CH3O)3B
	501	StatisticalClassification	
	502	CASNumber	121-43-7
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	Methylation of boric acid
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Energy consumptions and emissions are estimated. Infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	46
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

33.13 References

Frischknecht et al. 2007 Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Over-

- view and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.
- Gendorf 2000 Gendorf (2000) Umwelterklärung 2000, Werk Gendorf. Werk Gendorf, Burgkirchen as pdf-File under: <http://www.gendorf.de/pdf/umwelterklaerung2000.pdf>
- Kim 2003 Kim S., Overcash M.: Energy in chemical manufacturing processes: gate-to-gate information for life cycle assessment. In: Journal of Chemical Technology & Biotechnology vol. 78, no. 9: 995-1005(11). 2003
- Ullmanns 2004 Brotherton, R.: Boron Compounds. Published online: 2000. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

34 Water, ultrapure

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hirschier, EMPA, St. Gallen
 Last changes: 2007

34.1 Introduction

Water is an integral part of several “high-tech” manufacturing and processing industries. With severe processing requirements come stringent demands on all the raw materials and manufacturing components, included process water. Ultrapure water (UPW) is essentially a “high-tech” material as well. Consequently, significant technological efforts and expense are devoted to purifying water to severe standards.

Ultrapure water is produced in a series of separation units, the final step processing being primarily mixed-bed ion exchange (MBIE). MBIE is preceded by a range of stages where salts, biological contaminants, organic chemicals (usually collectively referred to as total organic carbon, or TOC), and dissolved gases are removed. These technologies include membrane processes (ultrafiltration, reverse osmosis RO, and pervaporation), adsorption typically on activated carbon for TOC and chlorine removal), disinfection e.g., by UV-light, degasification, or — most usually — a combination of these.

Standards for ultrapure water vary among industries, but UPW is typically defined as having electrolytic conductivity less than 0.1 $\mu\text{S}/\text{cm}$. For comparison, the theoretical conductivity of pure water is 0.054 $\mu\text{S}/\text{cm}$ at 25 °C. To be classified as ultrapure water, the ionic concentration must be less than 20 $\mu\text{g}/\text{L}$ (industry often uses parts per billion, ppb NaCl or CaCO_3 equivalent). The NaCl/ CaCO_3 equivalent of a specific ion is calculated by multiplying the actual concentration by the ratio of the ion equivalent weight and NaCl or CaCO_3 equivalent weights.

The most important chemical properties of ultrapure water and the input tap water used in this inventory are given here.

Tab. 34.1 Properties of ultrapure water and tap water

Component	Ultrapure water (Feodorenko 2004)	Tap water (Primas 2004)
Ca^{2+} (kg/m^3)	4.90E-05	4E-02
Mg^{2+} (kg/m^3)	1.00E-06	2.4E-02
Na^{2+} (kg/m^3)	5.60E-04	2.3E-02
K^+ (kg/m^3)	5.00E-06	3.9E-02
Cl^- (kg/m^3)	7.50E-03	3.5E-02
SO_4^- (kg/m^3)	9.50E-05	9.6E-02
NO_3^- (kg/m^3)	3.70E-05	6.2E-02
PO_4^- (kg/m^3)	4.75E-03	9.5E-02
Silica as SiO_2 (kg/m^3)	2.00E-03	6E-02
HCO_3^- (kg/m^3)	3.05E-03	6.1E-02
Free CO_2 (kg/m^3)	2.20E-03	4.4E-02

34.1.1 Use

Industries that require ultrapure water include:

Power generation

Steam cycle water chemistry is critical to sustained operation of nuclear and coal-fired power plants. Ultrapure water processing occurs in three major areas of the plant production of make-up water, condensate polishing, and reactor water clean-up. Each of these applications has specific requirements for

water quality and uses a range of different processes to achieve these requirements, including deep-bed ion exchange, powdered resin ion exchange, degasification and reverse osmosis.

Specialty materials processing

As the technology in the area of advanced materials becomes more and more complex, the purity of chemicals involved with their production and processing must also increase. Requirements for ultrapure water in the microchip and microelectronics industry are even more severe than for electrical power companies since contaminants, both ionic and particulate materials can effectively destroy microchip surfaces.

Biomedical and life science applications

The primary pharmaceutical applications for ultrapure water processing are: water for injection (WFI) and manufacturing of medicines. The separation requirements for pharmaceuticals are aimed at eliminating pyrogens (low molecular mass organic molecules — typical of membrane decay due to microbial decay that can cause fever. In addition, ultrapure water is used extensively as a diluent for dialysis. Another life science application of ion exchange includes water reuse on space missions.

Boiler feed water purification

Ultrapure water is also required as boiler feed water in chemical plants, such as ethylene and ammonia production. High purity water is also needed by the pulp and paper plants.

34.2 Production Technologies (Ullmann 2005)

Ion Exchange Resins

Ion exchange is required to obtain ultrapure *water* at the most stringent purity conditions, i.e., a resistivity > 18 MΩ/cm. Process schemes that include all other technologies discussed in this article, but exclude ion exchange can only reduce ionic concentrations to the µg/L level (5 – 9 MΩ/cm for reverse osmosis, 16 – 17 MΩ/cm for electrodeionization). As the resins become loaded with ions, they release ions back into solution in accordance with the law of mass action. Expressed for cation exchange resin in the hydrogen form and an arbitrary cation:

$$K_H^i = \left(\frac{C_H}{q_H} \right)^{z_i} \left(\frac{q_i}{C_i} \right)^{z_H}$$

where K_H^i is the selectivity of the resin for ion i with respect to hydrogen, C_H and C_i are the liquid phase concentrations of cations and protons, q_i and q_H are the resin phase concentrations, and z_i and z_H are the respective valence.

The use of ion exchange resins is also a two-edged sword, since at ng/L levels of impurities the resin itself becomes a primary source of contaminants. These contaminants include residual chemicals from resin regeneration — primarily sodium and chloride or sulfate ions — and degradation products from the ion exchange resins themselves — sulfonates from the cationic exchange resins and functional amines from anionic exchange resins. In addition, ion exchange resins are an expected source of low levels of particles and bacteria.

Mixed-Bed Ion Exchangers

A particular advantage of ion exchange resins is that both cationic and anionic resins can be used together to purify water. Cationic resins release hydrogen ions while anionic resins release hydroxide ions. The hydrogen and hydroxide ions combine on into water molecules. Mixed bed ion exchange is used to process boiler feedwater, make-up water, steam-cycle condensate or for final-stage polishing (removal of trace contaminants). With regard to boiler feedwater, make-up water, and applications

with less stringent UWP requirements, fixed-bed demineralizers followed by mixed-bed polishing units may be all the water treatment used.

Mixed-bed polishers are used in high pressure (> 3.55 MPa) feedwater applications, where coordinated phosphate or equilibrium phosphate chemistry is used as the boiler water treatment for pH control to prevent corrosion. In water used in coal-fired and nuclear power plants, microelectronic sites, and pharmaceutical applications a first stage MBIE produces pure water for additional water treatment.

The use of mixed-bed polishers accomplishes several goals. First, they usually reduce the total electrolyte concentration in make-up or boiler feedwater to nearly 0.04 to 0.09 mg/L and silica from 0.01 to 0.05 mg/L. Second, mixed-bed ion exchange reduces sodium and silica leakage that occurs at the beginning and end of each operating cycle to provide a more consistent, uniform feedwater quality. Third, the use of mixed-bed polishing provides protection against breakthrough of the primary demineralizer units. Mixed-bed polishers typically require infrequent regeneration; some units operate for several years without regeneration.

Mixed-bed demineralizers are usually monitored by observing the pH, conductivity, silica and sodium content.

The pH of the *water* exiting any primary mixed-bed demineralizer is typically 6.5 to 7.5, depending on the temperature and *water* chemistry. Whenever the pH drifts from normal operating conditions, the first action is to verify the accuracy of the sensor. If the sensor is functional, ionic breakthrough is occurring and the mixed bed polisher or fixed bed demineralizer should be regenerated.

Similarly, the conductivity of the mixed-bed unit may rise when breakthrough of the fixed-bed units occurs. When an increase of the product *water* conductivity or specific conductance occurs, the bed should be regenerated if needed.

Silica concentration in the effluent is a critical factor with mixed-bed demineralizers that provide boiler feedwater make-up to high pressure boilers because silica can volatilize at low concentrations in boiler *water* when operating at ≥ 10.4 MPa. The volatilized silica then deposits on turbine blades as the pressure is reduced across the turbine. Therefore, on-line monitoring of silica exiting the mixed bed is desirable when the product *water* is used for high-pressure boiler make-up.

Sodium analyzers are used to determine the breakthrough of sodium in mixed-bed demineralizers because sodium is usually the weakest held cation, and an increase in sodium effluent concentration indicates breakthrough is about to occur.

The separation of the cation and anion exchange resin is the most critical step in mixed-bed ion exchange regeneration. If the resins are separated poorly, some cationic beads are exposed to caustic and some anionic beads are exposed to acid. This cross-contamination results in higher equilibrium leakage — the initial baseline effluent concentration — and shorter service life.

The relative terminal settling velocities of the cation and anion exchange resins govern the separation of the resins. The terminal settling velocity of a spherical particle is directly related to its density and diameter, and the resins are manufactured to maximize the difference in settling velocity. Separation of the cation and anion exchange resin is accomplished with a backwash of *water* at a fluid velocity that is lower than the terminal settling velocity of the cation resin which is usually less than that of the anionic resin.

Conventionally manufactured ion exchange resin has a variable particle diameter and is classified by sifting the resin with fine screen meshes. Developments in resin manufacture have led to nearly uniform particle size, which enhances the separation of the resin types. However resins with uniform particle diameter may lead to operational complications. The increased settling velocity difference may lead to resin stratification before being returned to service.

Regeneration of a mixed-bed vessel requires cation and anion resin separation followed by the injection of acid and caustic to the cation and anion resin, respectively. The amount of regenerant chemical is determined by the volume of resin to be regenerated.

Regeneration intervals vary with the influent *water* quality and flow rate, regeneration of resins used in boiler feedwater at an ethylene plant may occur in 12 – 24 h increments, PWR nuclear regeneration cycles may typically be 30 d, while a polish mixed bed in a semiconductor plant may last a year or longer.

Typically two types of regeneration are performed for mixed beds: the in-situ regeneration and the external regeneration.

Homogeneous Beds

Homogeneous beds may contain several different types of resin, such as strong acid cation, weak acid cation, strong base anion, or weak base anion exchange resin. Demineralizers usually consist of cation units followed by anion units. The simplest and most common configuration is a strong acid cation exchange bed followed by a strong base anion exchange bed.

The function of the strong acid cation exchange unit is to remove the soluble cations (Ca^{2+} , Mg^{2+} , K^{+} , Na^{+} , Ba^{2+} , Sr^{2+}) from the feedwater stream. Divalent cations are more strongly held than monovalent cations. Na^{+} is often the first cation to leak through the strong acid cation exchanger because it is the weakest bonded species. As the cations are removed from the water, the resin releases hydrogen ions.

The strong base anion unit removes anions such as sulfate, chloride, alkalinity, and silica from the water and releases hydroxide ions into the water. Alkalinity is a measure of the buffering capacity of water. It is often associated with carbonic acid, but may include weak electrolytes. Sometimes it is defined as the concentration of OH^{-} ions. Silica is the weakest held anion, and it is usually the first one to leak from a strong base anion unit.

Two-bed systems with strong acid cation and strong base anion exchange resins typically generate water with a total electrolyte concentration of 2 to 3 mg/L, and a silica concentration of approximately 0.02 to 0.1 mg/L.

Demineralizers produce their best water quality when they are operated continuously, because the ions stratify in the resin bed depending on their respective valences and the corresponding selectivities during first of the resin. When demineralizers are shut down and subsequently restarted, the demineralized water eluting first usually contains an increased amount of impurities because the ions leak from the resin to the water and gradually diffuse throughout the length of the bed. Therefore, an additional rinse step is required before the demineralizer is put back into service.

Demineralizer trains consisting of strong acid cation exchange beds followed by strong base anion exchange beds are usually regenerated simultaneously and separately.

First, the anion resin is backwashed using cation product water. The backwash removes fines and any particulate material that has been filtered by the cation unit and settled on the top of the anion resin. This backwash step minimizes channeling that can cause short cation run lengths. In addition, the bed is re-classified and fines (e.g., fractured resin beads) are removed. Second, the anion unit is regenerated using 4 % aqueous sodium hydroxide solution. The dilution water must not contain Mg^{2+} or Ca^{2+} to prevent precipitation of these cations in the resin bed induced by the high pH.

The exchange capacity increases with increasing sodium hydroxide dosage and exposure time (or flow rate). Once the specified amount of sodium hydroxide is added to the dilution water, one to two bed volumes of dilution water are passed through the bed at the same flow rate. This is called the slow rinse and its purpose is to allow the entire regenerant dosage to pass through the bed at the designed flow rate. Next, the anion unit is fast rinsed at about 3.7 m/h. The purpose of the fast rinse is to remove the excess sodium hydroxide from the anion resin.

The most common acids used for regeneration of strong acid cation resin are sulfuric and hydrochloric acid. When sulfuric acid is employed as the regenerant, it is important to use a stepwise regenerant dosage, especially in water that contains a significant concentration of calcium ions to prevent calcium sulfate precipitation in the resin bed. When hydrochloric acid is used as the regenerant, a 10 % concentration is often maintained throughout the regeneration, precipitation of calcium salts not occur.

Membranes

Several membrane technologies are used in ultrapure water processing; reverse osmosis, nanofiltration, and ultrafiltration. The choice of membrane type depends on inlet water quality and required throughput. In general, the larger pore size ultrafiltration is used upstream in the process while RO is used to further treat processed water. Ultrafiltration helps minimize fouling of the more restrictive RO pores by not exposing them to a higher level of contaminants. If the feedwater does not produce excessive fouling of the RO, it is possible to use RO as an initial purification stage.

Nonporous membranes are available for specialized applications, such as pervaporation. Water dissolves into the membrane structure and passes through by concentration flux. A vacuum on the permeate stream generates the concentration gradient.

Reverse Osmosis

Several pressures and flow rates are normally monitored in a reverse osmosis system. In addition to these pressures and temperatures, the feed *water* temperature, the silt density index (SDI) and product and reject conductivities are monitored.

Perhaps the three most useful parameters for monitoring reverse osmosis operation are the normalized flow, percent salt rejection, and pressure drop (either the pressure drop across the individual element or that across the pressure vessel).

Normalized flow is used to monitor flow performance while correcting for change in *water* temperature. More *water* can pass through a reverse osmosis membrane when the *water* temperature is elevated because the pore size increases with increasing temperature.

Salt rejection is defined as the salt concentration in the feedwater minus the permeate concentration divided by the concentration in the feedwater. Salt rejection is usually expressed as a percentage.

The pressure drop per vessel should be limited to 500 kPa, and the pressure drop per element to 200 kPa. Excessive pressure drop can cause the membrane to compact and cause a permanent loss of flow. Periodic cleaning of the reverse osmosis membranes, or elements, should be done when the pressure drop begins to approach the limitations.

Reverse osmosis systems require cleaning when the normalized flow is reduced by 15 % or when the pressure drop per vessel increases to near the manufacturer's limits. This impaired performance can be due to particulate fouling, scale, organic fouling, and microbiological contamination.

Particulate fouling can occur when the pretreatment system does not remove the suspended solids present in the feedwater to the RO unit. Examples of particulate fouling include silt, particulate iron, and other corrosion products.

Scale occurs when salts exceed their solubility limit. This can occur as a result of many causes, including malfunction of pH control equipment a too high percent recovery of the RO unit, a too low addition or loss of antiscalant, or use of the wrong antiscalant.

Organic fouling occurs on RO membranes because the membrane surfaces are organic materials and have an affinity for other organic materials. Examples of common organic foulants include tannic acid and humic acid. Microbiological contamination can be either the slime of living organisms or the remains of dead organisms.

Ultrafiltration and Nanofiltration

Ultrafiltration and nanofiltration are similar in some respects to RO. The pore size of ultrafiltration and nanofiltration membranes is much larger, and they allow much higher flow rates. Both processes can be used effectively with UPW applications as preliminary separation equipment. Nanofiltration can remove larger molecule, particulate, and divalent species: all but simple elemental ions and low molecular mass organics. The cutoff size is about 0.001 μm , compared to 0.01 μm for ultrafiltration and 0.0001 μm for RO. Both technologies can play a significant role in upstream processing by reducing the load on downstream equipment. Nanofiltration may reduce the hardness in the feedwater supply.

Pressure drop is the primary monitoring variable for adequate functioning of membranes. Increasing pressure drop is a sign that the membrane is accumulating mass on the surface. In some cases downstream *water* samples are filtered and examined by microscope. The presence of fibers is an indication of potential membrane failure. Exposure to ppm levels of oxidants, such as chlorine, can degrade cellulose acetate membranes to the point that they fail to meet UWP standards.

Electrodeionization

Electrodeionization (EDI) is a relatively new separation technology that has found commercial application since the mid 1980s. EDI combines mixed-bed ion exchange with electrochemistry to split a feed stream into a product and concentrate stream.

The EDI unit works by placing a mixed-bed ion exchange resin between two special membranes that allow the migration of only cations or anions. Several mixed bed resin/membrane layers are placed between an anode and cathode, a direct current is applied between both electrodes. The influent is then passed through the mixed bed, where the cations migrate to the cathode through the cation membrane into the cation concentrate stream, while the anions migrate to the anode through the anion membrane into the anion concentrate stream. The concentrates are collected and led to the discharge effluent. *Water* is also electrolytically spit into H^+ and OH^- ions, thus providing a simultaneous regeneration mechanism for the cation and anion resin that does not require chemicals.

Electrodeionization units have several advantages; such as continuous operation without regeneration, no regeneration chemicals, and relatively small power requirements. Disadvantages of EDI units were initially high capital costs, mechanical/electrical complications, low flow rates, and susceptibility to fouling. Current EDI technology has equalized the capital costs to that of a mixed-bed, and flow capacity has been increased to be comparable to mixed-bed ion exchange columns for some applications.

34.3 System Characterisation

In this study the production of ultrapure water by electrodeionization is considered. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

34.4 Raw materials and auxiliaries

Tab. 34.2 shows values for the use of raw materials of the ultrapure water production process. As in the electrodeionization no regeneration chemicals are needed the only raw material considered is de-ionised water.

Tab. 34.2 Consumption of raw materials for the ultrapure water production

Input	This study (per kg ultrapure water)	Remarks
Tap Water (kg)	1.0056	calculated from the output and the emissions

34.5 Energy

In Fedore (2003) several values for energy consumption for the ultrapure water production are an given for electrodeionisation process. For this study an average value is used.

Tab. 34.3 Energy consumption for the ultrapure water production

Input	Fedore 2003 (per m ³ water)	This study (per kg water)
Electricity (kWh)	0.2-0.8	5E-04

34.6 Transportation

As the tap water is not transported, no transports are considered.

34.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of ultrapure water production plants. Therefore, in this study, the infrastructure is estimated based on the module "water treatment plant, deionisation, CH ". For this study, the estimated value is 9 E-11 units per kg water.

34.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air. No other emissions to air are considered. The removed carbon is calculated as CO₂ emission to air.

Tab. 34.4 Process emissions to air from the ultrapure water production

Output	Value	Remarks
Waste heat (MJ)	1.8E-03	calculated from electricity input
CO ₂ , to air (kg)	8.36E-05	calculated from removed carbon

34.9 Emissions to water

The removed ions are calculated as emissions to water.

Tab. 34.5 Process emissions to air from the ultrapure water production

Output	Value
Ca ²⁺ (kg/kg)	4E-05
Mg ²⁺ (kg/kg)	2.4E-05
Na ²⁺ (kg/kg)	2.24E-05
K ⁺ (kg/kg)	3.9E-05
Cl ⁻ (kg/kg)	2.75E-05
SO ₄ ⁻ (kg/kg)	9.59E-05
NO ₃ ⁻ (kg/kg)	6.2E-05
PO ₄ ⁻ (kg/kg)	9.03E-05
Si (kg/kg)	2.71E-05

34.10 Life cycle inventory of ultrapure water production and data quality considerations

Tab. 34.6 shows the life cycle inventory and the data quality indicators for the ultrapure water production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 34.6 Unit process raw data for the ultrapure water production

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	water, ultrapure, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
401										
662			Location				GLO			
493			InfrastructureProcess				0			
403			Unit				kg			
emission air, high population density emission water, river	-	0	water, ultrapure, at plant	GLO	0	kg	1.00E+0			
	5	-	tap water, at user	RER	0	kg	1.00E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	electricity, medium voltage, production	UCTE	0	kWh	5.00E-4	1	1.22	(2,2,1,3,3,3); Feodore 2003
	5	-	UCTE, at grid							
	5	-	water treatment plant, deionisation	CH	1	unit	9.00E-11	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Heat, waste	-	-	MJ	1.80E-3	1	1.22	(2,2,1,3,3,3); Calculated from the electricity input
	-	4	Carbon dioxide, fossil	-	-	kg	8.36E-5	1	1.22	(2,2,1,3,3,3); Calculated from removed carbon
	-	4	Calcium, ion	-	-	kg	4.00E-5	1	1.30	(4,5,na,na,na,na); Calculated from the removed Ca
	-	4	Magnesium	-	-	kg	2.40E-5	1	1.30	(4,5,na,na,na,na); Calculated from the removed Mg
	-	4	Sodium, ion	-	-	kg	2.24E-5	1	1.30	(4,5,na,na,na,na); Calculated from the removed Na
	-	4	Potassium, ion	-	-	kg	3.90E-5	1	1.30	(4,5,na,na,na,na); Calculated from the removed K
	-	4	Chloride	-	-	kg	2.75E-5	1	1.30	(4,5,na,na,na,na); Calculated from the removed Cl
	-	4	Sulfate	-	-	kg	9.59E-5	1	1.30	(4,5,na,na,na,na); Calculated from the removed SO4
	-	4	Nitrate	-	-	kg	6.20E-5	1	1.30	(4,5,na,na,na,na); Calculated from the removed NO3
	-	4	Phosphate	-	-	kg	9.03E-5	1	1.30	(4,5,na,na,na,na); Calculated from the removed PO4
	-	4	Silicon	-	-	kg	2.71E-5	1	1.30	(4,5,na,na,na,na); Calculated from the removed Si2

34.11 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 34.7 shows selected LCI results and the cumulative energy demand for the ultrapure water production.

Tab. 34.7 Selected LCI results and the cumulative energy demand for the ultrapure water production

Name		water, ultrapure, at plant	
Location	Unit	Unit	GLO kg 0
Infrastructure			
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	0.0
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	0.0
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.0
selected LCI results			
land occupation	resource	m2a	4.6E-5
CO2, fossil	air	kg	6.4E-4
NM VOC	air	kg	1.9E-7
nitrogen oxides	air	kg	1.1E-6
sulphur dioxide	air	kg	1.8E-6
particulates, <2.5 um	air	kg	1.6E-7
BOD	water	kg	3.1E-7
cadmium	soil	kg	2.4E-13

34.12 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	water, ultrapure, at plant
Geography	662	Location	GLO
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of ultrapure water including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Wasser, hochrein, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "water, ultrapure, at plant, GLO" is modelled for the production of ultrapure water in the world. Raw materials are modelled with a stoichiometric calculation. Emissions are calculated from the input properties. Infrastructure is estimated. Energy consumptions are calculated with literature data.
	494	InfrastructureIncluded	1
	495	Category	water supply
	496	SubCategory	production
	497	LocalCategory	Wasserversorgung
	498	LocalSubCategory	Bereitstellung
	499	Formula	H2O
	501	StatisticalClassification	
	502	CASNumber	7732-18-5
TimePeriod	601	StartDate	2000
	602	EndDate	2007
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	water electrodeionization
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Infrastructure is estimated.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

34.13 References

Fedorenko 2003

Fedorenko Viktor I.: Ultrapure water production using continuous electrodeionisation. In: Pharmaceutical Chemistry Journal Vol. 37, No. 3, 2003

Frischknecht et al. 2007

Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.

- Primas 2004 Primas, Alex: Water, deionized. In: Althaus H.-J., Chudacoff M., Hirsch R., Jungbluth N., Osses M. and Primas A. (2004) Life Cycle Inventories of Chemicals. Final report ecoinvent 2000 No. 8. EMPA Dübendorf, Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: www.ecoinvent.ch.
- Ullmanns 2004 Hussey, D., Foutch, G., Ward, M.: Water, ultrapure. Published online: 2002. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebaum D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>